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(54) **METHODS AND COMPOSITIONS RELATED  
TO THE NEXT GENERATION VACCINE**

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**C07K 14/245** (2006.01)

(52) **U.S. Cl.**

CPC ..... **C07K 14/28** (2013.01); **A61K 39/104**

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**14/21** (2013.01); **C07K 14/235** (2013.01);

**C07K 14/245** (2013.01); **C07K 2319/00**

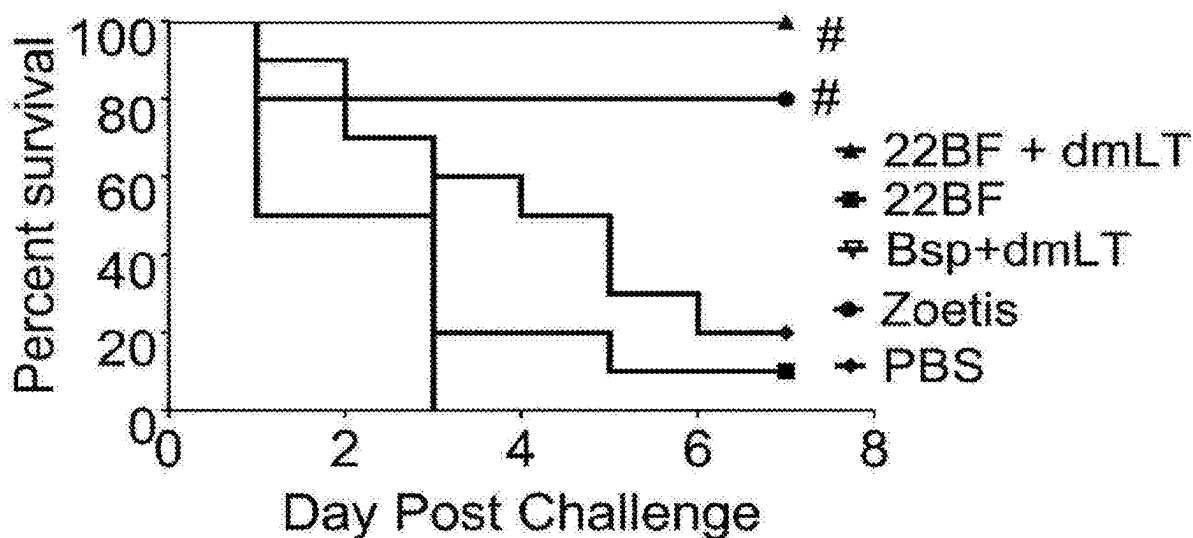
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**ABSTRACT**

Disclosed are compositions comprising a Gram negative  
needle tip protein and a translocator protein and methods of  
their use.

**Specification includes a Sequence Listing.**



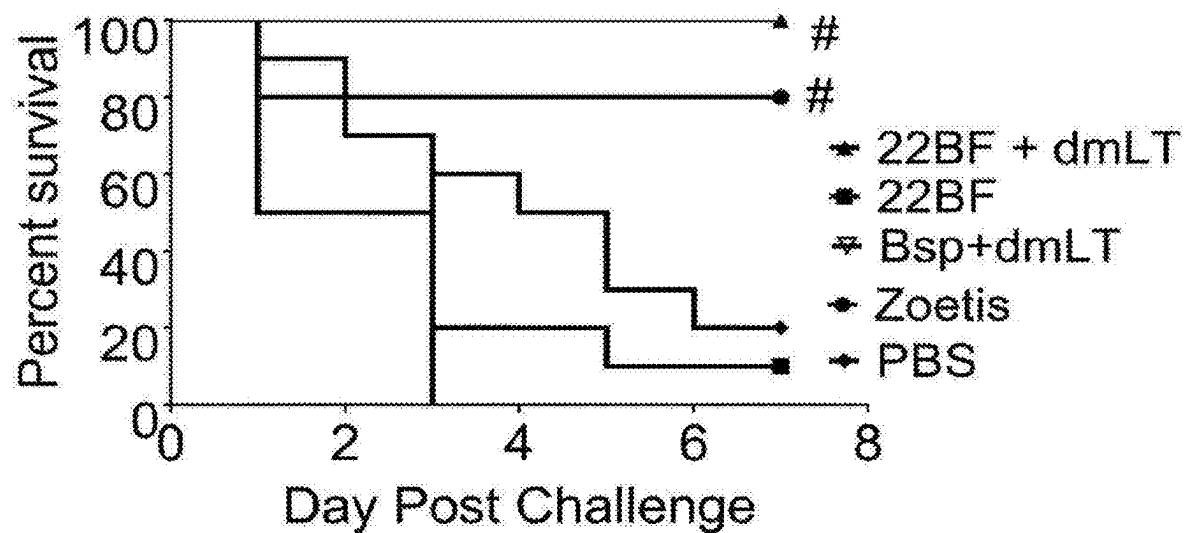


FIG. 1

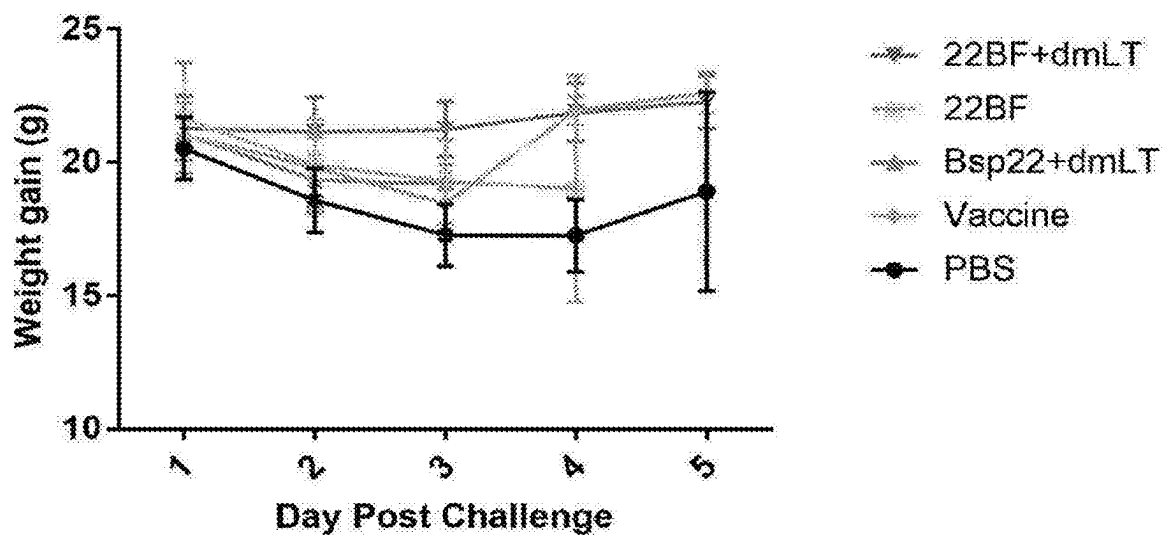


FIG. 2

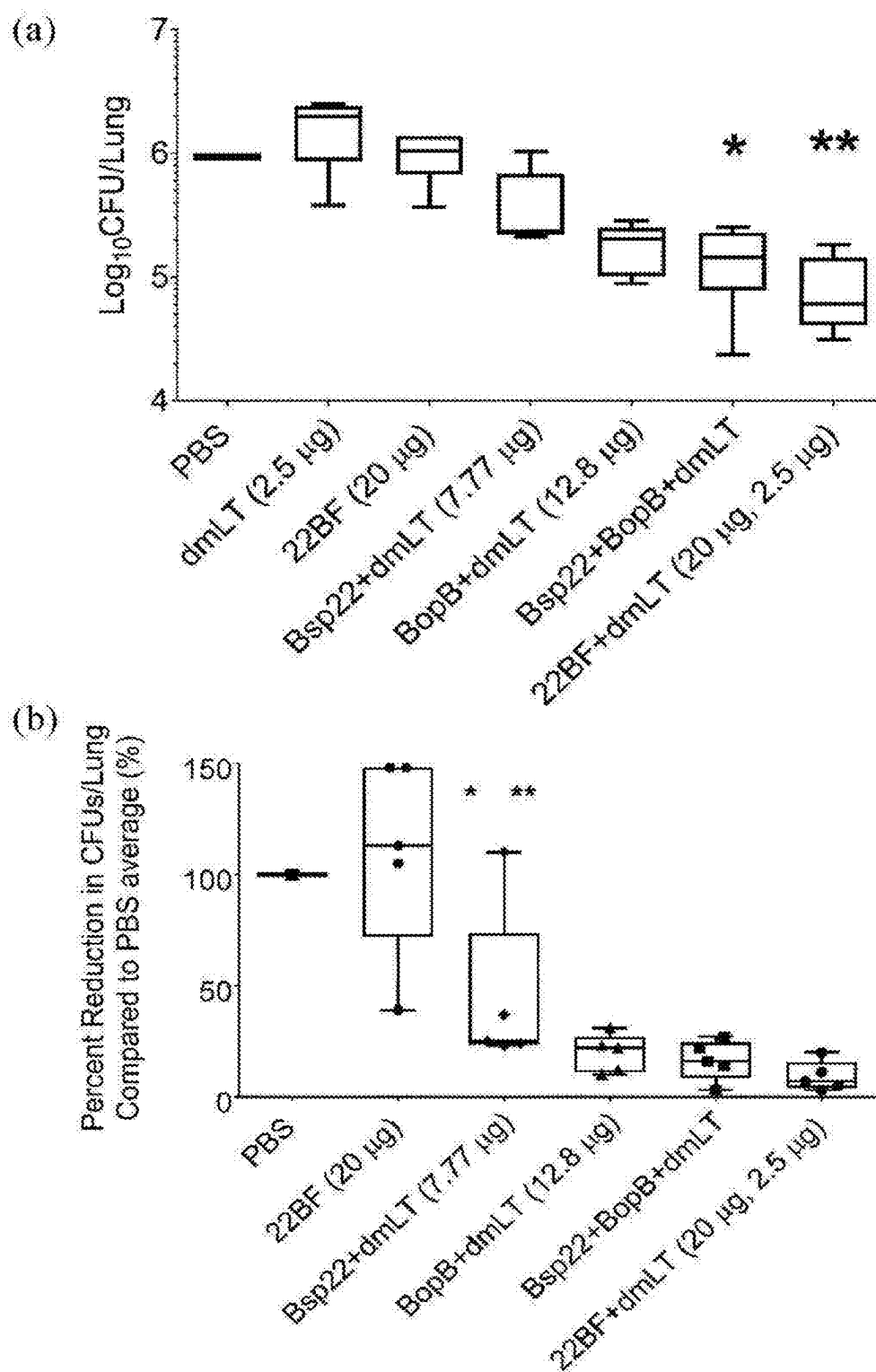
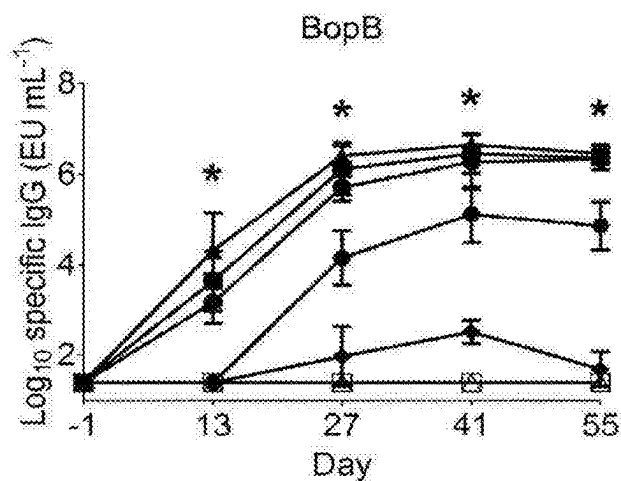
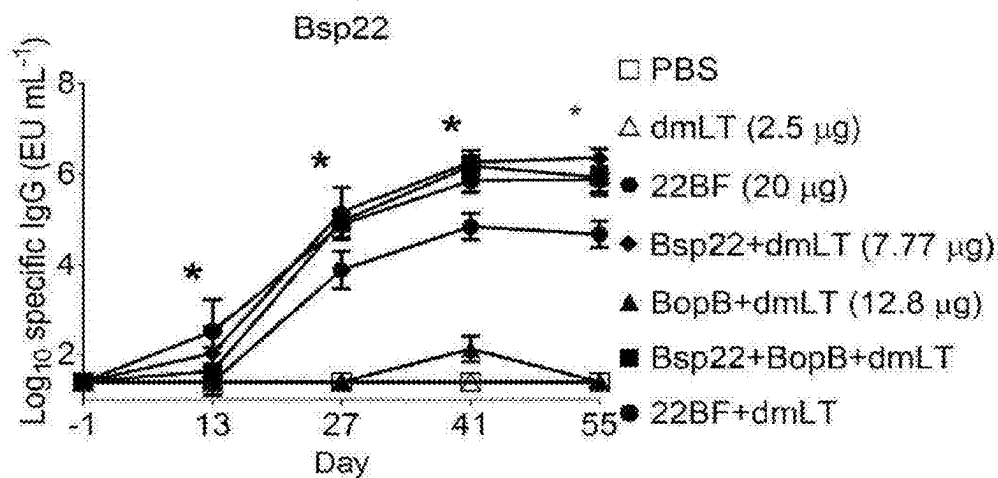


FIG. 3A and FIG. 3B

(a)



(b)



(c)

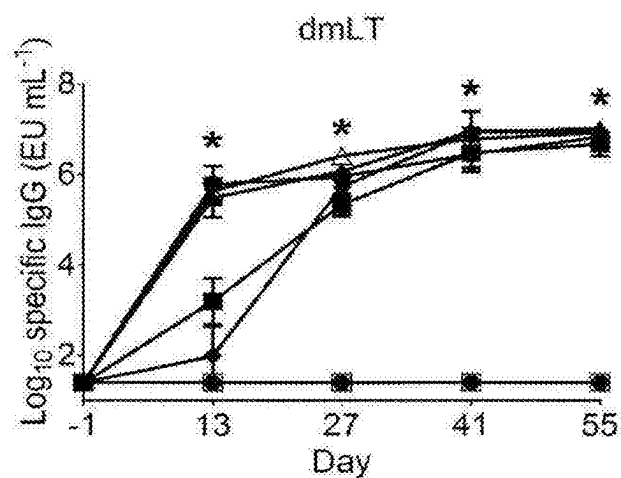


FIG. 4A, FIG. 4B, and FIG. 4C



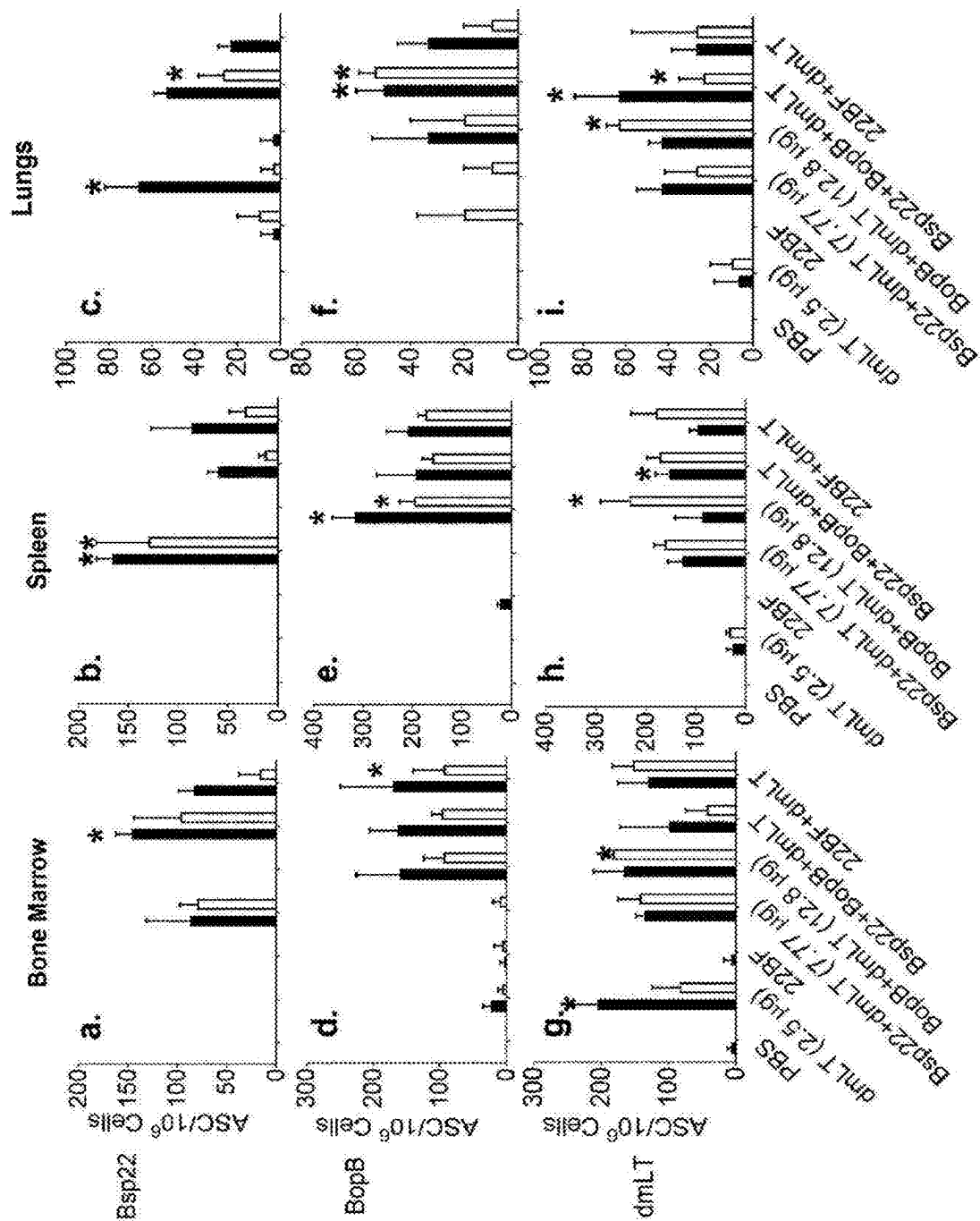


FIG. 5A, FIG. 5B, FIG. 5C, FIG. 5D, FIG. 5E, FIG. 5F, FIG. 5G, FIG. 5H, and FIG. 5I

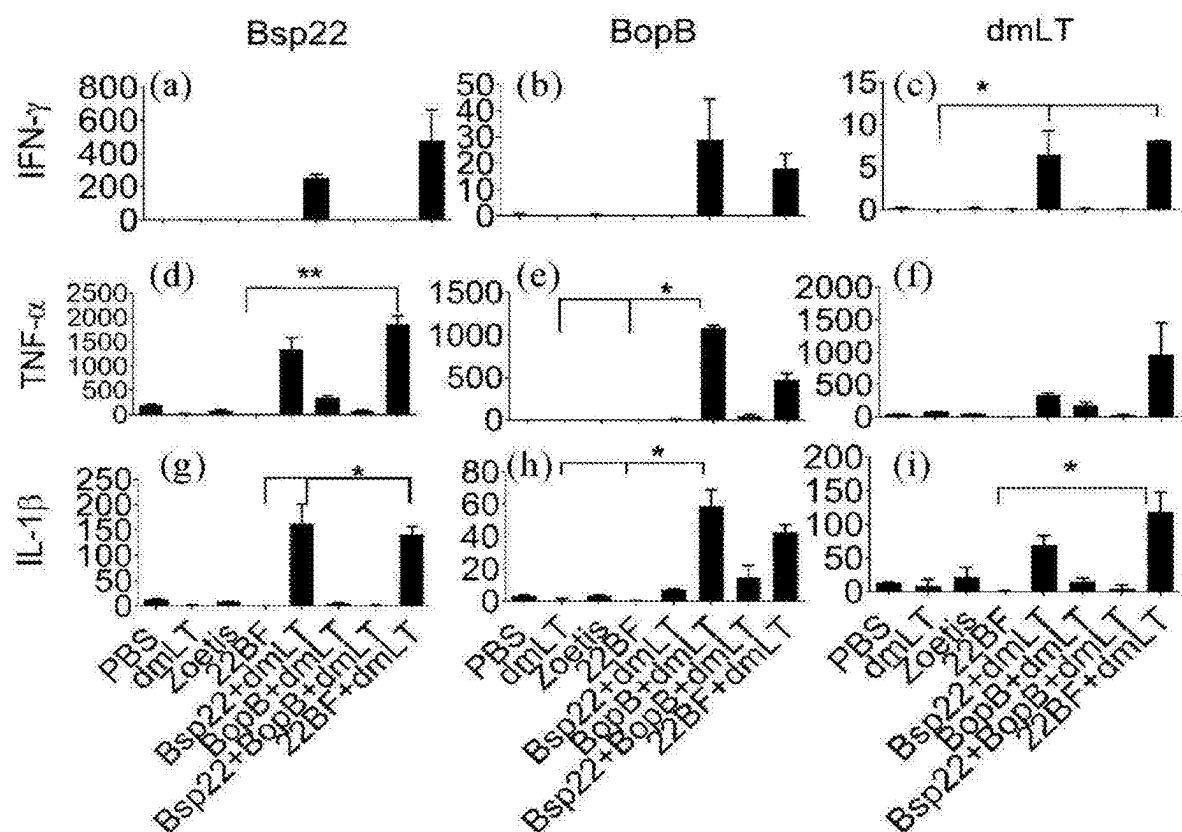


FIG. 6A, FIG. 6B, FIG. 6C, FIG. 6D, FIG. 6E, FIG. 6F, FIG. 6G,  
FIG. 6H, and FIG. 6I

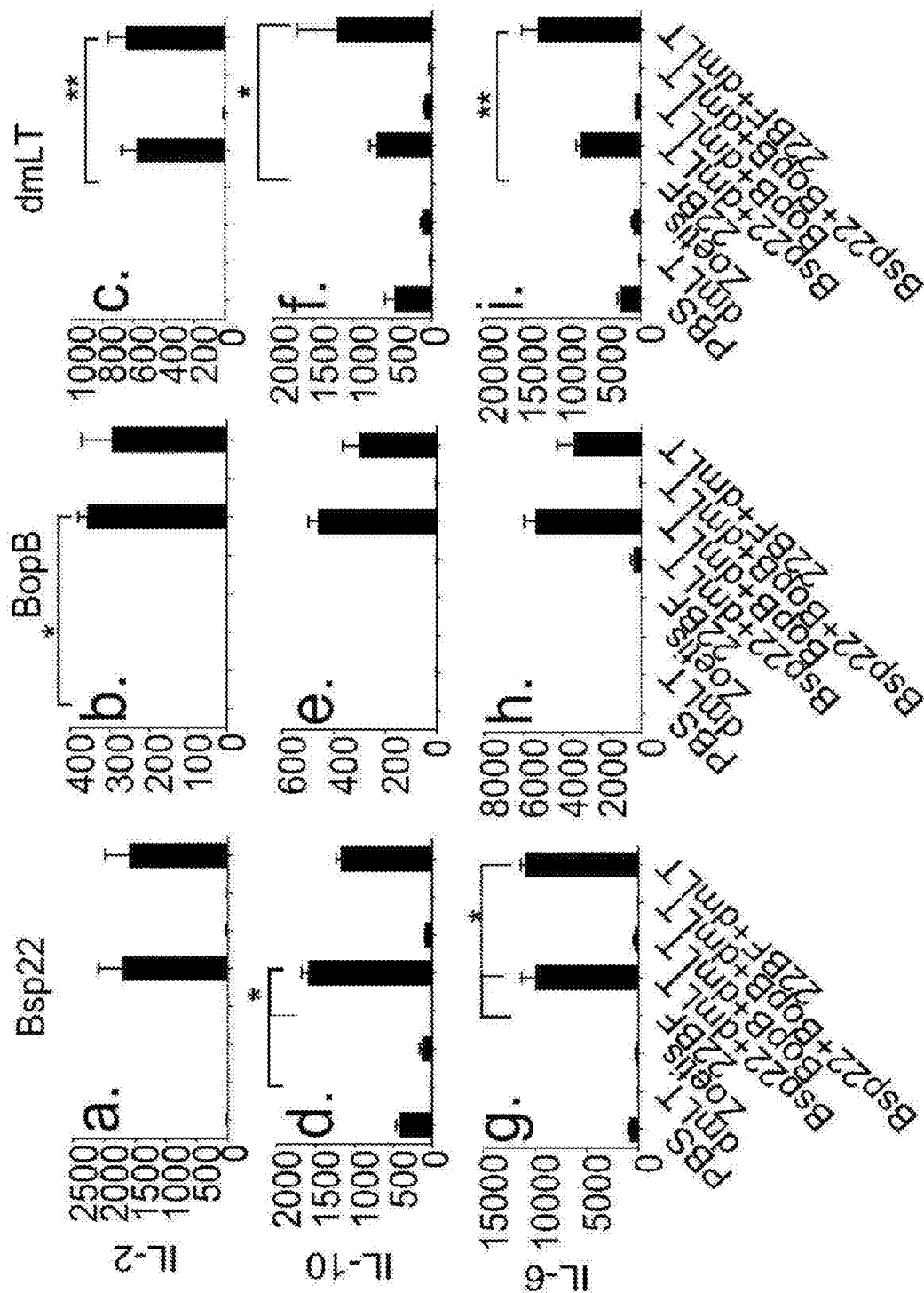


FIG. 7A, FIG. 7B, FIG. 7C, FIG. 7D, FIG. 7E, FIG. 7F, FIG. 7G, FIG. 7H, and FIG. 7I

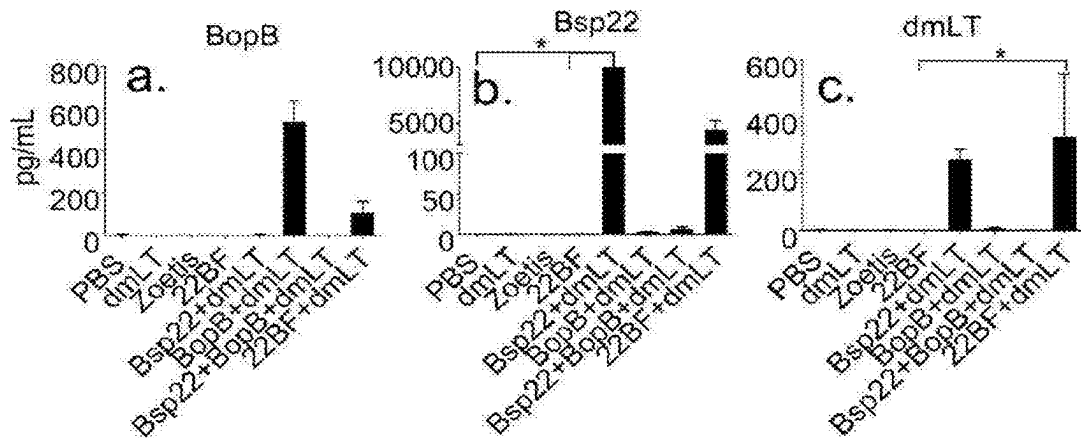


FIG. 8A, FIG. 8B, and FIG. 8C

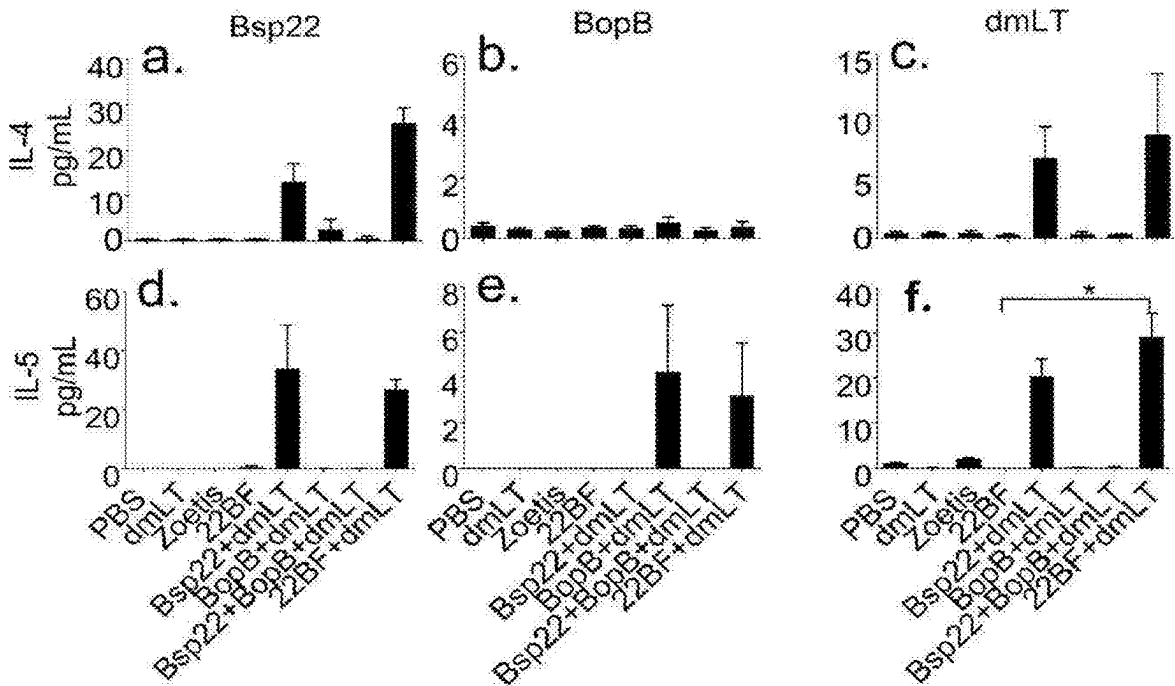


FIG. 9A, FIG. 9B, FIG. 9C, FIG. 9D, FIG. 9E, and FIG. 9F

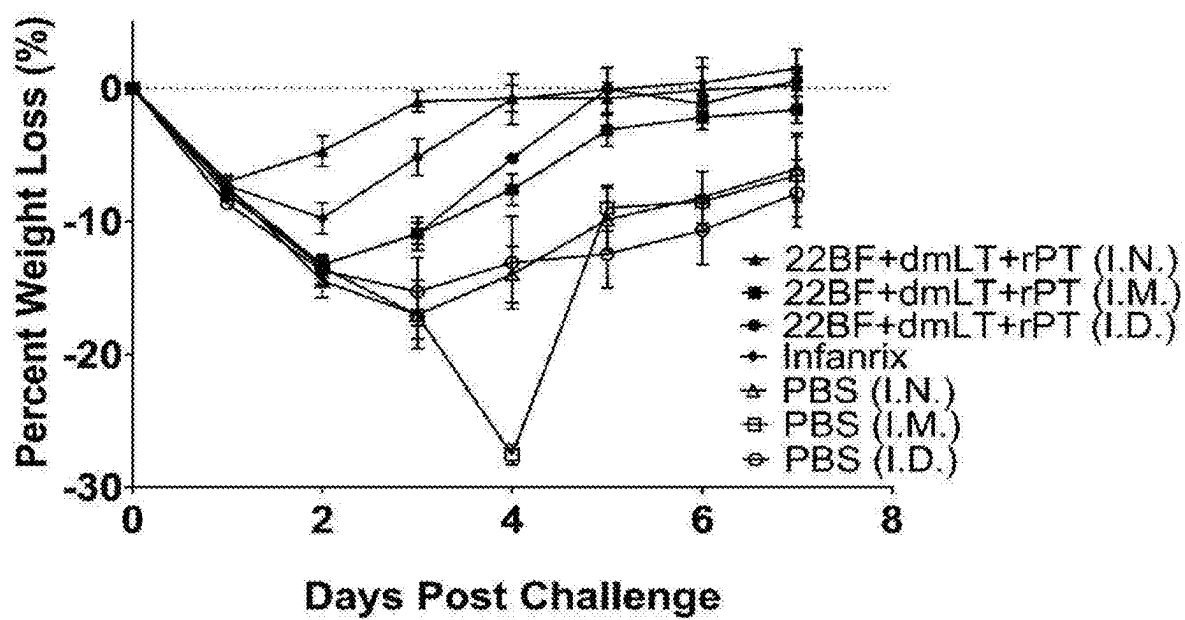


FIG. 10

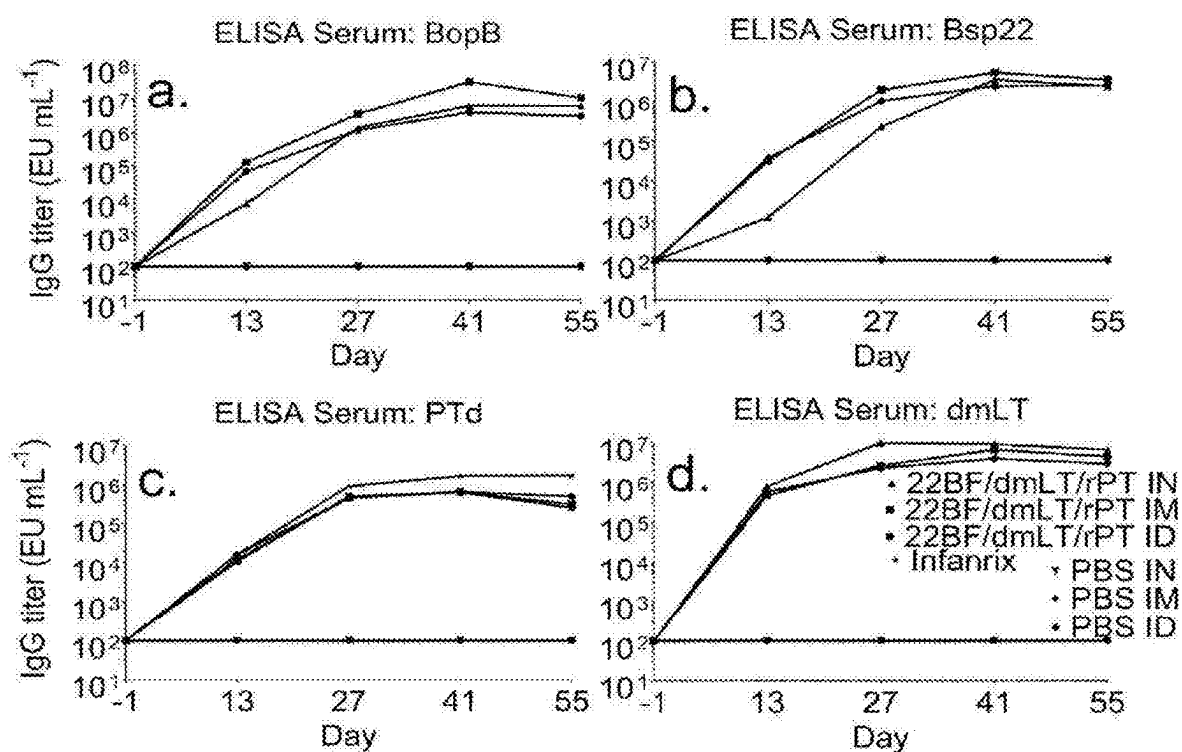


FIG. 11A, FIG. 11B, FIG. 11C, and FIG. 11D

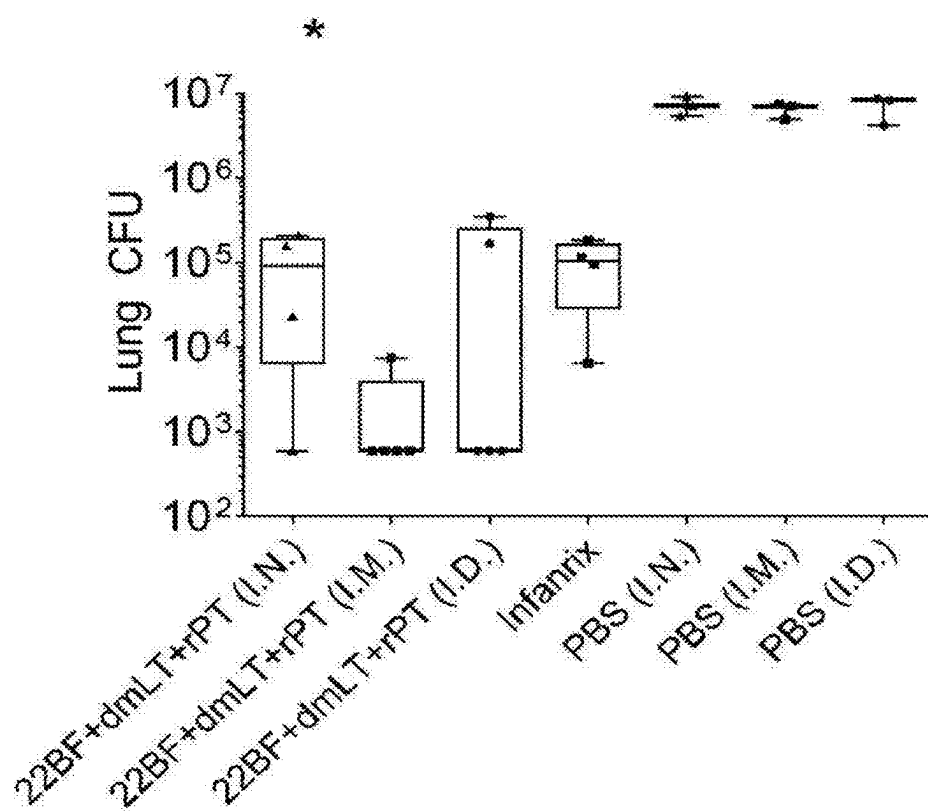


FIG. 12

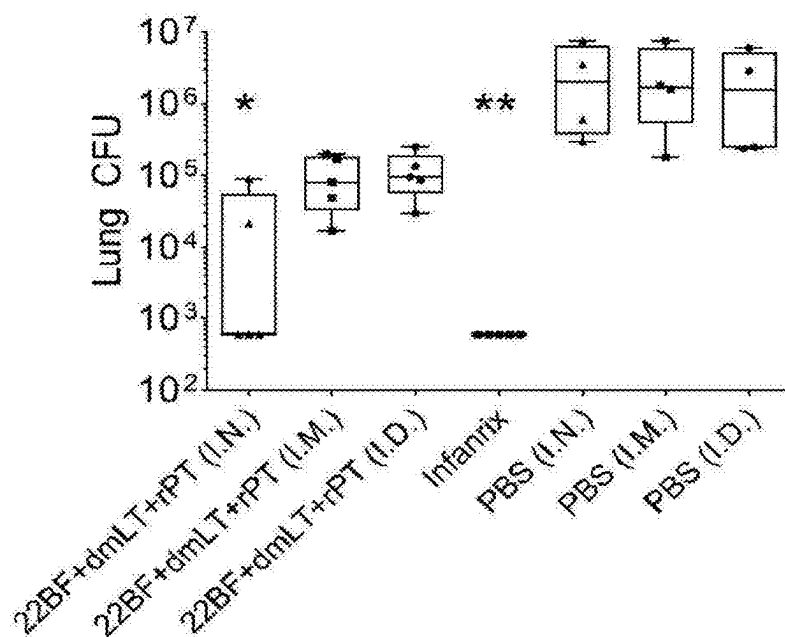


FIG. 13

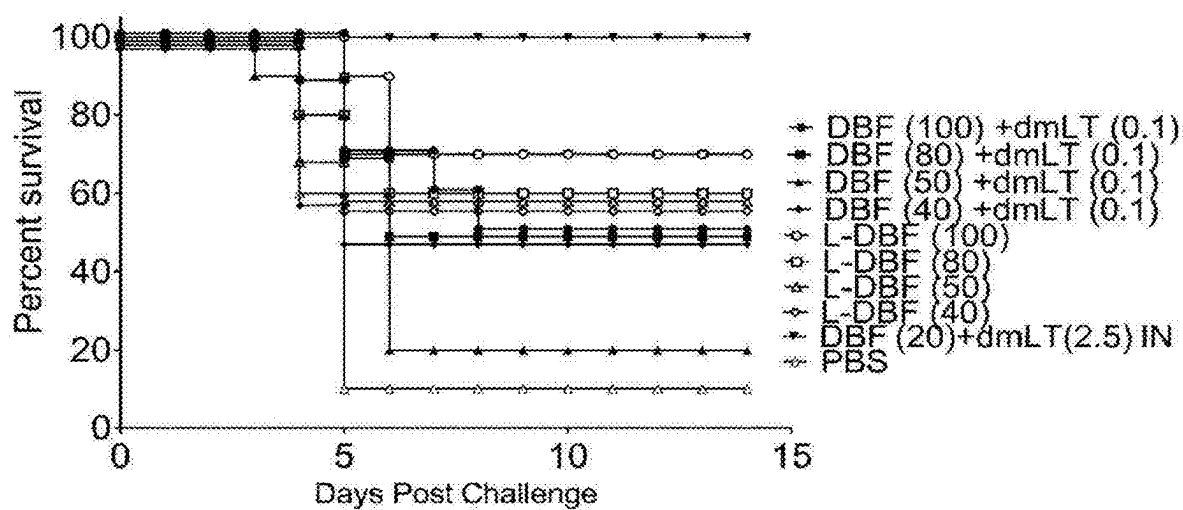


FIG. 14



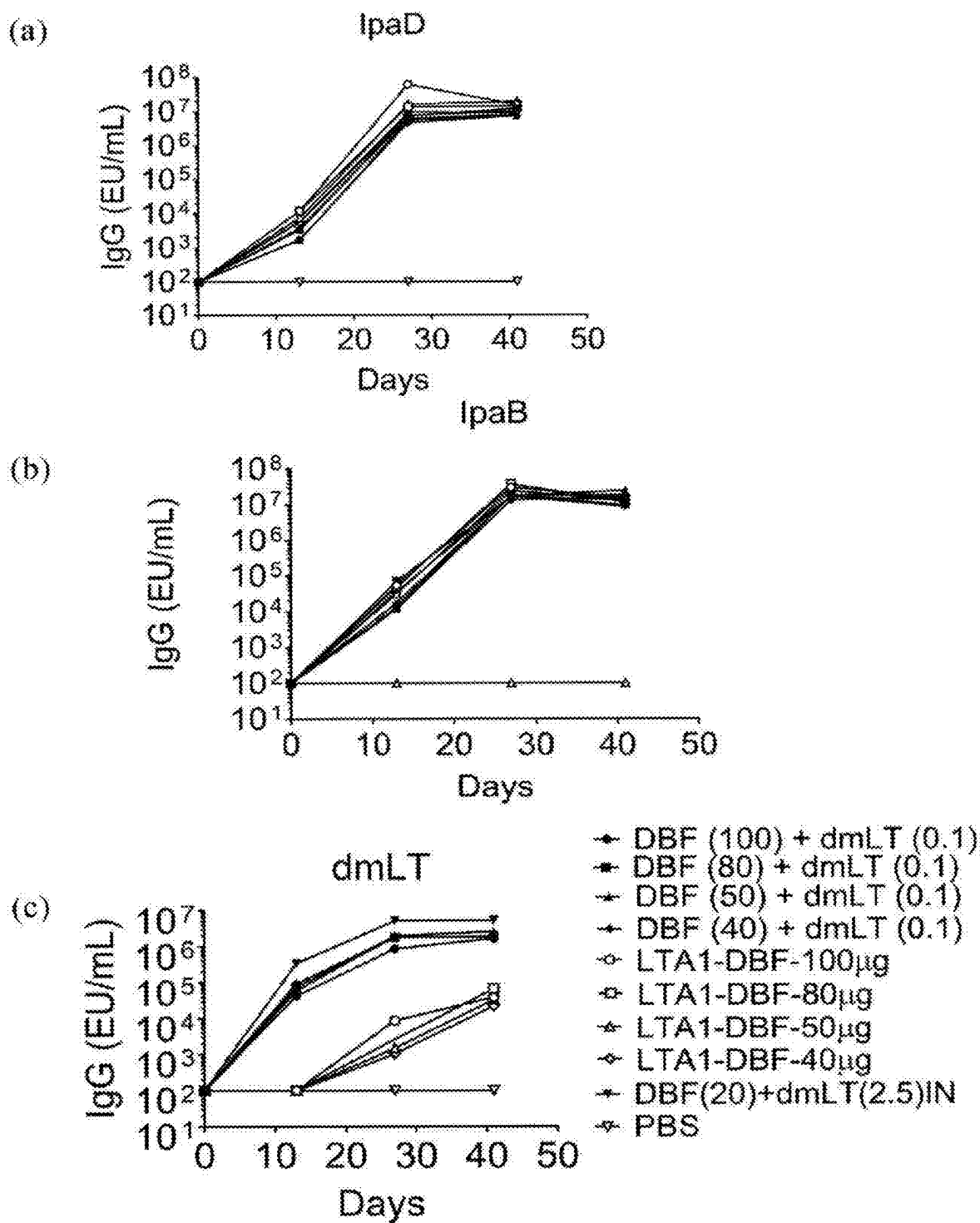


FIG. 15A, FIG. 15B, and FIG. 15C

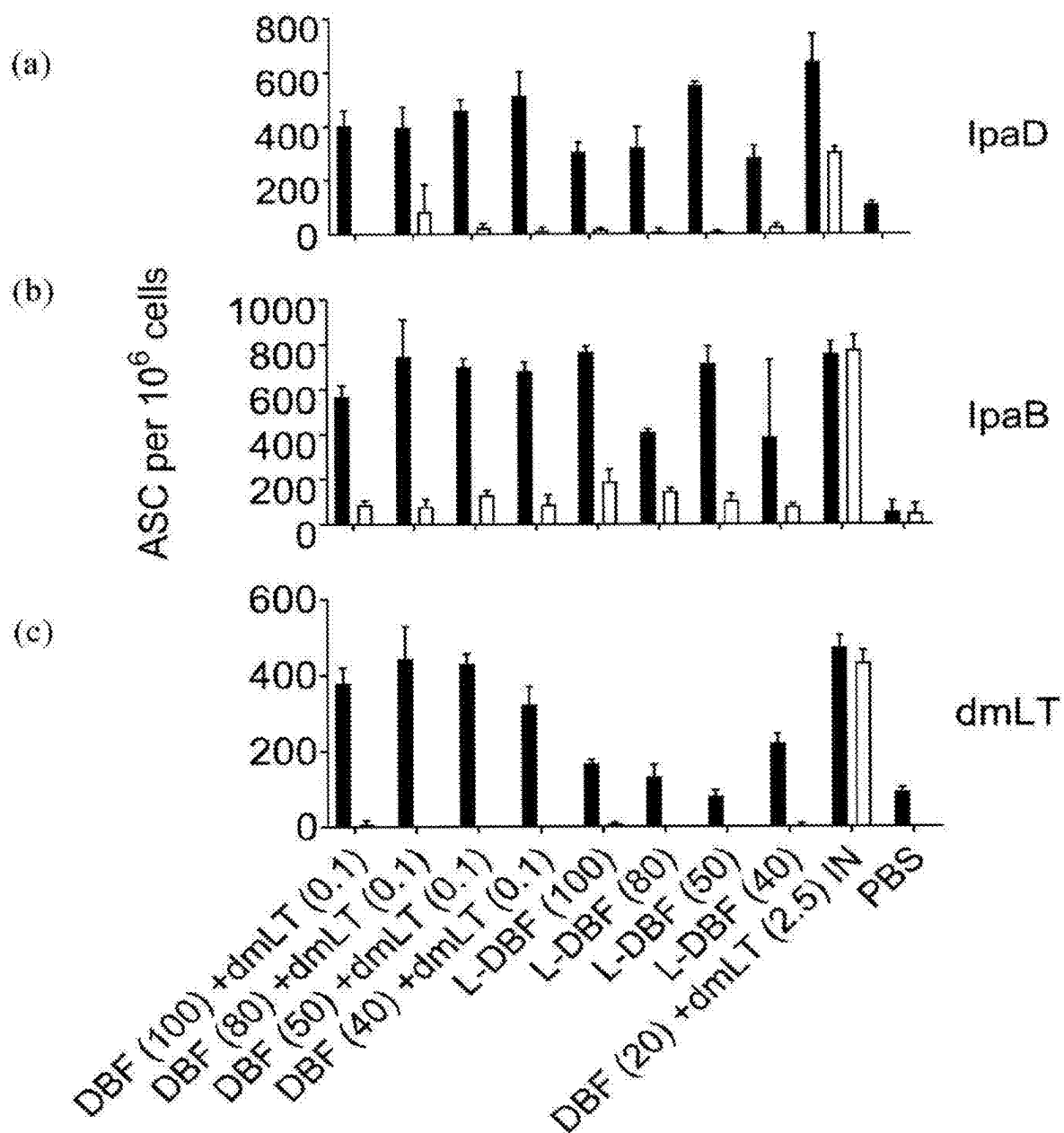


FIG. 16A, FIG. 16B, and FIG. 16C

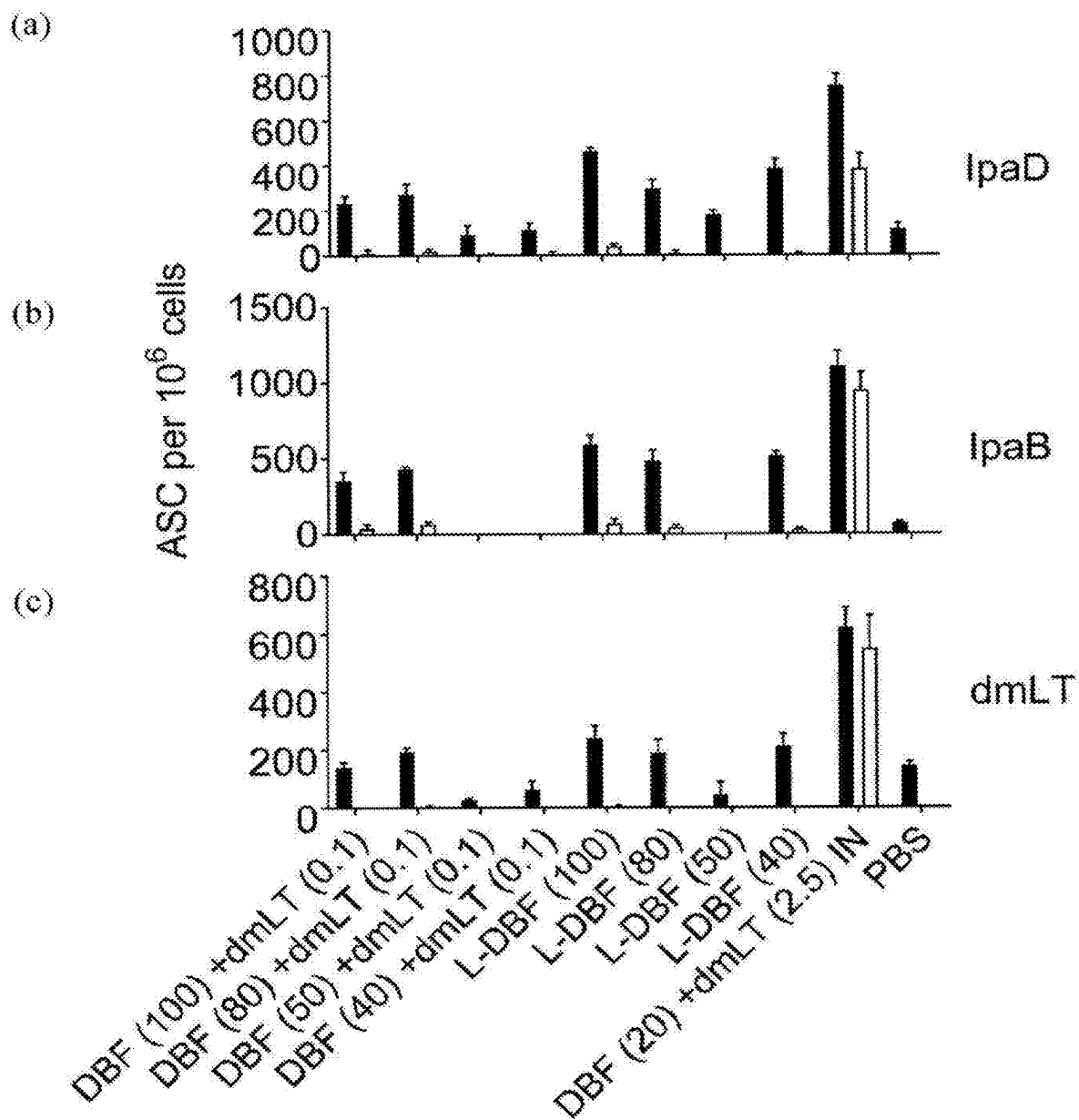


FIG. 17A, FIG. 17B, and FIG. 17C

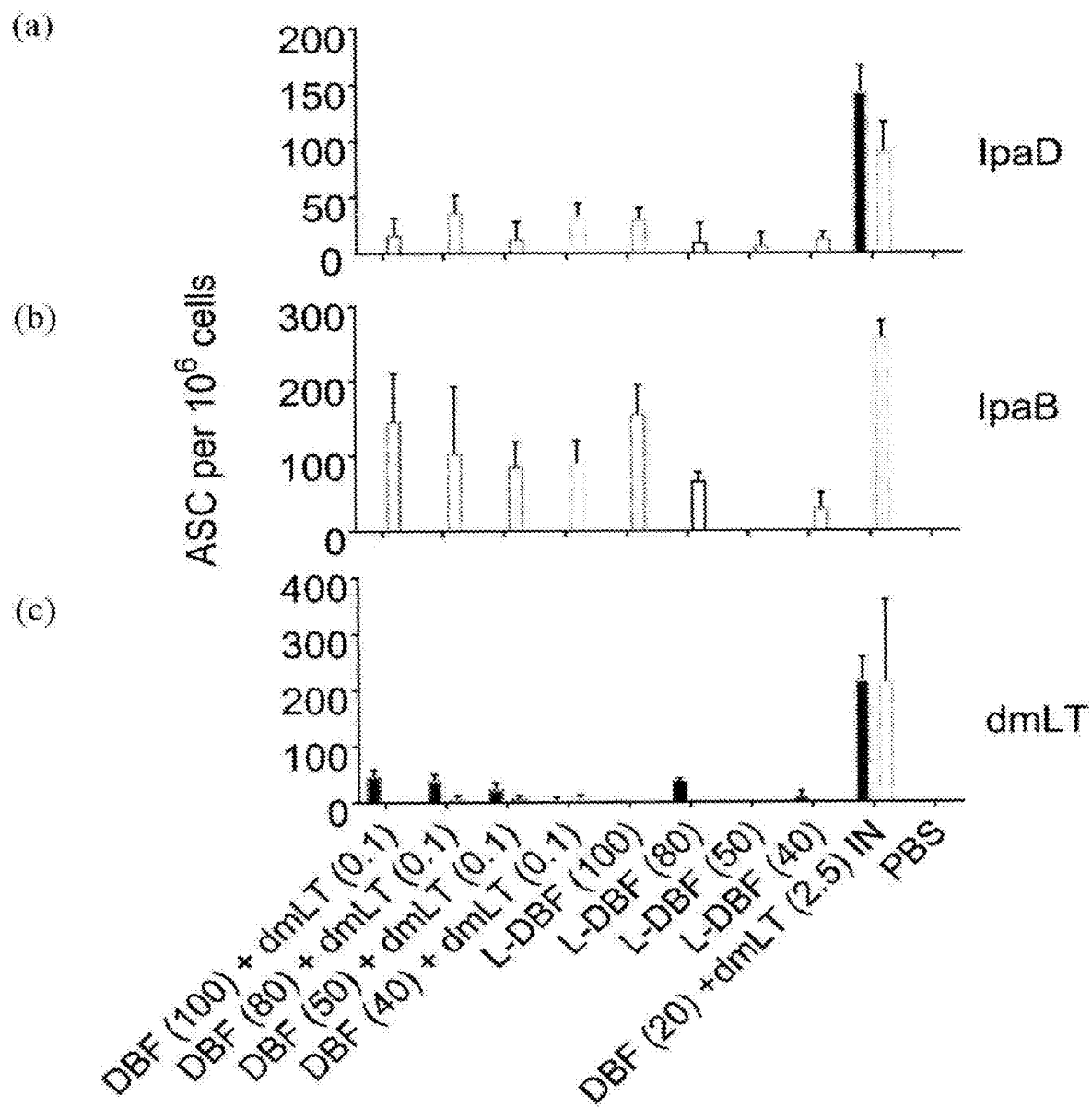


FIG. 18A, FIG. 18B, and FIG. 18C

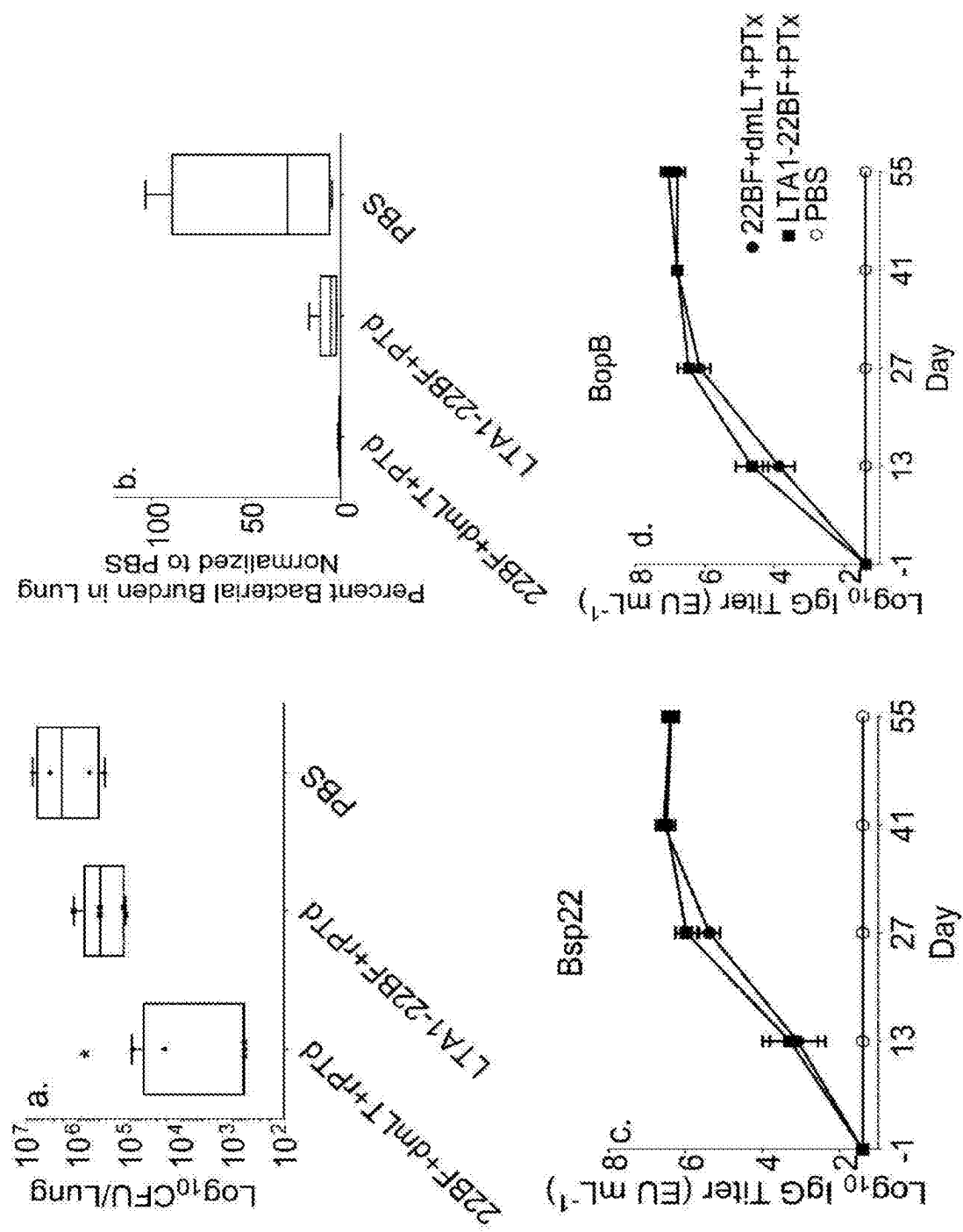


FIG. 19A, FIG. 19B, FIG. 19C and FIG. 19D

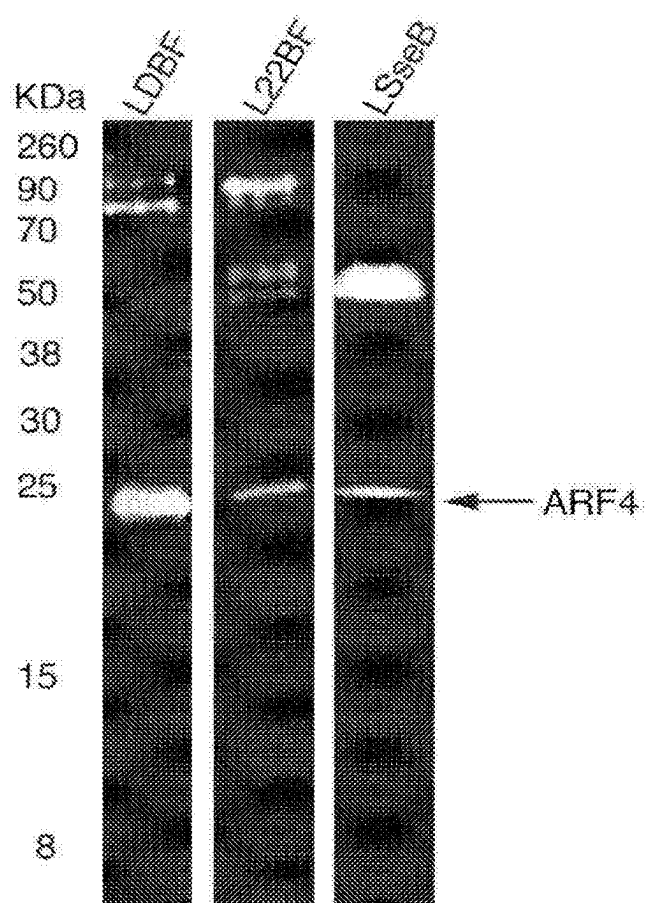


FIG. 20

## METHODS AND COMPOSITIONS RELATED TO THE NEXT GENERATION VACCINE

### I. CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] The present application claims the priority benefit of U.S. Provisional Patent Application Ser. No. 62/667,599, filed May 6, 2018, which is incorporated by reference in its entirety herein.

### II. BACKGROUND

[0002] *Bordetella pertussis* is a Gram-negative bacterial pathogen that causes pertussis, or whooping cough, a highly contagious, severe respiratory disease that is life threatening for infants and young children. This pathogen colonizes the trachea and secretes toxins that paralyze the cilia, which prevents clearance of mucus. Severe (paroxysmal), non-productive coughing fits are a result and attempts to acquire oxygen are manifested by the characteristic “whoop” upon gasps for air. The majority of deaths associated with pertussis are actually caused by secondary respiratory infections resulting from the inability to clear pulmonary secretions. In the 1940s a whole-cell pertussis (wP) vaccine was introduced that dramatically reduced the mortality caused by pertussis. Due to side effects attributed to the wP vaccine, a new acellular pertussis (aP) vaccine was developed and introduced in the US and other parts of the world in the 1990s. Although the aP vaccine has few side effects, its protective efficacy is lower than that of the wP vaccine. In 2012, which is considered the most recent major epidemic, 48,277 cases of pertussis were reported and, in 2015, 20,762 cases were reported. During the last 15 years, in addition to a greater overall incidence of pertussis, there is growing concern over the increase in the peak number of reported cases for each ensuing epidemic. In 2015, 45% of the 0.5 to 6-year-old children that contracted pertussis had been vaccinated with DTaP at least three times (with five vaccinations being optimal: 2, 4, 6, 15 months and one at 4-6 years). Additionally, there is evidence that selective pressure is causing *B. pertussis* to eliminate virulence factors that are components of the aP vaccine, further compromising the vaccine's efficacy. Taken together, a better vaccine is needed.

### III. SUMMARY

[0003] Disclosed are methods and compositions related to polypeptides comprising a fusion of the needle tip protein and translocator protein of a type III secretion apparatus (T3SA) from a type III secretion system (T3SS) of a Gram negative bacteria.

[0004] Disclosed herein are fusion polypeptides comprising a fusion of a needle tip protein (such as, for example, Bsp22, LcrV, BipD, PcrV, CT053, or CT668) or an antigenic fragment thereof and a translocator protein (such as, for example, BopB, YopB, BipB, PopB, CopB, or CopB2) or an antigenic fragment thereof from a Type III secretion system (T3SS) of a Gram negative bacteria (such as, for example, *Bordetella* spp., *Burkholderia* spp., *Chlamydia* spp., *Pseudomonas* spp., *Vibrio* spp. or *Yersinia* spp.); wherein the gram negative bacteria is not a *Salmonella enterica* or *Shigella* spp.

[0005] In one aspect, disclosed herein are fusion polypeptides, wherein the fusion polypeptide is arranged such that the needle tip protein is 5' of the translocator protein.

[0006] Also disclosed herein are fusion polypeptides of any preceding aspect, wherein the fusion further comprises an adjuvant such as, for example, Cholera Toxin or antigenic fragment thereof (such as, for example, CTA1) or double mutant labile toxin (dmLT) or an antigenic fragment thereof labile toxin (such as, for example, LTA1) from Enterotoxigenic *Escherichia coli*. In some aspect, the dmLT or fragment thereof can also be fused to the needle tip protein-translocator protein fusion at the 5' end.

[0007] In one aspect, disclosed herein are fusion polypeptides of any preceding aspect, wherein the fusion polypeptide further comprises pertussis toxoid (PTd).

[0008] Also disclosed herein are compositions comprising a T3SA needle tip protein (such as, for example, Bsp22, LcrV, BipD, PcrV, or CdsF) or an antigenic fragment thereof from a Gram negative bacteria (such as, for example, *Bordetella* spp., *Burkholderia* spp., *Chlamydia* spp., *Pseudomonas* spp., *Vibrio* spp., or *Yersinia* spp.) and a T3SA first translocator protein (such as, for example, BopB, YopB, BipB, PopB, or CopB/CopB2) or an antigenic fragment thereof from a Gram negative bacteria; wherein the gram negative bacteria is not a *Salmonella enterica* or *Shigella* spp. In one aspect, the composition can comprise the needle tip protein or fragment thereof and the translocator protein or fragment thereof as separate components or as a fusion polypeptide. Also disclosed herein are compositions of any preceding aspect, wherein the composition comprises an adjuvant (such as, for example, dmLT, LTA1, cholera toxin, or CTA1) and/or bacterial toxin protein such as a pertussis toxoid (PTd).

[0009] In one aspect, disclosed herein are vaccines comprising the fusion polypeptides or compositions of any preceding aspect. In some embodiments, the vaccine can further comprise an acellular gram negative vaccine or active components thereof. In one aspect, the vaccine can comprise pertussis toxoid (PTd).

[0010] Also disclosed herein are methods of treating, inhibiting, or preventing an infection of a Gram negative bacteria (such as, for example, *Bordetella* spp., *Burkholderia* spp., *Chlamydia* spp., *Pseudomonas* spp., *Vibrio* spp., or *Yersinia* spp.) in a subject comprising administering to the subject the fusion polypeptide, composition, or vaccine of any preceding aspect.

[0011] In one aspect, disclosed herein are methods of treating, inhibiting, or preventing an infection of a Gram negative bacteria of any preceding aspect, wherein the method further inhibits or prevents colony formation of the bacteria and/or transmission of the bacteria to another subject.

[0012] Also disclosed herein are methods of eliciting an immune response in a subject to a Gram negative bacteria (such as, for example, *Bordetella* spp., *Burkholderia* spp., *Chlamydia* spp., *Pseudomonas* spp., *Vibrio* spp., or *Yersinia* spp.) comprising administering to the subject the fusion polypeptide, composition, or vaccine of any preceding aspect. For example, disclosed herein are methods of eliciting an immune response against at least one Gram negative bacteria serovar in a subject in need thereof, comprising administering to the subject a composition comprising at least one needle tip protein or an antigenic fragment thereof and/or at least one translocator protein or an antigenic

fragment thereof; wherein said composition is administered in an amount sufficient to elicit an immune response to said at least one Gram negative bacteria serovar in said subject; and wherein the Gram negative bacteria is not a *Shigella* spp. or *Salmonella enterica*.

[0013] In one aspect, disclosed herein are methods of eliciting an immune response in a subject to a Gram negative bacteria of any preceding aspect, wherein the immune response provides sterilizing immunity.

#### IV. BRIEF DESCRIPTION OF THE DRAWINGS

[0014] The accompanying drawings, which are incorporated in and constitute a part of this specification, illustrate several embodiments and together with the description illustrate the disclosed compositions and methods.

[0015] FIG. 1 shows the protective efficacy of intranasally administered 22BF against *B. bronchiseptica* challenge. Mice (n=10) were vaccinated intranasally biweekly three times with the indicated formulation which contained 10 µg protein±dmLT. Zoetis vaccine was delivered subcutaneously on day 1 and 21 as per manufacturer's directions. Mice were challenged with  $1.3 \times 10^7$  *B. bronchiseptica* on day 56. BopB was not available at day 0. #P<0.05 compared to survival of mice vaccinated with PBS.

[0016] FIG. 2 shows the weight gain/loss of vaccinated mice during *B. bronchiseptica* challenge. Mice (same as above) were weighed daily in p.m. Note that the 22BF+dmLT mice gain weight and have small error bars.

[0017] FIG. 3A and FIG. 3B show the protective efficacy of 22BF+dmLT. Mice were vaccinated on days 0, 14, 28 and challenged on day 56 with a sublethal dose of *B. bronchiseptica*. FIG. 3A shows that on day 7 of the challenge, the CFU/lung were determined. \*P<0.05, \*\*P<0.01 when compared to dmLT. FIG. 3B shows the decrease in CFU compared to the 22BF average. \*P<0.05, \*\*P<0.01 when compared to 22BF.

[0018] FIGS. 4A, 4B, and 4C show the kinetics of IgG response. Blood was collected on days -1, 13, 27, 41, and 55. The kinetics of anti-Bsp22, -BopB and -dmLT IgG were assessed in all sera and shown in (4A) BopB, (4B) Bsp22, and (4C) dmLT. Typical logarithmic increases were seen. \*P value of <0.05 when comparing to PBS controls.

[0019] FIGS. 5A, 5B, 5C, 5D, 5E, 5F, 5G, 5H, and 5I show stimulation of antibody secreting cells from bone marrow, spleen and lungs. Bone marrow, spleens and lungs were collected on day 56. Single cell suspensions from 5 mice per group were stimulated in vitro Bsp22, BopB or dmLT. IgG (black) and IgA (white) ASC were measured by ELISpot. Bars represent mean ASC per  $10^6$  cells +SD from replicate wells. Data for bone marrow is shown in (4A), (5D), and (5G) for Bsp22, BopB, and dmLT, respectively. Data for spleen is shown in (5B), (5E), and (5H) for Bsp22, BopB, and dmLT, respectively. Data for lungs is shown in (5C), (5F), and (5I) for Bsp22, BopB, and dmLT, respectively.

[0020] FIGS. 6A, 6B, 6C, 6D, 6E, 6F, 6G, 6H, and 6I show Th1 cytokine secretion. Splenocytes were extracted from 5 mice of each group and incubated with Bsp22, BopB or dmLT. After 48 h, supernatants were collected and levels of cytokine secretion in response to specified antigen were then measured (in pg/ml) using an MSD cytokine detection plate. Each bar represents mean of triplicate wells ±S.D. Asterisk specified a P<0.05 when comparing specified groups. Data for IFN-γ is shown in (6A), (6B), and (6C) for Bsp22, BopB, and dmLT, respectively. Data for TNF-α is shown in (6D),

(6E), and (6F) for Bsp22, BopB, and dmLT, respectively. Data for IL-1B is shown in (6G), (6H), and (6I) for Bsp22, BopB, and dmLT, respectively.

[0021] FIGS. 7A, 7B, 7C, 7D, 7E, 7F, 7G, 7H, and 7I show Th1 cytokine secretion. Splenocytes were extracted from 5 mice of each group and incubated with Bsp22, BopB or dmLT. After 48 h, supernatants were collected and levels of cytokine secretion in response to specified antigen were then measured (in pg/ml) using an MSD cytokine detection plate. Each bar represents mean of triplicate wells ±S.D. Asterisk specified a P<0.05 when comparing specified groups. Data for IL-2 is shown in (7A), (7B), and (7C) for Bsp22, BopB, and dmLT, respectively. Data for IL-10 is shown in (7D), (7E), and (7F) for Bsp22, BopB, and dmLT, respectively. Data for IL-6 is shown in (7G), (7H), and (7I) for Bsp22, BopB, and dmLT, respectively.

[0022] FIGS. 8A, 8B, and 8C shows IL-17 secretion. Splenocytes were extracted from 5 mice of each group and incubated with Bsp22, BopB or dmLT. After 48 h, supernatants were collected and levels of IL-17 secretion in response to labeled antigen were then measured by the MSD® U-Plex Platform Multiplex Assay and the data is shown in (8A) BopB, (8B) Bsp22, and (8C) dmLT. Each bar represents the mean of triplicate wells ±S.D. Significance (Asterisk=P<0.05) was calculated for the comparison between labeled groups.

[0023] FIGS. 9A, 9B, 9C, 9D, 9E, and 9F show Th2 cytokine secretion. Splenocytes were extracted from 5 mice of each group and incubated with Bsp22, BopB or dmLT. After 48 h, supernatants were collected and levels of cytokine secretion in response to specified antigen were then measured (in pg/ml) using an MSD cytokine detection plate. Each bar represents mean of triplicate wells ±S.D. Asterisk specified a P<0.05 when comparing specified groups. Data for IL-4 is shown in (9A), (9B), and (9C) for Bsp22, BopB, and dmLT, respectively. Data for IL-5 is shown in (9D), (9E), and (9F) for Bsp22, BopB, and dmLT, respectively.

[0024] FIG. 10 shows the change in weight in percentage after infection with sublethal dosage of *B. pertussis* intranasally. There was an observable difference in weight loss between mice vaccinated with the 22BF+dmLT+PTd formulation and those that only received PBS. By Day 7 all mice aside from PBS treated mice had recovered to within 3% of pre-infection weight.

[0025] FIGS. 11A, 11B, 11C, and 11D show serum antibody responses to BopB, Bsp22, Pertussis Toxin Mutant, and dmLT. Mice were immunized on days 0, 14, and 28 with 22BF+PTd admixed with dmLT. Serum IgG antibodies specific for BopB, Bsp22, PTd, and dmLT were measured by ELISA and the data is shown in (11A) BopB, (11B) Bsp22, (11C) PTd, and (11D) dmLT. Data are the mean titers (EU ml<sup>-1</sup>) from group pools of animal samples. An asterisk indicates a P value of 0.05 when comparing vaccinated mice and the PBS controls. No responses were seen in the control mice that received PBS. Mice vaccinated with Infanrix only displayed a response against pertussis toxin mutant, which is part of its formulation.

[0026] FIG. 12 shows the Lung Colony forming units (CFU) from mice 3 days post intranasal infection. Mice vaccinated intranasally with 22BF+PTd and dmLT showed statistical (P<0.05) decreases in lung CFUs when compared to PBS treated mice. The mice vaccinated intradermally and mice vaccinated intramuscularly with 22BF+PTd and dmLT either showed sterilizing immunity, or no statistical decrease



in lung CFUs. Infanrix appeared to show a decrease in lung CFU, but this was not statistically significant ( $P>0.05$ ). ( $*=P<0.05$ , KW p-value=0.0003).

**[0027]** FIG. 13 shows the Lung Colony forming units (CFU) from mice 7 days post intranasal infection. Mice vaccinated intranasally with 22BF+PTd and dmLT showed statistical ( $P<0.05$ ) decreases in lung CFUs when compared to PBS treated mice, with 60% of the mice showing sterilizing immunity. The mice vaccinated intramuscularly or intradermally with 22BF+PTd and dmLT showed no statistical decrease in lung CFUs. Infanrix appeared to display sterilizing immunity with CFU measuring below the limit of detection. ( $*=P<0.05$ ,  $**P<0.01$ , KW p-value=0.0001).

**[0028]** FIG. 14 shows the protective efficacy of LTA1-DBF vs DBF+dmLT. Mice were vaccinated intramuscularly on days 0, 14 and 28 with the indicated  $\mu\text{g}$  of DBF+0.1  $\mu\text{g}$  dmLT or DBF equivalent of LTA1-DBF. The positive control was DBF+dmLT delivered intranasally. On day 56 the mice were challenged with *Shigella flexneri*. FIG. 14 indicates the percent survival of mice post infection with *Shigella flexneri*.

**[0029]** FIGS. 15A, 15B, and 15C show the kinetics of IgG response. Mice from FIG. 14 were bled prior to vaccination and on day 42. Sera were assessed for anti-IpaD, -IpaB and -dmLT IgG, and the data is shown in (15A) IpaD, (15Bb) IpaB, and (15C) dmLT. Differences in the IgG levels in mice vaccinated with dmLT vs. LTA1 are attributed to the recognition of the entire dmLT on the well.

**[0030]** FIGS. 16A, 16B, and 16C show the stimulation of antibody secreting cells from bone marrow. Bone marrow was collected on day 56. Single cell suspensions from 5 mice per group were stimulated in vitro IpaD, IpaB or dmLT. IgG (black) and IgA (white) ASC were measured by ELISpot, and the data is shown in (16A) IpaD, (16B) IpaB, and (16C) dmLT. Bars represent mean ASC per  $10^6$  cells +SD from replicate wells.

**[0031]** FIGS. 17A, 17B, and 17C show the stimulation of antibody secreting cells from spleen. Spleens were collected on day 56. Single cell suspensions from 5 mice per group were stimulated in vitro IpaD, IpaB or dmLT. IgG (black) and IgA (white) ASC were measured by ELISpot. Bars represent mean ASC per  $10^6$  cells +SD from replicate wells.

**[0032]** FIGS. 18A, 18B, and 18C show the stimulation of antibody secreting cells from lungs. Lungs were collected on day 56. Single cell suspensions from 5 mice per group were stimulated in vitro IpaD, IpaB or dmLT. IgG (black) and IgA (white) ASC were measured by ELISpot, and the data is shown in (18A) IpaD, (18B) IpaB, and (18C) dmLT. Bars represent mean ASC per  $10^6$  cells +SD from replicate wells.

**[0033]** FIGS. 19A, 19B, 19C, and 19D show the protective efficacy and IgG response kinetics of LTA1-22BF. Mice were vaccinated on days 0, 14, 28 and challenged on day 56 with a sublethal dose of *B. pertussis*. On day 7 of the challenge, the CFU/lung were determined. FIG. 19A shows the CFU/lung while FIG. 19B shows the decrease in CFU compared to the PBS average.  $*=P<0.05$  when compared to PBS. FIGS. 19C and 19D show the kinetics of the response of the anti-Bsp22 and anti-BopB IgG, respectively. No difference is seen between the mice vaccinated with 22BF+dmLT and LTA1-22BF.

**[0034]** FIG. 20 shows the ADPr activity of L-antigens. LTA1 was fused to DBF, 22BF or SseB. LTA1, however, must retain its ADP-ribosylation activity to maintain adjuvant activity. The ADPr of NAD<sup>+</sup> was biotin conjugated and

LTA1 transferred the biotin-ADPr moiety to ARF4. The biotin was then detected with Streptavidin-IR800. Lane 1: LTA1-DBF; 2: LTA1-22BF; 3: LTA1-SseB.

## V. DETAILED DESCRIPTION

**[0035]** Before the present compounds, compositions, articles, devices, and/or methods are disclosed and described, it is to be understood that they are not limited to specific synthetic methods or specific recombinant biotechnology methods unless otherwise specified, or to particular reagents unless otherwise specified, as such may, of course, vary. It is also to be understood that the terminology used herein is for the purpose of describing particular embodiments only and is not intended to be limiting.

### A. DEFINITIONS

**[0036]** As used in the specification and the appended claims, the singular forms “a,” “an” and “the” include plural referents unless the context clearly dictates otherwise. Thus, for example, reference to “a pharmaceutical carrier” includes mixtures of two or more such carriers, and the like.

**[0037]** Ranges can be expressed herein as from “about” one particular value, and/or to “about” another particular value. When such a range is expressed, another embodiment includes from the one particular value and/or to the other particular value. Similarly, when values are expressed as approximations, by use of the antecedent “about,” it will be understood that the particular value forms another embodiment. It will be further understood that the endpoints of each of the ranges are significant both in relation to the other endpoint, and independently of the other endpoint. It is also understood that there are a number of values disclosed herein, and that each value is also herein disclosed as “about” that particular value in addition to the value itself. For example, if the value “10” is disclosed, then “about 10” is also disclosed. It is also understood that when a value is disclosed that “less than or equal to” the value, “greater than or equal to the value” and possible ranges between values are also disclosed, as appropriately understood by the skilled artisan. For example, if the value “10” is disclosed the “less than or equal to 10” as well as “greater than or equal to 10” is also disclosed. It is also understood that the throughout the application, data is provided in a number of different formats, and that this data, represents endpoints and starting points, and ranges for any combination of the data points. For example, if a particular data point “10” and a particular data point 15 are disclosed, it is understood that greater than, greater than or equal to, less than, less than or equal to, and equal to 10 and 15 are considered disclosed as well as between 10 and 15. It is also understood that each unit between two particular units are also disclosed. For example, if 10 and 15 are disclosed, then 11, 12, 13, and 14 are also disclosed.

**[0038]** The needle tip protein and/or translocator proteins or antigenic portions thereof disclosed herein are used to elicit an immune response in subjects to whom they are administered. By “elicit an immune response”, “induces or enhances an immune response”, or “stimulates an immune response” which are used interchangeably herein, is meant that the subject mounts one or both of an innate and/or an adaptive immune reaction against antigenic determinants of the proteins or antigenic portions thereof that are administered. Preferably a statistically measurable induction or

increase in an immune response over a control sample to which the needle tip protein and/or translocator proteins or antigenic portions thereof disclosed herein has not been administered. Preferably the induction or enhancement of the immune response results in a prophylactic or therapeutic response in a subject. In particular, the adaptive immune reaction entails production of e.g. B and T cell lymphocytes and antibodies specific for binding and forming complexes with the antigenic determinants. In some embodiments, the proteins and/or antigenic fragments thereof elicit a protective immune response in the subject, i.e. administration of one or more of the proteins and/or antigenic portions thereof results in an immune response that is protective against later challenge by the disease causing organism itself, either preventing infection altogether, or lessening the impact of infection by decreasing disease symptoms that would otherwise occur, had the subject not been vaccinated as described herein.

**[0039]** “Vaccine” as used herein is a preparation that stimulates an immune response that produces immunity against particular antigens, e.g. Gram negative bacteria. Vaccines may be administered prophylactically (for example, to prevent or inhibit the establishment of an infection) or therapeutically to inhibit, reduce, or treat an established infection, or to ameliorate the effects or symptoms of an infection. Vaccines may contain, but are not limited to, live, attenuated infectious material such as viruses or bacteria, and dead or inactivated organisms or purified products derived therefrom. A vaccine can be administered by injection, orally, or by inhalation. Injections may be, but are not limited to, subcutaneous (sc), intramuscular (im), intraperitoneal (ip), intradermal (id) or intravenous (iv).

**[0040]** “Optional” or “optionally” means that the subsequently described event or circumstance may or may not occur, and that the description includes instances where said event or circumstance occurs and instances where it does not.

**[0041]** The term “subject” refers to any individual who is the target of administration or treatment. The subject can be a vertebrate, for example, a mammal. In one aspect, the subject can be human, non-human primate, bovine, equine, porcine, canine, or feline. The subject can also be a guinea pig, rat, hamster, rabbit, mouse, or mole. Thus, the subject can be a human or veterinary patient. The term “patient” refers to a subject under the treatment of a clinician, e.g., physician or veterinarian.

**[0042]** Throughout this application, various publications are referenced. The disclosures of these publications in their entireties are hereby incorporated by reference into this application in order to more fully describe the state of the art to which this pertains. The references disclosed are also individually and specifically incorporated by reference herein for the material contained in them that is discussed in the sentence in which the reference is relied upon.

## B. COMPOSITIONS

**[0043]** Disclosed are the components to be used to prepare the disclosed compositions as well as the compositions themselves to be used within the methods disclosed herein. These and other materials are disclosed herein, and it is understood that when combinations, subsets, interactions, groups, etc. of these materials are disclosed that while specific reference of each various individual and collective

combinations and permutations of these compounds may not be explicitly disclosed, each is specifically contemplated and described herein. For example, if a particular needle tip protein (such as, for example, IpaD, SipD, SseB, Bsp22, LcrV, BipD, PcrV, CT053, or CT668), translocator protein (such as, for example, IpaB, SipB, SseC, BopB, YopB, BipB, PopB, CopB, or CopB2), or fusion polypeptide thereof (such as, for example, 22BF, BurkF, PaF, YerF, CT053-CopB, CT053-CopB2, CT668-CopB, or CT668-CopB2) is disclosed and discussed and a number of modifications that can be made to a number of molecules including the needle tip protein (such as, for example, IpaD, SipD, SseB, Bsp22, LcrV, BipD, PcrV, CT053, or CT668), translocator protein (such as, for example, IpaB, SipB, SseC, BopB, YopB, BipB, PopB, CopB, or CopB2), or fusion polypeptide thereof (such as, for example, 22BF, BurkF, PaF, YerF, CT053-CopB, CT053-CopB2, CT668-CopB, or CT668-CopB2) are discussed, specifically contemplated is each and every combination and permutation of needle tip protein such as, for example, IpaD, SipD, SseB, Bsp22, LcrV, BipD, PcrV, CT053, or CT668), translocator protein (such as, for example, IpaB, SipB, SseC, BopB, YopB, BipB, PopB, CopB, or CopB2), or fusion polypeptide thereof (such as, for example, DBF, S1, S2, 22BF, BurkF, PaF, YerF, CT053-CopB, CT053-CopB2, CT668-CopB, or CT668-CopB2) and the modifications that are possible unless specifically indicated to the contrary. Thus, if a class of molecules A, B, and C are disclosed as well as a class of molecules D, E, and F and an example of a combination molecule, A-D is disclosed, then even if each is not individually recited each is individually and collectively contemplated meaning combinations, A-E, A-F, B-D, B-E, B-F, C-D, C-E, and C-F are considered disclosed. Likewise, any subset or combination of these is also disclosed. Thus, for example, the sub-group of A-E, B-F, and C-E would be considered disclosed. This concept applies to all aspects of this application including, but not limited to, steps in methods of making and using the disclosed compositions. Thus, if there are a variety of additional steps that can be performed it is understood that each of these additional steps can be performed with any specific embodiment or combination of embodiments of the disclosed methods.

**[0044]** To infect a host, *B. pertussis* uses an arsenal of well-characterized virulence factors. These factors include pertussis toxin (PT), adenylate cyclase toxin (ACT), the type III secretion system (T3SS), tracheal cytotoxin (TCT), dermonecrotic toxin (DNT), filamentous hemagglutinin (FHA), pertactin (PRN), and lipooligosaccharide (LOS). Current aP vaccines are comprised of PT, FHA, PRN, and the fimbrial proteins in varying proportions, but not necessarily all four proteins. Though the aP vaccine causes fewer adverse reactions than the wP vaccine, it is not as efficacious. This same situation exists for other pathogenic Gram negative bacteria. Accordingly, disclosed herein are fusion polypeptides from a Type III secretion system (T3SS) of a Gram negative bacteria (such as, for example, *Shigella* spp., *Salmonella enterica*, *Bordetella* spp. (such as, for example *B. pertussis* and/or *B. bronchiseptica*), *Burkholderia* spp. (such as, for example, *B. cepacia*, *B. mallei*, and/or *B. pseudomallei*), *Chlamydia* spp. (such as, for example, *C. trachomatis*), *Pseudomonas* spp., *Vibrio* spp. or *Yersinia* spp.) comprising a polypeptide of needle tip protein (such as, for example, IpaD, SipD, SseB, Bsp22, LcrV, BipD, PcrV, CT053, or CT668) or an antigenic fragment thereof and polypeptides of

a translocator protein (such as, for example, IpaB, SipB, SseC, BopB, YopB, BipB, PopB, CopB, or CopB2) or an antigenic fragment thereof. In some aspect, the fusion polypeptide does not comprise a needle tip protein polypeptide or translocator polypeptide from a *Shigella* spp. (IpaD and IpaB) or a *Salmonella* spp. (such as, for example, *S. enterica*) (SipD, SseB, SipB, and SseC). It is recognized and herein contemplated that the disclosed polypeptides can be separate components of a composition or more preferably a fusion construct. By a “fusion polypeptide” is meant a peptide, polypeptide, or protein that is translated from a single, contiguous nucleic acid molecule, and which comprises sequences from at least two different proteins or antigenic regions thereof. Typically, the individual sequences are joined via a linker or spacer sequence of e.g. from about 2 to about 20 amino acids, usually from about 2 to about 10 amino acids. The amino acids in linking sequences are typically uncharged and the linker sequence usually does not exhibit secondary or tertiary structure, but does allow the fused protein/peptide segments to adopt functional secondary, tertiary, etc. conformations. One such exemplary fusion polypeptide includes Bsp22 (as set forth in SEQ ID NO: 4 and encoded by the nucleic acid sequence as set forth in SEQ ID NO: 3) and BopB (as set forth in SEQ ID NO: 6 and encoded by the nucleic acid sequence as set forth in SEQ ID NO: 5). The amino acid sequence of this chimera (i.e., 22BF) is set forth in SEQ ID NO: 2. The chimera may be encoded by any suitable nucleic acid sequence, e.g. the exemplary nucleic acid sequence depicted in SEQ ID NO: 1.

**[0045]** Thus, in one aspect, disclosed herein are fusion polypeptides comprising a fusion of a needle tip protein (such as, for example, IpaD, SipD, SseB, Bsp22, LcrV, BipD, PcrV, CT053, or CT668) or an antigenic fragment thereof and a translocator protein (such as, for example, IpaB, SipB, SseC, BopB, YopB, BipB, PopB, CopB, or CopB2) or an antigenic fragment thereof from a Type III secretion system (T3SS) of a Gram negative bacteria (such as, for example, *Bordetella* spp., *Burkholderia* spp., *Chlamydia* spp., *Pseudomonas* spp., *Vibrio* spp. or *Yersinia* spp. For example, the fusion polypeptide can comprise a fusion of the *Shigella* spp. needle tip protein (IpaD) and first translocator protein (IpaB) or fragments thereof (the fusion referred to as DBF), *Salmonella* spp. (such as, for example, *S. enterica*) SPI-1 needle tip protein (SipD) (as set forth in SEQ ID NO: 52 and encoded by the nucleic acid sequence as set forth in SEQ ID NO: 51) and translocator protein (SipB) (as set forth in SEQ ID NO: 54 and encoded by the nucleic acid sequence as set forth in SEQ ID NO: 53) or fragments thereof (the fusion referred to as S1) (as set forth in SEQ ID NO: 56 and encoded by the nucleic acid sequence as set forth in SEQ ID NO: 55), *Salmonella* spp. (such as, for example, *S. enterica*) SPI-2 needle tip protein (SseB) (as set forth in SEQ ID NO: 62 and encoded by the nucleic acid sequence as set forth in SEQ ID NO: 61) and translocator protein (SseC) (as set forth in SEQ ID NO: 64 and encoded by the nucleic acid sequence as set forth in SEQ ID NO: 63) or fragments thereof (the fusion referred to as S2) (as set forth in SEQ ID NO: 66 and encoded by the nucleic acid sequence as set forth in SEQ ID NO: 65), *Bordetella* spp. needle-tip protein (Bsp22) and translocator protein (BopB), or fragments thereof (the fusion referred to as 22BF); a fusion of the *Yersinia* spp. needle-tip protein (LcrV) (as set forth in SEQ ID NO: 42 and encoded by the nucleic acid

sequence as set forth in SEQ ID NO: 41) and translocator protein (YopB) (as set forth in SEQ ID NO: 44 and encoded by the nucleic acid sequence as set forth in SEQ ID NO: 43), or fragments thereof (the fusion referred to as YerF) (as set forth in SEQ ID NO: 46 and encoded by the nucleic acid sequence as set forth in SEQ ID NO: 45); a fusion of the *Burkholderia* spp. needle-tip protein (BipD) (as set forth in SEQ ID NO: 22 and encoded by the nucleic acid sequence as set forth in SEQ ID NO: 21) and translocator protein (BipB) (as set forth in SEQ ID NO: 24 and encoded by the nucleic acid sequence as set forth in SEQ ID NO: 23), or fragments thereof (the fusion referred to as BurkF) (as set forth in SEQ ID NO: 26 and encoded by the nucleic acid sequence as set forth in SEQ ID NO: 25); a fusion of the *Pseudomonas* spp. needle-tip protein (PcrV) (as set forth in SEQ ID NO: 32 and encoded by the nucleic acid sequence as set forth in SEQ ID NO: 31) and translocator protein (PopB) (as set forth in SEQ ID NO: 34 and encoded by the nucleic acid sequence as set forth in SEQ ID NO: 33), or fragments thereof (the fusion referred to as PaF) (as set forth in SEQ ID NO: 36 and encoded by the nucleic acid sequence as set forth in SEQ ID NO: 35); and/or a fusion of the *Chlamydia* spp. needle-tip protein CT053 (as set forth in SEQ ID NO: 74 and encoded by the nucleic acid sequence as set forth in SEQ ID NO: 73) or CT668 (as set forth in SEQ ID NO: 84 and encoded by the nucleic acid sequence as set forth in SEQ ID NO: 83) and translocator protein (CopB) (as set forth in SEQ ID NO: 76 and encoded by the nucleic acid sequence as set forth in SEQ ID NO: 75) or CopB2 (as set forth in SEQ ID NO: 92 and encoded by the nucleic acid sequence as set forth in SEQ ID NO: 91)), or fragments thereof such fusions including but not limited to CT053-CopB (the CT053-CopB fusion as set forth in SEQ ID NO: 78 and encoded by the nucleic acid sequence as set forth in SEQ ID NO: 77), CT668-CopB (the CT668-CopB fusion as set forth in SEQ ID NO: 86 and encoded by the nucleic acid sequence as set forth in SEQ ID NO: 85), CT053-CopB2 (the CT053-CopB2 fusion as set forth in SEQ ID NO: 94 and encoded by the nucleic acid sequence as set forth in SEQ ID NO: 93), and CT053-CopB2 (the CT668-CopB2 fusion as set forth in SEQ ID NO: 100 and encoded by the nucleic acid sequence as set forth in SEQ ID NO: 99) and collectively each fusion being referred to as ChlamF. In one aspect, the fusion polypeptide does not comprise any needle tip protein or translocator protein or fragment thereof from a *Salmonella* spp. or a *Shigella* spp. Accordingly, disclosed herein are fusion polypeptides comprising a fusion of a needle tip protein (such as, for example, Bsp22, LcrV, BipD, PcrV, CT053, or CT668) or an antigenic fragment thereof and a translocator protein (such as, for example, BopB, YopB, BipB, PopB, CopB, or CopB2) or an antigenic fragment thereof from a Type III secretion system (T3SS) of a Gram negative bacteria (such as, for example, *Bordetella* spp., *Burkholderia* spp., *Chlamydia* spp., *Pseudomonas* spp., *Vibrio* spp. or *Yersinia* spp., wherein the gram negative bacteria is not a *Salmonella enterica* or *Shigella* spp.

**[0046]** It is understood and herein contemplated that the arrangement of the polypeptides in a fusion construct can have significant impact on the antigenicity of the fusion construct. Accordingly, in one aspect, disclosed herein are fusion polypeptides, wherein the fusion polypeptide is arranged such that the needle tip protein is 5' of the translocator protein.

[0047] The present invention provides compositions for use in eliciting an immune response and/or vaccinating an individual against Gram negative bacterial infection, and/or against disease symptoms caused by Gram negative bacterial infection. The compositions include one or more substantially purified proteins, polypeptides or antigenic regions thereof as described herein, or substantially purified nucleic acid sequences (e.g. DNA cDNA, RNA, etc.) encoding such proteins, polypeptides or antigenic regions thereof, and a pharmacologically suitable/compatible carrier. By “substantially purified” is meant that the molecule is largely free of other organic molecules, cellular debris, solvents, etc. when tested using standard techniques known to those of skill in the art (e.g. gel electrophoresis, column chromatography, sequencing, mass spectroscopy, etc.). For example, the molecule is generally at least about 50, 55, 60, 65, 70, or 75% pure by wt %, and preferably is at least about 80, 85, 90, 95% or more preferably pure (e.g. 96, 97, 98, 99 or even 100% pure). The preparation of proteins, polypeptides, and peptides as described herein is well-known to those in the art, and includes, for example, recombinant preparation; isolation from a natural source; chemical synthesis; etc. The purification of proteinaceous materials is also known. However, specific exemplary methods for preparing the vaccinating agents utilized in the practice of the invention are described in detail in the Examples section below.

[0048] In addition, the composition may contain adjuvants, many of which are known in the art. For example, adjuvants suitable for use in the invention include but are not limited to: bacterial or microbial derivatives such as non-toxic derivatives of enterobacterial lipopolysaccharide (LPS), Lipid A derivatives, immunostimulatory oligonucleotides and ADP-ribosylating toxins and detoxified derivatives thereof. Non-toxic derivatives of LPS include monophosphoryl lipid A (MPL) and 3-O-deacylated MPL (3dMPL). 3dMPL is a mixture of three de-O-acylated monophosphoryl lipid A with 4, 5 or 6 acylated chains. A preferred non-toxic derivative of LPS is 3 De-O-acylated monophosphoryl lipid A. Other non-toxic LPS derivatives include monophosphoryl lipid A mimics, such as aminoalkyl glucosamine phosphate derivatives, e.g. RC-529.

[0049] Lipid A derivatives include derivatives of lipid A from *Escherichia coli* such as OM-174. Immunostimulatory oligonucleotides suitable for use as adjuvants in the invention include nucleotide sequences containing a CpG motif (a dinucleotide sequence containing an unmethylated cytosine linked by a phosphate bond to a guanosine). Double-stranded RNAs and oligonucleotides containing palindromic or poly(dG) sequences have also been shown to be immunostimulatory. The CpG's can include nucleotide modifications/analogues such as phosphorothioate modifications and can be double-stranded or single-stranded, e.g. replacement of guanosine with 2'-deoxy-7-deazaguanosine. The CpG sequence may include, for example, the motif GTCGTT or TTCGTT. The CpG sequence may be specific for inducing a Th1 immune response, such as a CpG-A ODN, or it may be more specific for inducing a B cell response, such as a CpG-B ODN, CpG-A and CpG-B ODNs. Preferably, the CpG is a CpG-A ODN. Preferably, the CpG oligonucleotide is constructed so that the 5' end is accessible for receptor recognition. Optionally, two CpG oligonucleotide sequences may be attached at their 3' ends to form “immunomers”.

[0050] Bacterial ADP-ribosylating toxins and detoxified derivatives thereof may be used as adjuvants in the inven-

tion. Preferably, the protein is derived from *E. coli* (e.g. *E. coli* heat labile enterotoxin (“LT”), cholera (“CT”)(Table 1), or pertussis (“PT”).

TABLE 1

| Cholera Toxin (CTA1) subunits and sequences |                             |               |
|---------------------------------------------|-----------------------------|---------------|
| Subunit                                     | DNA sequence                | AA sequence   |
| Subunit A                                   | ATGGTAAAGATAATATTGTGTTTTT   | MVKIIFVFFIFL  |
|                                             | TATTTTCTATCATCATTTTCATATG   | SSFSYANDDKLY  |
|                                             | CAAATGATGATAAGTTATATCGGGCA  | RADSRPPDEIKQ  |
|                                             | GATTCTAGACCTCCTGATGAAATAAA  | SGGLMPRGQSEY  |
|                                             | GCAGTCAGGTGGTCTTATGCCAAGAG  | FDRGTQMNINLY  |
|                                             | GACAGAGTGAGTACTTTGACCGAGGT  | DHARGTQTGFVR  |
|                                             | ACTCAAATGAATATCAACCTTTATGA  | HDDGYVSTISL   |
|                                             | TCATGCAAGAGGAAGCTCAGACGGGAT | RSAPHLVGTILS  |
|                                             | TTGTTAGGCACGATGATGGATATGTT  | GHSTYYIYVIAT  |
|                                             | TCCACCTCAATTAGTTTGGAGAAGTGC | APNMFNVNDVLG  |
|                                             | CCACTTAGTGGGTCAAATATATTTGT  | AYSHPHDEQVS   |
|                                             | CTGGTCATTCTACTTATATATATAT   | ALGGIPYSQIYG  |
|                                             | GTTATAGCCACTGCACCAACATGTT   | WYRVHFGVLDEQ  |
|                                             | TAACGTTAATGATGTATTAGGGGCAT  | LHNRNGYRDRYY  |
|                                             | ACAGTCCTCATCCAGATGAACAAGAA  | SNLDIAPAADGY  |
|                                             | GTTTCTGCTTTAGGTGGGATTCCTATA | GLAGFPPEHRAW  |
|                                             | CTCCCAAATATATGGATGGTATCGAG  | REEPWIIHAPPG  |
|                                             | TTCATTTTGGGGTGCTTGATGAACAA  | CGNAPRSSMSNT  |
|                                             | TTACATCGTAATAGGGGCTACAGAGA  | CDEKTSQSLGVKF |
|                                             | TAGATATTACAGTAAGTTAGATATTG  | LDEYQSKVKRQI  |
|                                             | CTCCAGCAGCAGATGGTTATGGATTG  | FSGYQSDIDHTN  |
|                                             | GCAGGTTTCCCTCCGGAGCATAGAGC  | RIKDEL        |
|                                             | TTGGAGGGAAGAGCCGTGGATTTCATC | (SEQ ID       |
|                                             | ATGCACCGCCGGGTTGTGGGAATGCT  | NO: 116)      |
|                                             | CCAAGATCATCGATCAGTAATACTTG  |               |
|                                             | CGATGAAAAACCCAAAGTCTAGGTG   |               |
|                                             | TAAAATTCCTTGACGAATACCAATCT  |               |
|                                             | AAAGTTAAAGACAAATATTTTCAGG   |               |
|                                             | CTATCAATCTGATATTGATACACATA  |               |
|                                             | ATAGAATTAAGGATGAATTATGA     |               |
|                                             | (SEQ ID NO: 115)            |               |
| Subunit B                                   | ATGATTAAATTAAAATTTGGTGTTTT  | MIKLKFGVFFTV  |
|                                             | TTTTACAGTTTTACTATCTTCAGCAT  | LLSSAYAHGTPQ  |
|                                             | ATGCACATGGAACACCTCAAATATAT  | NITDLCAEYHNT  |
|                                             | ACTGATTTGTGTGAGAAATACCACAA  | QIYTLNDKIFSY  |
|                                             | CACACAAATATATACGCTAAATGATA  | TESLAGKREMAI  |
|                                             | AGATATTTTCGTATACAGAAATCTCTA | ITFKNGAIFQVE  |
|                                             | GCTGGAAAAAGAGAGATGGCTATCAT  | VPGSQHIDSQKK  |
|                                             | TACTTTTAAGAATGGTGCAATTTTTC  | AIERMKDTLRIA  |
|                                             | AAGTAGAAGTACCAGGTAGTCAACAT  | YLTEAKVEKLCV  |
|                                             | ATAGATTACAAAAAAGCGGATTGA    | WNNKTPHAIAAI  |
|                                             | AAGGATGAAGGATACCTGAGGATTG   | SMAN          |
|                                             | CATATCTTACTGAAGCTAAAGTCGAA  | (SEQ ID       |
|                                             | AAGTTATGTGTATGGAATAATAAAAC  | NO: 118)      |
|                                             | GCCTCATGCGATTGCCGCAATTAGTA  |               |
|                                             | TGGCAATTA                   |               |
|                                             | (SEQ ID NO: 117)            |               |

[0051] The toxin or toxoid is preferably in the form of a holotoxin, comprising both A and B subunits. Preferably, the A subunit contains a detoxifying mutation; preferably the B subunit is not mutated. More preferably, the adjuvant is a detoxified LT mutant such as LT-5 K63, LT-R72, and LT-G192. The use of ADP-ribosylating toxins and detoxified derivatives thereof, particularly LT-K63 and LT-R72, is known. Such adjuvants are described, for example, in issued U.S. Pat. No. 8,039,007 (the complete contents of which is hereby incorporated by reference in entirety). Various interleukins may also be used as adjuvants to increase the immune response in a subject. In preferred embodiments, the adjuvant is a mucosal adjuvant such as, for example, the double mutant heat-labile toxin (dmLT) as set forth in SEQ ID NOS: 113 and 114) from enterotoxigenic *E. coli* or the

active moiety thereof known as LTA1 (as set forth in SEQ ID NO: 13 and encoded by the nucleic acid sequence as set forth in SEQ ID NO: 12) and encoded by nor cholera toxin or the active moiety thereof known as CTA1. Accordingly, disclosed herein are fusion polypeptides of any preceding aspect, wherein the fusion further comprises an adjuvant such as, for example, double mutant labile toxin (dmLT) or an antigenic fragment thereof (such as, for example, LTA1 or CTA1) from Enterotoxigenic *Escherichia coli*. In some aspect, the dmLT or fragment thereof can also be fused to the needle tip protein-translocator protein fusion at the 5' end. For example, specifically disclosed herein are LTA1-DBF, LTA1-S1 (as set forth in SEQ ID NO: 57 and encoded by the nucleic acid sequence as set forth in SEQ ID NO: 58), LTA1-S2 (as set forth in SEQ ID NO: 68 and encoded by the nucleic acid sequence as set forth in SEQ ID NO: 67), LTA1-SseB (as set forth in SEQ ID NO: 70 and encoded by the nucleic acid sequence as set forth in SEQ ID NO: 69), LTA1-22BF (as set forth in SEQ ID NO: 18 and encoded by the nucleic acid sequence as set forth in SEQ ID NO: 17), LTA1-BurkF (as set forth in SEQ ID NO: 28 and encoded by the nucleic acid sequence as set forth in SEQ ID NO: 27), LTA1-CT668-CopB (as set forth in SEQ ID NO: 88 and encoded by the nucleic acid sequence as set forth in SEQ ID NO: 87), LTA1-CT668-CopB2 (as set forth in SEQ ID NO: 102 and encoded by the nucleic acid sequence as set forth in SEQ ID NO: 101), LTA1-CT053-CopB (as set forth in SEQ ID NO: 80 and encoded by the nucleic acid sequence as set forth in SEQ ID NO: 79), LTA1-CT053-CopB2 (as set forth in SEQ ID NO: 96 and encoded by the nucleic acid sequence as set forth in SEQ ID NO: 95), LTA1-PaF (as set forth in SEQ ID NO: 38 and encoded by the nucleic acid sequence as set forth in SEQ ID NO: 37), and LTA1-YerF (as set forth in SEQ ID NO: 48 and encoded by the nucleic acid sequence as set forth in SEQ ID NO: 47).

**[0052]** Whooping cough still causes significant mortality and morbidity in children all over the world. It also continues to be a problem in adults whose immunity has waned. Herein is disclosed a strong candidate for a new protective vaccine based on research on the T3SS proteins and resulting subunit vaccines, including the vaccine against shigellosis. It is demonstrated herein that the vaccine has 100% protective efficacy against *B. bronchiseptica* using 22BF+dmLT. While this is a remarkable step forward, examined herein is the immune response and the protective efficacy of 22BF+dmLT±PTd against *B. pertussis*. The vaccine can also be taken a step further by eliciting sterilizing immunity so that the *B. pertussis* transmission chain can be broken.

**[0053]** Originally, the mechanism of protection against *B. pertussis*, an extracellular organism, was thought to be the humoral immune response, however, cell-mediated immunity has been found to also be important for protection with bacterial clearance mediated by Th1 and Th17 cells. By measuring cytokines corresponding to specific immune pathways, Ross et al. concluded that the wP vaccine promotes Th1 and Th17 responses while the aP vaccine elicits a mix of Th1 and Th2 responses. These differences likely account for the increased protection seen for the wP vaccine. A study in a baboon model compared wP vaccines with an aP vaccine and confirmed that the wP elicits a Th1/Th17 response while the aP vaccine elicits a Th1/Th2 response. Moreover, these studies found that aP does not prevent colonization or transmission of *B. pertussis*, even in asymptomatic subjects. Thus, the current pertussis resurgence

could be due, in part, to the ability of the aP vaccine to protect the host against the overt symptoms of the disease while not preventing colonization and the resulting transmission of *B. pertussis* to susceptible children. Furthermore, protection of newborns against pertussis via aP or wP is problematic due not only to possible side effects but also because newborns lack the ability to mount a vaccine-induced Th1 response elicited through the requisite antigen presentation and T-cell activation. Although it has been shown, in some cases, that neonatal immunization can prime the immune system for subsequent booster vaccinations, the development of a protective pertussis vaccine for infants remains a need.

**[0054]** As noted above, the current aP vaccine does not provide sterilizing immunity. That is, the aP vaccine protects the immunized host, but does not stop colonization and transmission of the *Bordetella* spp. In one aspect, disclosed herein are fusion polypeptides of any preceding aspect, wherein the composition or fusion polypeptide further comprises an acellular Gram negative vaccine component (such as, for example, the acellular pertussis vaccine (aP) component pertussis toxoid (PTd)).

**[0055]** Pertussis toxin (PTX) is produced by *Bordetella pertussis*, the bacterium responsible for whooping cough. Pertussis toxin is a multi-component protein composed of six non-covalently bound subunits ranging in molecular weight from approximately about 9 kDa to about 28 kDa. These subunits are designated as S1, S2, S3, S4 and S5 and occur in native pertussis toxin in a ratio of 1:1:1:2:1, where the subunit S4 is present in two copies. The largest subunit S1, also called the A protomer, is responsible for the ADP-ribosyltransferase activity. List Labs produces Pertussis Toxin Mutant R9K, E129A (both in the S1 subunit), a genetically inactivated mutant of pertussis toxin, which has a modified sequence encoding the enzyme subunit (Table 2). Virulence of this pertussis mutant is reduced relative to that found with the wild type.

TABLE 2

| Pertussis Toxin Mutant R9K, E129A |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                              |
|-----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Subunit                           | DNA sequence                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | AA sequence                                                                                                                                                                                                                                                                                                                                                                                  |
| Subunit 1                         | ATGCGTTGCACTCGGGCAATTCGCCA<br>AACCGCAGAAGCAGGCTGGGTGAGCT<br>GGCTGGCGATTCTTCCGCTCACGGCG<br>CCCGTGACTTCGCCGGCATGGGCCGA<br>CGATCCTCCCGCCACCGTATACCGCT<br>ATGACTCCCGCCCGCGGAGGACGTT<br>TTCCAGAACGGATTACGGCGTGGGG<br>AAACAACGACAATGTGCTCGACCATC<br>TGACCGGACGTTCTGCCAGGTCCGC<br>AGCAGCAACAGCGCTTTCGTCTCCAC<br>CAGCAGCAGCCGGCGCTATACCGAGG<br>TCTATCTCGAACATCGCATGCAGGAA<br>GCGGTCGAGGCCGAACGGCGCGGAG<br>GGGCACCGGCCACTTCATCGGCTACA<br>TCTACGAAGTCCGCGCCGACACCAAT<br>TTCTACGGCGCGCCAGCTCGTACTT<br>CGAATACGTCGACACTTATGGCGACA<br>ATGCCGGCCGTATCCTCGCGGCGCG<br>CTGGCCACCTACCAGAGCGAATATCT<br>GGCACACCGGCGCATTCGCCCGAAA<br>ACATCCGAGGTAACCGGGGTCTAT<br>CACACGGCATCACGGCGAGACCAC<br>GACCACGGAGTATTCACCGCTCGCT<br>ACGTCAGCCAGCAGACTCGCGCCAAT<br>CCCAACCCCTACACATCGCAAGGTC | MRCTRAIRQTAR<br>TGWLTWLAILAV<br>TAPVTSFPAWADD<br>PPATVYRYDSRP<br>PEDVFQNGFTAW<br>GNNDNVLDHLTG<br>RSCQVGSNSAF<br>VSTSSRRRYTEV<br>YLEHRMQEAVEA<br>ERAGRGTHGFIG<br>YIYEVNRADNNFY<br>GAASSYFEYVDT<br>YGDNAGRILAGA<br>LATYQSEYLAHR<br>RIPPENIRRVTR<br>VYHNGITGETTT<br>TEYSNARYVSSQ<br>TRANPNPYTSRR<br>SVASIVGTLVRM<br>APVIGACMARQA<br>ESSEMAAWSER<br>AGEAMVLVYVES<br>IAYSF<br>(SEQ ID<br>NO: 104) |

TABLE 2-continued

| Pertussis Toxin Mutant R9K, E129A |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                  |
|-----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Subunit                           | DNA sequence                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | AA sequence                                                                                                                                                                                                                                                                                                                      |
|                                   | CGTAGCGTCGATCGTCGGCACATTGG<br>TGCGCATGGCGCCGGTGATAGGCGCT<br>TGCATGGCGCGGCAGGCCGAAAGCTC<br>CGAGGCCATGGCAGCCTGGTCCGAAC<br>GCGCCGGCGAGGCGATGGTTCTCGTG<br>TACTACGAAAGCATCGCGATTCTGTT<br>CTAG<br>(SEQ ID NO: 103)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                  |
| Subunit 2                         | ATGCCGATCGACCGAAGACGCTCTG<br>CCATCTCCTGTGCTGTTCTGCGGTTGG<br>CCCTCTCGGATCTCACGTGCGCGCG<br>GCCTCCACGCCAGGCATCGTCATTCC<br>GCCGACGGAACAGATATACCCAGCATG<br>GCAGCCCCATTGAGACGCTGCGCGAAC<br>AAGACCCGTGCCCTGACCGTGGCGGA<br>ATTGCGCGCGCAGCGGCGATCTGACGAG<br>AGTACTCTCGGTCATGTGACGCGCGCG<br>TGGTCAATATTTGCGCTCTACGATGG<br>CACCTATCTCGCGCGGCAATATGGCG<br>GCGTGATCAAGACGGAACACCCGCG<br>GGCGCATTTCGACCTGAAAACGACGTT<br>CTGCATCATGACACGCGCAATACGG<br>GTCAACCCGCAACGGATCACTACTAC<br>AGCAACGTCACCGCCACTCGCCTGCT<br>CTCCAGCACCAACAGCAGGCTATGCG<br>CGGTCTTCGTGAGAAGCGGGCAACCG<br>GTCATTGGCGCCTGCACACGCCCGTA<br>TGACGGCAAGTACTGGAGCATGTACA<br>GCCGCTGCGGAAAAATGCTTTACTCT<br>ATCTACGTGGCGGCATCTCCGTACG<br>CGTCCATGTCAGCAAGGAAGAACAGT<br>ATTACGACTATGAGGACGCAACGTTCT<br>GAGACTTACGCCCTTACCGGCATCTC<br>CATCTGCAATCTCGGATGCTCCCTTAT<br>GCTGA<br>(SEQ ID NO: 105) | MPIDRKTLCHELL<br>SVLPALLGSHV<br>ARASTPGIVIPP<br>QEQTQHGPGYQ<br>RCANKTRALTVA<br>ELRGSDDLQEYL<br>RHVTRGWSIFAL<br>YDGYTLGGEYGG<br>VIKDGTPGGAFFD<br>LKTTFCIMTTRN<br>TGQPATDHYYSN<br>VTATRLLSSTNS<br>RLCAVFRVRSQGP<br>VIGACTSPYDGG<br>YWSMYSLRLKML<br>YLIYVAGISVRV<br>HVSKEEQYYDYE<br>DATFETYALTGI<br>SICNPGSSLLC<br>(SEQ ID NO: 106) |
| Subunit 3                         | ATGCTGATCAACAACAAGAAGCTGCT<br>TCATCACATTCTGCCCATCCTGGTGC<br>TCGCCCTGCTGGGCATGCGCACGGCC<br>CAGGCCGTTGCGCCAGGCATCGTCAT<br>CCCGCCGAAGGCACTGTTACCCAAAC<br>AGGGCGGCGCCTATGGACGCTGCCCG<br>AACCGAAACCGCGCCTTGACCGTGGC<br>CGAACTGCGCGGCAACGCCGAATTGC<br>AGACGTATTTGCGCCAGATAACGCCCC<br>GGCTGGTCCATATACGCTCTCTATGA<br>CGGTACGTACCTGGGCCAGGCGTACG<br>GCGGCATCATCAAGGACGCGCCGCCA<br>GGCGCGGGTTCTATTATCGCGAAAC<br>TTTCTGCATCACGACCATATACAAGA<br>CCGGGCAACCGGCTGCGGATCACTAC<br>TACAGCAAGGTACCGGCCACGCGCCT<br>GCTCGCCAGCACCAACAGCAGGCTGT<br>GCGCGGTATTCGTGAGGACGGGCAA<br>TCGGTCACTCGGAGCCTGCGCCAGGCC<br>GTATGAAGGCAGGTACAGAGACATGT<br>ACGACGCGCTGCGGCGCCTGCTGTAC<br>ATGATCTATATGTCCGGCCTTGCCGT<br>ACGCGTCCACGTCAAGCAAGGAAGAGC<br>AGTATTACGACTACGAGGACGCCACA<br>TTCCAGACCTATGCCCTCACCGGCAT<br>TTCCCTCTGCAACCCGGCAGCGTCGA<br>TATGCTGA<br>(SEQ ID NO: 107) | MLINNKKLLHHI<br>LPILVLLALLGMR<br>TAQAVAPGIVIP<br>PKALFTQQGGAY<br>GRCPNGTRALT<br>AELRGNALQTY<br>LRQITPGWSIYG<br>LYDGYTLGQAYG<br>GIIKDAPPGAGF<br>IYRETFICITTY<br>KTGQPAADHYYS<br>KVTATRLLASTN<br>SRLCAVFRVDGQ<br>SVIGACASPYEG<br>RYRDMYDALRRL<br>LYMIYMSGLAVR<br>VHVSKEEQYYDY<br>EDATFQTYALTG<br>ISLCNPAASIC<br>(SEQ ID NO: 108)   |
| Subunit 4                         | ATGCTGAGACGCTTCCCCACTCGAAC<br>CACCGCCCCGGGACAGGGCGGCGCC<br>GGCGGTGCGCGGTGCGCGCCTGGCG<br>TGGTTGCTGGCATCCGGCGCGATGAC<br>GCATCTTTCCCGCGCCTGGCCGACG<br>TTCTTATGTGCTGTTGAAGACCAAT                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | MLRRFPTRTTAP<br>GQGGARRSRVRA<br>LAWLLASGAMTH<br>LSPALADVPYVL<br>VKTNNVTSVAM<br>KPYEVTPTRLV                                                                                                                                                                                                                                       |

TABLE 2-continued

| Pertussis Toxic Mutant R9K, E129A |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                          |
|-----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Subunit                           | DNA sequence                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | AA sequence                                                                                                                                                                                              |
| Subunit 5                         | ATGGTGGTACCAGCGTAGCCATGAA<br>GCCGTATGAAGTACACCCGACGCGCA<br>TGCTGGTCTGCGGCATCGCCGCCAAA<br>CTGGGCGCCGCGGCGAGCCCGGA<br>CGCGCACGTGCGGTTCTGCTTCGGCA<br>AGGATCTCAAGCGTCCCGGACAGT<br>CCCATGGAAGTCATGTTGCGCGCCGT<br>CTTCATGCAACAACGGCCGCTGCGCA<br>TGTTTCTGGGTCCCAAGCAACTCACT<br>TTCGAAGCAAGCCGCGCTCGAAT<br>GATCCGGATGGTGAATGACAGCGGCA<br>AGCAGGATTGCCCTGA<br>(SEQ ID NO: 109)                                                                                                                     | CGIAAKLGAAAS<br>SPDAHVPFCFGK<br>LKRPGSSPMEVM<br>LRVFMQDQRP<br>RMFLGPKQLTFE<br>GKPALELIRMVE<br>CSGKQDCP<br>(SEQ ID<br>NO: 110)                                                                            |
| Subunit 5                         | ATGACGCGGCAAGCAGGATTGCCCT<br>GAAGGCGAACCCATGCATACCATCG<br>CATCCATCCTGTTGTCGCTGCTCGGC<br>ATATACAGCCCGGCTGACGTGCGCGG<br>CTTGCCGACCCATCTGTACAAGAACT<br>TCACTGTCCAGGAGCTGGCCCTTGAA<br>CTGAAGGGCAAGAATCAGGAGTTCTG<br>CCTGACCGCCTTCATGTGCGGCAGAA<br>GCCTGGTCCGGCGGTGCTGTCGAC<br>GCGGGACACGAGCACGACGTGGTT<br>CGACACCATGCTTGGCTTGGCATAT<br>CCGCGTATGCGCTCAAGAGCCGGATC<br>GCGCTGACGGTGGAGACTCGCCGTA<br>TCCGGGCACTCCCGGCGATCTGCTCG<br>AACTGCAGATCTGCCGCTCAACGGA<br>TATTGCGAATGA<br>(SEQ ID NO: 111) | MQRQAGLPLKAN<br>PMHTIASILLSV<br>LGIYSPADVAGL<br>PTHLYKNFTVQE<br>LALKLKGNQEF<br>CLTAFMSGRSLV<br>RACLSADAGHEHD<br>TWFDTMLGFAIS<br>AYALKSRIALTV<br>EDSPYPGTPGDL<br>LELQICPLNGYC<br>E<br>(SEQ ID<br>NO: 112) |

[0056] It is understood and herein contemplated that the disclosed polypeptides, adjuvants, and acellular vaccine components for use in eliciting an immune response or for treating, inhibiting, or preventing a Gram negative bacterial infection can be administered in compositions such as vaccines as individual polypeptides or as a fusion construct or a combination thereof. Thus, in one aspect, disclosed herein are compositions comprising a T3SA needle tip protein (such as, for example, Bsp22, LcrV, BipD, PcrV, CT053, or CT668) or an antigenic fragment thereof from a Gram negative bacteria (such as, for example, *Bordetella* spp., *Burkholderia* spp., *Chlamydia* spp., *Pseudomonas* spp., *Vibrio* spp. or *Yersinia* spp.) and a T3SA translocator protein (such as, for example, BopB, YopB, BipB, PopB, CopB, or CopB2) or an antigenic fragment thereof from a Gram negative bacteria; wherein the gram negative bacteria is not a *Salmonella enterica* or *Shigella* spp. In one aspect, the composition can comprise the needle tip protein or fragment thereof and the translocator protein or fragment thereof as separate components or as a fusion polypeptide. Also disclosed herein are compositions of any preceding aspect, wherein the composition comprises an adjuvant (such as, for example, cholera toxin, CTA1, dmlT, or LTA1) and/or bacterial toxin protein, such as a pertussis toxoid (PTd). Thus, in one aspect, disclosed herein are vaccines comprising any of the peptides, polypeptides, proteins, fusion peptides, fusion polypeptides, fusion proteins, or compositions disclosed herein. In some embodiments, the vaccine can further comprise an acellular gram negative vaccine or active components thereof.

#### 1. Sequence Similarities

[0057] It is understood that as discussed herein the use of the terms homology and identity mean the same thing as

similarity. Thus, for example, if the use of the word homology is used between two non-natural sequences it is understood that this is not necessarily indicating an evolutionary relationship between these two sequences, but rather is looking at the similarity or relatedness between their nucleic acid sequences. Many of the methods for determining homology between two evolutionarily related molecules are routinely applied to any two or more nucleic acids or proteins for the purpose of measuring sequence similarity regardless of whether they are evolutionarily related or not.

**[0058]** In general, it is understood that one way to define any known variants and derivatives or those that might arise, of the disclosed genes and proteins herein, is through defining the variants and derivatives in terms of homology to specific known sequences. This identity of particular sequences disclosed herein is also discussed elsewhere herein. In general, variants of genes and proteins herein disclosed (such as, for example, Bsp22, LcrV, BipD, PcrV, CT053, CT668, BopB, YopB, BipB, PopB, CopB, CopB2, 22BF, BurkF, PaF, YerF, CT053-CopB, CT053-CopB2, CT668-CopB, or CT668-CopB2) typically have at least about 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, or 99 percent homology to the stated sequence or the native sequence. Those of skill in the art readily understand how to determine the homology of two proteins or nucleic acids, such as genes. For example, the homology can be calculated after aligning the two sequences so that the homology is at its highest level.

**[0059]** Another way of calculating homology can be performed by published algorithms. Optimal alignment of sequences for comparison may be conducted by the local homology algorithm of Smith and Waterman *Adv. Appl. Math.* 2:482 (1981), by the homology alignment algorithm of Needleman and Wunsch, *J. Mol. Biol.* 48:443 (1970), by the search for similarity method of Pearson and Lipman, *Proc. Natl. Acad. Sci. U.S.A.* 85:2444 (1988), by computerized implementations of these algorithms (GAP, BEST-FIT, FASTA, and TFASTA in the Wisconsin Genetics Software Package, Genetics Computer Group, 575 Science Dr., Madison, WI), or by inspection.

**[0060]** It is understood that any of the methods typically can be used and that in certain instances the results of these various methods may differ, but the skilled artisan understands if identity is found with at least one of these methods, the sequences would be said to have the stated identity, and be disclosed herein.

**[0061]** For example, as used herein, a sequence recited as having a particular percent homology to another sequence refers to sequences that have the recited homology as calculated by any one or more of the calculation methods described above. For example, a first sequence has 80 percent homology, as defined herein, to a second sequence if the first sequence is calculated to have 80 percent homology to the second sequence using the Zuker calculation method even if the first sequence does not have 80 percent homology to the second sequence as calculated by any of the other calculation methods. As another example, a first sequence has 80 percent homology, as defined herein, to a second sequence if the first sequence is calculated to have 80 percent homology to the second sequence using both the Zuker calculation method and the Pearson and Lipman calculation method even if the first sequence does not have 80 percent homology to the second sequence as calculated

by the Smith and Waterman calculation method, the Needleman and Wunsch calculation method, the Jaeger calculation methods, or any of the other calculation methods. As yet another example, a first sequence has 80 percent homology, as defined herein, to a second sequence if the first sequence is calculated to have 80 percent homology to the second sequence using each of calculation methods (although, in practice, the different calculation methods will often result in different calculated homology percentages).

## 2. Nucleic Acids

**[0062]** There are a variety of molecules disclosed herein that are nucleic acid based, including for example the nucleic acids that encode, for example Bsp22, LcrV, BipD, PcrV, CT053, CT668, BopB, YopB, BipB, PopB, CopB, CopB2, 22BF, BurkF, PaF, YerF, CT053-CopB, CT053-CopB2, CT668-CopB, or CT668-CopB2 or antigenic fragments thereof, as well as various functional nucleic acids. The disclosed nucleic acids are made up of for example, nucleotides, nucleotide analogs, or nucleotide substitutes. Non-limiting examples of these and other molecules are discussed herein. It is understood that for example, when a vector is expressed in a cell, that the expressed mRNA will typically be made up of A, C, G, and U. Likewise, it is understood that if, for example, an antisense molecule is introduced into a cell or cell environment through for example exogenous delivery, it is advantageous that the antisense molecule be made up of nucleotide analogs that reduce the degradation of the antisense molecule in the cellular environment.

### a) Nucleotides and Related Molecules

**[0063]** A nucleotide is a molecule that contains a base moiety, a sugar moiety and a phosphate moiety. Nucleotides can be linked together through their phosphate moieties and sugar moieties creating an internucleoside linkage. The base moiety of a nucleotide can be adenine-9-yl (A), cytosine-1-yl (C), guanine-9-yl (G), uracil-1-yl (U), and thymine-1-yl (T). The sugar moiety of a nucleotide is a ribose or a deoxyribose. The phosphate moiety of a nucleotide is pentavalent phosphate. A non-limiting example of a nucleotide would be 3'-AMP (3'-adenosine monophosphate) or 5'-GMP (5'-guanosine monophosphate). There are many varieties of these types of molecules available in the art and available herein.

**[0064]** A nucleotide analog is a nucleotide which contains some type of modification to either the base, sugar, or phosphate moieties. Modifications to nucleotides are well known in the art and would include for example, 5-methylcytosine (5-me-C), 5-hydroxymethyl cytosine, xanthine, hypoxanthine, and 2-aminoadenine, as well as modifications at the sugar or phosphate moieties. There are many varieties of these types of molecules available in the art and available herein.

**[0065]** Nucleotide substitutes are molecules having similar functional properties to nucleotides, but which do not contain a phosphate moiety, such as peptide nucleic acid (PNA). Nucleotide substitutes are molecules that will recognize nucleic acids in a Watson-Crick or Hoogsteen manner, but which are linked together through a moiety other than a phosphate moiety. Nucleotide substitutes are able to conform to a double helix type structure when interacting

with the appropriate target nucleic acid. There are many varieties of these types of molecules available in the art and available herein.

**[0066]** It is also possible to link other types of molecules (conjugates) to nucleotides or nucleotide analogs to enhance for example, cellular uptake. Conjugates can be chemically linked to the nucleotide or nucleotide analogs. Such conjugates include but are not limited to lipid moieties such as a cholesterol moiety. (Letsinger et al., *Proc. Natl. Acad. Sci. USA*, 1989, 86, 6553-6556). There are many varieties of these types of molecules available in the art and available herein.

**[0067]** A Watson-Crick interaction is at least one interaction with the Watson-Crick face of a nucleotide, nucleotide analog, or nucleotide substitute. The Watson-Crick face of a nucleotide, nucleotide analog, or nucleotide substitute includes the C2, N1, and C6 positions of a purine based nucleotide, nucleotide analog, or nucleotide substitute and the C2, N3, C4 positions of a pyrimidine based nucleotide, nucleotide analog, or nucleotide substitute.

**[0068]** A Hoogsteen interaction is the interaction that takes place on the Hoogsteen face of a nucleotide or nucleotide analog, which is exposed in the major groove of duplex DNA. The Hoogsteen face includes the N7 position and reactive groups (NH2 or O) at the C6 position of purine nucleotides.

#### b) Sequences

**[0069]** There are a variety of sequences related to the protein molecules involved in the signaling pathways disclosed herein, for example Bsp22, LcrV, BipD, PcrV, CT053, CT668, BopB, YopB, BipB, PopB, CopB, CopB2, 22BF, BurkF, PaF, YerF, CT053-CopB, CT053-CopB2, CT668-CopB, or CT668-CopB2, or any of the nucleic acids disclosed herein for making Bsp22, LcrV, BipD, PcrV, CT053, CT668, BopB, YopB, BipB, PopB, CopB, CopB2, 22BF, BurkF, PaF, YerF, CT053-CopB, CT053-CopB2, CT668-CopB, or CT668-CopB2, all of which are encoded by nucleic acids or are nucleic acids. The sequences for the human analogs of these genes, as well as other analogs, and alleles of these genes, and splice variants and other types of variants, are available in a variety of protein and gene databases, including GENBANK®. Those of skill in the art understand how to resolve sequence discrepancies and differences and to adjust the compositions and methods relating to a particular sequence to other related sequences. Primers and/or probes can be designed for any given sequence given the information disclosed herein and known in the art.

### 3. Nucleic Acid Delivery

**[0070]** In the methods described above which include the administration and uptake of exogenous DNA into the cells of a subject (i.e., gene transduction or transfection), the disclosed nucleic acids can be in the form of naked DNA or RNA, or the nucleic acids can be in a vector for delivering the nucleic acids to the cells, whereby the antibody-encoding DNA fragment is under the transcriptional regulation of a promoter, as would be well understood by one of ordinary skill in the art. The vector can be a commercially available preparation, such as an adenovirus vector (Quantum Biotechnologies, Inc. (Laval, Quebec, Canada). Delivery of the nucleic acid or vector to cells can be via a variety of mechanisms. As one example, delivery can be via a lipo-

some, using commercially available liposome preparations such as LIPOFECTIN, LIPOFECTAMINE (GIBCO-BRL, Inc., Gaithersburg, MD), SUPERFECT (Qiagen, Inc. Hilden, Germany) and TRANSFECTAM (Promega Biotec, Inc., Madison, WI), as well as other liposomes developed according to procedures standard in the art. In addition, the disclosed nucleic acid or vector can be delivered in vivo by electroporation, the technology for which is available from Genetronics, Inc. (San Diego, CA) as well as by means of a SONOPORATION machine (ImaRx Pharmaceutical Corp., Tucson, AZ).

**[0071]** As one example, vector delivery can be via a viral system, such as a retroviral vector system which can package a recombinant retroviral genome (see e.g., Pastan et al., *Proc. Natl. Acad. Sci. U.S.A.* 85:4486, 1988; Miller et al., *Mol. Cell. Biol.* 6:2895, 1986). The recombinant retrovirus can then be used to infect and thereby deliver to the infected cells nucleic acid encoding a broadly neutralizing antibody (or active fragment thereof). The exact method of introducing the altered nucleic acid into mammalian cells is, of course, not limited to the use of retroviral vectors. Other techniques are widely available for this procedure including the use of adenoviral vectors (Mitani et al., *Hum. Gene Ther.* 5:941-948, 1994), adeno-associated viral (AAV) vectors (Goodman et al., *Blood* 84:1492-1500, 1994), lentiviral vectors (Naidini et al., *Science* 272:263-267, 1996), pseudotyped retroviral vectors (Agrawal et al., *Exper. Hematol.* 24:738-747, 1996). Physical transduction techniques can also be used, such as liposome delivery and receptor-mediated and other endocytosis mechanisms (see, for example, Schwartzenberger et al., *Blood* 87:472-478, 1996). This disclosed compositions and methods can be used in conjunction with any of these or other commonly used gene transfer methods.

**[0072]** As one example, if the antibody-encoding nucleic acid is delivered to the cells of a subject in an adenovirus vector, the dosage for administration of adenovirus to humans can range from about  $10^7$  to about  $10^9$  plaque forming units (pfu) per injection but can be as high as about  $10^{12}$  pfu per injection (Crystal, *Hum. Gene Ther.* 8:985-1001, 1997; Alvarez and Curiel, *Hum. Gene Ther.* 8:597-613, 1997). A subject can receive a single injection, or, if additional injections are necessary, they can be repeated at six month intervals (or other appropriate time intervals, as determined by the skilled practitioner) for an indefinite period and/or until the efficacy of the treatment has been established.

**[0073]** Parenteral administration of the nucleic acid or vector, if used, is generally characterized by injection. Injectables can be prepared in conventional forms, either as liquid solutions or suspensions, solid forms suitable for solution of suspension in liquid prior to injection, or as emulsions. A more recently revised approach for parenteral administration involves use of a slow release or sustained release system such that a constant dosage is maintained. For additional discussion of suitable formulations and various routes of administration of therapeutic compounds, see, e.g., *Remington: The Science and Practice of Pharmacy* (19th ed.) ed. A. R. Gennaro, Mack Publishing Company, Easton, PA 1995.

### 4. Delivery of the Compositions to Cells

**[0074]** There are a number of compositions and methods which can be used to deliver nucleic acids to cells, either in



vitro or in vivo. These methods and compositions can largely be broken down into two classes: viral based delivery systems and non-viral based delivery systems. For example, the nucleic acids can be delivered through a number of direct delivery systems such as, electroporation, lipofection, calcium phosphate precipitation, plasmids, viral vectors, viral nucleic acids, phage nucleic acids, phages, cosmids, or via transfer of genetic material in cells or carriers such as cationic liposomes. Appropriate means for transfection, including viral vectors, chemical transfectants, or physico-mechanical methods such as electroporation and direct diffusion of DNA, are described by, for example, Wolff, J. A., et al., *Science*, 247, 1465-1468, (1990); and Wolff, J. A. *Nature*, 352, 815-818, (1991). Such methods are well known in the art and readily adaptable for use with the compositions and methods described herein. In certain cases, the methods will be modified to specifically function with large DNA molecules. Further, these methods can be used to target certain diseases and cell populations by using the targeting characteristics of the carrier.

#### a) Nucleic Acid Based Delivery Systems

**[0075]** Transfer vectors can be any nucleotide construction used to deliver genes into cells (e.g., a plasmid), or as part of a general strategy to deliver genes, e.g., as part of recombinant retrovirus or adenovirus (Ram et al. *Cancer Res.* 53:83-88, (1993)).

**[0076]** As used herein, plasmid or viral vectors are agents that transport the disclosed nucleic acids, such as 22BF into the cell without degradation and include a promoter yielding expression of the gene in the cells into which it is delivered. Viral vectors are, for example, Adenovirus, Adeno-associated virus, Herpes virus, Vaccinia virus, Polio virus, AIDS virus, neuronal trophic virus, Sindbis and other RNA viruses, including these viruses with the HIV backbone. Also preferred are any viral families which share the properties of these viruses which make them suitable for use as vectors. Retroviruses include Murine Maloney Leukemia virus, MMLV, and retroviruses that express the desirable properties of MMLV as a vector. Retroviral vectors are able to carry a larger genetic payload, i.e., a transgene or marker gene, than other viral vectors, and for this reason are a commonly used vector. However, they are not as useful in non-proliferating cells. Adenovirus vectors are relatively stable and easy to work with, have high titers, and can be delivered in aerosol formulation, and can transfect non-dividing cells. Pox viral vectors are large and have several sites for inserting genes, they are thermostable and can be stored at room temperature. A preferred embodiment is a viral vector which has been engineered so as to suppress the immune response of the host organism, elicited by the viral antigens. Preferred vectors of this type will carry coding regions for Interleukin 8 or 10.

**[0077]** Viral vectors can have higher transaction (ability to introduce genes) abilities than chemical or physical methods to introduce genes into cells. Typically, viral vectors contain, nonstructural early genes, structural late genes, an RNA polymerase III transcript, inverted terminal repeats necessary for replication and encapsidation, and promoters to control the transcription and replication of the viral genome. When engineered as vectors, viruses typically have one or more of the early genes removed and a gene or gene/promotor cassette is inserted into the viral genome in place of the removed viral DNA. Constructs of this type can carry

up to about 8 kb of foreign genetic material. The necessary functions of the removed early genes are typically supplied by cell lines which have been engineered to express the gene products of the early genes in trans.

#### (1) Retroviral Vectors

**[0078]** A retrovirus is an animal virus belonging to the virus family of Retroviridae, including any types, subfamilies, genus, or tropisms. Retroviral vectors, in general, are described by Verma, I. M., *Retroviral vectors for gene transfer*.

**[0079]** A retrovirus is essentially a package which has packed into it nucleic acid cargo. The nucleic acid cargo carries with it a packaging signal, which ensures that the replicated daughter molecules will be efficiently packaged within the package coat. In addition to the package signal, there are a number of molecules which are needed in cis, for the replication, and packaging of the replicated virus. Typically a retroviral genome, contains the gag, pol, and env genes which are involved in the making of the protein coat. It is the gag, pol, and env genes which are typically replaced by the foreign DNA that it is to be transferred to the target cell. Retrovirus vectors typically contain a packaging signal for incorporation into the package coat, a sequence which signals the start of the gag transcription unit, elements necessary for reverse transcription, including a primer binding site to bind the tRNA primer of reverse transcription, terminal repeat sequences that guide the switch of RNA strands during DNA synthesis, a purine rich sequence 5' to the 3' LTR that serve as the priming site for the synthesis of the second strand of DNA synthesis, and specific sequences near the ends of the LTRs that enable the insertion of the DNA state of the retrovirus to insert into the host genome. The removal of the gag, pol, and env genes allows for about 8 kb of foreign sequence to be inserted into the viral genome, become reverse transcribed, and upon replication be packaged into a new retroviral particle. This amount of nucleic acid is sufficient for the delivery of a one to many genes depending on the size of each transcript. It is preferable to include either positive or negative selectable markers along with other genes in the insert.

**[0080]** Since the replication machinery and packaging proteins in most retroviral vectors have been removed (gag, pol, and env), the vectors are typically generated by placing them into a packaging cell line. A packaging cell line is a cell line which has been transfected or transformed with a retrovirus that contains the replication and packaging machinery, but lacks any packaging signal. When the vector carrying the DNA of choice is transfected into these cell lines, the vector containing the gene of interest is replicated and packaged into new retroviral particles, by the machinery provided in cis by the helper cell. The genomes for the machinery are not packaged because they lack the necessary signals.

#### (2) Adenoviral Vectors

**[0081]** The construction of replication-defective adenoviruses has been described (Berkner et al., *J. Virology* 61:1213-1220 (1987); Massie et al., *Mol. Cell. Biol.* 6:2872-2883 (1986); Haj-Ahmad et al., *J. Virology* 57:267-274 (1986); Davidson et al., *J. Virology* 61:1226-1239 (1987); Zhang "Generation and identification of recombinant adenovirus by liposome-mediated transfection and PCR analysis"

*BioTechniques* 15:868-872 (1993)). The benefit of the use of these viruses as vectors is that they are limited in the extent to which they can spread to other cell types, since they can replicate within an initial infected cell, but are unable to form new infectious viral particles. Recombinant adenoviruses have been shown to achieve high efficiency gene transfer after direct, in vivo delivery to airway epithelium, hepatocytes, vascular endothelium, CNS parenchyma and a number of other tissue sites (Morsy, *J. Clin. Invest.* 92:1580-1586 (1993); Kirshenbaum, *J. Clin. Invest.* 92:381-387 (1993); Roessler, *J. Clin. Invest.* 92:1085-1092 (1993); Moullier, *Nature Genetics* 4:154-159 (1993); La Salle, *Science* 259:988-990 (1993); Gomez-Foix, *J. Biol. Chem.* 267:25129-25134 (1992); Rich, *Human Gene Therapy* 4:461-476 (1993); Zabner, *Nature Genetics* 6:75-83 (1994); Guzman, *Circulation Research* 73:1201-1207 (1993); Bout, *Human Gene Therapy* 5:3-10 (1994); Zabner, *Cell* 75:207-216 (1993); Caillaud, *Eur. J. Neuroscience* 5:1287-1291 (1993); and Ragot, *J. Gen. Virology* 74:501-507 (1993)). Recombinant adenoviruses achieve gene transduction by binding to specific cell surface receptors, after which the virus is internalized by receptor-mediated endocytosis, in the same manner as wild type or replication-defective adenovirus (Chardonnet and Dales, *Virology* 40:462-477 (1970); Brown and Burlingham, *J. Virology* 12:386-396 (1973); Svensson and Persson, *J. Virology* 55:442-449 (1985); Seth, et al., *J. Virol.* 51:650-655 (1984); Seth, et al., *Mol. Cell. Biol.* 4:1528-1533 (1984); Varga et al., *J. Virology* 65:6061-6070 (1991); Wickham et al., *Cell* 73:309-319 (1993)).

**[0082]** A viral vector can be one based on an adenovirus which has had the E1 gene removed and these virions are generated in a cell line such as the human 293 cell line. In another preferred embodiment both the E1 and E3 genes are removed from the adenovirus genome.

#### 5. Adeno-Associated Viral Vectors

**[0083]** Another type of viral vector is based on an adeno-associated virus (AAV). This defective parvovirus is a preferred vector because it can infect many cell types and is nonpathogenic to humans. AAV type vectors can transport about 4 to 5 kb and wild type AAV is known to stably insert into chromosome 19. Vectors which contain this site specific integration property are preferred. An especially preferred embodiment of this type of vector is the P4.1 C vector produced by Avigen, San Francisco, CA, which can contain the herpes simplex virus thymidine kinase gene, HSV-tk, and/or a marker gene, such as the gene encoding the green fluorescent protein, GFP.

**[0084]** In another type of AAV virus, the AAV contains a pair of inverted terminal repeats (ITRs) which flank at least one cassette containing a promoter which directs cell-specific expression operably linked to a heterologous gene. Heterologous in this context refers to any nucleotide sequence or gene which is not native to the AAV or B19 parvovirus.

**[0085]** Typically the AAV and B19 coding regions have been deleted, resulting in a safe, noncytotoxic vector. The AAV ITRs, or modifications thereof, confer infectivity and site-specific integration, but not cytotoxicity, and the promoter directs cell-specific expression. U.S. Pat. No. 6,261,834 is herein incorporated by reference for material related to the AAV vector.

**[0086]** The disclosed vectors thus provide DNA molecules which are capable of integration into a mammalian chromosome without substantial toxicity.

**[0087]** The inserted genes in viral and retroviral usually contain promoters, and/or enhancers to help control the expression of the desired gene product. A promoter is generally a sequence or sequences of DNA that function when in a relatively fixed location in regard to the transcription start site. A promoter contains core elements required for basic interaction of RNA polymerase and transcription factors, and may contain upstream elements and response elements.

#### a) Large Payload Viral Vectors

**[0088]** Molecular genetic experiments with large human herpesviruses have provided a means whereby large heterologous DNA fragments can be cloned, propagated and established in cells permissive for infection with herpesviruses (Sun et al., *Nature genetics* 8:33-41, 1994; Cotter and Robertson, *Curr Opin Mol Ther* 5:633-644, 1999). These large DNA viruses (herpes simplex virus (HSV) and Epstein-Barr virus (EBV)), have the potential to deliver fragments of human heterologous DNA >150 kb to specific cells. EBV recombinants can maintain large pieces of DNA in the infected B-cells as episomal DNA. Individual clones carried human genomic inserts up to 330 kb appeared genetically stable. The maintenance of these episomes requires a specific EBV nuclear protein, EBNA1, constitutively expressed during infection with EBV. Additionally, these vectors can be used for transfection, where large amounts of protein can be generated transiently in vitro. Herpesvirus amplicon systems are also being used to package pieces of DNA >220 kb and to infect cells that can stably maintain DNA as episomes.

**[0089]** Other useful systems include, for example, replicating and host-restricted non-replicating vaccinia virus vectors.

#### b) Non-Nucleic Acid Based Systems

**[0090]** The disclosed compositions can be delivered to the target cells in a variety of ways. For example, the compositions can be delivered through electroporation, or through lipofection, or through calcium phosphate precipitation. The delivery mechanism chosen will depend in part on the type of cell targeted and whether the delivery is occurring for example in vivo or in vitro.

**[0091]** Thus, the compositions can comprise, in addition to the disclosed needle tip protein-translocator protein fusion (such as, for example, 22BF) or vectors for example, lipids such as liposomes, such as cationic liposomes (e.g., DOTMA, DOPE, DC-cholesterol) or anionic liposomes. Liposomes can further comprise proteins to facilitate targeting a particular cell, if desired. Administration of a composition comprising a compound and a cationic liposome can be administered to the blood afferent to a target organ or inhaled into the respiratory tract to target cells of the respiratory tract. Regarding liposomes, see, e.g., Brigham et al. *Am. J. Resp. Cell. Mol. Biol.* 1:95-100 (1989); Felgner et al. *Proc. Natl. Acad. Sci USA* 84:7413-7417 (1987); U.S. Pat. No. 4,897,355. Furthermore, the compound can be administered as a component of a microcapsule that can be targeted to specific cell types, such as macrophages, or

where the diffusion of the compound or delivery of the compound from the microcapsule is designed for a specific rate or dosage.

**[0092]** In the methods described above which include the administration and uptake of exogenous DNA into the cells of a subject (i.e., gene transduction or transfection), delivery of the compositions to cells can be via a variety of mechanisms. As one example, delivery can be via a liposome, using commercially available liposome preparations such as LIPOFECTIN, LIPOFECTAMINE (GIBCO-BRL, Inc., Gaithersburg, MD), SUPERFECT (Qiagen, Inc. Hilden, Germany) and TRANSFECTAM (Promega Biotec, Inc., Madison, WI), as well as other liposomes developed according to procedures standard in the art. In addition, the disclosed nucleic acid or vector can be delivered in vivo by electroporation, the technology for which is available from Genetronics, Inc. (San Diego, CA) as well as by means of a SONOPORATION machine (ImaRx Pharmaceutical Corp., Tucson, AZ).

**[0093]** The materials may be in solution, suspension (for example, incorporated into microparticles, liposomes, or cells). These may be targeted to a particular cell type via antibodies, receptors, or receptor ligands. The following references are examples of the use of this technology to target specific proteins to tumor tissue (Senter, et al., *Bioconjugate Chem.*, 2:447-451, (1991); Bagshawe, K. D., *Br. J. Cancer*, 60:275-281, (1989); Bagshawe, et al., *Br. J. Cancer*, 58:700-703, (1988); Senter, et al., *Bioconjugate Chem.*, 4:3-9, (1993); Battelli, et al., *Cancer Immunol. Immunother.*, 35:421-425, (1992); Pietersz and McKenzie, *Immunolog. Reviews*, 129:57-80, (1992); and Roffler, et al., *Biochem. Pharmacol.*, 42:2062-2065, (1991)). These techniques can be used for a variety of other specific cell types. Vehicles such as "stealth" and other antibody conjugated liposomes (including lipid mediated drug targeting to colonic carcinoma), receptor mediated targeting of DNA through cell specific ligands, lymphocyte directed tumor targeting, and highly specific therapeutic retroviral targeting of murine glioma cells in vivo. The following references are examples of the use of this technology to target specific proteins to tumor tissue (Hughes et al., *Cancer Research*, 49:6214-6220, (1989); and Litzinger and Huang, *Biochimica et Biophysica Acta*, 1104:179-187, (1992)). In general, receptors are involved in pathways of endocytosis, either constitutive or ligand induced. These receptors cluster in clathrin-coated pits, enter the cell via clathrin-coated vesicles, pass through an acidified endosome in which the receptors are sorted, and then either recycle to the cell surface, become stored intracellularly, or are degraded in lysosomes. The internalization pathways serve a variety of functions, such as nutrient uptake, removal of activated proteins, clearance of macromolecules, opportunistic entry of viruses and toxins, dissociation and degradation of ligand, and receptor-level regulation. Many receptors follow more than one intracellular pathway, depending on the cell type, receptor concentration, type of ligand, ligand valency, and ligand concentration. Molecular and cellular mechanisms of receptor-mediated endocytosis has been reviewed (Brown and Greene, *DNA and Cell Biology* 10:6, 399-409 (1991)).

**[0094]** Nucleic acids that are delivered to cells which are to be integrated into the host cell genome, typically contain integration sequences. These sequences are often viral related sequences, particularly when viral based systems are used. These viral integration systems can also be incorpo-

rated into nucleic acids which are to be delivered using a non-nucleic acid based system of deliver, such as a liposome, so that the nucleic acid contained in the delivery system can become integrated into the host genome.

**[0095]** Other general techniques for integration into the host genome include, for example, systems designed to promote homologous recombination with the host genome. These systems typically rely on sequence flanking the nucleic acid to be expressed that has enough homology with a target sequence within the host cell genome that recombination between the vector nucleic acid and the target nucleic acid takes place, causing the delivered nucleic acid to be integrated into the host genome. These systems and the methods necessary to promote homologous recombination are known to those of skill in the art.

#### c) In Vivo/Ex Vivo

**[0096]** As described above, the compositions can be administered in a pharmaceutically acceptable carrier and can be delivered to the subject's cells in vivo and/or ex vivo by a variety of mechanisms well known in the art (e.g., uptake of naked DNA, liposome fusion, intramuscular injection of DNA via a gene gun, endocytosis and the like).

**[0097]** If ex vivo methods are employed, cells or tissues can be removed and maintained outside the body according to standard protocols well known in the art. The compositions can be introduced into the cells via any gene transfer mechanism, such as, for example, calcium phosphate mediated gene delivery, electroporation, microinjection or proteoliposomes. The transduced cells can then be infused (e.g., in a pharmaceutically acceptable carrier) or homotopically transplanted back into the subject per standard methods for the cell or tissue type. Standard methods are known for transplantation or infusion of various cells into a subject.

### 6. Expression Systems

**[0098]** The nucleic acids that are delivered to cells typically contain expression controlling systems. For example, the inserted genes in viral and retroviral systems usually contain promoters, and/or enhancers to help control the expression of the desired gene product. A promoter is generally a sequence or sequences of DNA that function when in a relatively fixed location in regard to the transcription start site. A promoter contains core elements required for basic interaction of RNA polymerase and transcription factors, and may contain upstream elements and response elements.

#### a) Viral Promoters and Enhancers

**[0099]** Preferred promoters controlling transcription from vectors in mammalian host cells may be obtained from various sources, for example, the genomes of viruses such as: polyoma, Simian Virus 40 (SV40), adenovirus, retroviruses, hepatitis-B virus and most preferably cytomegalovirus, or from heterologous mammalian promoters, e.g. beta actin promoter. The early and late promoters of the SV40 virus are conveniently obtained as an SV40 restriction fragment which also contains the SV40 viral origin of replication (Fiers et al., *Nature*, 273:113 (1978)). The immediate early promoter of the human cytomegalovirus is conveniently obtained as a HindIII E restriction fragment

(Greenway, P. J. et al., *Gene* 18:355-360 (1982)). Of course, promoters from the host cell or related species also are useful herein.

**[0100]** Enhancer generally refers to a sequence of DNA that functions at no fixed distance from the transcription start site and can be either 5' (Laimins, L. et al., *Proc. Natl. Acad. Sci.* 78:993 (1981)) or 3' (Lusky, M. L., et al., *Mol. Cell Bio.* 3:1108 (1983)) to the transcription unit. Furthermore, enhancers can be within an intron (Banerji, J. L. et al., *Cell* 33:729 (1983)) as well as within the coding sequence itself (Osborne, T. F., et al., *Mol. Cell Bio.* 4:1293 (1984)). They are usually between 10 and 300 bp in length, and they function in cis. Enhancers function to increase transcription from nearby promoters. Enhancers also often contain response elements that mediate the regulation of transcription. Promoters can also contain response elements that mediate the regulation of transcription. Enhancers often determine the regulation of expression of a gene. While many enhancer sequences are now known from mammalian genes (globin, elastase, albumin,  $\alpha$ -fetoprotein and insulin), typically one will use an enhancer from a eukaryotic cell virus for general expression. Preferred examples are the SV40 enhancer on the late side of the replication origin (bp 100-270), the cytomegalovirus early promoter enhancer, the polyoma enhancer on the late side of the replication origin, and adenovirus enhancers.

**[0101]** The promotor and/or enhancer may be specifically activated either by light or specific chemical events which trigger their function. Systems can be regulated by reagents such as tetracycline and dexamethasone. There are also ways to enhance viral vector gene expression by exposure to irradiation, such as gamma irradiation, or alkylating chemotherapy drugs.

**[0102]** In certain embodiments the promoter and/or enhancer region can act as a constitutive promoter and/or enhancer to maximize expression of the region of the transcription unit to be transcribed. In certain constructs the promoter and/or enhancer region be active in all eukaryotic cell types, even if it is only expressed in a particular type of cell at a particular time. A preferred promoter of this type is the CMV promoter (650 bases). Other preferred promoters are SV40 promoters, cytomegalovirus (full length promoter), and retroviral vector LTR.

**[0103]** It has been shown that all specific regulatory elements can be cloned and used to construct expression vectors that are selectively expressed in specific cell types such as melanoma cells. The glial fibrillary acidic protein (GFAP) promoter has been used to selectively express genes in cells of glial origin.

**[0104]** Expression vectors used in eukaryotic host cells (yeast, fungi, insect, plant, animal, human or nucleated cells) may also contain sequences necessary for the termination of transcription which may affect mRNA expression. These regions are transcribed as polyadenylated segments in the untranslated portion of the mRNA encoding tissue factor protein. The 3' untranslated regions also include transcription termination sites. It is preferred that the transcription unit also contains a polyadenylation region. One benefit of this region is that it increases the likelihood that the transcribed unit will be processed and transported like mRNA. The identification and use of polyadenylation signals in expression constructs is well established. It is preferred that homologous polyadenylation signals be used in the transgene constructs. In certain transcription units, the polyade-

nylation region is derived from the SV40 early polyadenylation signal and consists of about 400 bases. It is also preferred that the transcribed units contain other standard sequences alone or in combination with the above sequences improve expression from, or stability of, the construct.

## b) Markers

**[0105]** The viral vectors can include nucleic acid sequence encoding a marker product. This marker product is used to determine if the gene has been delivered to the cell and once delivered is being expressed. Preferred marker genes are the *E. Coli* lacZ gene, which encodes  $\beta$ -galactosidase, and green fluorescent protein.

**[0106]** In some embodiments the marker may be a selectable marker. Examples of suitable selectable markers for mammalian cells are dihydrofolate reductase (DHFR), thymidine kinase, neomycin, neomycin analog G418, hygromycin, and puromycin. When such selectable markers are successfully transferred into a mammalian host cell, the transformed mammalian host cell can survive if placed under selective pressure. There are two widely used distinct categories of selective regimes. The first category is based on a cell's metabolism and the use of a mutant cell line which lacks the ability to grow independent of a supplemented media. Two examples are: CHO DHFR-cells and mouse LTK-cells. These cells lack the ability to grow without the addition of such nutrients as thymidine or hypoxanthine. Because these cells lack certain genes necessary for a complete nucleotide synthesis pathway, they cannot survive unless the missing nucleotides are provided in a supplemented media. An alternative to supplementing the media is to introduce an intact DHFR or TK gene into cells lacking the respective genes, thus altering their growth requirements. Individual cells which were not transformed with the DHFR or TK gene will not be capable of survival in non-supplemented media.

**[0107]** The second category is dominant selection which refers to a selection scheme used in any cell type and does not require the use of a mutant cell line. These schemes typically use a drug to arrest growth of a host cell. Those cells which have a novel gene would express a protein conveying drug resistance and would survive the selection. Examples of such dominant selection use the drugs neomycin, (Southern P. and Berg, P., *J. Molec. Appl. Genet.* 1:327 (1982)), mycophenolic acid, (Mulligan, R. C. and Berg, P., *Science* 209:1422 (1980)) or hygromycin, (Sugden, B. et al., *Mol. Cell. Biol.* 5:410-413 (1985)). The three examples employ bacterial genes under eukaryotic control to convey resistance to the appropriate drug G418 or neomycin (geneticin), xgpt (mycophenolic acid) or hygromycin, respectively. Others include the neomycin analog G418 and puromycin.

## 7. Peptides

### a) Protein Variants

**[0108]** As discussed herein there are numerous variants of the needle tip protein-translocator protein fusion (such as, for example, Bsp22, LcrV, BipD, PcrV, CT053, CT668, BopB, YopB, BipB, PopB, CopB, CopB2, 22BF, BurkF, PaF, YerF, CT053-CopB, CT053-CopB2, CT668-CopB, or CT668-CopB2) that are known and herein contemplated. In addition, to the known functional strain variants there are

derivatives of the needle tip protein and translocator protein which also function in the disclosed methods and compositions. Protein variants and derivatives are well understood to those of skill in the art and can involve amino acid sequence modifications. For example, amino acid sequence modifications typically fall into one or more of three classes: substitutional, insertional or deletional variants. Insertions include amino and/or carboxyl terminal fusions as well as intrasequence insertions of single or multiple amino acid residues. Insertions ordinarily will be smaller insertions than those of amino or carboxyl terminal fusions, for example, on the order of one to four residues. Immunogenic fusion protein derivatives, such as those described in the examples, are made by fusing a polypeptide sufficiently large to confer immunogenicity to the target sequence by cross-linking in vitro or by recombinant cell culture transformed with DNA encoding the fusion. Deletions are characterized by the removal of one or more amino acid residues from the protein sequence. Typically, no more than from about 2 to about 6 residues are deleted at any one site within the protein molecule. These variants ordinarily are prepared by site specific mutagenesis of nucleotides in the DNA encoding the protein, thereby producing DNA encoding the variant, and thereafter expressing the DNA in recombinant cell culture. Techniques for making substitution mutations at predetermined sites in DNA having a known sequence are well known, for example M13 primer mutagenesis and PCR mutagenesis. Amino acid substitutions are typically of single residues, but can occur at a number of different locations at once; insertions usually will be on the order of from about 1 to about 10 amino acid residues; and deletions will range from about 1 to about 30 residues. Deletions or insertions preferably are made in adjacent pairs, i.e. a deletion of 2 residues or insertion of 2 residues. Substitutions, deletions, insertions or any combination thereof may be combined to arrive at a final construct. The mutations must not place the sequence out of reading frame and preferably will not create complementary regions that could produce secondary mRNA structure. Substitutional variants are those in which at least one residue has been removed and a different residue inserted in its place. Such substitutions generally are made in accordance with the following Tables 3 and 4 and are referred to as conservative substitutions.

TABLE 3

| Amino Acid Abbreviations |               |   |
|--------------------------|---------------|---|
| Amino Acid               | Abbreviations |   |
| Alanine                  | Ala           | A |
| Allosoleucine            | AlIe          |   |
| Arginine                 | Arg           | R |
| Asparagine               | Asn           | N |
| Aspartic acid            | Asp           | D |
| Cysteine                 | Cys           | C |
| Glutamic acid            | Glu           | E |
| Glutamine                | Gln           | Q |
| Glycine                  | Gly           | G |
| Histidine                | His           | H |
| Isoleucine               | Ile           | I |
| Leucine                  | Leu           | L |
| Lysine                   | Lys           | K |
| Phenylalanine            | Phe           | F |
| Proline                  | Pro           | P |
| Pyroglutamic acid        | pGlu          |   |
| Serine                   | Ser           | S |
| Threonine                | Thr           | T |

TABLE 3-continued

| Amino Acid Abbreviations |               |   |
|--------------------------|---------------|---|
| Amino Acid               | Abbreviations |   |
| Tyrosine                 | Tyr           | Y |
| Tryptophan               | Trp           | W |
| Valine                   | Val           | V |

TABLE 4

| Amino Acid Substitutions |                                                                    |
|--------------------------|--------------------------------------------------------------------|
| Original Residue         | Exemplary Conservative Substitutions, others are known in the art. |
| Ala                      | Ser                                                                |
| Arg                      | Lys; Gln                                                           |
| Asn                      | Gln; His                                                           |
| Asp                      | Glu                                                                |
| Cys                      | Ser                                                                |
| Gln                      | Asn, Lys                                                           |
| Glu                      | Asp                                                                |
| Gly                      | Pro                                                                |
| His                      | Asn; Gln                                                           |
| Ile                      | Leu; Val                                                           |
| Leu                      | Ile; Val                                                           |
| Lys                      | Arg; Gln                                                           |
| Met                      | Leu; Ile                                                           |
| Phe                      | Met; Leu; Tyr                                                      |
| Ser                      | Thr                                                                |
| Thr                      | Ser                                                                |
| Trp                      | Tyr                                                                |
| Tyr                      | Trp; Phe                                                           |
| Val                      | Ile; Leu                                                           |

[0109] Substantial changes in function or immunological identity are made by selecting substitutions that are less conservative than those in Table 4, i.e., selecting residues that differ more significantly in their effect on maintaining (a) the structure of the polypeptide backbone in the area of the substitution, for example as a sheet or helical conformation, (b) the charge or hydrophobicity of the molecule at the target site, or (c) the bulk of the side chain. The substitutions which in general are expected to produce the greatest changes in the protein properties will be those in which (a) a hydrophilic residue, e.g. seryl or threonyl, is substituted for (or by) a hydrophobic residue, e.g. leucyl, isoleucyl, phenylalanyl, valyl or alanyl; (b) a cysteine or proline is substituted for (or by) any other residue; (c) a residue having an electropositive side chain, e.g., lysyl, arginyl, or histidyl, is substituted for (or by) an electronegative residue, e.g., glutamyl or aspartyl; (d) a residue having a bulky side chain, e.g., phenylalanine, is substituted for (or by) one not having a side chain, e.g., glycine, in this case, or (c) by increasing the number of sites for sulfation and/or glycosylation.

[0110] For example, the replacement of one amino acid residue with another that is biologically and/or chemically similar is known to those skilled in the art as a conservative substitution. For example, a conservative substitution would be replacing one hydrophobic residue for another, or one polar residue for another. The substitutions include combinations such as, for example, Gly, Ala; Val, Ile, Leu; Asp, Glu; Asn, Gln; Ser, Thr; Lys, Arg; and Phe, Tyr. Such conservatively substituted variations of each explicitly disclosed sequence are included within the mosaic polypeptides provided herein.

[0111] Substitutional or deletional mutagenesis can be employed to insert sites for N-glycosylation (Asn-X-Thr/Ser) or O-glycosylation (Ser or Thr). Deletions of cysteine or other labile residues also may be desirable. Deletions or substitutions of potential proteolysis sites, e.g. Arg, is accomplished for example by deleting one of the basic residues or substituting one by glutamyl or histidyl residues.

[0112] Certain post-translational derivatizations are the result of the action of recombinant host cells on the expressed polypeptide. Glutamyl and asparaginyl residues are frequently post-translationally deamidated to the corresponding glutamyl and asparyl residues. Alternatively, these residues are deamidated under mildly acidic conditions. Other post-translational modifications include hydroxylation of proline and lysine, phosphorylation of hydroxyl groups of seryl or threonyl residues, methylation of the o-amino groups of lysine, arginine, and histidine side chains (T. E. Creighton, *Proteins: Structure and Molecular Properties*, W. H. Freeman & Co., San Francisco pp 79-86 [1983]), acetylation of the N-terminal amine and, in some instances, amidation of the C-terminal carboxyl.

[0113] It is understood that one way to define the variants and derivatives of the disclosed proteins herein is through defining the variants and derivatives in terms of homology/identity to specific known sequences. For example, SEQ ID NO: 1 sets forth a particular sequence of *Bordetella* needle tip protein-translocator protein fusion (22BF) and SEQ ID NO: 2 sets forth a particular sequence of a 22BF fusion protein. Specifically disclosed are variants of these and other proteins herein disclosed which have at least, 70% or 75% or 80% or 85% or 90% or 95% homology to the stated sequence. Those of skill in the art readily understand how to determine the homology of two proteins. For example, the homology can be calculated after aligning the two sequences so that the homology is at its highest level.

[0114] Another way of calculating homology can be performed by published algorithms. Optimal alignment of sequences for comparison may be conducted by the local homology algorithm of Smith and Waterman *Adv. Appl. Math.* 2:482 (1981), by the homology alignment algorithm of Needleman and Wunsch, *J. Mol. Biol.* 48:443 (1970), by the search for similarity method of Pearson and Lipman, *Proc. Natl. Acad. Sci. U.S.A.* 85:2444 (1988), by computerized implementations of these algorithms (GAP, BEST-FIT, FASTA, and TFASTA in the Wisconsin Genetics Software Package, Genetics Computer Group, 575 Science Dr., Madison, WI), or by inspection.

[0115] The same types of homology can be obtained for nucleic acids by for example the algorithms disclosed in Zuker, M. *Science* 244:48-52, 1989, Jaeger et al. *Proc. Natl. Acad. Sci. USA* 86:7706-7710, 1989, Jaeger et al. *Methods Enzymol.* 183:281-306, 1989.

[0116] It is understood that the description of conservative mutations and homology can be combined together in any combination, such as embodiments that have at least 70% homology to a particular sequence wherein the variants are conservative mutations.

[0117] As this specification discusses various proteins and protein sequences it is understood that the nucleic acids that can encode those protein sequences are also disclosed. This would include all degenerate sequences related to a specific protein sequence, i.e. all nucleic acids having a sequence that encodes one particular protein sequence as well as all

nucleic acids, including degenerate nucleic acids, encoding the disclosed variants and derivatives of the protein sequences. Thus, while each particular nucleic acid sequence may not be written out herein, it is understood that each and every sequence is in fact disclosed and described herein through the disclosed protein sequence. For example, one of the many nucleic acid sequences that can encode the protein sequence set forth in SEQ ID NO: 2 is set forth in SEQ ID NO: 1. It is understood that for this mutation all of the nucleic acid sequences that encode this particular derivative of the 22BF are also disclosed. It is also understood that while no amino acid sequence indicates what particular DNA sequence encodes that protein within an organism, where particular variants of a disclosed protein are disclosed herein, the known nucleic acid sequence that encodes that protein in the particular needle tip protein-translocator protein fusion (such as, for example, 22BF) from which that protein arises is also known and herein disclosed and described.

[0118] It is understood that there are numerous amino acid and peptide analogs which can be incorporated into the disclosed compositions. For example, there are numerous D amino acids or amino acids which have a different functional substituent than the amino acids shown in Table 3 and Table 4. The opposite stereo isomers of naturally occurring peptides are disclosed, as well as the stereo isomers of peptide analogs. These amino acids can readily be incorporated into polypeptide chains by charging tRNA molecules with the amino acid of choice and engineering genetic constructs that utilize, for example, amber codons, to insert the analog amino acid into a peptide chain in a site specific way.

[0119] Molecules can be produced that resemble peptides, but which are not connected via a natural peptide linkage. For example, linkages for amino acids or amino acid analogs can include  $-\text{CH}_2\text{NH}-$ ,  $-\text{CH}_2\text{S}-$ ,  $-\text{CH}_2-\text{CH}_2-$ ,  $-\text{CH}=\text{CH}-$  (cis and trans),  $-\text{COCH}_2-$ ,  $-\text{CH}(\text{OH})\text{CH}_2-$ , and  $-\text{CHH}_2\text{SO}-$  (These and others can be found in Spatola, A. F. in *Chemistry and Biochemistry of Amino Acids, Peptides, and Proteins*, B. Weinstein, eds., Marcel Dekker, New York, p. 267 (1983); Spatola, A. F., *Vega Data* (March 1983), Vol. 1, Issue 3, Peptide Backbone Modifications (general review); Morley, *Trends Pharm Sci* (1980) pp. 463-468; Hudson, D. et al., *Int J Pept Prot Res* 14:177-185 (1979) ( $-\text{CH}_2\text{NH}-$ ,  $-\text{CH}_2\text{CH}_2-$ ); Spatola et al. *Life Sci* 38:1243-1249 (1986) ( $-\text{CH}_2-\text{S}-$ ); Hann *J. Chem. Soc Perkin Trans. I* 307-314 (1982) ( $-\text{CH}=\text{CH}-$ , cis and trans); Almquist et al. *J. Med. Chem.* 23:1392-1398 (1980) ( $-\text{COCH}_2-$ ); Jennings-White et al. *Tetrahedron Lett* 23:2533 (1982) ( $-\text{COCH}_2-$ ); Szelke et al. *European Appln*, EP 45665 CA (1982): 97:39405 (1982) ( $-\text{CH}(\text{OH})\text{CH}_2-$ ); Holladay et al. *Tetrahedron. Lett* 24:4401-4404 (1983) ( $-\text{C}(\text{OH})\text{CH}_2-$ ); and Hruby *Life Sci* 31:189-199 (1982) ( $-\text{CH}_2-\text{S}-$ ); each of which is incorporated herein by reference. A particularly preferred non-peptide linkage is  $-\text{CH}_2\text{NH}-$ . It is understood that peptide analogs can have more than one atom between the bond atoms, such as  $\beta$ -alanine,  $\gamma$ -aminobutyric acid, and the like.

[0120] Amino acid analogs and analogs and peptide analogs often have enhanced or desirable properties, such as, more economical production, greater chemical stability, enhanced pharmacological properties (half-life, absorption, potency, efficacy, etc.), altered specificity (e.g., a broad-spectrum of biological activities), reduced antigenicity, and others.

[0121] D-amino acids can be used to generate more stable peptides, because D amino acids are not recognized by peptidases and such. Systematic substitution of one or more amino acids of a consensus sequence with a D-amino acid of the same type (e.g., D-lysine in place of L-lysine) can be used to generate more stable peptides. Cysteine residues can be used to cyclize or attach two or more peptides together. This can be beneficial to constrain peptides into particular conformations.

#### 8. Pharmaceutical Carriers/Delivery of Pharmaceutical Products

[0122] As described above, the compositions can also be administered in vivo in a pharmaceutically acceptable carrier. By “pharmaceutically acceptable” is meant a material that is not biologically or otherwise undesirable, i.e., the material may be administered to a subject, along with the nucleic acid or vector, without causing any undesirable biological effects or interacting in a deleterious manner with any of the other components of the pharmaceutical composition in which it is contained. The carrier would naturally be selected to minimize any degradation of the active ingredient and to minimize any adverse side effects in the subject, as would be well known to one of skill in the art.

[0123] The compositions may be administered orally, parenterally (e.g., intravenously), by intramuscular injection, by intraperitoneal injection, transdermally, extracorporeally, topically or the like, including topical intranasal administration or administration by inhalant. As used herein, “topical intranasal administration” means delivery of the compositions into the nose and nasal passages through one or both of the nares and can comprise delivery by a spraying mechanism or droplet mechanism, or through aerosolization of the nucleic acid or vector. Administration of the compositions by inhalant can be through the nose or mouth via delivery by a spraying or droplet mechanism. Delivery can also be directly to any area of the respiratory system (e.g., lungs) via intubation. The exact amount of the compositions required will vary from subject to subject, depending on the species, age, weight and general condition of the subject, the severity of the allergic disorder being treated, the particular nucleic acid or vector used, its mode of administration and the like. Thus, it is not possible to specify an exact amount for every composition. However, an appropriate amount can be determined by one of ordinary skill in the art using only routine experimentation given the teachings herein.

[0124] Parenteral administration of the composition, if used, is generally characterized by injection. Injectables can be prepared in conventional forms, either as liquid solutions or suspensions, solid forms suitable for solution of suspension in liquid prior to injection, or as emulsions. A more recently revised approach for parenteral administration involves use of a slow release or sustained release system such that a constant dosage is maintained. See, e.g., U.S. Pat. No. 3,610,795, which is incorporated by reference herein.

[0125] The materials may be in solution, suspension (for example, incorporated into microparticles, liposomes, or cells). These may be targeted to a particular cell type via antibodies, receptors, or receptor ligands. The following references are examples of the use of this technology to target specific proteins to tumor tissue (Senter, et al., *Bioconjugate Chem.*, 2:447-451, (1991); Bagshawe, K. D., *Br. J. Cancer*, 60:275-281, (1989); Bagshawe, et al., *Br. J. Cancer*, 58:700-703, (1988); Senter, et al., *Bioconjugate*

*Chem.*, 4:3-9, (1993); Battelli, et al., *Cancer Immunol. Immunother.*, 35:421-425, (1992); Pietersz and McKenzie, *Immunolog. Reviews*, 129:57-80, (1992); and Roffler, et al., *Biochem. Pharmacol.*, 42:2062-2065, (1991)). Vehicles such as “stealth” and other antibody conjugated liposomes (including lipid mediated drug targeting to colonic carcinoma), receptor mediated targeting of DNA through cell specific ligands, lymphocyte directed tumor targeting, and highly specific therapeutic retroviral targeting of murine glioma cells in vivo. The following references are examples of the use of this technology to target specific proteins to tumor tissue (Hughes et al., *Cancer Research*, 49:6214-6220, (1989); and Litzinger and Huang, *Biochimica et Biophysica Acta*, 1104:179-187, (1992)). In general, receptors are involved in pathways of endocytosis, either constitutive or ligand induced. These receptors cluster in clathrin-coated pits, enter the cell via clathrin-coated vesicles, pass through an acidified endosome in which the receptors are sorted, and then either recycle to the cell surface, become stored intracellularly, or are degraded in lysosomes. The internalization pathways serve a variety of functions, such as nutrient uptake, removal of activated proteins, clearance of macromolecules, opportunistic entry of viruses and toxins, dissociation and degradation of ligand, and receptor-level regulation. Many receptors follow more than one intracellular pathway, depending on the cell type, receptor concentration, type of ligand, ligand valency, and ligand concentration. Molecular and cellular mechanisms of receptor-mediated endocytosis has been reviewed (Brown and Greene, *DNA and Cell Biology* 10:6, 399-409 (1991)).

#### a) Pharmaceutically Acceptable Carriers

[0126] The compositions, including antibodies, can be used therapeutically in combination with a pharmaceutically acceptable carrier.

[0127] Suitable carriers and their formulations are described in *Remington: The Science and Practice of Pharmacy* (19th ed.) ed. A. R. Gennaro, Mack Publishing Company, Easton, PA 1995. Typically, an appropriate amount of a pharmaceutically-acceptable salt is used in the formulation to render the formulation isotonic. Examples of the pharmaceutically-acceptable carrier include, but are not limited to, saline, Ringer's solution and dextrose solution. The pH of the solution is preferably from about 5 to about 8, more preferably from about 7 to about 7.6, and most preferably about 7.5. Further carriers include sustained release preparations such as semipermeable matrices of solid hydrophobic polymers containing the antibody, which matrices are in the form of shaped articles, e.g., films, liposomes or microparticles. It will be apparent to those persons skilled in the art that certain carriers may be more preferable depending upon, for instance, the route of administration and concentration of composition being administered.

[0128] Pharmaceutical carriers are known to those skilled in the art. These most typically would be standard carriers for administration of drugs to humans, including solutions such as sterile water, saline, and buffered solutions at physiological pH. The compositions can be administered intramuscularly or subcutaneously. Other compounds will be administered according to standard procedures used by those skilled in the art.

[0129] Pharmaceutical compositions may include carriers, thickeners, diluents, buffers, preservatives, surface active agents and the like in addition to the molecule of choice.

Pharmaceutical compositions may also include one or more active ingredients such as antimicrobial agents, anti-inflammatory agents, anesthetics, and the like.

**[0130]** The pharmaceutical composition may be administered in a number of ways depending on whether local or systemic treatment is desired, and on the area to be treated. Administration may be topically (including ophthalmically, vaginally, rectally, intranasally), orally, by inhalation, or parenterally, for example by intravenous drip, subcutaneous, intraperitoneal or intramuscular injection. The disclosed antibodies can be administered intravenously, intraperitoneally, intramuscularly, subcutaneously, intracavity, or transdermally.

**[0131]** Preparations for parenteral administration include sterile aqueous or non-aqueous solutions, suspensions, and emulsions. Examples of non-aqueous solvents are propylene glycol, polyethylene glycol, vegetable oils such as olive oil, and injectable organic esters such as ethyl oleate. Aqueous carriers include water, alcoholic/aqueous solutions, emulsions or suspensions, including saline and buffered media. Parenteral vehicles include sodium chloride solution, Ringer's dextrose, dextrose and sodium chloride, lactated Ringer's, or fixed oils. Intravenous vehicles include fluid and nutrient replenishers, electrolyte replenishers (such as those based on Ringer's dextrose), and the like. Preservatives and other additives may also be present such as, for example, antimicrobials, anti-oxidants, chelating agents, and inert gases and the like.

**[0132]** Formulations for topical administration may include ointments, lotions, creams, gels, drops, suppositories, sprays, liquids and powders. Conventional pharmaceutical carriers, aqueous, powder or oily bases, thickeners and the like may be necessary or desirable.

**[0133]** Compositions for oral administration include powders or granules, suspensions or solutions in water or non-aqueous media, capsules, sachets, or tablets. Thickeners, flavorings, diluents, emulsifiers, dispersing aids or binders may be desirable.

**[0134]** Some of the compositions may potentially be administered as a pharmaceutically acceptable acid- or base-addition salt, formed by reaction with inorganic acids such as hydrochloric acid, hydrobromic acid, perchloric acid, nitric acid, thiocyanic acid, sulfuric acid, and phosphoric acid, and organic acids such as formic acid, acetic acid, propionic acid, glycolic acid, lactic acid, pyruvic acid, oxalic acid, malonic acid, succinic acid, maleic acid, and fumaric acid, or by reaction with an inorganic base such as sodium hydroxide, ammonium hydroxide, potassium hydroxide, and organic bases such as mono-, di-, tri-alkyl and aryl amines and substituted ethanolamines.

#### b) Therapeutic Uses

**[0135]** Effective dosages and schedules for administering the compositions may be determined empirically, and making such determinations is within the skill in the art. The dosage ranges for the administration of the compositions are those large enough to produce the desired effect in which the symptoms of the disorder are effected. The dosage should not be so large as to cause adverse side effects, such as unwanted cross-reactions, anaphylactic reactions, and the like. Generally, the dosage will vary with the age, condition, sex and extent of the disease in the patient, route of administration, or whether other drugs are included in the regimen, and can be determined by one of skill in the art.

The dosage can be adjusted by the individual physician in the event of any counterindications. Dosage can vary, and can be administered in one or more dose administrations daily, for one or several days. Guidance can be found in the literature for appropriate dosages for given classes of pharmaceutical products. For example, guidance in selecting appropriate doses for antibodies can be found in the literature on therapeutic uses of antibodies, e.g., *Handbook of Monoclonal Antibodies*, Ferrone et al., eds., Noyes Publications, Park Ridge, N.J., (1985) ch. 22 and pp. 303-357; Smith et al., *Antibodies in Human Diagnosis and Therapy*, Haber et al., eds., Raven Press, New York (1977) pp. 365-389. A typical daily dosage of the antibody used alone might range from about 1 µg/kg to up to 100 mg/kg of body weight or more per day, depending on the factors mentioned above.

**[0136]** In a preferred embodiment, the amount of protein that is administered per dose of vaccine is in the range of from about 0.0001 to about 1000 µg/kg. In one embodiment, the amount is in the range of from about 0.001 to about 1000 µg/kg of body weight of the recipient. In one embodiment, the amount is in the range of from about 0.01 to about 1000 µg/kg of body weight of the recipient. In one embodiment, the amount is in the range of from about 0.01 to about 100 µg/kg of body weight of the recipient. Those of skill in the art will recognize that the precise dosage may vary from situation to situation and from patient to patient, depending on e.g. age, gender, overall health, various genetic factors, and other variables known to those of skill in the art. Dosages are typically determined e.g. in the course of animal and/or human clinical trials as conducted by skilled medical personnel, e.g. physicians or veterinarians.

#### C. METHODS OF USING THE COMPOSITIONS

**[0137]** Herein, the protective efficacy of the *Bordetella* spp. tip/translocator fusion, 22BF, is examined against lethal lung challenge and with complete (sterilizing) clearance of colonizing bacteria. Unlike some components of the current aP vaccine, Bsp22 and BopB are required for infection and are not mutable since they must be retained structurally and functionally within the context of a large nanomachine residing within the *Bordetella* cell envelope. Furthermore, targeting the *Bordetella* T3SA renders the pathogen less able to fight off the host innate and adaptive immune responses. Regardless of whether 22BF is protective alone or when used with components of the current aP vaccine, the innovation of this high risk, high reward investigation lies in whether this subunit vaccine can elicit sterilizing immunity and thereby prevent the colonization that results in host to host transmission. It has been reported that Bsp22 (a component of the 22BF fusion vaccine) does not elicit a serum antibody response in humans during the course of natural infection and is not a protective antigen in mice. Nevertheless, as shown herein, protective and sterilizing immunity can be obtained with the compositions disclosed herein.

**[0138]** Thus, in one aspect, disclosed herein are methods of eliciting an immune response in a subject to a Gram negative bacteria (such as, for example, *Bordetella* spp., *Burkholderia* spp., *Chlamydia* spp., *Pseudomonas* spp., *Vibrio* spp. Enteropathogenic or Enterohemorrhagic *E. coli* or *Yersinia* spp.) comprising administering to the subject the fusion polypeptides, compositions, or vaccines disclosed herein. Accordingly, in one aspect, disclosed herein are methods of eliciting an immune response against at least one



Gram negative bacteria serovar in a subject in need thereof, comprising administering to the subject a composition comprising at least one needle tip protein or a fragment thereof and/or at least one translocator protein or a fragment thereof; wherein said composition is administered in an amount sufficient to elicit an immune response to said at least one Gram negative bacteria serovar in said subject; and wherein the Gram negative bacteria is not a *Shigella* spp. or *Salmonella* spp. In one aspect, the immune response elicited provides sterilizing immunity to the infectious bacterium.

**[0139]** As can be appreciated by the skilled artisan, the methods of eliciting an immune response can be used for the purpose of treating, inhibiting, or preventing an infection of a Gram negative bacteria (such as, for example, *Bordetella* spp., *Burkholderia* spp., *Chlamydia* spp., *Pseudomonas* spp., *Vibrio* spp. Enteropathogenic or Enterohemorrhagic *E. coli* or *Yersinia* spp.). Thus, in one aspect, disclosed herein are methods of treating, inhibiting, or preventing an infection of a Gram negative bacteria in a subject comprising administering to the subject a therapeutic amount of any of the fusion polypeptides, compositions, or vaccines disclosed herein. As one goal of any vaccine is not only to prevent infection or reducing the severity of disease in the individual receiving the vaccine, but also to prevent further transmission of the infectious agent (sterilizing immunity), it is understood and herein contemplated that the disclosed methods of treatment, inhibition, or preventing an infection can further comprise inhibiting and/or preventing colony formation of the bacteria and/or transmission of the bacteria to another subject.

**[0140]** The term “therapeutically effective” refers to the amount of the composition used that is of sufficient quantity to ameliorate one or more causes or symptoms of a disease or disorder. Such amelioration only requires a reduction or alteration, not necessarily elimination.

**[0141]** The term “treatment” refers to the medical management of a patient with the intent to cure, ameliorate, stabilize, or prevent a disease, pathological condition, or disorder. This term includes active treatment, that is, treatment directed specifically toward the improvement of a disease, pathological condition, or disorder, and also includes causal treatment, that is, treatment directed toward removal of the cause of the associated disease, pathological condition, or disorder. In addition, this term includes palliative treatment, that is, treatment designed for the relief of symptoms rather than the curing of the disease, pathological condition, or disorder; preventative treatment, that is, treatment directed to minimizing or partially or completely inhibiting the development of the associated disease, pathological condition, or disorder; and supportive treatment, that is, treatment employed to supplement another specific therapy directed toward the improvement of the associated disease, pathological condition, or disorder.

**[0142]** The term “inhibit” refers to a decrease in an activity, response, condition, disease, or other biological parameter. This can include, but is not limited to, the complete ablation of the activity, response, condition, or disease. This can also include, for example, a 10% reduction in the activity, response, condition, or disease as compared to the native or control level. Thus, the reduction can be a 10, 20, 30, 40, 50, 60, 70, 80, 90, 100%, or any amount of reduction in between as compared to native or control levels.

**[0143]** By “reduce” or other forms of the word, such as “reducing” or “reduction,” is meant lowering of an event or

characteristic (e.g., tumor growth). It is understood that this is typically in relation to some standard or expected value, in other words it is relative, but that it is not always necessary for the standard or relative value to be referred to. For example, “reduces tumor growth” means reducing the rate of growth of a tumor relative to a standard or a control.

**[0144]** By “prevent” or other forms of the word, such as “preventing” or “prevention,” is meant to stop a particular event or characteristic, to stabilize or delay the development or progression of a particular event or characteristic, or to minimize the chances that a particular event or characteristic will occur. Prevent does not require comparison to a control as it is typically more absolute than, for example, reduce. As used herein, something could be reduced but not prevented, but something that is reduced could also be prevented. Likewise, something could be prevented but not reduced, but something that is prevented could also be reduced. It is understood that where reduce or prevent are used, unless specifically indicated otherwise, the use of the other word is also expressly disclosed.

#### D. EXAMPLES

**[0145]** The following examples are put forth so as to provide those of ordinary skill in the art with a complete disclosure and description of how the compounds, compositions, articles, devices and/or methods claimed herein are made and evaluated, and are intended to be purely exemplary and are not intended to limit the disclosure. Efforts have been made to ensure accuracy with respect to numbers (e.g., amounts, temperature, etc.), but some errors and deviations should be accounted for. Unless indicated otherwise, parts are parts by weight, temperature is in ° C. or is at ambient temperature, and pressure is at or near atmospheric.

##### 1. Example 1

a) Use of a T3SS Needle Tip/Translocator Protein Fusion as a Protective Antigen Against *B. pertussis*:

**[0146]** The dominant antigen eliciting protection against Gram-negative pathogens is LPS, which confers O-antigen serotype specificity. The initial project focused on *Shigella*, however, there are at least 58 distinct *Shigella* serotypes. This reduces broad-spectrum efficacy for live, attenuated and whole, killed vaccines, which tend to be somatic antigen driven. IpaD and IpaB are the surface-localized needle tip and first translocator proteins of the T3SA, respectively. They are essential for virulence in all *Shigella*, are >98% conserved across all *Shigella* species, and provide serotype-independent protection. When IpaD+IpaB+dmLT was given IN to mice, the formulation was about 80-90% protective against lethal challenge by homologous and heterologous *Shigella* spp. (Table 5). To reduce the production cost, IpaD and IpaB were genetically fused to make DBF. Not only did the DBF provide protection against lethal challenge, it also unexpectedly increased the cell-mediated immunity, most notably the IL-17 and IFN- $\gamma$  responses. When the *Salmonella enterica* tip and first translocator proteins of the T3SSs of SPI-1 and SPI-2 (*Salmonella* Pathogenicity Islands 1 and 2) were fused, about 70% protection against lethal challenge by two *S. enterica* serovars was observed when both fusion proteins were administered simultaneously (Table 5).

TABLE 5

| Protective efficacy of tip/translocator fusions<br>against challenge by appropriate pathogen. |                |                                               |
|-----------------------------------------------------------------------------------------------|----------------|-----------------------------------------------|
| Vaccine                                                                                       | Protection (%) | Pathogen                                      |
| DBF + dmLT                                                                                    | 80-90          | <i>S. flexneri</i> , <i>S. sonnei</i>         |
| S1S2 + MPL (IM)                                                                               | 70             | <i>S. Typhimurium</i> , <i>S. Enteritidis</i> |
| 22BF + dmLT                                                                                   | 100            | <i>B. bronchiseptica</i>                      |

Mice (n = 10) were vaccinated 3 times biweekly with the vaccine and then challenged with indicated pathogen at day 56. Protection is indicated as percent survival after 21 days.

[0147] *B. bronchiseptica* and *B. pertussis* infections have been shown to also require a T3SS for virulence. To determine if a subunit of the T3SS could be used to confer protection, *B. pertussis* Bsp22 and BopB, the T3SS needle tip and first translocator proteins, which are 98% conserved with those of *B. bronchiseptica* were genetically fused. Mice were vaccinated biweekly, three times with 22BF, 22BF+dmLT, Bsp22+dmLT, or PBS. One group was also vaccinated twice with the Zoetis canine vaccine that is a killed cellular extract of *B. bronchiseptica*. The 22BF+dmLT exhibited 100% protection against *B. bronchiseptica* in a mouse lethal pulmonary model, while the commercial vaccine provided 80% protection (FIG. 1). Moreover, the mice vaccinated with 22BF+dmLT did not show signs of illness and they actually gained weight throughout the challenge (FIG. 2). In contrast, the mice vaccinated with the Zoetis vaccine lost weight and then regained it, however, they continued to display lower health scores than the mice vaccinated with 22BF+dmLT for the remainder of the experiment (FIG. 2).

[0148] Next, the protective efficacy of the 22BF fusion with and without dmLT and in comparison to the needle tip protein Bsp22 and translocator protein BopB and the effect of fusion of dmLT relative to concurrent administration was examined. Mice were vaccinated IN on days 0, 14, 28 and challenged with *Bordetella bronchiseptica* on day 56. On day 7 mice were sacrificed and the colony forming units (CFU)/lung were determined (FIG. 3). 22BF+dmLT reduced the *B. bronchiseptica* CFU lung burden by 96.5%. Blood and fecal samples were collected on days-1, 13, 27, 41, and 55. The kinetics of IgG were assessed. Typical logarithmic increases were seen (FIG. 4) regardless of the antigen on the well. No IgA was detected in fecal samples of any of the mice. Antibody secreting cells in the bone marrow, spleen, and lungs (FIG. 5) were stimulated with Bsp22, BopB and dmLT. All organs from mice that had been vaccinated with the single proteins or double proteins or fusion, as long as the dmLT was present, exhibited some level of IgG or IgA ASCs

[0149] While protection is important, the key to a next generation subunit vaccine against *Bordetella* spp. must be that it exhibits sterilizing immunity (to prevent carriage). The lungs of the mice that survived the challenge of FIG. 2 were examined to look for sterilizing immunity. All of the remaining 22BF and PBS vaccinated mice exhibited CFU/lung. In contrast, the 22BF+dmLT vaccinated mice demonstrated 38% sterilizing immunity while only 12% of the mice vaccinated with the Zoetis vaccine demonstrated sterilizing immunity. Thus, the 22BF represents an innovative new antigen to protect against *B. bronchiseptica* and *B. pertussis* due to the conservation of these antigens.

[0150] It is shown herein that the use of a broad, serotype-independent subunit vaccine against *Shigella* spp. and *S.*

*enterica* serotypes. These vaccines are based on the fusion of the T3SA tip and first translocator proteins, which are highly conserved within a given bacterial genus. The DBF has also been shown to protect monkeys from severe diarrhea and *S. enterica* S1S2 fusions are protective in a bovine calf model. A genetic fusion of the *B. pertussis* T3SA tip/translocator system, 22BF, was generated which protects 100% of the mice challenged with a lethal dose of the heterologous *B. bronchiseptica*. Furthermore, 22BF elicited 38% sterilizing immunity while the commercial vaccine provided 12%. Herein, not only was the *B. bronchiseptica* experiment repeated, but also it demonstrated that the 22BF+dmLT protects against *B. pertussis*. *B. bronchiseptica* was chosen as the first challenge pathogen since it lacks pertussis toxin (PT), which could conceivably compromise adequate assessment of the 22BF. PT is an especially important toxin secreted by *B. pertussis* that can significantly affect the health of the host. Shiga toxin was also considered when DBF vaccinated mice were challenged with *S. dysenteriae*, however, DBF protected 40% of those vaccinated mice while the individual proteins did not. Furthermore, due to the disease progression of *S. dysenteriae*, mice killed by the Shiga toxin versus those succumbing to the bacterial infection (based on health score immediately preceding death) were distinguished. 22BF provides some level of protection against *B. pertussis*, which can be boosted by the addition of pertussis toxoid (PTd). At the end of this study, it was demonstrated that 22BF protects against both *B. bronchiseptica* and *B. pertussis*. Results for two vaccination routes and doses are obtained. PTd is required in the protection against *B. pertussis* was shown herein. Finally, it is demonstrated that 22BF+dmLT elicits >90, if not 100%, sterilizing immunity.

b) Cytokine Assay

[0151] Cytokines were collected from splenocytes collected on Day 55. 100  $\mu$ L of homogenized splenocytes were seeded at a concentration of  $5 \times 10^6$  cells/mL onto a flat-bottom 96-well plate. 100  $\mu$ L of BopB, Bsp22, or dmLT at a concentration of 20  $\mu$ g/mL was added onto cells (bringing final protein concentration to 10  $\mu$ g/mL). Plates were incubated at 37° C. and 5% CO<sub>2</sub> for 48 hours. After incubation, plates were centrifuged at 1600 rpm at 4° C. for seven minutes. Supernatants were collected and stored at -20° C. until analysis. Cytokines were analyzed using the MSD® U-Plex Platform Multiplex Assay for the following cytokines for the following immune responses: Th1: IFN- $\gamma$ , TNF- $\alpha$ , IL-1B, IL-2, IL-6, IL-10; (FIGS. 6 and 7) Th17: IL-17A; (FIG. 8) Th2: IL-4, IL-5 (FIG. 9).

2. Example 2: Assess the Respective Humoral and Cell-Mediated Immune Responses Elicited by 22BF+dmLT Delivered IN and IM

[0152] IN delivery of 20  $\mu$ g 22BF+2.5  $\mu$ g dmLT protects 100% of vaccinated mice from death and provide 38% sterilizing immunity in the lungs following *B. bronchiseptica* challenge. 50  $\mu$ g or 15  $\mu$ g 22BF IN is delivered and then in parallel deliver 100  $\mu$ g or 20  $\mu$ g 22BF IM or ID to assess the humoral and cell mediated responses in each case. Based on previous work, it is known that some of these dose/route combinations do not protect mice while others offer protection.

[0153] First the effect of the route of administration of 22BF following challenge was assessed. Group 1 is a 22BF+2.5 µg dmLT IN, group 2 is 22BF+2.5 µg dmLT IM, and group 3 is a 22BF+2.5 µg dmLT IN. For each administrative route, PBS vaccinated controls were included (Groups 4, 5, and 6). Group 7 (n=10) is vaccinated subcutaneously on day 1 and 21 with the Zoetis vaccine. After day 56 following initial vaccination, mice were challenged with a sublethal dosage of *B. pertussis* intranasally. There was an observable difference in weight loss between mice vaccinated with the 22BF+dmLT+PTd formulation and those that only received PBS. By Day 7 all mice aside from PBS treated mice had recovered to within 3% of pre infection weight.

[0154] The immune responses can be compared with the protective efficacy and potentially define a protective correlate in the mouse model. The correlate is useful in the development of the 22BF vaccine, making the necessary adjustments when it is translated to humans. The systemic immune response can be assessed by measuring serum IgG against BopB, Bsp22, and dmLT, as well as the mucosal immune response by assessing IgA in fecal pellets. Cytokine secretion is assessed in splenocytes from vaccinated mice.

[0155] This experiment can be performed in two parts. The IN route can use a high dose of 50 µg and low dose of 15 µg. The IM route uses a high dose of 100 µg and a low dose of 40 µg. For each route, three groups of female C57BL/6 mice (10/group) are vaccinated on days 0, 14 and 28. Group 1 is a 22BF+2.5 µg dmLT IN, group 2 is 22BF+2.5 µg dmLT IM, and group 3 is a 22BF+2.5 µg dmLT IN. For each administrative route, PBS vaccinated controls were included (Groups 4, 5, and 6). Group 7 (n=10) is vaccinated subcutaneously on day 1 and 21 with the Zoetis vaccine. Blood and fecal pellets are collected on days -1, 13, 27, 41 and 55 to assess mucosal and systemic humoral responses. Individual samples are tested for the presence of anti-Bsp22, -BopB, -dmLT, -PTd IgG and IgA antibodies by ELISA. Mice were immunized on days 0, 14, and 28 with 22BF+PTd admixed with dmLT. Serum IgG antibodies specific for BopB, Bsp22, PTd, and dmLT were measured by ELISA. and IFN-γ/IL-17A secreting cells by ELISpot and cytokine secretion using Multi array assays. BAL is collected to measure IgG and IgA responses. GraphPad Prism 5.04 can be used for graphics and statistical comparisons. Differences were analyzed using t test or ANOVA where appropriate. A P value of less than 0.05 is considered significant for all comparisons.

[0156] For these experiments, serum IgG levels are >10<sup>5</sup> EU/ml, antibody secreting cells at >50 IgG ASC/10<sup>6</sup> cells or >20 IgA ASC/10<sup>6</sup> cells, and cytokine secreting cells at >50 IFN-γ/10<sup>6</sup> cells and IL-17/10<sup>6</sup> cells. There can also be unique systemic and mucosal humoral immune responses from mice immunized via the IN and IM routes. 50 µg IN was chosen to facilitate an increase in sterilizing immunity. The 100 µg IM dose was based on prior findings. Antibody secreting cells specific for both proteins are detected in both mucosal and memory compartments. Finally, the full profile of cytokine secretion elicited by the vaccine can demonstrate a dose and administration route dependence. Thus, these two routes (each with a high and low dose) are expected to give rise to unique immune response profiles.

### 3. Example 3: Determine the Protective Efficacy of 22BF+dmLT Against *B. bronchiseptica* and *B. pertussis* Challenge

[0157] As discussed above, it is demonstrated herein that initial protective efficacy of 22BF+dmLT against *B. bronchiseptica* challenge. Insight is gained into the immune responses elicited by two doses of 22BF delivered IN and IM. Here, mice can be vaccinated and challenged with *B. bronchiseptica* and *B. pertussis*. In addition to assessing protective efficacy and sterilizing immunity of the 22BF+dmLT as well as the requirement for PTd, a protective correlate can be established for use. This method was used to identify a protective correlate associated with DBF protection of mice against *Shigella*. This, however, prior to the present disclosure has never been determined for such a vaccine type against an extracellular pathogen.

a) Assess Protective Efficacy of the 22BF+dmLT Delivered IN and IM Against *B. Bronchiseptica* Challenge Using the Mouse Lung Model with Two Challenge Doses—a Lethal Dose to Assess Survival and a Sublethal Dose to Assess Sterilizing Immunity.

[0158] Protective efficacy of 22BF+dmLT delivered IN, IM, and ID against a *B. bronchiseptica* challenge can be assessed in the mouse lung model. A high dose of *B. bronchiseptica* can be administered initially to assess protection via the lethal dose. In a second trial, a lower dose can be used to assess sterilizing immunity.

[0159] Experimental Details: Mice (20/group) are vaccinated IN on days 0, 14 and 28. Serum, and stool samples are collected as described above to measure specific antibody responses to confirm that immune responses are comparable to those obtained above. For bacterial challenges, 10 mice are challenged on day 56 with 1×10<sup>7</sup> *B. bronchiseptica* (lethal dose) and 10 animals are challenged with 1×10<sup>6</sup> *B. bronchiseptica* (sub-lethal dose). The mouse experiment can be repeated with vaccination occurring by the IM route. Survival can be plotted and a Log-rank test used to evaluate the differences. A P value of less than 0.05 is considered significant for all comparisons. Association of protective efficacy and markers of humoral and cellular immunity can be assessed with logistic regression models (see FIG. 1).

[0160] With respect to the IN vaccinated mice, at both doses, some level of protection is shown in the lethal lung model. The mice vaccinated with 50 µg are protected with complete sterilizing immunity. The 15 µg dose gives >90% protection with a moderate level of sterilizing immunity. Similarly, 100 µg 22BF+dmLT delivered IM has a high level of protection as well as sterilizing immunity, but perhaps not 100%, but greater than 70% protection. The 40 µg dose shows minimal protection. With these results, a protective correlate for *B. bronchiseptica* can be predicted, as long as the immune responses were above the levels anticipated.

b) Assess Protective Efficacy of the 22BF+dmLT±PTd Delivered IN, IM, and ID Against *B. Pertussis* Challenge Using the Mouse Lung Model with Two Challenge Doses—a Lethal Dose to Assess Survival and a Sublethal Dose to Assess Sterilizing Immunity.

[0161] The ultimate test of the 22BF formulation is the protective efficacy against *B. pertussis*. Here, the protective efficacy of 22BF+dmLT is tested with a focus on a *B. pertussis* challenge using the mouse lung model. Vaccinations occur IN with PTd. Furthermore, a high dose of *B. pertussis* is used initially to assess protection via the lethal lung model.

[0162] Mice (10/group) are vaccinated IN on days 0, 14 and 28. Serum, and stool samples are collected, as described above to measure specific antibody responses to confirm a comparable immune response. For bacterial challenges, all mice can be challenged on day 56 with  $1 \times 10^7$  *B. pertussis* (lethal dose). The experiment can be repeated using IM route and ID route again with the most protective vaccine and challenge with a lethal dose and a sub-lethal dose to assess protection and sterilizing immunity. Survival can be plotted and a Log-rank test used to evaluate the differences. A P value of less than 0.05 is considered significant for all comparisons.

[0163] PTd can additionally be administered for protection against *B. pertussis* and to prevent the cellular damage associated with PT as well as increase sterilizing immunity. Mice can be vaccinated IN, IM, or ID with 22BF+PTd and dmLT and challenged with *B. pertussis*. Lung CFU were measured at day 3 (FIG. 12) and day 7 (FIG. 13) post challenge. As with the predicted *B. bronchiseptica* results, 100% protection and sterilizing immunity is obtained with 50  $\mu$ g 22BF+dmLT+PTd delivered IN with reduced protection for the 15  $\mu$ g dose delivered IN. Similarly, the 100  $\mu$ g 22BF+dmLT+PTd delivered IM and ID achieves some significant level of protection, but sterilizing immunity is limited though could be greater at higher dosage by day 7 post challenge. Antibodies are important, but the impact of cytokines cannot be ignored.

#### 4. Example 4: LTA-1 Fusion

[0164] LTA1 is the active moiety of lethal toxin from Enterotoxigenic *E. coli* (ETEC). The activity of the LTA1 is required for the adjuvant activity of dmLT. The double mutants are in the region usually targeted by a protease to allow A1 to traffic to the cytoplasm of intestinal cells to cause the secretory diarrhea. Without the protease the LT still has some activation of cAMP. Likewise, LTA1 remains active.

##### a) LTA1-Fusions:

[0165] The LTA1-fusions were expressed in a manner similar to the fusion alone. The LTA1 sequence was inserted 5' to the start of the each fusion. Some of the LTA1-fusions required a small linker between the LTA1 and fusion in order for protein production to occur. LTA1-DBF, LTA1-S1, LTA1-S2, LTA1-SseB, LTA1-22BF, LTA1-BurkF, and LTA1-PaF were produced. One of the assays that appear to be required for adjuvant activity is the ability to ADP-ribosylate ARF4. The ADP-ribosylation assay was performed with the LTA1-fusions. In the assay, ADPr was biotin conjugated and when mixed with LTA1 and rARF4, the LTA1 transferred the biot-ADPr to rARF4. The biotin was then detected with Streptavidin-IR800 (FIG. 20).

##### b) LTA1-Fusion Protective Efficacy:

[0166] Mice were vaccinated parenterally with LTA1-DBF or DBF+dmLT (FIG. 14). Although the DBF+dmLT delivered IN elicited complete protection, this formulation cannot be given to humans since the dmLT can travel the olfactory nerve to the brain causing Bell's palsy. Thus, parenteral routes or other mucosal routes are required. DBF (100  $\mu$ g)+dmLT (0.1  $\mu$ g) delivered IM was only 50% protective while LTA1-DBF at a concentration of 100  $\mu$ g DBF equivalent was 70% protective. The lower dilutions, regard-

less of formulation, exhibited less protective efficacy. Similarly, LTA1-22BF+PTd elicited significant protective efficacy at a concentration of 80  $\mu$ g of 22BF equivalents (FIG. 19A-D)

##### c) LTA1-DBF Immune Response:

[0167] When the kinetics of the IgG titer was examined, responses against IpaD and IpaB were essentially the same. Mice from FIG. 14 were bled prior to vaccination and on day 42. Sera were assessed for anti-IpaD, -IpaB and -dmLT IgG. The lower IgG titers of the LTA1 samples can be attributed to the lower recognition of the dmLT on the well for the LTA1 samples vs the samples from mice vaccinated with dmLT (FIG. 15). No IgA was detected in the fecal samples of the mice vaccinated IM, but was detectable in mice vaccinated IN with DBF+dmLT. Antibody-secreting cells (ASCs) present in the bone marrow (FIG. 16), spleen (FIG. 17), and lungs (FIG. 18) were also stimulated with IpaD, IpaB and dmLT. In each compartment, anti-IpaD, anti-IpaB, and anti-dmLT IgG ASCs could be detected. Interestingly, IgA ASCs could also be detected in the bone marrow against all dilutions of LTA1-DBF and resemble a curve similar to the dose escalation. A similar phenomenon was seen in the IgA ASCs from the lungs, but less pronounced.

##### d) LTA1-DBF Purification.

[0168] The yield of LTA1-DBF was very low. Therefore, a linker was inserted in the DNA sequence between LTA1 and DBF to encode GSAAS (Seq. ID No. 14). The mother plasmid was Novagen's pACYCDuet-1. The translocator for each fusion cannot be made without its cognate chaperone. Therefore, the complex of LTA1-DBF/Histag-IpgC (IPG chaperone comprises the nucleic acid sequence as set forth in SEQ ID NO: 10 which encodes the amino acid sequence as set forth in SEQ ID NO: 11) was produced from the plasmid PACYC-His-IpgC-LTA1-GSAAS-DBF where the ipgC gene was inserted into the BamHI/HindIII sites allowing for expression of His-tag IpgC and LTA1-GSAAS-DBF (nucleic acid sequence as set forth in SEQ ID NO: 15 and amino acid sequence as set forth in SEQ ID NO: 16)) was inserted at the NdeI-XhoI site. The DBF sequence had a 3' stop codon prior to the XhoI restriction site.

[0169] pACYC-His-IpgC-LTA1-GSAAS-DBF was transformed into Tuner cells. A small overnight culture of LB+Chloramphenicol (Cm) that had been inoculated with the freezer stock of the cells was transferred to 8 L TB, and grown at 37° C. until OD=1-1.5, add 0.5 mM IPTG with 20  $\mu$ g/liter AEBBSF, 16 C overnight, harvested at 4000 rpm for 15 min at 4° C., and resuspended in IMAC binding buffer. The cells were frozen at -80° C. until ready for purification. After thawing the suspension was sonicated at 70% amplitude for 3-4 min, 15 s on, 30 s off, clarified by centrifugation at 13000 rpm for 30 min at 4° C. and decanted to obtain supernatant.

[0170] IMAC purification with 5 ml NINTA FF crude column on AKTA was as follows: (1) equilibrate column with 5CV binding buffer (20 mM Tris, 500 mM NaCl, 5 mM Imidazole pH 7.9), (2) load supernatant on column, collect FT in outlet1, (3) wash with binding buffer for 30CV, (4) elute with linear 0-60% elution buffer (20 mM Tris, 500 mM NaCl, 500 mM Imidazole pH 7.9) for 10CV, (5) elute with

60% elution buffer for 2CV, (6) wash column with 100% elution buffer for 3CV, (7) re-equilibrate column with 5CV binding buffer for 5CV.

**[0171]** HIC purification of the protein was as follows: Dilute pooled fraction into equal volume of 2×HIC binding buffer (50 mM Sodium Phosphate (dibasic), 1M Ammonium Sulfate, pH 7.0). Purify with 5 ml HIC Phenyl HP column: (1) equilibrate column with 5CV binding buffer, (2) load diluted sample on column, collect FT in outlet1, (3) wash with binding buffer for 5CV, (4) elute with linear 0-100% elution buffer (5 mM Sodium Phosphate (dibasic), pH 7.0) for 40CV, (6) elute with 100% elution buffer for 6CV, (7) wash column with 100% elution buffer for 3CV, (8) re-equilibrate column with binding buffer for 5CV.

**[0172]** Pooled fractions were dialyzed in 4 L Q binding buffer for 2 hrs, exchanged buffer, and then dialyzed overnight.

**[0173]** Purification using a 5 mL Q FF columns on AKTA was as follows: (1) equilibrate column with 5CV binding buffer (50 mM Tris, pH 8.0), (2) load dialyzed sample on column, collect FT in outlet1, (3) wash with binding buffer for 5CV, (4) elute with linear 0-30% elution buffer (50 mM Tris, 1M NaCl, pH 8.0) for 20CV, (6) elute with 100% elution buffer for 5CV, (7) wash column with 100% elution buffer for 3CV, (8) re-equilibrate column with binding buffer for 5CV.

**[0174]** To facilitate final IMAC purification 8×IMAC binding buffer (NO Imidazole) was added to pooled fractions to obtain 1× and then LDAO to 0.05% was added.

**[0175]** Purification by LDAO IMAC using 5 ml NINTA FF was as follows: (1) equilibrate column with 5CV LDAO (20 mM Tris, 500 mM NaCl, 0.05% LDAO pH 7.9) binding buffer, (2) load supernatant on column, fractionate FT, (3) wash with binding buffer for 5CV, fractionate, (4) wash with 3% LDAO elution buffer (20 mM Tris, 500 mM NaCl, 500 mM Imidazole, 0.005% LDAO pH 7.9) for 5CV, fractionate (5) elute with 6% LDAO elution buffer for 6.65CV, fractionate (6) elute with 100% LDAO elution buffer for 5CV, (8) re-equilibrate column with 5CV binding buffer for 5CV.

**[0176]** Pooled samples were dialyzed in 4 L PBS+0.005% LDAO, exchanged buffer after 2 hrs, and then dialyzed overnight.

#### e) Protective Efficacy of LTA1-22BF

**[0177]** The initial assessment of the protective efficacy of the LTA1-22BF is presented here and demonstrated that LTA1-22BF+rPT reduced the CFU lung burden by 99.8% while the 22BF+dmLT+rPT reduced it to 99.98% (FIG. 19 (a)). Non-toxic pertussis toxin was added to the groups that prior to challenge by *B. pertussis* to offset the damage that pertussis toxin would cause and negatively impact the challenge. The kinetics of the IgG response is also shown where no differences are seen between the two groups (FIG. 19 (b)).

#### f) LTA1-22BF Purification

**[0178]** The mother plasmid was Novagen's pACYCDuet-1. The translocator for each fusion cannot be made without its cognate chaperone. Therefore, the complex of LTA1-22BF/Histag-BcrHI is produced from the plasmid pACYC-His-BcrHI-LTA1-22BF where the bcrHI gene (as set forth in SEQ ID NO: 7 with a histidine tag and encodes the amino acid sequence as set forth in SEQ ID NO: 8, the sequence

minus the his-tag set forth in SEQ ID NO: 9) is inserted into the BamHI/HindIII sites allowing for expression of His-tag BcrHI and 22BF is inserted at the NdeI-XhoI site. The 22BF sequence has a 3' stop codon prior to the XhoI restriction site.

**[0179]** pACYC-His-BcrHI-22BF was transformed into Tuner cells. A small overnight culture of LB+Chloramphenicol (Cm) that had been inoculated with the freezer stock of the cells was transferred to 8 L TB, grown at 37° C. until OD=1-1.5, added 0.5 mM IPTG with 20 ug/liter AEBSF, 16° C. overnight, harvested at 4000 rpm for 15 min at 4 C, and resuspended in IMAC binding buffer. The cells were frozen at -80 until ready for purification. After thawing, the suspension was sonicated at 70% amplitude for 3-4 min, 15 s on, 30 s off, clarified by centrifugation at 13000 rpm for 30 min at 4° C., and decanted to obtain supernatant.

**[0180]** IMAC purification with 5 ml NiNTA FF crude column on AKTA was as follows: (1) equilibrate column with 5CV binding buffer (IMAC elution buffer: 20 mM Tris, 500 mM NaCl, 500 mM Imidazole pH 7.9), (2) load supernatant on column, collect FT in outlet1, (3) wash with binding buffer for 30CV, (4) elute with linear 0-60% elution buffer for 10CV, (5) elute with 60% elution buffer for 2CV, (6) wash column with 100% elution buffer for 3CV, (7) re-equilibrate column with 5CV binding buffer (IMAC binding buffer: 20 mM Tris, 500 mM NaCl, 5 mM Imidazole pH 7.9) for 5CV.

**[0181]** Diluted pooled fractions 20× into Q binding buffer Q binding buffer: 50 mM Tris, pH 8.0). Purification with 3×5 mL Q FF columns on AKTA was as follows: (1) equilibrate column with 6CV binding buffer, (2) load dialyzed sample on column, collect FT in outlet1, (3) wash with binding buffer for 12CV, (4) elute with 15% elution buffer (Q elution buffer: 50 mM Tris, 1M NaCl, pH 8.0) for 6CV, (5) elute with linear 15-40% elution buffer for 34CV, (6) elute with 100% elution buffer for 6CV, (7) wash column with 100% elution buffer for 3CV, (8) re-equilibrate column with binding buffer for 6CV.

**[0182]** To facilitate final IMAC purification 8×IMAC binding buffer (NO Imidazole) was added to pooled fractions to obtain 1× and then LDAO to 0.05% was added. Purification by LDAO IMAC using 5 ml NiNTA FF was as follows: (1) equilibrate column with 5CV LDAO binding buffer (LDAO IMAC binding buffer: 20 mM Tris, 500 mM NaCl, 0.05% LDAO pH 7.9), (2) load supernatant on column, fractionate FT, (3) wash with binding buffer for 5CV, fractionate, (4) wash with 3% LDAO elution buffer (LDAO IMAC elution buffer: 20 mM Tris, 500 mM NaCl, 500 mM Imidazole, 0.005% LDAO pH 7.9) for 5CV, fractionate (5) elute with 6% LDAO elution buffer for 6.65CV, fractionate (6) elute with 100% LDAO elution buffer for 5CV, (8) re-equilibrate column with 5CV binding buffer for 5CV.

**[0183]** Pooled samples were dialyzed in 4 L PBS+0.005% LDAO, exchanged buffer after 2 hrs, and then dialyzed overnight.

#### g) LTA1-BurkF Purification

**[0184]** The mother plasmid was Novagen's pACYCDuet-1. The translocator for each fusion cannot be made without its cognate chaperone. Therefore, the complex of LTA1-BurkF/Histag-BicA (SEQ ID NOs: 28 and 20) was produced from the plasmid pACYC-His-BicA-LTA1-BurkF (as set forth in SEQ ID NO: 20 and encoded by the nucleic acid

sequence as set forth in SEQ ID NO: 19) wherein the bica gene was inserted into the BamHI/HindIII sites allowing for expression of His-tag BicA and LTA1-BurkF was inserted at the NdeI-XhoI site. The BurkF sequence had a 3' stop codon prior to the XhoI restriction site.

**[0185]** pACYC-His-BicA-LTA1-BurkF was transformed into Tuner cells. A small overnight culture of LB+Chloramphenicol (Cm) that had been inoculated with the freezer stock of the cells was transferred to 8 L TB, grown at 37° C. until OD=1-1.5, added 0.5 mM IPTG with 20 ug/liter AEBSE, 16° C. overnight, harvested at 4000 rpm for 15 min at 4 C, and resuspended in IMAC binding buffer. The cells were frozen at -80 until ready for purification. After thawing the suspension was sonicated at 70% amplitude for 3-4 min, 15 s on, 30 s off, Clarified by centrifugation at 13000 rpm for 30 min at 4° C., and decanted to obtain supernatant.

**[0186]** IMAC purification with 5 ml NiNTA FF crude column on AKTA was as follows: (1) equilibrate column with 5CV binding buffer (20 mM Tris, 500 mM NaCl, 5 mM Imidazole pH 7.9), (2) load supernatant on column, collect FT in outlet1, (3) wash with binding buffer for 30CV, (4) elute with linear 0-60% elution buffer (20 mM Tris, 500 mM NaCl, 500 mM Imidazole pH 7.9) for 10CV, (5) elute with 60% elution buffer for 2CV, (6) wash column with 100% elution buffer for 3CV, (7) re-equilibrate column with 5CV binding buffer for 5CV.

**[0187]** The pooled fractions were diluted 20× into Q binding buffer (50 mM Tris, pH 8.0). Purification using 3×5 mL Q FF columns on AKTA was as follows: (1) equilibrate column with 6CV binding buffer, (2) load dialyzed sample on column, collect FT in outlet1, (3) wash with binding buffer for 12CV, (4) elute with 15% elution buffer (50 mM Tris, 1M NaCl, pH 8.0) for 6CV, (5) elute with linear 15-40% elution buffer for 34CV, (6) elute with 100% elution buffer for 6CV, (7) wash column with 100% elution buffer for 3CV, (8) re-equilibrate column with binding buffer for 6CV.

**[0188]** To facilitate final IMAC purification add 8×IMAC binding buffer (NO Imidazole) to pooled fractions to obtain 1× and then LDAO to 0.05% was added:

**[0189]** Purification by LDAO IMAC using 5 ml NiNTA FF was as follows: (1) equilibrate column with 5CV LDAO binding buffer (20 mM Tris, 500 mM NaCl, 0.05% LDAO pH 7.9), (2) load supernatant on column, fractionate FT, (3) wash with binding buffer for 5CV, fractionate, (4) wash with 3% LDAO elution buffer (20 mM Tris, 500 mM NaCl, 500 mM Imidazole, 0.005% LDAO pH 7.9) for 5CV, fractionate (5) elute with 6% LDAO elution buffer for 6.65CV, fractionate (6) elute with 100% LDAO elution buffer for 5CV, (8) re-equilibrate column with 5CV binding buffer for 5CV.

**[0190]** Pooled samples were dialyzed in 4 L PBS+0.005% LDAO, exchanged buffer after 2 hrs, and then dialyzed overnight.

#### h) LTA1-PaF

**[0191]** The PaF+dmLT vaccinated mice exhibited 100% survival with 44% sterilizing immunity against Pa challenge in a mouse lethal pulmonary model, while the PBS vaccinated mice exhibited 60% survival but all had >108-9 CFU/lung.

#### i) LTA1-PaF Purification

**[0192]** The mother plasmid is Novagen's pACYCDuet-1. The translocator for each fusion cannot be made without its

cognate chaperone. Therefore, the complex of LTA1-PaF/Histag-PcrHI was produced from the plasmid pACYC-His-PcrHI-LTA1-PaF where the brcHI gene was inserted into the BamHI/HindIII sites allowing for expression of His-tag PcrHI (as set forth in SEQ ID NO: 30 and encoded by the nucleic acid sequence as set forth in SEQ ID NO: 29) and LTA1-PaF (as set forth in SEQ ID NO: 38 and encoded by the nucleic acid sequence as set forth in SEQ ID NO: 37) which was inserted at the NdeI-XhoI site. The PaF sequence had a 3' stop codon prior to the XhoI restriction site. The purification of LTA1-PaF was the same of for LTA1-22BF.

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## F. SEQUENCES

## 22BF Nucleotide Sequence

SEQ ID NO: 1

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22BF Amino Acid Sequence

SEQ ID NO: 2

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Bsp22 Nucleotide Sequence

SEQ ID NO: 3

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Bsp22 Amino Acid Sequence

SEQ ID NO: 4

MTIDLVSLTSQAGGLQGIDLKSMIDIQLMVYVQGRRAELLTAQMQTQAEV  
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## BopB Nucleotide Sequence

SEQ ID NO: 5

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## BopB Amino Acid Sequence

SEQ ID NO: 6

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## His-BcrH1 chaperone with histidine tag nucleotide sequence

SEQ ID NO: 7

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## His-BcrH1 with histidine tag chaperone amino acid sequence

SEQ ID NO: 8

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BcrH1 amino acid sequence

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IpgC Chaperone of DBF nucleic acid sequence

SEQ ID NO: 10

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IpgC Chaperone of DBF Amino Acid sequence

SEQ ID NO: 11

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LTA1 nucleic acid sequence

SEQ ID NO: 12

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LTA1 amino acid sequence

SEQ ID NO: 13

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HPYEQEVSA LGGIPYSQIYGWYRVNFGVIDERLHRNREYDRYYRNLNIAPAEDGYR

LAGFPDPDHQAWREEPWIHAPQCGNSSR

GSAAS Linker amino acid sequence

SEQ ID NO: 14

GSAAS

LTA1-GSAAS-DBF (IpaD-LE-IpaB) nucleic acid sequence

SEQ ID NO: 15

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LTA1-GSAAS-IpaD-LE-IpaB (DBF) Amino Acid sequence

SEQ ID NO: 16

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LTA1-22BF nucleic acid sequence

SEQ ID NO: 17

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tcacggctcaaatgcagacccaggccgaagtggcgcagaaggccaatgaacgcacatggcgcagctcaacgaggtcctgtccgcgct  
gtcccgggccaaggccgagtttccgcccacatccgaagcgggcgacacccatcccggtgggacaaaccagaaggtcagccggat  
cgaggttcctctcaatgatgcgctgcgcgctgcccgcctgacgggcatgttcgaagcgcgcgatggccaagtgaacgccccggcg  
gccggggtacgcaggtcgtaacggcacggcgctcatggccggttccacgacctataaggaactcgaaagtgcctacaccaccgta  
aaggggatgctggatacggcgctccaatacgcaacagatggacatgatcaggctgcaggccgcagcaacaagcgcaacgaggtt  
tcgaggtcatgaccaacaccgagaagcggcgacgcgacctgaacagttccatcaccaacaacatgcgcaagcttatgaccgtcatga  
gtacgacctatccacagccccgagcggcgccgcgcttgccgcgctctcgcatagatatgcgggcaccggagccccgggagtgccgg  
cgaaggcgccggcatcctggcgccggtagcacgctggctctggcgggcgccggcgcttccagcgtcacctcgctgcgc  
accgcgcccgtcctggatccgccagtgcgcgatctcagccccgcgacttgccgcgacctgctgcgcgctcttgcatccaggcggt  
ggacgggcagttggccacggcgcgagaaacctgcaggacgcgcaagtcaaggcgaagcagaacaccaggccagctcgac

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aagctggacgcatgggtttcggaaaggccgaaggagccgagagcaagggatggctgagcaaggtgttcggctggatcggcaaggtg  
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tgctcagcgcatggcactggtcagcgccgtgacatcgctggccgaccagatatcgcaagaggcgggagcccgctatcagcct  
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gccagcgatgccaccgggggtacatagccatggcgatgtccatcgtggcgcgatcgcggtcgcccgatcaatgccgcccgtg  
cagccggcgcggttagcgcttcggcgatcaagggggcctgggatcgggccgcccggtagccaccaggctccttcaagggggta  
cggcagtggcgcaaggcggtcgcggtgctgagtgccatcgatcgcaaacaggccgatctcctggtcgcccgaagcggtatc  
ggcgcgagcctgacaaaactcgggggcgccatggagcgtgagggcgacgatatcaagaagatcctggctcaattcgacgagggc  
ctatcacatgatcgcaagatgatcagcgatattggcgagtacgcacagccaggtcagcgccaacctcgggcgccgcccaggcggtg  
tagCTCGAG

LTA1-22BF Amino acid sequence

SEQ ID NO: 18

MDNGDRLYRADSRPPDEIKRSGGLMPRGHNEYFDRGTQMNINLYDHARGTQ  
TGFVRYDDGYVSTSLSLRSAHLAQQSILSGYSTYYIYVIATAPNMFNVNDVLGVYSP  
HPYEQEVSA LGGIPYSQIYGWYRVNFGVIDERLHRNREYDRYRNLNIAPAEDGYR  
LAGFPDPHQAWREEPWIHAPQGCGNSSRMTIDLGVSLTSQAGGLQGIDLKSMDIQT  
LMVYVQGRRAELLTAQMQTQAEVVQKANERMAQLNEVLSALSRAKAEPFPPNPKPG  
DTIPGWDNQKVSRIEVPLNDALRAAGLTGMFEARDGQVTAPGGRGTQVVGNGTGM  
AGSTTYKELESAYTTVKMLDTASNTQQMDMIRLQAASNKRNEAFEVMTNTEKRR  
SDLNSSITNMRLKMTVMSTTISTAPSGAALAPSRIDMRAPEPGSAGEGAGILAPVTT  
LALAAGRPAFPASPRLTAPVLDPPVRDLSPADLADLLRVLSRAVDGQLATARENL  
QDAQVKAKQNTQAQLDKLDKDAWFRKAEEAESKGWLSKVFVGWIGKVLAVVASALAV  
GFAAVASVATGAAATPMLLLSGMALVSAVTSLADQISQEAGGPPI SLGGFLSGLAGR  
LLTALGVDQSQADQIAKIVAGLAVPVVLLIEPQMLGEMAQGVARLAGASDATAGYI  
AMAMSI VAAIAVA AINAAGTAGAGSASAIKGAWDRAAAVATQVLQGGTAVAQGG  
VGVSMAVDRKQADLLVADKADLAASLTKLRAAMEREADDIKKILAQFDEAYHMIA  
KMISDMASTHSQVSANLGRRAV\*

His-BicA (Chaperone of BurkF) nucleic acid sequence

SEQ ID NO: 19

ATGGGCAGCAGCCATCACCATCATCACCACAGCCAGGATCCGATGACGCA  
ACGCGACGTGAACATAGACGACATCGAGGCGCAGGAAATGGCGGCGGCGCTGCT  
GGACGCGGTCCAGAACGGCGCGACGCTGAAGACCTGCATCAGGTGCCGCGAGGA  
CCTGATGGACGGCATCTATGCGTTCGCGTACCGCTTCTACCAGCAGGGGCGGCTC  
GACGACGCGGAGGTGTTCTTCGCTTTCTGCGCATCTACGACTTCTACAACGCCG  
AATACGCGATGGGGCTCGCGCGGTGTGCCAGTTGAAGAAGGAGTACGCGCGGG  
CGATCGATCTGTATGCACTCGCGTATTCGCTGTGAAGGACGACCACCGGCCGAT  
GTTCCACACCGCCAATGCCATCTGCTGATGGGCAAGGCGGCGCTCGCGCGGCG  
CTGCTTCGCGATCGTCTGAGCGCTCGCGCGACGAGCGCCTCGCGCAGAAGGC  
GCAGTCTATCTCGACGGGCTCGACGAAGTGGGCGCCGACGCGGCGCCCGCATC  
CGCCGGGAACGACCACTGAGCGGCCGC

-continued

His-BicA (Chaperone of BurkF) Amino acid sequence

SEQ ID NO: 20

MGSSHHHHHSQDPMTQRDVNIDDI EAQEMAAALLDAVQNGATLKDHLHQV  
PQDLMDGIYAFAYRFYQQGRLLDAEVFFRFLRIYDFYNAEYAMGLAAVCQLKKEY  
ARAILDYLALAYSLSKDDHRPMFHTGQCHLLMGKAALARRCFGI VVERS RDERLAQK  
AQSYLDGLDEVGADAAPASAGNDH-

BipD nucleic acid sequence

SEQ ID NO: 21

CATATGAACATGCACGTGGACATGGGTCTGCGCTGACCGTTCGTGATTG  
GCCGGCGCTGGAGGCGCTGGCGAAAACCATGCCGGCGGATGCGGGTGC GCGTGC  
GATGACCGATGATGACCTGCGTGC GCGCGGTGTGGACCGTCTGTTCGGAGCA  
GAAGCTGGGTGCGGCGATTGATGAATTCGCGAGCCTGCGTCTGCCGGATCGTATC  
GACGGTCTTTCTGTGGATGGCCGTCTGCGAACCTGACCGTTTTTGATGATGCGC  
GTGTTGCGGTTCTGTGGTCATGCGCGTGC GCAACGTAACCTGCTGGAGCGTCTGGA  
GACCGAAGTCTGGGTGGCACCCTGGATACCGCGGTGACGAAGGTGGCATTCA  
GCCGGACCCGATCCTGCAAGGCCCTGGTGGATGTTATCGGTGAGGCAAAAGCGA  
TATTGACGCGTACGCGACCATCGTGGAAGGTCTGACCAAGTATTTTCAAAGCGTG  
GCGGACGTTATGAGCAAACTGCAGGATTACATTAGCGCGAAGGATGACAAAAAC  
ATGAAGATCGACGGTGGCAAGATCAAAGCGCTGATTCAGCAAGTGATCGACCAC  
CTGCCGACCATGCAGCTGCCGAAGGGTGC GCGATATTGCGCGTTGGCGTAAAGAG  
CTGGGCGACGCGGTTAGCATCAGCGATAGCGGTGTGGTTACCATTAACCCGGAC  
AAACTGATCAAGATGCGTGATAGCCTGCCCGCGGATGGCACCGTTTGGGATACC  
GCGCGTTACCAAGCGTGGAACACCGCGTTCAGCGGTGAGAAAGGCCAGCATCCG  
GAACGTCTGCGGATGCGCGTCTGTAATATAGCCACCAGAACGCAACTTTGAT  
AACCTGGTGAAGGTTCTGAGCGGTGCGATTAGCACCTGACCGACACCCAGAGC  
TATCTGCAAATC

BipD amino acid sequence

SEQ ID NO: 22

MMNHVDMGRALTVRDWPAL EALAKTMPADAGARAMTDDDLRAAGVDRR  
VPEQLGAAIDEFASRLPDRIDGRFVDGRRANLTVFDDARVAVRGHARAQRNLLE  
RLETELLGGTLDTAGDEGGIQPDPI LQGLVDVIGQGKSDIDAYATIVEGLTKYFQSV A  
DVMSKLQDYISAKDDKNMKIDGGKI KALIQQVIDHLPTMQLPKGADIARWRKELGD  
AVSISDSGVVTINPDKLIKMRDSLPPDGTVWD TARYQAWNTAFSGQKGQHPERRAD  
ARRKYSHQNSNFDNLVKVLSGAISTLTDTQSYLQI

BipB nucleic acid sequence

SEQ ID NO: 23

ATGAGCAGCGGTGTTCAAGGTGGCCGGCGGCGAACGCGAACGCGTACC  
AGACCCACCCGCTGCGTGATGCGGCGAGCGCGCTGGGCACCTGAGCCCGCAGG  
CGTATGTGGATGTGGTTAGCGCGGCGCAACGTAACCTCCTGGAGCGTATGAGCC  
AACTGGCGAGCGAACAGTGC GATGCGCAACCGGCGGCGCATGATGCGCGTCTGG  
ATGATCGTCCGCGCTGCGTGC GCGCAGGAACGTGACGCGCGCGCGCTGGGTG  
CGAGCGATACCGGTAGCCGTGCGAGCGGTGCGGCGAACTGACCGAGCTGCTGG  
GTGTGCTGATGAGCGTTATTAGCGCGAGCAGCCTGGACGAACGAAGCAACGTA

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GCGATATCTGGAACCAGATGAGCAAAGCGGCGCAAGACAACCTGAGCCGTCTGA  
GCGATGCGTTTCAGCGTGCGACCGACGAGGCGAAAGCGGCGGCGATGCGGCGG  
AACAGGCGGCGGCGGCGGCGAAGCAAGCGGGTGCAGGACGCGAAAGCGGCGGAT  
GCGGCGGGTGGATGCGGCGCAAAAACGTTACGATGACGCGGTTAAGCAGGGCCTG  
CCGGATGACCGTCTGCAAAGCCTGAAAGCGGCGCTGGAGCAGGCGCGTCAGCAA  
GCGGGTGATGCGCATGGTCGTGCGGATGCGCTGCAGGCGGATGCGACCAAGAAA  
CTGGACGCGGCGAGCGCGCTGGCGACCCCAAGCGCGTGCCTGCGAACAGCAAGTG  
GATGACGCGGTTAACCAGGCGACCCAGCAATATGGTGCGAGCGCGAGCCTGCGT  
ACCCCGCAAGCCCGCGTCTGAGCGGTGCGGCGGAGCTGACCGCGGTGCTGGGC  
AAGCTGCAGGAACTGATTAGCAGCGGCAACGTTAAAGAGCTGGAAGCAAGCA  
GAAACTGTTACCCGAGATGCAAGCGAAGCGTGAGGCGGAACTGCAAAAGAAAA  
GCGACGAATATCAGGCGCAAGTGAAGAAAGCGGAGGAAATGCAGAAAACGATG  
GGTTGCATCGGCAAGATTGTGGGTTGGGTTATTACCGCGGTTAGCTTTGCGGCGG  
CGGCGTTTACCGGTGGCGCGAGCCTGGCGCTGGCGGCGGTGGGCTGGCGCTGG  
CGGTTGGTGACGAGATTAGCCGTGCGACCAACCGGTGAGCTTCATGGACAAGC  
TGATGCGAGCCGTTATGGATGCGATCCTGAAACCGCTGATGGAGATGATTAGCA  
GCCTGATCACCAGGCGCTGGTTGCGTGCGGCGTTGATCAGCAAAAAGCGGAAC  
TGCGGGGTGCGATTCTGGGTGCGGTTGTTACCGGTGTGGCGCTGGTTGCGGCGGC  
GTTTGTGGTGCGAGCGCGGTGAAAGCGGTTGCGAGCAAGGTTATCGACGCGAT  
GGCGGGTCAGCTGACCAAGCTGATGGATAGCGCGATTGGCAAAATGCTGGTGCA  
ACTGATCGAGAAATTCAGCGAAAAGAGCGGTCTGCAGGCGCTGGGTAGCCGTAC  
CGCGACCGCGATGACCCGTATGCGTCGTGCGATTGGCGTTGAGCGAAGGAAGA  
CGGTATGCTGCTGGCGAACCGTTTGA AAAAGCGGGCACCGTGATGAACGTTGG  
TAACCAAGTGAGCCAAGCGGCGGGTGGCATTGTGGTTGGCGTTGAGCGTGCGAA  
AGCGATGGGTCTGCTGGCGGATGTGAAAGAAGCGATGTATGACATCAAGCTGCT  
GGGTGATCTGCTGAAACAGGCGGTGGACGCGTTTGCGGAGCACACCCTGTTCT  
GGCGCAACTGATGCAGCAAATGAGCGATGCGGGCGAAATGCAGACCAGCACCG  
GCAAGCTGATCCTGCGTAACGCGCGTGCGGTTTAAGGATCC

BipB amino acid sequence

MSSGVQGGPAANANAYQTHPLRDAASALGTLSPQAYVDVVSAAQRNFLER  
MSQLASEQCDQAHDARLDDR PALRAPQERDAPPLGASDTGSRASGAAKLTELL  
GVLMSVISASSLDELKQRS DIWNQMSKAAQDNLSRLSDAFQRTDEAKAADAAEQ  
AAAAAKQAGADAKAADA AVDAAQKRYDDAVKQGLPDDR LQSLKAALQARQQA  
GDAHGRADALQADATKKLDAASALATQARACEQQVDDAVNQATQQYGASASLRT  
PQSPRLSGAAELTAVLGKLQELISSGNVKELESKQKLFTEMQAKREAELOKKSDEYQ  
AQVKKAEEMQKTMGCIGKIVGWVITAVSF AAAAFTGGASLALAAVGLALAVGDEIS  
RATTGVSFMDKLMQPVMDAILKPLMEMISSLITKALVACGV DQQKAELAGAILGAV  
VTGVALVAAAFVGASAVKAVASKVIDAMAGQLTKLMDSAIGKMLVQLIEKFSEKS  
GLQALGSRTATAMTRMRAIGVEAKEDGMLLANRFEKAGTVMNVGNQV SQAAGG

SEQ ID NO: 24

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IVVGVERAKAMGLLADVKEAMYDIKLLGDLLKQAVDAFAEHNRLVLAQLMQQMSD

AGEMQTSTGKLILRNARAV

BurkF nucleic acid sequence

SEQ ID NO: 25

CATATGAACATGCACGTGGACATGGGTCTGCGCTGACCGTTCGTGATTG  
GCCGGCGCTGGAGGCGCTGGCGAAAACCATGCCGGCGGATGCGGGTGC GCGTGC  
GATGACCGATGATGACCTGCGTGC GCGCGGGTGTGGACCGTCTGTTCGGAGCA  
GAAGCTGGGTGCGGCGATTGATGAATTCGCGAGCCTGCGTCTGCCGGATCGTATC  
GACGGTCTTTTCGTGGATGGCCGTCTGCGAACCTGACCGTTTTTGATGATGCGC  
GTGTTGCGGTTCTGTGGTCATGCGCGTGC GCAACGTAACTGCTGGAGCGTCTGGA  
GACCGAATGCTGGGTGGCACCTGGATACCGCGGTGACGAAGGTGGCATTCA  
GCCGGACCCGATCCTGCAAGGCCCTGGTGGATGTTATCGGTGAGGGCAAAGCGA  
TATTGACGCGTACGCGACCATCGTGGAAGGTCTGACCAAGTATTTTCAAAGCGTG  
GCGGACGTTATGAGCAAATGCAGGATTACATTAGCGCGAAGGATGACAAAAAC  
ATGAAGATCGACGGTGGCAAGATCAAAGCGCTGATTGACCAAGTGATCGACCAC  
CTGCCGACCATGCAGCTGCCGAAGGGTGC GGTATTGCGCGTTGGCGTAAAGAG  
CTGGGCGACGCGGTTAGCATCAGCGATAGCGGTGTGGTTACCATTAACCCGGAC  
AAACTGATCAAGATGCGTGATAGCCTGCCCGCGGATGGCACCGTTTGGGATACC  
GCGGTTTACCAAGCGTGGAACACCGCGTTCAGCGGTGAGAAAGGCCAGCATCCG  
GAACGTCGTGCGGATGCGCGTCGTAAATATAGCCACCAGAACAGCAACTTTGAT  
AACCTGGTGAAGGTTCTGAGCGGTGCGATTAGCACCTGACCGACACCCAGAGC  
TATCTGCAAAATCAAGCTTATGAGCAGCGGTGTTCAAGGTGCCCCGGCGCGAAC  
GCGAAGCGGTACAGAGCCACCCGCTGCGTGATGCGGCGAGCGCGCTGGGCACC  
CTGAGCCCCGAGCGTATGTGGATGTGGTTAGCGCGGCGCAACGTAATTCCTG  
GAGCGTATGAGCCAACCTGGCGAGCGAACAGTGCATGCGCAACCGGCGGCGCAT  
GATGCGCGCTCTGGATGATCGTCCGGCGCTGCGTGC GCGCAGGAACGTGACGCG  
CCGCCGCTGGGTGCGAGCGATACCGGTAGCCGTGCGAGCGGTGCGGCGAACTG  
ACCGAGCTGCTGGGTGTGCTGATGAGCGTTATTAGCGCGAGCAGCCTGGACGAA  
CTGAAGCAACGTAGCGATATCTGGAACAGATGAGCAAAGCGGCGCAAGACAA  
CCTGAGCCGTCTGAGCGATGCGTTTCAGCGTGC GACCGAGGCGAAAGCGGC  
GGCGGATGCGGCGGAACAGGCGGCGGCGGCGGCGAAGCAAGCGGTGCGGACG  
CGAAAGCGGCGGATGCGGCGGTGGATGCGGCGCAAAAACGTTACGATGACGCG  
GTTAAGCAGGGCCTGCCGGATGACCGTCTGCAAAGCCTGAAAGCGGCGCTGGAG  
CAGGCGCGTCAGCAAGCGGGTATGCGCATGGTCTGTGCGGATGCGCTGCAGGCG  
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GTGCGAACAGCAAGTGGATGACGCGGTTAACAGGCGACCCAGCAATATGGTGC  
GAGCGCGAGCCTGCGTACCCGCAAAGCCCGCTCTGAGCGGTGCGGCGGAGCT  
GACCGCGGTGCTGGGCAAGCTGCAGGAAGTATTAGCAGCGGCAACGTTAAAGA  
GCTGGAAAGCAAGCAGAACTGTTACCGGATGCAAGCGAAGCGTGAGGCGG  
AACTGCAAAAGAAAAGCGACGAATATCAGGCGCAAGTGAAGAAAGCGGAGAA



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ATGCAGAAAACGATGGGTTGCATCGGCAAGATTGTGGGTTGGGTTATTACCGCG  
GTTAGCTTTGCGGCGGCGCGTTTACCGGTGGCGCGAGCCTGGCGCTGGCGGCG  
GTGGGCTGGCGCTGGCGGTTGGTGACGAGATTAGCCGTGCGACACCGGTGTG  
AGCTTCATGGACAAGCTGATGCAGCCGGTTATGGATGCGATCCTGAAACCGCTG  
ATGGAGATGATTAGCAGCCTGATCACCAAGGCGCTGGTTGCGTGCGGCGTTGAT  
CAGCAAAAAGCGGAAGTGGCGGGTGCATTCTGGGTGCGGTTGTTACCGGTGTG  
GCGCTGGTTGCGGCGGCGTTTGTGGTGCGAGCGCGTGAAAGCGGTTGCGAGC  
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GGCAAAATGCTGGTGCAACTGATCGAGAAATTCAGCGAAAAGAGCGGTCTGCAG  
GCGCTGGGTAGCCGTACCGCGACCGCGATGACCCGTATGCGTCGTGCGATTGGC  
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ACCGTGATGAACGTTGGTAACCAAGTGAGCCAAGCGGCGGGTGGCATTGTGGTT  
GGCGTTGAGCGTGCGAAAGCGATGGGTCTGCTGGCGGATGTGAAAGAAGCGATG  
TATGACATCAAGCTGCTGGGTGATCTGCTGAAACAGGCGGTGGACGCGTTTGC  
GAGCACACCGGTGTTCTGGCGCAACTGATGCAGCAAATGAGCGATGCGGGCGAA  
ATGCAGACCAGCACCGGCAAGCTGATCCTGCGTAACGCGCGTGCGGTTTAAGGA  
TCC

BurkF amino acid sequence

SEQ ID NO: 26

MMNHVDMGRALTVRDWPALAKTMPADAGARAMTDDDLRAAGVDRR  
VPEQKLGAIDEFASLRLPDRIDGRFVDGRRANLTVFDDARVAVRGHARAQRNLLE  
RLETELLGGTLDTAGDEGGIQPDPIQLGLVDVIGQKSDIDAYATIVEGLTKYFQSV  
DVMSKLQDYISAKDDKNMKIDGGIKALIQQVIDHLPTMQLPKGADIARWRKELGD  
AVSISDSGVVTINPDKLIKMRDSLPPDGTVWDTARYQAWNTAFSGQKGHPERRAD  
ARRKYSHQNSNFDNLVKVLSGAISTLTDTSYLQIKLMSSGVQGGPAANANAYQTH  
PLRDAASALGTLSPQAYVDVVSAAQRNFLERMSQLASEQCDAPAAHDARLDDRP  
ALRAPQERDAPPLGASDTGSRASGAAKLTELLGVLMVISASSLDELKQRSDIWNQM  
SKAAQDNLRLSDAFQATDEAKAADAEEQAAAAKQAGADAKAADAADVAA  
QKRYDDAVKQGLPDDRLQSLKAALQARQQAGDAHGRADALQADATKKLDAASA  
LATQARACEQQVDDAVNQATQQYGASASLRTPQSPRLSGAAELTAVLGKLQELISS  
GNVKELESKQLFTEMQAKREAEQKKSDEYQAQVKKAEEMQKTMGCIGKIVGW  
VITAVSFAAAFTGGASLALAAVGLALAVGDEISRATTGVSFMDKLMQPVMDAILK  
PLMEMISSLITKALVACGVDQQKAELAGAILGAVVTGVALVAAAPVGASAVKAVAS  
KVIDAMAGQLTKLMSAIGKMLVQLIEKFSEKSLQALGSRTATAMTRMRAIGVE  
AKEDGMLLANRFEKAGTVMNVGNQVSQAAGGIVGVVERAKAMGLLADVKEAMY  
DIKLLGDLKQAVDAFAEHNRLVLAQLMQQMSDAGEMQTSTGKLIILRNARAV

LTA1-BurkF nucleic acid sequence

SEQ ID NO: 27

CATatggacaatggcgatcggttataccgtgccgactcgcgctccccagatgagattaacgtagcgggtgggttaatgcc  
acgtgggcacaatgagatattttgaccgtggaacacagatgaacattaacctttacgatcatgccgtgggaccagaccgggtttgtcc  
gttatgatgacgggtatgtagtagacgagttgtccttacgctccgcacaccttgcgggacaaagtattttatcaggctacagcacatattac

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atttatgtgatcgccactgccccaaacatgttcaatgtgaacgatgtgttgggggtttacagcccccatccatgatgaacaagaagtctcg  
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ccgcgatcgctaactaccgtaacttgaacattgcacctgccgaggacggetatcgtttagcgggattccccccgatcatcaggcgctgg  
cgtgaggaaccgtggatccatcacgcccctcaggggtgcgggaacagtagtcgcCATATGAACATGCACGTGGA  
CATGGGTCGTGCGCTGACCGTTTCGTGATTGGCCGGCGCTGGAGGCGCTGGCGAA  
AACCATGCCGCGGATGCGGGTGCGGTGCGATGACCGATGATGACCTGCGTGC  
GGCGGTGTGGACCGTCGTGTTCCGGAGCAGAAGCTGGGTGCGGCGATTGATGA  
ATTGCGGAGCCTGCGTCTGCCGGATCGTATCGACGGTCGTTTCGTGGATGGCCGT  
CGTGCGAACCTGACCGTTTTTGTATGATGCGCGTGTGCGGTTTCGTGGTCATGCGC  
GTGCGCAACGTAACCTGCTGGAGCGTCTGGAGACCGAACTGCTGGGTGGCACC  
TGGATACCGCGGTGACGAAGGTGGCATTACGCCGACCCGATCCTGCAAGGCC  
TGGTGGATGTTATCGGTGAGGCAAAAGCGATATTGACGCGTACGCGACCATCG  
TGGAAGGTCTGACCAAGTATTTTCAAAGCGTGGCGGACGTTATGAGCAAACTGC  
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ATCAAAGCGCTGATTACGCAAGTGATCGACCACCTGCCGACCATGCAGCTGCCG  
AAGGGTGCGGATATTGCGCGTTGGCGTAAAGAGCTGGGCGACGCGGTTAGCATC  
AGCGATAGCGGTGTGGTTACCATTAAACCCGACAACTGATCAAGATGCGTGAT  
AGCCTGCCGCCGGATGGCACCGTTTGGGATACCGCGCTTACCAAGCGTGGAAC  
ACCGCGTTAGCGGTGAGAAAGGCCAGCATCCGGAACGTCGTGCGGATGCGCGT  
CGTAAATATAGCCACCAGAACAGCAACTTTGATAACCTGGTGAAGGTTCTGAGC  
GGTGCGATTAGCACCTGACCGACACCCAGAGCTATCTGCAAATCAAGCTTATG  
AGCAGCGGTGTTCAAGGTGGCCCGCGCGAAGCGAAGCGGTACAGACCCAC  
CCGCTGCGTGATGCGGCGAGCGCGCTGGGCACCCCTGAGCCCGCAGGCGTATGTG  
GATGTGGTTAGCGCGCGCAACGTAACCTTCCTGGAGCGTATGAGCCAACCTGGCG  
AGCGAAGTGTGCGATGCGCAACCGCGCGCATGATGCGCGTCTGGATGATCGT  
CCGCGCTGCGTGCGCCGAGGAACGTGACGCGCCGCGCTGGGTGCGAGCGAT  
ACCGGTAGCCGTGCGAGCGGTGCGGCGAACTGACCGAGCTGCTGGGTGTGCTG  
ATGAGCGTTATTAGCGCGAGCAGCCTGGACGAACTGAAGCAACGTAGCGATATC  
TGGAACAGATGAGCAAAGCGCGCAAGACAACCTGAGCCGTCTGAGCGATGCG  
TTTCAGCGTGCGACCGACGAGGCGAAAGCGGCGGCGATGCGGCGGAACAGGC  
GGCGGCGGCGGCAAGCAAGCGGTGCGGACGCGAAGCGGCGGATGCGGCGG  
TGGATGCGGCGCAAAACGTTACGATGACGCGGTTAAGCAGGGCCTGCCGGATG  
ACCGTCTGCAAAGCCTGAAAGCGGCGCTGGAGCAGGCGCGTCAGCAAGCGGGTG  
ATGCGCATGTCGTGCGGATGCGCTGACGGCGGATGCGACCAAGAACTGGACG  
CGCGAGCGCGCTGGCGACCCAAGCGCGTGCGTGCGAACAGCAAGTGGATGACG  
CGGTTAACAGGCGACCCAGCAATATGGTGCGAGCGCGAGCCTGCGTACCCGCG  
AAAGCCCGCGTCTGAGCGGTGCGGCGGAGCTGACCGCGGTGCTGGGCAAGCTGC  
AGGAACTGATTAGCAGCGGCAACGTTAAAGAGCTGGAAAGCAAGCAGAACTG  
TTCACCGAGATGCAAGCGAAGCGTGAGGCGGAACTGCAAAGAAAAGCGACGA

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ATATCAGGCGCAAGTGAAGAAAGCGGAGGAAATGCAGAAAACGATGGGTTGCA  
TCGGCAAGATTGTGGGTGGGTTATTACCGCGGTTAGCTTTGCGGCGGCGCGTT  
TACCGGTGGCGCGAGCCTGGCGCTGGCGGCGGTGGGCTGGCGCTGGCGGTTGG  
TGACGAGATTAGCCGTGCGACCAACCGGTGTGAGCTTCATGGACAAGCTGATGCA  
GCCGGTTATGGATGCGATCCTGAAACCGCTGATGGAGATGATTAGCAGCCTGAT  
CACCAAGGCGCTGGTTGCGTGCGGCGTTGATCAGCAAAAAGCGGAAC TGCGGG  
TGCGATTCTGGGTGCGGTTGTTACCGGTGTGGCGCTGGTTGCGGCGGCGTTTGT  
GGTGCGAGCGCGGTGAAAGCGGTTGCGAGCAAGGTTATCGACGCGATGGCGGGT  
CAGCTGACCAAGCTGATGGATAGCGGATTGGCAAAATGCTGGTGCAACTGATC  
GAGAAATT CAGCGAAAAGAGCGGTCTGCAGGCGCTGGGTAGCCGTACCGCGACC  
GCGATGACCCGTATGCGTCGTGCGATTGGCGTTGAGGCGAAGGAAGACGGTATG  
CTGCTGGCGAACCCTTTTGAAAAGCGGGCACCGTGATGAACGTTGGTAACCAA  
GTGAGCCAAGCGGCGGGTGGCATTGTGTTGCGGTTGAGCGTGCGAAAGCGATG  
GGTCTGCTGGCGGATGTGAAAGAAGCGATGTATGACATCAAGCTGCTGGGTGAT  
CTGCTGAAACAGGCGGTGGACGCGTTTGCGGAGCACAACTGTTCTGGCGCAA  
CTGATGCAGCAAAATGAGCGATGCGGGCGAAATGCAGACCAGCACCGGCAAGCTG  
ATCCTGCGTAACGCGCGTGCGGTTTAAGGATCC

LTA1-BurkF Amino acid sequence

SEQ ID NO: 28

MDNGDRLYRADSRPPEIKRSGGLMPRGHNEYFDRGTQMNINLYDHARGTQ  
TGFVRYDDGYVSTSLSLRSAHLAQSSILSGYSTYYIYVIATAPNMFNVNDVLGVYSP  
HPYEQEVSA LGGIPYSQIYGWYRVNFGVIDERLHRNREYDRYYRN LNIAPAEDGYR  
LAGFPDPDHQAWREEPWIIHAPQCGNSSRMNMHVDMGRALTVRDWPAL EALAKT  
MPADAGARAMTDDDLRAAGVDRRVPEQKLGA AIDEFASLRLPDRIDGRFVDGRR A  
NLTVFDDARVAVRGHARAQRNLLERLETELLGGTLDTAGDEGGIQDPPI LQGLVDVI  
GQKSDIDAYATIVEGLTKYFQSVADVMSKLQDYISAKDDKNMKIDGGKIKALIQQ  
VIDHLPTMQLPKGADIARWRKELGDAVSISDSGVVTINPDKLIKMRDSLPPDGTVWD  
TARYQAWN TAFSGQKGQHPERRADARRKYSHQNSNFDNLVKVLSGAISTLTDTQSY  
LQIKLMSSGVQGGPAANANAYQTHPLRDAASALGTLSPQAYVDVVSA AQRNFLER  
MSQLASEQCD AQPAAHDARLDDRPALRAPQERDAPPLGASDTGSRASGA AKLTELL  
GVLMSVISASSLDELKQRSDIWNQMSKAAQDNLSRLSDAFQ RATDEAKAAADAEQ  
AAAAAKQAGADAKAADA AVDAAQKRYDDAVKQGLPDDR LQSLKAAL EQRQQA  
GDAHGRADALQADATKKLDAASALATQARACEQQVDDAVNQATQQY GASASLRT  
PQSPRLSGAELTAVLGKLQELISSGNVKELESKQKLFTEMQAKREAE LQKKSDEYQ  
AQVKKAEEMQKTMGCIGKIVGWVITAVSF AAAFTGGASLALAAVGLALAVGDEIS  
RATTGVSFMDKLMQPVMDAILKPLMEMISSLITKALVACGVDQQAELAGAILGAV  
VTGVALVAAAFVGASAVKAVASKVIDAMAGQLTKLMDSAIGKMLVQLIEKFSEKS  
GLQALGSR TATAMTRMRAIGVEAKEDGMLLANRFEKAGTVMNVGNQVSQAAGG  
IVVGV ERAKAMGLLADVKEAMYDIKLLGDLLKQAVDAFAEHN RVLAQLMQQMSD  
AGEMQTSTGKLILRNARAV -

-continued

His-PcrH (Chaperone of PaF) nucleic acid sequence

SEQ ID NO: 29

ATGGGCAGCAGCCATCACCATCATCACCACAGCCAGGATCCGATGAACCA  
GCCGACCCCTTCCGACACCGACCAGCAACAGGCGCTGGAGGCCTTCTGCGCGA  
CGGCGGCACCCTGGCGATGCTTCGCGGACTCAGCGAGGACACCCTGGAGCAGCT  
CTATGCGCTGGGCTTCAACCAGTACCAGGCGGGCAAGTGGGACGACGCGCAGAA  
GATCTTCCAGGCACGTGTCATGCTCGACCACTACGACGCCCGCTACTTTCTCGGC  
CTGGGCGCCTGCCGCCAGTCCCTCGGTCTCTATGAACAGGCCCTGCAGAGCTACA  
GCTACGGCGCGCTGATGGACATCAACGAGCCGCGCTTCCCTTCCATGCCGCCGA  
GTGCCACCTGCAACTGGGTGATCTCGACGGAGCCGAGAGTGGCTTCTACTCGGCC  
CGGGCCCTGGCCGCGCACAGCCGCGCACGAGGCCCTGGCCGCGCTGCCGGC  
GCCATGTTGGAAGCCGTAACCGCGAGAAAGGATCGAGCCTATGAATCCGATAAC  
GCTTGAAAGCTT

His-PcrH (Chaperone of PaF) Amino acid sequence

SEQ ID NO: 30

MGSSHHHHHSQDPMNQPTPSDQQALEAFLRDGGTLAMLRGLSEDTLE  
QLYALGFNQYQAGKWDDAQKIFQALCMLDHYDARYFLGLGACRQSLGLYEQALQS  
YSYGALMDINEPRFPFHAAEHLQLGLDLGAESGFYSARALAAAQPAHEALAARAG  
AMLEAVTARKDRAYESDNA-

PcrV nucleic acid sequence

SEQ ID NO: 31

CATATGGAAGTCAGAAACCTTAATGCCGCTCGCGAGCTGTTCTCTGGACGA  
GCTCCTGGCCGCTCGGCGGCGCTGCCAGTGCCGAGCAGGAGGAACGTGCTGGC  
CCTGTTGCGCAGCGAGCGGATCGTGCTGGCCACGCCGCCAGCCGCTGAGCGA  
GGCGCAAGTGCTCAAGGCGCTCGCCTGGTTGCTCGCGCCAATCCGTCGCGCCT  
CCGGGGCAGGGCCTCGAGGTACTCCGCGAAGTCCTGCAGGCACGTCCGCAGCCC  
GGTGCGCAGTGGGATCTGCGTGAGTTCCTGGTGTGCGCCTATTTAGCCTGCACG  
GGCGTCTCGACGAGGATGTCATCGGTGTCTACAAGGATGTCCTGCAGACCCAGG  
ACGGCAAGCGCAAGGCGCTGCTCGACGAGCTCAAGGCGCTGACCGCGGAGTTGA  
AGGTCTACAGCGTGATCCAGTCGAGATCAACGCCGCGCTGTCGGCCAGGCAGG  
GCATCAGGATCGACGCTGGCGGTATCGATCTGGTCGACCCACGCTATATGGCTA  
TGCCGTCGCGGATCCCAGGTGGAAGGACAGCCCCGAGTATGCGCTGCTGAGCAA  
TCTGGATACCTTCAGCGGCAAGCTGTCGATCAAGGATTTTCTAGCGGCTCGCCG  
AAGCAGAGCGGGAACTCAAGGCCCTCAGCGATGAGTACCCCTTCGAGAAGGAC  
AACAACCCGGTCGGCAATTTCCGCCACCACGGTGAGCGACCGCTCGCGTCCGCTG  
AACGACAAGGTCAACGAGAAGACCACCCTGCTCAACGACACCAGCTCCCGCTAC  
AACTCGGCGGTGAGGCGCTCAACCGCTTCATCCAGAAATACGACAGCGTCCTG  
AGCGACATTCTCAGCGCGATC

PcrV amino acid sequence

SEQ ID NO: 32

MEVRNLNAARELFLDELLAASAPASAEQEELLALLRSERIVLAHAGQPLSEA  
QVLKALAWLLAANPSAPPQGLEVLREVLQARRQPGAQWDLREFLVSAFYSLHGRL  
DEDVIGVYKDLQTDGKRKALLDELKALTAEKVVSVIQSINAALSARQGIRIDA

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GGIDLVDPTLYGYAVGDPWKDSPEYALLSNLDTFSGKLSIKDFLSGSPKQSGELKG  
LSDEYPPFEKDNPNVGNFATTVSDRSRPLNDKVNEKTLLNDTSSRYNSAVEALNRFI  
QKYDSVLSDILSAI

PopB nucleic acid sequence

SEQ ID NO: 33

ATGAACCCGATTACGCTGGAACGTGCTGGTCTGCCGTATGGTGTGCCGA  
TGCTGGTGACATCCCGGCTCTGGGTCGCCCCGGTCGCACGTGATGTGGAAGTCTG  
CGTGTGTAACGTCTGGCAGCACCGGCAGCTGCAAGCGCATCTGGCACCGGTGTC  
GCTCTGACGCCGCCGTCTGCAGCAAGTCAGCAACGTCTGGAAGTTGCTAACCGC  
GCGGAAATTGCCCTCACTGGTCCAGGCAGTGGGTGAAGACGTGGGTCTGGCACGT  
CAAGTGGTTCTGGCAGGTGCATCGACCCTGCTGAGCGCAGGTCTGATGTGCCCGC  
AGGCGTTCGAAATTGAACTGGCCAAAATCACCGGCGAAGTTGAAAATCAGCAGA  
AAAACTGAACTGACGGAATCGAACAGGCCCGTAAACAGAACCTGCAAAAA  
ATGGAAGATAACCAAGCAAAAAATCCGCGAATCGGAAGAAGCTGCCAAGAAGC  
GCAGAAAAGCGGCTGGCCGCAAAAATTTTGGTTGGATTTCTGCTATCGCGAGT  
ATTATCGTGGGTGCAATCATGGTTGCAACCGGTGTCGGTGCTGCAGCAGGTGCAC  
TGATGATTGCTGGCGGTGTCATGGGTGTCGTGAGTCAGTCCGTGCAGCAAGCAGC  
TGCGGATGGTCTGATCTCAAAGAAGTGATGGAAAACTGGGCCCGGCCCTGAT  
GGGTATTGAAATGGCCGTGGCACTGCTGGCCGAGTTGTCTCCTTGGTGGTTCA  
GCAGTTGGTGGTCTGGCACGTCTGGGTGCAAAAAATCGGCGGTAAAGCTGCGGAA  
ATGACGGCATCCCTGGCTTCAAAGTGGCAGACCTGGGCGGTAAATTCGGCTCTC  
TGCGGGGCCAGTCACTGTGCGCATAGCCTGAAACTGGGTGTGCAAGTTTCTGATCT  
GACCCCTGGACGTTGCAACGGCGCGCACAGGCTACGCACAGTGGTTTTCAAGC  
GAAAGCTGCGAATCGTCAGGCCGATGTTCAAGAATCCCGTGCAGACCTGACCAC  
GCTGCAGGGTGTCAATGAACGTCTGAAAGAAGAACTGAGCCGCATGCTGGAAGC  
CTTTCAGGAAATTATGGAACGCATCTTCGCAATGCTGCAAGCGAAAGGCGAAAC  
CCTGCACAATCTGTCTTCCCGTCCGGCGGCTATCTGAGGATCC

PopB amino acid sequence

SEQ ID NO: 34

MNPITLERAGLPYGVADAGDIPALGRPVARDVESLRVERLAAPAAASASGTG  
VALTPPSAASQQRLEVANRAEIASLVQAVGEDVGLARQVVLGASTLLSAGLMSPQ  
AFEIELAKITGEVENQQKKLKLTEIEQARKQNLQKMDNQKIRESEEAKEAQKSG  
LAAKIFGWISAIASIIVGAIMVATGVGAAAGALMIAGGVMGVVSQSVQQAADGLIS  
KEVMEKLGPALMGIEMAVALLAAVVSFPGSAVGGGLARLGAKIGKAAEMTASLAS  
KVADLGKKFGSLAGQSLSHSLKLGQVSDLTLDVANGAAQATHSGFQAKAANRQA  
DVQESRADLTTLQGVIERLKEELSRMLEAFQEIMERIFAMLQAKGETLHNLSSRPAAI

PaF nucleic acid sequence

SEQ ID NO: 35

CATATGGAAGTCAGAAACCTTAATGCCGCTCGCGAGCTGTTCTTGACGA  
GCTCCTGGCCGCTCGGCGGCCTGCCAGTCCCGAGCAGGAGGAAGTGTGGC  
CCTGTTGCGCAGCGAGCGGATCGTGCTGGCCACGCCGGCCAGCCGCTGAGCGA  
GGCGCAAGTGCTCAAGGCGCTCGCCTGGTTGCTCGCGGCCAATCCGTCCGCGCCT

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CCGGGGCAGGGCCTCGAGGTACTCCGCGAAGTCCTGCAGGCACGTCCGGCAGCCC  
GGTGCGCAGTGGGATCTGCGTGAGTTCCTGGTGTGGCCTATTTTCAGCCTGCACG  
GGCGTCTCGACGAGGATGTCATCGGTGTCTACAAGGATGTCTGCAGACCCAGG  
ACGGCAAGCGCAAGGCGCTGCTCGACGAGCTCAAGGCGCTGACCGCGGAGTTGA  
AGGTCTACAGCGTGATCCAGTCGCAGATCAACGCCGCGCTGTCGGCCAGGCAGG  
GCATCAGGATCGACGCTGGCGGTATCGATCTGGTCGACCCACGCTATATGGCTA  
TGCCGTGGCGATCCCAGGTGGAAGGACAGCCCCGAGTATGCGCTGCTGAGCAA  
TCTGGATACCTTCAGCGGCAAGCTGTCGATCAAGGATTTTCTCAGCGGCTCGCCG  
AAGCAGAGCGGGAACTCAAGGGCCTCAGCGATGAGTACCCCTTCGAGAAGGAC  
AACAACCCGGTCGGCAATTTCCGCCACCACGGTGAGCGACCGCTCGCGTCCGCTG  
AACGACAAGGTCAACGAGAAGACCACCTGCTCAACGACACCAGCTCCCGCTAC  
AACTCGGCGGTGAGGCGCTCAACCGCTTCATCCAGAAATACGACAGCGTCTTG  
AGCGACATTTCTCAGCGCGATCGGATCCATGAACCCGATTACGCTGGAACGTGCT  
GGTCTGCCGTATGGTGTGCCGATGCTGGTGACATCCCGGCTCTGGGTGCCCCGG  
TCGCACGTGATGTGGAAGTCTGCGTGTGAACGTCTGGCAGCACCGGCAGCTG  
CAAGCGCATCTGGCACCGGTGTCGCTCTGACGCCCGCTCTGCAGCAAGTCAGC  
AACGTCTGGAAGTTGCTAACCGCGCGGAAATTGCCTCACTGGTCCAGGCAGTGG  
GTGAAGACGTGGGTCTGGCACGTCAAGTGGTTCTGGCAGGTGCATCGACCCTGCT  
GAGCGCAGGTCTGATGTGCCGCGAGGCGTTGAAAATTGAAGTGGCAGAAATCAC  
CGCGGAAGTTGAAAATCAGCAGAAAAAACTGAACTGACGGAAATCGAACAGG  
CCCGTAAACAGAACCTGCAAAAAATGGAAGATAACCAGCAAAAAATCCGCGAA  
TCGGAAGAAGCTGCGAAAGAAGCGCAGAAAAAGCGCCTGGCCGCAAAAATTTT  
GGTTGGATTTCTGCTATCGCGAGTATTATCGTGGGTGCAATCATGGTTGCAACCG  
GTGTCCGTGCTGCAGCAGGTGCACTGATGATTGCTGGCGGTGTCTGGGTGTCTGT  
GAGTCAGTCCGTGCAGCAAGCAGCTGCGGATGGTCTGATCTCAAAGAAGTGAT  
GGAAAACTGGGCCCGGCCCTGATGGGTATTGAAATGGCCGTGGCACTGCTGGC  
CGCAGTTGTCTCCTTGGTGGTTCAGCAGTGGTGGTCTGGCACGTCTGGGTGCA  
AAAAATCGGCGGTAAAGCTGCGGAAATGACGGCATCCCTGGCTTCAAAGTGGCA  
GACCTGGGCGGTAAATTCGGCTCTCTGGCGGGCAGTCACTGTGCGATAGCCTGA  
AACTGGGTGTGCAAGTTTCTGATCTGACCTGGACGTTGCAACGGCGCCGACA  
GGCTACGCACAGTGGTTTTCAAGCGAAAGCTGCGAATCGTCAGGCCGATGTTCA  
AGAATCCCGTGCAGACCTGACCACGCTGCAGGGTGTATTGAACGTCTGAAAGA  
AGAACTGAGCCGATGCTGGAAGCCTTTCAGGAAATTATGGAACGCATCTTCGC  
AATGCTGCAAGCGAAAGCGAAACCTGCACAATCTGTCTTCCCGTCCGGCGGC  
TATCTGAGGATCC

PaF amino acid sequence

MEVRNLNAARELFLDELLAASAPASAEQEELLALLRSERIVLAHAGQPLSEA  
QVLKALAWLLAANPSAPPQGQLEVLREVLQARRQPGAQWDLREFLVSAFYSLHGRL  
DEDVIGVYKDLQTDGKRKALLDELKALTAEKLVYSVQSQINAALSARQGIRIDA

SEQ ID NO: 36

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GGIDLVDPTLYGYAVGDPRWKDSPEYALLSNLDTFSGKLSIKDFLSGSPKQSGELKG  
LSDEYPPFEKDNPNVGNFATTVSDRSRPLNDKVNEKTTLLNDTSSRYNSAVEALNRFI  
QKYDSVLSIDLISAIGSMNPFITLERAGLPYGVADAGDIPALGRPVARDVESLRVERLAA  
PAAASASGTGVALTPPSAASQQRLEVANRAEIASLVQAVGEDVGLARQVVLGAST  
LLSAGLMSPQAFIEELAKITGEVENQQKKLKLTEIEQARKQNLQKMEDNQKIRESE  
EAAKEAQKSGLAAKIFGWISAIASIIVGAIMVATGVGAAAGALMIAGGVMGVVSQS  
VQQAADGLISKEVMEKLGPALMGIEMAVALLAAVVSFPGGSVAVGLARLGAKIGG  
KAAEMTASLASKVADLGGKFGSLAGQSLSHSLKLGVSDDLTLVDVANGAAQATHS  
GPQAKAANRQADVQESRADLTTLQGVIERLKEELSRMLEAFQEIMERIFAMLQAKG  
ETLHNLSSRPAAI

LTA1-PaF nucleic acid sequence

SEQ ID NO: 37

CATatggacaaatggcgatcggttataccgtgccgactcgcgtccccagatgagattaaacgtagcggtaggggtaatgcc  
acgtgggcacaatgagtagttttgacgtggaacacagatgaacattaacctttacgatcatgccgtgggacccagaccgggtttgtcc  
gttatgatgacgggtatgttagtagcaggtttgtccttacgctccgcacaccttgccgggacaaagtattttatcaggctacagcacatattac  
atztatgtgatcgccactgccccaaacatgttcaatgtgaacgatgtgttgggggtttacagcccccatccatgaacaagaagtctcg  
gccccgtgggggatcccatatagccagatttatgggttggtaccgcgtaaatttgggtgatgtgaacgtttgcatcgtaaccgtgaata  
ccgcatcgctactaccgtaacttgaacattgcacctgccgaggacggctatcggttagcgggattcccaccgatcatcaggcgtgg  
cgtgaggaaacgtggatccatcacgcccctcaggggtgcgggaacagtagtcgcCATATGGAAGTCAGAAACCT  
TAATGCCGCTCGCGAGCTGTTCTTGACGAGCTCCTGGCCGCTCGGCGGCGCCT  
GCCAGTGCCGAGCAGGAGGAACGTGCTGGCCCTGTTGCGCAGCAGCGGATCGTG  
CTGGCCACGCGGCCAGCCGCTGAGCGAGGCGCAAGTGCTCAAGGCGCTCGCC  
TGGTTGCTCGCGGCCAATCCGTCGCGCCTCCGGGCGAGGCGCTCGAGGTACTCC  
GCGAAGTCTGCGAGGCAGTCGCGCAGCCCGTGCGCAGTGGGATCTGCGTGAGT  
TCCTGGTGTCGGCCTATTTTCAGCCTGCACGGGCGTCTCGACGAGGATGTCATCGG  
TGTCTACAAGGATGTCCTGCAGACCCAGGACGGCAAGCGCAAGGCGCTGCTCGA  
CGAGCTCAAGGCGCTGACCGCGAGTTGAAGTCTACAGCGTGATCCAGTCGCA  
GATCAACGCCGCGCTGTGCGCCAGGCAGGCATCAGGATCGACGCTGGCGGTAT  
CGATCTGGTCGACCCACGCTATATGGCTATGCCGTGCGCGATCCAGGTGGAAG  
GACAGCCCCGAGTATGCGCTGCTGAGCAATCTGGATACCTTCAGCGGCAAGCTG  
TCGATCAAGGATTTTCTCAGCGGCTCGCCGAAGCAGAGCGGGGAACCTCAAGGGC  
CTCAGCGATGAGTACCCCTTCGAGAAGGACAACAACCCGGTCGGCAATTCGCC  
ACCACGGTGAGCGACCGCTCGCGTCCGCTGAACGACAAGGTCAACGAGAAGACC  
ACCTTGCTCAACGACACCAGCTCCCGCTACAACCTCGGCGGTGAGGCGCTCAAC  
CGCTTCATCCAGAAATACGACAGCGTCTGAGCGACATTCTCAGCGGATCGGAT  
CCATGAACCCGATTACGCTGGAACGTGCTGGTCTGCCGTATGGTGTGCGGATGC  
TGGTGACATCCCGGCTCTGGGTGCGCCGCTCGCACGTGATGTGGAAGTCTGCGT  
GTTGAACGTCTGGCAGCACCGGCGAGCTGCAAGCGCATCTGGCACCGGTGTCGCT  
CTGACGCCCGCTCTGCGCAAGTCAGCAACGTCTGGAAGTTGCTAACCGCGCG  
GAAATGCCTCACTGGTCCAGGCAGTGGGTGAAGACGTGGGTCTGGCACGTCAA

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GTGGTTCTGGCAGGTGCATCGACCCTGCTGAGCGCAGGTCTGATGTCGCCGCAGG  
CGTTCGAAATTGAACTGGCCAAAATCACCGCGAAGTTGAAAATCAGCAGAAAA  
AACTGAACTGACGGAAATCGAACAGGCCCGTAAACAGAACCTGCAAAAAATG  
GAAGATAACCAGCAAAAAATCCGCGAATCGGAAGAAGCTGCGAAAAGAAGCGCA  
GAAAAGCGGCCTGGCCGCAAAAATTTTGGTTGGATTCTGCTATCGCGAGTATT  
ATCGTGGGTGCAATCATGGTTGCAACCGGTGTCGGTGTGTCAGCAGGTGCACTG  
ATGATTGCTGGCGGTGTATGGGTGTCGTGAGTCAGTCCGTGAGCAAGCAGCTG  
CGGATGGTCTGATCTCAAAGAAGTGATGGAAAACTGGGCCCGGCCCTGATGG  
GTATTGAAATGGCCGTGGCACTGCTGGCCGAGTTGTCTCCTTTGGTGGTTCAGC  
AGTTGGTGGTCTGGCACGTCTGGGTGCAAAAATCGGCGGTAAAGCTGCGGAAAT  
GACGGCATCCCTGGCTTCAAAGTGGCAGACCTGGGCGGTAAATTCGGCTCTCTG  
GCGGCCAGTCACTGTCGCATAGCCTGAACTGGGTGTGCAAGTTCTGATCTGA  
CCCTGGAGCTTGCAAACGGCGCCGCACAGGTACGCACAGTGGTTTCAAGCGA  
AAGCTGCGAATCGTCAGGCCGATGTTCAAGAATCCCGTGACACCTGACCACGC  
TGCAGGGTGTCAATTGAACGTCTGAAAGAAGAACTGAGCCGCATGCTGGAAGCCT  
TTCAGGAAATTATGGAACGCATCTTCGCAATGCTGCAAGCGAAAGGCGAAACCC  
TGCACAATCTGTCTCCCGTCCGGCGGCTATCTGAGGATCC

LTA1-PaF Amino acid sequence

SEQ ID NO: 38

MDNGDRLYRADSRPPDEIKRSGGLMPRGHNEYFDRGTQMNINLYDHARGTQ  
TGFVRYDDGYVSTSLSLRSAHLAQQSILSGYSTYYIYVIATAPNMFNVNDVLGVYSP  
HPYEQEVSA LGGIPYSQIYGWYRVNFGVIDERLHRNREYDRYYRNLNIIAPAEDGYR  
LAGFPDPDQAWREEPWIHHAPQCGNSRMEVRNLNAARELFLDELLAASAAPASA  
EQEELLALLRSEIRIVLAHAGQPLSEAQVLKALAWLLAANPSAPPQGLEVLREVLQA  
RRQPGAQWDLREFLVSAFYSLHGRLEDVIGVYKDVLTQDGKRAKLLDELKALT  
AELKVYSVQSQINAALSARQGIRIDAGGIDLVDPTLYGYAVGDPRWKDSPEYALLS  
NLDTFSGKLSIKDPLSGSPKQSGELKGLSDEYPFEKDNNPVGNFATTVSDRSRPLNDK  
VNEKTTLLNDTSSRYNSAVEALNRFIQYDSVLSLILSAIGSMNPITLERAGLPYGVA  
DAGDIPALGRPVARDVESLRVERLAAPAAASASGTGVALTPPSAASQQRLEVANRAE  
IASLVQAVGEDVGLARQVVLGASTLLSAGLMSPQAFETELAKITGEVENQQKKLKL  
TEIEQARKQNLQKMEDNQKIRESEEAKEAQKSGLAAKIFGWISAIASIIVGAIMVA  
TGVGAAAGALMIAGGVMGVVSQSVQQAADGLISKEVMEKLGPAALMGIEMAVAL  
LAAVVSFGGS AVGGLARLGA KIGKAAEMTASLASKVADLGGKFGSLAGQSLSHSL  
KLG VQVSDLTLDVANGAAQATHSGFQAKAANRQADVQESRADLTTLQGVIERLKE  
ELSRMLEAFQEIMERIFAMLQAKGETLHNLSSRPAAI

SyncD (Chaperone for YerF) nucleic acid sequence

SEQ ID NO: 39

ATGGGCAGCAGCCATCACCATCATCACACAGCCAGGATCCGatgcaacaagag  
acgacagacactcaagaataccagctggcaatggaatccttcctaaaaggagggggaactatcgccatgctcaacgaaatttcaagt  
acacttttagagcaactctactctcttgcgtttaaccaataccagtcaggaaaatacaggatgctcacaaggctctttcaagctctctgtgtg  
ctagaccactatgattcacgtttctcttttagggctaggcgcttgctgctcaagccatggggcaatacgacttagcgattcatagctacagcta



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tggcgccataatggatataaaagaacctcggttttccggtttcatgctgccgaatgtttactgcaaaaggagagcttgctgaagcagaaag  
tggtctgttcttggctcaagagcttatcgagacaaacctgagtttaaggagctttccaccgaggttagctcaatgttagaagcaattaaat  
tgaaaaaggagatggaacatgagtgcggttgataacccatga**AAAGCTT**

SycD (Chaperone for YerF) Amino acid sequence

SEQ ID NO: 40

MGSSHHHHHSSGLVPRGSHMQQETTDTEYQLAMESFLKGGGTIAMLNEIS  
SDTLEQLYSLAFNQYQSGKYEDAHKVFQALCVLDHYDSRFGLGLGACRQAMGQYD  
LAIHSYSYGAIMDIKEPRFPFHAAECLLQKGELAEAESGLFLAQELIADKPEFKELSTR  
VSSMLEAIKLLKEMEHECVDNP-

LcrV amino acid sequence

SEQ ID NO: 41

CATATGATTAGAGCCTACGAACAAACCCACAACATTTTATTGAGGATCT  
AGAAAAAGTTAGGGTGAACAACCTACTGGTCATGGTTCTTCAGTTTGTAGAAGA  
ATTGGTTCAGTTAGTCAAAGATAAAATATAGATATTTCCATTAAATATGATCCC  
AGAAAAGATTGCGAGGTTTTTGC CAATAGAGTAATTACTGATGATATCGAATTGC  
TCAAGAAAATCCTAGCTTATTTCTACCCGAGGATGCCATTCTTAAAGCGGTCA  
TTATGACAACCAACTGCAAAATGGCATCAAGCGAGTAAAGAGTTCCTTGAATC  
ATCGCCGAATACACAATGGGAATTGCGGCGTTCATGGCAGTAATGCATTTCTCT  
TTAACCGCCGATCGTATCGATGATGATATTTTGAAAGTGATTGTTGATTCAATGA  
ATCATCATGGTGATGCCCGTAGCAAGTTGCGTGAAGAATTAGCTGAGCTTACCGC  
CGAATTAAAGATTTATTTCAGTTATTCAAGCCGAAATTAATAAGCATCTGTCTAGT  
AGTGGCACCATAAATATCCATGATAAATCCATTAATCTCATGGATAAAAAATTTAT  
ATGGTTATACAGATGAAGAGATTTTAAAGCCAGCGCAGAGTACAAAATTCCTCG  
AGAAAATGCCCTCAAACCACCATTGAGGTGGATGGGAGCGAGAAAAAATAGTCT  
CGATAAAGGACTTTCTTGGAAGTGAGAATAAAAGAACCGGGCGTTGGGTAATC  
TGAAAACTCATACTCTTATAATAAGATAATAATGAATTATCTCACTTTGCCAC  
CACCTGCTCGGATAAGTCCAGGCCGCTCAACGACTTGGTTAGCCAAAAACAAC  
TCAGCTGTCTGATATTACATCACGTTTTAATTCAGCTATTGAAGCACTGAACCGTT  
TCATTAGAAATATGATTGAGTGATGCAACGCTCTGCTAGATGACAGCTCTGGTAA  
A

LcrV amino acid sequence

SEQ ID NO: 42

MIRAYEQNPQHFIEDLEKVRVEQLTGHGSSVLEELVQLVKDKNIDISIKYDPR  
KDSEVFANRVITDDIELLKKILAYFLPEDAILKGGHYDNQLQNGIKRVKEFLESSPNT  
QWELRAFMAVMHFSLTADRIDDDILKVIVDSMNHHGDARSKLREELAEELTAELKIYS  
VIQAEINKHLSSSGTINIHDKSINLMDKNLYGYTDEEIFKASAEYKILEKMPQTTIQVD  
GSEKKIVSIKDFLGSSENKRTGALGNLKNYSYNKDNNELSHFATTCDSKSRPLNDLV  
SQKTTQLSDITSRFNSAIEALNRFIQKYDSVMQRLDDTSGK

YopB nucleic acid sequence

SEQ ID NO: 43

ATGAGTGCGTTGATAACCCATGATCGCTCAACGCCAGTAAGTGAAGTCT  
ACTTCCCTACGTCGAGACACCAGCGCCCGCCCCCTTCAGACTCAACAAGTCGCG  
GGAGAACTGAAGGATAAAATGGTGGGTGAGTTCTCAGGGCGTACAGCTCCCT

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GCACCACTAGCAGTGGTTGCCAGCCAAGTCACTGAAGGACAACAGCAAGAAATC  
ACTAAATTATTGGAGTCGGTCACCCGCGGCACGGCAGGATCTCAACTGATATCA  
AATTATGTTTCAGTGCTAACGAATTTTACGCTCGCTTCACCTGATACATTTGAGAT  
TGAGTTAGGTAAGCTAGTTTCTAATTTAGAAGAAGTACGCAAAGACATAAAAAAT  
CGCTGATATTAGCGTCTTCATGAACAAAACATGAAGAAAATTGAAGAGAATCA  
AGAGAAAATCAAAGAAACAGAAGAGAATGCCAAGCAAGTCAAGAAATCCGGCA  
TGGCATCAAAGATTTTGGCTGGCTCAGCGCCATAGCCTCAGTGGTTATCGGTGC  
CATCATGGTGGCCTCAGGGGTAGGAGCCGTTGCCGGTGCAATGATGATTGCCTCA  
GGCGTAATTGGGATGGCGAATATGGCTGTGAAACAAGCGCGGAAGATGGCCTG  
ATATCCCAAGAGGCAATGCAAGTATTAGGGCCGATACTCACTGCGATTGAAGTC  
GCATTGACTGTAGTTTCAACCGTAATGACCTTTGGCGGTTCCGGCACTAAAATGCC  
TGGCTGATATTGGCGCAAAACTCGGTGCTAACACCGCAAGTCTTGCTGCTAAAGG  
AGCCGAGTTTTCGGCCAAAGTTGCCCAAATTCGACAGGCATATCAAACACTGTC  
GGGAATGCAGTGACTAAATTAGGGGGCAGTTTGGTAGTTTAAACAATGAGCCAT  
GTAATCCGTACAGGATCACAGGCAACACAAGTCGCCGTTGGTGTGGGCAGCGGA  
ATAACTCAGACCATCAATAATAAAAAACAAGCTGATTTACAACATAATAACGCT  
GATTTGGCCTTGAACAAGGCAGACATGGCAGCGTTACAAAGTATTATTGACCGA  
CTCAAAGAAGAGTTATCCCATTTGTCAGAGTCACATCAACAAGTATGGAAGT  
ATTTTCCAGATGATTAATGCAAAAGGTGACATGCTGCATAATTTGGCCGGCAGAC  
CCCATACTGTTTAAGGTACC

YopB amino acid sequence

SEQ ID NO: 44

MSALITHDRSTPVTGSLLPYVETPAPAPLQTQQVAGELKDKNGGVSSQGVQL  
PAPLAVVASQVTEGQQQEITKLLESVTRGTAGSQLISNYVSVLTNFTLASPDTFEIELG  
KLVSNLEEVKDKIKIADIQRLHEQNMKKIEENQEIKETEENAKQVKKSGMASKIFG  
WLSAIASVVGAIMVASGVGAVAGAMMIASGVIGMANMAVKQAAEDGLISQEAMQ  
VLGPILTAIEVALTVVSTVMTFGGSALKCLADIGAKLGANTASLAAKGAEPsAKVAQ  
ISTGISNTVGNVTKLGSFGLTMSHVIRTGSQATQVAVGVSGITQTINNKKQADL  
QHNNADLALNKADMAALQSIIDRLKEELSHLSESHQQVMELIFQMINAKGDMHLNL  
AGRPHTV

YerF nucleic acid sequence

SEQ ID NO: 45

CATATGATTAGAGCCTACGAACAAAACCCACAACATTTTATTGAGGATCT  
AGAAAAAGTTAGGGTGAACAACCTTACTGGTCATGGTTCTTCAGTTTGTAGAAGA  
ATTGTTTCAGTTAGTCAAAGATAAAAATATAGATATTTCCATTAAATATGATCCC  
AGAAAAGATTTCGAGGTTTTTGCCAATAGAGTAATTACTGATGATATCGAATTGC  
TCAAGAAAATCCTAGCTTATTTTCTACCCGAGGATGCCATTCTTAAAGGCGGTCA  
TTATGACAACCAACTGCAAAATGGCATCAAGCGAGTAAAGAGTTCTTGAATC  
ATCGCCGAATACACAATGGGAATTGCGGCGTTTCATGGCAGTAATGCATTCTCT  
TTAACCGCCGATCGTATCGATGATGATATTTTGAAAGTGATTGTTGATTCAATGA  
ATCATCATGGTGATGCCCGTAGCAAGTTGCGTGAAGAATTAGCTGAGCTTACCGC

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CGAATTAAAGATTATTTCAGTTATTCAAGCCGAAATTAATAAGCATCTGTCTAGT  
AGTGGCACCATAAATATCCATGATAAATCCATTAATCTCATGGATAAAAAATTTAT  
ATGGTTATACAGATGAAGAGATTTTAAAGCCAGCGCAGAGTACAAAATTCCTCG  
AGAAAATGCCTCAAACCACCATT CAGGTGGATGGGAGCGAGAAAAAATAGTCT  
CGATAAAGGACTTTCTTGGAGTGAGAATAAAGAACC GGGCGTTGGGTAATC  
TGAAAACTCATACTCTTATAATAAGATAATAATGAATTATCTCACTTTGCCAC  
CACCTGCTCGGATAAGTCCAGGCCGCTCAACGACTTGGTTAGCCAAAAACAAC  
TCAGCTGTCTGATATTACATCACGTTTTAATTCAGCTATTGAAGCACTGAACCGTT  
TCATTAGAAAATATGATT CAGTGATGCAACGTCTGCTAGATGACACGTCTGGTAA  
AGGATCCATGAGTGCGTTGATAACCCATGATCGCTCAACGCCAGTAAGTGAAG  
TCTACTTCCCTACGTCGAGACACCAGCGCCCGCCCCCTTCAGACTCAACAAGTC  
GCGGGAGAACTGAAGGATAAAAATGGTGGGGTGAGTTCTCAGGGCGTACAGCTC  
CCTGCACCCTAGCAGTGGTTGCCAGCCAAGTCACTGAAGGACAACAGCAAGAA  
ATCACTAAATTATTGGAGTCGGTCACCCCGGCACGGCAGGATCTCAACTGATAT  
CAAATTATGTTTCAGTGCTAACGAATTTTACGCTCGCTTCACCTGATACATTTGAG  
ATTGAGTTAGGTAAGCTAGTTTCTAATTTAGAAGAAGTACGCAAGACATAAAA  
ATCGCTGATATT CAGCGTCTTCATGAACAAAACATGAAGAAAATTGAAGAGAAT  
CAAGAGAAAAATCAAAGAAACAGAAGAGAAATGCCAAGCAAGTCAAGAAATCCGG  
CATGGCATCAAAGATTTTGGCTGGCTCAGCGCCATAGCCTCAGTGGTTATCGGT  
GCCATCATGGTGGCCTCAGGGGTAGGAGCCGTTGCCGGTGCAATGATGATTGCCT  
CAGGCGTAATTGGGATGGCGAATATGGCTGTGAAACAAGCGCGGAAGATGGCC  
TGATATCCCAAGAGGCAATGCAAGTATTAGGGCCGATACTCACTGCGATTGAAG  
TCGCATTGACTGTAGTTTCAACCGTAATGACCTTTGGCGGTTCCGGCACTAAAATG  
CCTGGCTGATATTGGCGCAAACTCGGTGCTAACACCGCAAGTCTTGCTGCTAAA  
GGAGCCGAGTTTTCGGCCAAAGTTGCCCAAATTTGACAGGCATATCAAACT  
GTCCGGGAATGCAGTGACTAAATTAGGGGCGAGTTTGGTAGTTTAAACATGAGC  
CATGTAATCCGTACAGGATCACAGGCAACACAAGTCGCCGTTGGTGTGGGCAGC  
GGAATAACTCAGACCATCAATAATAAAAAACAAGCTGATTTACAACATAATAAC  
GCTGATTTGGCCTTGAACAAGGCAGACATGGCAGCGTTACAAAGTATTATTGACC  
GACTCAAAGAAGAGTTATCCATTTGT CAGAGTCACATCAACAAGTATGGAAC  
TGATTTTCAGATGATTAATGCAAAAGGTGACATGCTGCATAATTGGCCGCGAG  
ACCCCATACTGTTTAAGGTACC

YerF amino acid sequence

MIRAYEQNPQHFIEDLEKVRVEQLTGHGSSVLEELVQLVKDKNIDISIKYDPR  
KDSEVFANRVITDDIELLKKILAYFLPEDAILKGGHYDNQLQNGIKRVKEFLESSPNT  
QWELRAFMAVMHFSLTADRIDDDILKVIIVDSMNHGHDARSKLREELAELETAEIKIYS  
VIQAEINKHLSSSGTINIHDKSINLMDKNLYGYTDEEIFKASAEYKILEKMPQTTIQVD  
GSEKKIVSIKDFLGSSENKRTGALGNLKNYSYNKDNNELSHFATTCSDKSRPLNDLV  
SQKTTQLSDITSRFNSAIEALNRFIQKYDSVMQRLDDTSGKGSMSALITHDRSTPVT

SEQ ID NO: 46

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GSLLPYVETPAPAPLQTTQQVAGELKDKNGGVSSQGVQLPAPLAVVASQVTEGQQQE  
ITKLESVTRGTAGSQLISNYVSVLTNFTLASPDTFEIELGKLVSNL EEVRKDIKIADIQ  
RLHEQNMKKI EENQEKI KETEENAKQVKKSGMASKIFGWLSAIASVVIGAIMVASGV  
GAVAGAMMIASGVIGMANMAVKQAAEDGLISQEQVLGPILTAEI VALTVVSTV  
MTFGGSALKCLADIGAKLGANTASLAAKGAEFSAKVAQISTGISNTVGNVTKLGGS  
FGSLTMSHVIRTGSQATQVAVGVGSGITQTINNKKQADLQHNNADLALNKADMAA  
LQSIIDRLKEELSHLSESHQQVMELIFQMINAKGDMHLHNLAGRPHTV

LTA1-YerF nucleic acid sequence

SEQ ID NO: 47

CATatggacaatggcgatcggttataccgtgccgactcgcgtccccagatgagattaaacgtagcggtagggttaatgcc  
acgtgggcacaatgagatattttgacgtggaacacagatgaacattaacctttacgatcatgcccgtgggacccagaccgggtttgtcc  
gttatgatgacgggtatgttagtacgagtttgtccttacgctccgcacaccttgccgggacaaagtattttatcaggctacagcacatattac  
atttatgtgatcgccactgccccaaacatgttcaatgtgaacgatgtgttgggggtttacagcccccatccatgaacaagaagctctcg  
gcccttgggggatcccatatagccagatttatggttggtagccgcgtaaattttgggtgatgatgaacgtttgcatcgtaaccgtgaata  
ccgcatcgctactaccgtaacttgaacattgcacctgccgaggacggctatcggttagcgggattcccaccgatcatcaggcgtgg  
cgtgaggaaccgtggatccatcacgccctcaggggtgcgggaacagtagtcgcCATATGATTAGAGCCTACGA  
ACAAAACCCACAACATTTATTGAGGATCTAGAAAAAGTTAGGGTGGAACAAC  
TACTGGTCATGGTTCTTCAGTTT TAGAAGAATTGGTTCAGTTAGTCAAAGATAAA  
AATATAGATATTTCCATTAAATATGATCCAGAAAAGATTCCGAGGTTTTTGCCA  
ATAGAGTAATTACTGATGATATCGAATTGCTCAAGAAAATCCTAGCTTATTTTCT  
ACCCGAGGATGCCATTCTTAAGGCGGTCATTATGACAACCAACTGCAAAATGG  
CATCAAGCGAGTAAAAGAGTTCCTTGAATCATCGCCGAATACACAATGGGAATT  
GCGGGCGTTCATGGCAGTAATGCATTTCTCTTAAACCGCCGATCGTATCGATGAT  
GATATTTTGAAAGTGATTGTTGATTCAATGAATCATCATGGTGATGCCCGTAGCA  
AGTTGCGTGAAGAATTAGCTGAGCTTACCGCCGAATTAAAGATTTATTCAGTTAT  
TCAAGCCGAAATTAATAAGCATCTGTCTAGTAGTGGCACCATAAATATCCATGAT  
AAATCCATTAATCTCATGGATAAAAAATTTATATGGTTATACAGATGAAGAGATTT  
TTAAAGCCAGCGCAGAGTACAAAATTCGAGAAAAATGCCTCAAACCACTTC  
AGGTGGATGGGAGCGAGAAAAAATAGTCTCGATAAAGGACTTCTTGGAAGTG  
AGAATAAAAGAACCGGGCGTTGGGTAATCTGAAAACTCATACTCTTATAATA  
AAGATAATAATGAATTATCTCACTTTGCCACCACCTGCTCGGATAAGTCCAGGCC  
GCTCAACGACTTGGTTAGCCAAAAACAACCTCAGCTGTCTGATATTACATCACGT  
TTTAATTCAGCTATTGAAGCACTGAACCGTTTCATTAGAAATATGATTTCAGTGA  
TGCAACGTCTGCTAGATGACACGTCTGGTAAAGGATCCATGAGTGC GTTGATAAC  
CCATGATCGCTCAACGCCAGTAACTGGAAGTCTACTTCCCTACGTCGAGACACCA  
GCGCCCCCCCCCTTCAGACTCAACAAGTCGCGGAGAACTGAAGGATAAAAAAT  
GGTGGGTGAGTTCTCAGGGCGTACAGCTCCCTGCACCACTAGCAGTGGTTGCCA  
GCCAAGTCACTGAAGGACAACAGCAAGAAATCACTAAATTATTGGAGTCGGTCA  
CCCGCGCACGGCAGGATCTCAACTGATATCAAATTATGTTTCAGTGCTAACGAA  
TTTTACGCTCGCTTACCTGATACATTTGAGATTGAGTAGGTAAGCTAGTTTCTA

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ATTAGAAGAAGTACGCAAAGACATAAAAAATCGCTGATATTCAGCGTCTTCATG  
AACAAAACATGAAGAAAATTGAAGAGAATCAAGAGAAAATCAAAGAAACAGAA  
GAGAATGCCAAGCAAGTCAAGAAATCCGGCATGGCATCAAGATTTTTGGCTGG  
CTCAGCGCCATAGCCTCAGTGGTTATCGGTGCCATCATGGTGGCCTCAGGGGTAG  
GAGCCGTTGCCGGTGCAATGATGATTGCCTCAGGCGTAATTGGGATGGCGAATA  
TGGCTGTGAAACAAGCGGCGAAGATGGCCTGATATCCAAGAGGCAATGCAAG  
TATTAGGGCCGATACTACTGCGATTGAAGTCGCATTGACTGTAGTTTCAACCGT  
AATGACCTTTGGCGGTTCCGGCACTAAAAATGCCTGGCTGATATTGGCGCAAACTC  
GGTGCTAACACCGCAAGTCTTGCTGCTAAAGGAGCCGAGTTTTCGGCCAAGTTG  
CCCAAATTTCGACAGGCATATCAAACACTGTCGGGAATGCAGTGACTAAATTAG  
GGGGCAGTTTTGGTAGTTTAAACATGAGCCATGTAATCCGTACAGGATCACAGG  
CAACACAAGTCGCCGTTGGTGTGGGCAGCGGAATAACTCAGACCATCAATAATA  
AAAAACAAGCTGATTACAACATAATAACGCTGATTGGCCTTGAACAAGGCAG  
ACATGGCAGCGTTACAAAGTATTATTGACCGACTCAAAGAAGAGTTATCCCATTT  
GTCAGAGTCACATCAACAAGTATGGAAGTATTTCCAGATGATTAATGCAA  
AGGTGACATGCTGCATAATTTGGCCGGCAGCCCCATACTGTTTAAGGTACC

LTAl-YerF Amino acid sequence

SEQ ID NO: 48

MDNGDRLYRADSRPPEIKRSGGLMPRGHNEYFDRGTQMNINLYDHARGTQ  
TGFVRYDDGYVSTSLSLSAHLAQQSILSGYSTYYIYVIATAPNMFNVNDVLGVYSP  
HPYEQEVSA LGGIPYSQIYGWYRVNFGVIDERLHRNREYRDRYYRNLNIAPAEDGYR  
LAGFPDPDHQAWREEPWIHHAPQCGNSSRMIRAYEQNPQHFIEDLEKVRVEQLTGH  
GSSVLEELVQLVKDKNIDISIKYDPRKDSEVFANRVITDDIELLKKILAYFLPEDAILK  
GGHYDNQLQNGIKRVKEFLESSPNTQWELRAPMAVMHFSLTADRIDDDILKIVIDS  
MNHHDARSKLREELAE LTAELKIYSVIQAEINKHLSSSGTINIHDKSINLMDKNLYG  
YTDEEIFKSAEYKILEKMPQTTIQVDGSEKIVSIKDFLGSENKRTGALGNLKNYS  
YNKDNNELSHFATTCSDKSRLNDLVSQKTTQLSDITSRFNSAIEALNRFIQKYDSVM  
QRLDDTSGKGMSALITHDRSTPVTGSLLPYVETPAPAPLQTTQVAGELKDKNGV  
SSQGVQLPAPLAVVASQVTEGQQQEITKLESVTRGTAGSQLISNYVSVLTNFTLASP  
DTFEIELGKLVSNLEEVRKDIKIADIQRLHEQNMKKIEENQEKIKETEENAKQVKKSG  
MASKIFGWLSAISVVIGAIMVASGVGAVAGAMMIASGVIGMANMAVKQAAEDGL  
ISQEQMQLGPILTAIEVALTVVSTVMTFPGGSALKCLADIGAKLGANTASLAAGAE  
FSAKVAQISTGISNTVGNVTKLGGSPGSLTMSHVIRTGSQATQVAVGVGSGITQTIN  
NKKQADLQHNNADLALNKADMAALQSIIDRLKEELSHLSESHQQVMELIFQMINAK  
GDMLHNLAGRPHTV-

His-SicA chaperone for S1 nucleic sequence

SEQ ID NO: 49

ATGGGCAGCAGCCATCACCATCATCACCACAGCCAGGATCCGatggactaccaga  
acaacgtcagcgaagaacgtgttcggaatgatattgggatgccgttagtgaaggcgccacgctaaaagacgttcaggaatccctca  
agatatgatggacggtttatatgctcatgcttatgagttttataaccagggacgactggatgaagctgagacgttctttcgtttcttatgcatt  
tatgatttttacaatccccgattacacatgggactggcggcagtagtgcgaactgaaaaacaatttcagaaagcatgtgacctttatgcag

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tagcggttacgttacttaaaaatgattatcgccccgtttttttaccgggcagtgtaattattaatgcgtaaggcagcaaaagccagacagt  
gttttgaacttgatcaatgaacgtactgaagatgagtcctcgccggcctaaagcggttggtctatctggaggcgctaaaaacggcgagaca  
gagcagcacagcgagcaggagaaggagtaaAAGCTT

His-SicA chaperone for S1 Amino acid sequence

SEQ ID NO: 50

MGSSHHHHHSQDPMYQNNVSEERVAEMIWDVASEGATLKDVHGIPQDM  
MDGLYAHAYEFYNQGRLEAETFFRFLCIYDFYNPDYTMGLAAVCQLKKQFQKAC  
DLYAVAFTHLLKNDYRPVFFTGQCQLLMRKAARKARQCFELVNTERTEDESLRAKALVY  
LEALKTAETEQHSEQEKE\*

SipD nucleic acid sequence

SEQ ID NO: 51

atgcttaaatattcaaaattattccgcttctcctcatccggggatcggtgcccgaacggcgccagactccctcgccgagcgagc  
acgtcgagactgccgtggtaccgtctaccacagaacatcgcggtacagatatcatttcattatcgaggcggtactaaaatccaccag  
gcacagcagacgctgcagtcacgcgcccgatctctgaagagaataatgacgagcgccagctggcgccgagcagttgaccagca  
gcctgaatgcgctggcggaagtccggcggtgcattatccgcagacaaaatgagaacctgcggagcgcttttctgcgccgacgtcg  
ccttatttagcgcttcgcctatggcgcgagccgagaaacaccatttctgatgctgagatttgggatatggtttcccaaatatcgcggcata  
ggtgacagctatctggcgctttatgaaaacgttgctgcagctctataccgatttttatcaggccttcagtgatattctttccaaaatgggagg  
ctggttattaccaggaaggacggttaataccgttaagctagatgttacctcactcaaaaatgatttaaacagtttagtcaataaataatca  
aataaacagtaataaccgctttatttccagcgagtcaggcagcggttaagtagccactgaagcgaagcgagacagtggtcagtc  
gaattgaatttaccgaatagctgcctgaaatcttatggatccggttatgtcgtcaccggttgatctgacgccattacaaaaatgggttcagga  
tattgatggtttaggcgccgggaaaagactcaaaactcgaaatggataacgcgcaaatatcaagcctggcagtcgggttttaaacgc  
caggaagaaaatgaaaccacattacagacgctgacgcgcaaaatagcaatgccaatcattgtacgacaacctggtaaaagtgc  
gagcagtagcagataagtagcagcctggaaccgcaaaagcttctctgcaagga

SipD amino acid sequence

SEQ ID NO: 52

MLNIQNSASPHPGIVAERPQTPSASEHVETAVVPSTTEHRGTDIISLSQAATKI  
HQAQQLQSTPPISEENNDERTLARQQLTSSLNALAKSGVSLSAEQNENLRSAFSAPT  
SALFSASPMAQPRTTISDAEIWDMVSQNISAIGDSYLGVEENVVAVYTDFYQAFSDIL  
SKMGWLLPGKDGNVTKLDVTSKNDLNSLVNKYNQINSNTVLFPQSGSGVKVA  
TEAEARQWLSELNLPNSCLKSYGSGYVTVDLTPLQKMQDIDGLGAPGKDSKLEM  
DNAKYQAWQSGFKAQEENMKTLTTLTQKYSNANSLYDNLVKVLSSTISSSLETAK  
SFLQG

SipB nucleic acid sequence

SEQ ID NO: 53

atggttaaatgacgcaagtagcattagccgtagcggatatacccaaatccgcgcctcgctgaggcggttttgaaggcggt  
cgtaagaacacgggactttttaaagcgccggataaagcttttaagatgtggtggcaacgaaagcgggcgaccttaagccggaaca  
aagtcggcgagagcgctattaatacgggtgggtctaaagccgcctacggacgcgcgcgggaaaaactctccagcgaagggaatt  
gacattactgcttgcaagttaatgacctactggcgatgtttcgctgtctcaactggagtcctgctggcggtatggcaggcgatgatt  
gagtcacaaaaagagatggggattcaggtatcgaaagaattccagacggctctgggagaggctcaggaggcgacggatctctatga  
agccagtatcaaaaagacggataccgccaaagagtggttatgacgctgacgacaaaaaactgacgcaggcgcaaaataaattgcaatc  
gctggaccggcgtagccccggtatgcacaagctgaagcccggtagaacaggccggaaaagaagcgacagaggcgaaagagg  
ccttagataaggccacggatgcgacggttaaaagcaggcacagacgcaaaagcgaaagccgagaaagcgataaacattctgaccaa  
attccagggaacggctaatagcgcctctcagaatcaggtttccagggtgagcaggataaatctgtcaaatgtcgccgcctcactatgc

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tcattggccatgtttattgagattgtgggcaaaaatacggagaagcctgcaaacgatcttgcgcttttcaacgccttgagggaaggg  
cgtcaggcggagatggaaaagaatcggtgaattccaggagagacgcgcaaacgcgaggaaaacgaaccgcattatgggatgta  
tcgggaaagtctcggcgcgctgctaaccattgtcagcgttggtggccgctgttttacgggtggggcgagctctggcgctggctgcggt  
gggacttgcggtaatgggtggccgatgaaattgtgaaggcggcgacgggagtgctgtttattcagcaggcgctaaaccgattatgga  
gcatgtgctgaagccgttaatggagctgattggcaaggcgattaccaaacgcgtggaaggattaggcgctcgataagaaaacggcag  
agatggccggcagcattgttggtgcgattgtgcgcgctattgccatggtggcggtcatttgtggtggtcgagttgtcgggaaaggcgc  
ggcggcgaaaactgggtaacgcgctgagcaaatgatggcgcaaacgattaaagaagttggtgcctaacgtgctgaaacagttggcgc  
aaaacggcagcaaacctctttaccaggggatgcaacgtattactagcggctcgggtaatgtgggtagcaagatgggcctgcaaacga  
atgccttaagtaagagctggttaggtaataccctaaataaagtggcggtgggcattggaagtcacgaataccgcagcccagtcagccg  
gtggtgttgccgagggcgctattttataaaaatgccagcgaggcgcttgctgattttatgctcgccgcttttgccatggatcagattcagca  
gtggcttaaacatccgtagaaatatttggtgaaaaaccagaaggtaacggcggaactgcaaaaagccatgtctctgcggtacagcaa  
aatgcggatgcttcgcggtttttattctgcgccagagtcgcgcataa

SipB amino acid sequence

SEQ ID NO: 54

MVNDASSISRSGYTQNPRLAEAAFEQVRKNTDFLKAADKAFKDVVATKAGD  
LKAGTKSGESAINTVGLKPPTDAAREKLSSEGQLTLLGKMLTLLGDVSLSQLESRL  
AVWQAMIESQKEMGIQVSKFQ TALGEAQEATDLYEASIKKTD TAKSVYDAATKKL  
TQAQNKQLSLDPADPGYAQAEEAVEQAGKEATEAKEALDKATDATVKAGTDAKA  
KAEKADNII LTKFQGTANAASQNQVSQGEQDNLSNVARLTMLMAMFIEIVGKNTEES  
LQNDLALFNALQEGRQAEMEKKSAEFQEETRKAETNRIMGICIGKVLGALLTIVSVV  
AAVFTGGASLALAAVGLAVMVADEIVKAATGVSFIIQALNPIMEHVLKPLMELIGK  
AITKALEGLGVDKKTAE MAGSIVGAIVAAIAMVAVIVVAVVGKAAAKLGNALS  
KMMGETIKKLVPNVLKQLAQNGSKLFTQGMQRITSGLG NVGSKMGLQTNALSKEL  
VGNTLNKVALGMEVTNTAAQSAGGVAEGVFIKNASEALDFMLARFAMDQIQOWL  
KQSVEIFGENQKVTAELQKAMSSAVQQNADASRFILRQSRA

S1 nucleic acid sequence

SEQ ID NO: 55

atgcttaataattcaaaattattccgcttctcctcatccggggatcgttgccgaacggcgcgagactccctcggcgagcgagc  
acgtcgagactgccgtggtaccgtctaccacagaacatcgcggtacagatatcatttcattatcgcaggcggctactaaaatccaccag  
gcacagcagacgctgcagtcacgcgcccgatctctgaagagaataatgacgagcgcacgctggcgcgccagcagttgaccagca  
gcctgaatgcgctggcgaagtccggcggtgcattatccgcagacaaaatgagaacctgcggagcgcttttctgcgccgacgtcgg  
ccttatttagcgcttcgcctatggcgcgagcgcagacaaccatttctgatgctgagatttgggatatggtttcccaaaatatatcggcgata  
ggtgacagctatctggcggtttatgaaaacgttgctgcagctctataccgatttttatcaggccttcagtgatattctttccaaaatgggagg  
ctggttattaccaggttaaggacggttaataccggttaagctagatgttacctcactcaaaaatgatttaaacagtttagtcaataaataaatca  
aataaacagtaataaccggtttttatccagcgcagtcaggcagcgcggttaagtagccactgaagcggaagcgagacagtggtcagtc  
gaattgaattttaccgaatagctgcctgaaatcttatggatccggttatgtcgctcaccgttgatctgacgccattacaaaaatgggttcagga  
tattgatggtttaggcgcgccgggaaaagactcaaaactcgaaatggataacgccaatatcaagcctggcagtcgggttttaagcg  
caggaagaaaatataaaaaccacattacagacgctgacgcaaaaatatagcaatgccaattcattgtacgacaacctggtaaaagtgc  
gagcagtagcataagtagcagcctggaaaccgccaaaagcttcctgcaaggagtcgacatggtaaatgacgcaagtagcattagcc  
gtagcggatatacccaaaatccgcgctcgtgaggcggttttgaggcggttcgtgaagaacacggactttttaaagcggcgataa  
agcttttaagatgtggtggcaacgaaagcggcgaccttaagccggaacaaagtcggcgagagcgctattaatacggtggtgtc

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aaagccgcctacggacgccgccgggaaaaactctccagcgaaggccaattgacattactgcttggcaagttaatgaccctactggg  
cgatgtttcgctgtctcaactggagtcctcgtctggcggtatggcaggcgatgattgagtcacaaaaagagatggggattcaggtatcga  
aagaattccagacggctctgggagaggctcaggaggcgacggatctctatgaagccagtatcaaaaagacggataccgccaaagagt  
gtttatgacgctgcgacaaaaactgacgcaggcgcaaaataaattgcaatcgctggacccggctgaccccggtatgcacaagct  
gaagccgcggtagaacaggccggaaaaagaagcgacagaggcgaaagaggccttagataaggccacggatgcgacggttaaagc  
aggcacagacgccaaagcgaaagccgagaaaagcgataaacattctgaccaaattccagggaacggctaatgccgcctctcagaatc  
agggttccagggtgagcaggataaatctgtcaaatgtcgccgcctcactatgctcatggccatgtttattgagattgtgggcaaaaatac  
ggaagaaagcctgcaaaacgatcttgcgcttttcaacgccttgacggaagggcgtcaggcgagatggaaaagaatcggtgaatt  
ccaggaagagacgcgcaaaagccgaggaacgaaccgcattatgggatgtatcggaagtcctcgccgcgctgctaaccattgtca  
gcgttggtggccgctgttttacgggtggggcgagtcctggcgctggctgcggtgggacttgcggtaatggtggcgatgaaatgtgaa  
ggcgcgacgggagtgctgctttattcagcaggcgctaaaccgcattatggagcatgtgctgaagccgttaaggagctgattggcaag  
gcgattaccaaagcgctggaaggattaggcgctcgataagaaaaaggcagagatggccggcagcattgttggtgcgattgtcgccgct  
attgccatgggtggcgctcatgtggtggtcgcatgtcggaagggcgcgccggcgaaactgggtaacgcgctgagcaaaatgat  
ggcgaaacgattaagaagttggtgcctaacgtgctgaacacgttggtgcgcaaaacggcagcaaaactctttaccaggggatgcaac  
gtattactagcggctctgggtaattgtgggtagcaagatgggcctgcaaacgaatgccttaagtaagagctggtaggtaataccctaaat  
aaagtgccgttgggcatggaagtacgaataccgcagcccgatcagccggtggtgttgccgaggcgctatttattaaaaatgccagc  
gaggcgcttgctgattttatgctcgcccgcttttgccatggatcagattcagcagtggttaacaatccgtagaaaatatttgggtgaaaacc  
agaaggtaacggcggaactgcaaaaagccatgtcttctcgcggtacagcaaaatgcggatgcttcgcgcttttattctgcgccagagtcg  
cgcataa

S1 amino acid sequence

SEQ ID NO: 56

MLNIQNYASPHPGIVAERPQTPSASEHVETAVVPSTTEHRGTDIISLSQAATKI  
HQAQQTLQSTPPISEENNDERTLARQQLTSSLNALAKSGVLSAEQENENLRSFAFSAPT  
SALFSASPMAQPRTTISDAEIWDMVSNISAIIGDSYLGVIENVVAVYTDFYQAFSDIL  
SKMGGWLLPGKDGNVTKLDVTSKNDLNSLVNKYNQINSNTVLFPAQSGSGVKVA  
TEAEARQWLSELNLPNSCLKSYGSGYVVTVDLTPLQKMQDIDGLGAPGKDSKLEM  
DNAKYQAWQSGFKAQEENMKTTLTQKYSNANSYDNLVKVLSSTISSLETAK  
SFLQGVDMVNDASSISRSGYTQNPRLAEAAFEGVRKNTDFLKAADKAFKDVVATKA  
GDLKAGTKSGESAINTVGLKPPTDAAREKLSSEGQLTLLGKMLTLGDSLSQLES  
RLAVWQAMIESQKEMGIQVSKEFQTALGEAQEATDLYEASIKKTDATAKSVYDAATK  
KLTQAQNKQLSLDPADPGYAQAEAAVEQAGKEATEKEALDKATDATVKAGTDA  
KAKAEKADNILTQFQGTANAASQNVVSQGEQDNLNVARLTMLMAMFIEIVGKNTE  
ESLQNDLALFNALQEGRQAEMEKSAEFQEETRKAETNRIMGCIGKVLGALLTIVS  
VVAAVFTGGASLALAAVGLAVMVADEIVKAATGVSFQQALNPIMEHVLKPLMELI  
GKAITKALEGLVDKKTAEAGSIVGAIVAAIAMVAVIVVVAVVGKAAAKLGNA  
LSKMMGETIKKLPVNLKQLAQNGSKLFTQGMQRITSGLGNVSGKMLQTNALSKE  
LVGNTLNKVALGMEVTNTAAQSAGGVAEGVFIKNASEALDFMLARFAMDQIQQW  
LKQSVEIFGENQKVTAEKQKAMSSAVQQNADSRFILRQSR



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LTA1-GSAAS-S1 Nucleic acid sequence

SEQ ID NO: 57

CATatggacaatggcgatcggtttataccgtgccgactcgcggtccccagatgagattaaacgtagcgggtgggttaatgcc  
acgtgggcacaaatgagatattttgacgtggaacacagatgaacattaaacctttacgatcatgccgtgggaccagaccgggtttgtcc  
gttatgatgacgggtatgttagtacgagtttgccttacgctccgcacaccttgctgggacaaagtattttatcaggctacagcacatattac  
atztatgtgatcgccactgccccaaacatgttcaatgtgaacgatgtgttgggggtttacagccccatccatgaacaagaagtctcg  
gcccttgggggatcccatatagccagatttatgggtgggtaccgcgtaaattttgggtgtgattgatgaacgtttgcatcgtaaccgtgaata  
ccgcatcgctactaccgtaacttgaacattgcacctgccgaggacggctatcggttagcgggattccccccgatcatcaggcgtgg  
cgtgaggaaccggtggatccatcacgccctcaggggtgcgggaacagtagtcgcggtccgcggcatccatgcttaatttcaaatt  
attccgcttctcctcatccggggatcggtgcgaacggccgcagactccctcggcgagcgagcacgtcgagactgccgtggtaccgt  
ctaccacagaacatcgcggtacagatatcatttcattatcgaggcggtactaaaaatccaccaggcacagcagcgtcgactcaac  
gccaccgatctctgaagagaataatgacgagcgacgctggcgccagcagttgaccagcagcctgaatgcgctggcgaagtcc  
ggcgtgtcattatccgcagaacaaaatgagaacctgcggagcgcggttttctgcgcgacgtcgcccttatttagcgttcgcctatggc  
gcagccgagaacaaccatttctgatgctgagatttgggatatgggtttcccaaatatatacgcgataggtagacgtatctggcggttat  
gaaaacgttgtcgagctataccgatttttatcaggccttcagtgtatcttccaaaatgggaggctggttattaccaggttaaggacgg  
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cagcgcatcaggcagcggttaaaagtagccactgaagcgaagcgagacagtggtcagtgaaatgaatttaccgaatagctgc  
ctgaaatcttatggatccggttatgtcgtcacccgttgatctgacgccattacaaaaatggttcaggatatttgatgggttagggcgccgg  
gaaaagactcaaaactcgaaatggataacgccaaatatacaagcctggcagtcgggttttaagcgcaggaagaaaatataaaaacca  
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ctggaacccgcaaaaagcttcttgcaaggagtcgacatggtaaatgacgcaagtagcattagccgtagcggatatacccaaaatccg  
cgctcgctgaggcggttttgaaggcggtcgtgaagaacacggactttttaaagcggcgataaagcttttaagatgtggtggcaac  
gaaagcggcgaccttaaagcgggaacaaagtccggcgagagcgctattaatacggtgggtctaaagccgcctacggacgccgc  
cgggaaaaactctccagcaagggcaattgacattactgcttggaagttaatgacctactggcgatgttctcgctgtctcaactgga  
gtctcgctggcggtatggcaggcgatgattgagtcacaaaaagagatggggattcagggtatcgaaagaattccagacggctctggg  
agaggctcaggaggcgacggatctctatgaagccagtatcaaaaagacggataccgccaagagtggttatgacgctgcgacaaaa  
aactgacgcagcgcaaaaataattgcaatcgctggaccggctgacccggctatgcacaagctgaagccgggtagaacaggc  
cggaaaagaagcgacagaggcgaaagagcgcttagataaggccacggatgcgacggttaaagcaggcacagacgcaaaagcga  
aagccgagaaagcgataacattctgaccaaattccagggaacggctaatgccgctctcagaatcagggttccaggggtgagcagg  
ataatctgtcaaatgtcgccgcctcactatgctcatggccatgtttattgagattgtgggcaaaaatcagggaagaaagcctgcaaaacg  
atcttgcgcttttcaacgccttgcaagaaagggcgctcaggcgagatggaaaagaaatcggtgaattccagggaagagacgcgcaaa  
gccgaggaacgaaccgcatatgggatgtatcgggaaagtcctcggcgcgctgctaaccattgtcagcgttgtggcgctgttttac  
cgggtggggcagtgctggcgctggctgcgggtgggacttgcggtaatggtggccgatgaaattgtgaaggcgcgacgggagtgctgt  
ttattcagcaggcgctaaaccgattatggagcatgtgctgaagccggttaattggagctgattggcaaggcgattaccaagcgctgga  
aggattagcgctcgataagaaaacggcagagatggccggcagcattgttgggtgcgattgtcgccgctattgccatggtggcggtcatt  
gtggtggtgcgagtgctgggaaagggcgggcggaactgggtaacgcgctgagcaaaatgatggcgaaacgattaaagaagt  
tggtgcctaacgtgctgaaacagttggcgcaaaacggcagcaaaactcttaccaggggatgcaacgtattactagcggctctgggtaa  
tgtgggtagcaagatggcgctgcaaacgaatgccttaagtaagagctggtaggtaataccctaataaaagtggcgttgggcgatggaa

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gtcacgaataccgcagccagtcagccggtggtgttgcgagggcgctatttattaaaaatgccagcgaggcgcttgctgattttatgct  
cgcccggttttgccatggatcagattcagcagtggttaacaatccgtagaaatatttggtgaaaaccagaaggtaacggcggaactgc  
aaaaagccatgtctctcgcggtacagcaaatgcggatgcttcgcggtttattctgcgccagagtcgcgcataaCTCGAG

LTA1-GSAAS-S1 Amino acid sequence

SEQ ID NO: 58

MDNGDRLYRADSRPPDEIKRSGGLMPRGHNEYFDRGTQMNINLYDHARGTQ  
TGFVRYDDGYVSTSLSLRSAHLAQQSILSGYSTYYIYVIATAPNMFNVNDVLGVYSP  
HPYEQEVSA LGGIPYSQIYGWYRVNFGVIDERLHRNREYRDYRNLNIAPAEDGYR  
LAGFPDPDHQAWREEPWIHHAPQGCN SSRGSAASMLNIQNYASPHPGIVAERPQTP  
SASEHVETAVVPSTTEHRGTDIIISLSQAATKIHQAQQLQSTPPISEENNDERTLARQQ  
LTSSLNALAKSGVSLSAEQNENLRSAFSAPTSALFSASPMAQPRTTISDAEIWDMVSLQ  
NISAIGDSYLVYENNVAVYTDIFYQAFSDILSKMGGWLLPGKDGN TVKLDVTS LKN  
DLNSLVNKYNQINSNTVLFPAQSGSGVKVATEAEARQWLSELNLPNSCLKSYSGSY  
VVTVDLTLPLQKMQDIDGLGAPGKDSKLEMDNAKYQAWQSGFKAQEENMKTTLQ  
TLTQKYSNANSYLDNLVKVLSSTISSSLETAKSFLQGVDMVNDASSISRSGYTQNPRL  
AEA AFE GVRKNTDFLKAADKAFKDVVATKAGDLKAGTKSGESAIN TVGLKPPTDA  
AREKLSSEGQLTLLLGKMLTLLGDVLSQLESRLAVWQAMIESQKEMGIQVSKEFQT  
ALGEAQEATDLYEASIKKTD TAKSVYDAATKKLTQAQNKLSLDPADPGYAQAEA  
AVEQAGKEATEAKEALDKATDATVKAGTDAKAKAEKADNILT K FQGTANAASQNQ  
VSQGEQDNLNSVARLTMLMAMFIEIVGKNT EESLQNDLALFNALQEGRQAEMEKK  
AEFQEETRKAETNRIMGCIGKVLGALLTIVSVVAAVFTGGASLALAAVGLAVMVA  
DEIVKAATGVSPFIQQALNPIMEHVLKPLMELIGKAITKALEGLGVDKKTAEMAGSIV  
GAIVAAIAMVAVIVVAVVGKGA AAKLGNALS KMMGETIKKLVPNVLKQLAQNGS  
KLFTQGMQRITSGLGNVGSKMGLQTNALSKELVGNTLNKVALGMEVTNTAAQSAG  
GVAEGVFIKNASEALADFMLARFAMDQIQWLKQSV EIFGENQKVTAELQKAMSSA  
VQQNADASRFILRQSRA\*

His-SscA chaperone for S2 nucleic acid sequence

SEQ ID NO: 59

ATGGGCAGCAGCCATCACCATCATCACCACAGCCAGGATCCGatgaaaaagac  
ccgaccctacaacaggcacatgacacgatgcggtttttccggcggtggcggtcgctgcgctatgttgttgatgacgatgttacacagcc  
gcttaatactctgtatcgctatgccacgcagccttatggaggtaaaagaattcgccggcgagcgcgactttttcaattgctgacgatatat  
gatgcctggtcatttgactactggtttcggttaggggaatgctgccaggctcaaaaacattggggggaagcgatatacgcttatggacg  
cgcgccacaaaataagattgatgcgccgcaggcgccatggggccgcagcggaatgctatctcgcggtgtgataacgtctgttatgcaatc  
aaagcggttaaaggccggtgctgctatttgcggcgaggtcagtgaaatcaaatctccgacagcggtgcagaaaagatgttacagcaac  
tttctgacaggagctaaAGCTT

His-SscA chaperone for S2 Amino acid sequence

SEQ ID NO: 60

MGSSHHHHHSQDPMKDP TLQQAHD TMRFFRRGSLRMLLDDDV TQPLN  
TLRYATQLMEVKEFAGAARLFQLLTIYDAWSFDYWFRLGECCQAQKHWEAIIYA  
YGRAAQIKIDAPQAPWAAAECYLACDNVCYAIKALKAVVRICGEVSEHQILRQRAE  
KMLQQLSDRS\*

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SseB nucleic acid sequence

SEQ ID NO: 61

Atgtcttcaggaacatcttatggggaagtcaaaaccctattgtgtttaaaaatagcttcggcgtcagcaacgctgataccgg  
gagccaggatgacttatcccagcaaaatccgtttgccgaagggtatgggtgttttgcctattctccttatggttattcaggctatcgcaataa  
taattttattgaagtcagaagaacgctgaacgtgccagaaataccaggaaaagtcaaatgagatggatgaggtgattgctaagca  
gccccagggtatgctaaaaccaagaggaggtgcctgaggatgtaattaaatcatgcgtgataatggattctcatcgatggatgac  
cattgatgattatatggctaataatggcgatcatgggaagctggataaagggtggctacaggcgatcaaagcggccttggataatgacg  
ccaaccggaataccgatcttatgagtcaggggcagataacaattcaaaaaatgtctcaggagcttaacgctgtccttaccgaactgaca  
gggcttatcagtaagtgggggaaatttccagtatgatagcgcagaaaaacgtactca

SseB amino acid sequence

SEQ ID NO: 62

MSSGNILWGSQNPIVFKNSFGVSNADTGSQDDLQONPFAEGYGVLLILLMVI  
QAIANNKFIEVQKNAERARNTQEKSNEMDEVIKAAKGDATKKEVPEDVIKYMED  
NGILIDGMTIDDMYAKYGDHGLDKGLQAIKALDNDANRNTDMSQGQITIQK  
MSQELNAVLTQLTGLISKWGEISSMIAQKTYK

SseC nucleic acid sequence

SEQ ID NO: 63

atgaatcgaattcacagtaaatagcgacagcgccgcaggagtaaccgccttaacacatcatcacttaagcaatgtcagttgcg  
tttctcgggttcgctgggaaagcgccagcatcgtgtgaattctacttttggcgatggcaacgcgcgctgtctgctatccgggaaaatta  
gtcttcaggaggcaagcaatgcgttgaaagcaactgcttgatgcgtaccgggaaatcataagcgtccatcattgctgactttttgcaga  
ccaatcccgcggttttatcaatgatgatgacgtcattaatactcaacgtctttggtaataacgctcaatcgttatgccaacagcttgagcgg  
gcaactgaggtgcaaaatgcattacgtaataagcaggtaaaggagatcaggagcagatccagaaagcgatagagcaggaggataa  
agcgcgtaaaagcgggtatttttggcgctatttttgactggattaccggcatatttgaaacgctgattggcgccctaaaagttgtggaaggtt  
ttctgtccggaatcccgcagaaatggctagcggcgtagcttatatggccgcaggttggtgcaggaaatggtaaaagccggagccgaaa  
cggcaatgatgtgcgggtgctgaccacgatacctgtcaggcaattattgacgtgacaagtaagattcaatttggttggaagcgcgtcgcg  
ctggcactggatgtttccagattggcgcgtgttttattggcgacgagaggttatctggcgacgctgcaaaagtgttgactccggttttg  
gcgaggaagtgggttagcgctatggtaggtgcaggggaagcagaaatagaggagttggctgaaagtgtggcgaagaagtgcgcga  
aagtttttccaaacaatttgagcgcgttgaaactgaaatggctatggcgaaatgagatggcagaggaggtgcccaggttttctcgtaacgt  
agaaaaataatagacgcgaagcgcgggaaaaagctttacgaaagaggggtgaaagcaatggcaaaagaagcggcgaagaagc  
cctggaaaaatgtgtgcaagaaggtggaagtctctgttaaaaaatccgtaataaagttctctcaatatgttcaaaaaatcctgtatg  
ccttactgagggattgttcatttaaggtcttacaggctatcagatgtgcaaccgagggcgccagtcagatgaatactggcatggttaaca  
cagaaaaagcggaagatcgaaaagaaaatagagcaattaataactcagcaacggtttctggatttcataatgcaacaaacagaaaaacca  
gaaaaagatagaacaaaaacgcttagaggagctttataaggggagcgggtgccgcgcttagagatgtattagataaccattgatcactata  
gtagcgttcaggcgagaatagctggctatcgcgcttaa

SseC amino acid sequence

SEQ ID NO: 64

MNRIHSNSDSAAGVTALTHHLSNVSCVSSGLGKRQHRVNSTFGDGNAC  
LLSGKISLQEQASNALQLLDAVPGNHKRPSLPDFLQTNPAVLSMMMTSLILNVFGNN  
AQSLCQQLERATEVQNALRNKQVKEYQEQIKAI EQEDKARKAGIFGAI FDWITGIFE  
TVIGALKVVEGFLSGNPAEMASGVAYMAAGCAGMVKAGAETAMMCGADHDTQ  
AIIDVTSKIQFGCEAVALALDVFQIGRAFMATRGLSGAAKVLDSGFGEVVERVMG  
AGEAEIEELAEKFGEEVSESFQFEPLEREMAMANEMAEAEFSRNVENNMTRSA  
GKSFTKEGVKAMAKEAAKEALEKCVQEGGKFLKKPRNKVLFNMFKKILYALLRD

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CSFKGLQAIRCATEGASQMNTGMVNTEKAKIEKKIEQLITQQRFLDFIMQQTENQKKI

EQKRLEELYKSGAALRDVLDTIDHYSSVQARIAGYRA

S2 nucleic acid sequence

SEQ ID NO: 65

atgtcttcaggaacatcttatggggaagtcaaaaccctattgtgtttaaaaatagcttcggcgctcagcaacgctgataccgg  
gagccaggatgacttatcccagcaaaatccggtttgccgaagggtatggtgttttgccttattctccttatggttatcaggctatcgcaataa  
taaatattattgaagtcagaagaacgctgaacgtgccagaaataccaggaaaagtcaaagagatggatgaggtgattgctaaagca  
gccaaaggggatgctaaaaccaagaggaggtgcctgaggatgtaattaaatacatgcgtgataatggtattctcatcgatggtatgac  
cattgatgattataggctaaataggcgatcatgggaagctggataaaggtggcctacaggcgatcaaagcggctttggataatgacg  
ccaaccggaataccgatcttatgagtcaggggcagataacaattcaaaaaatgtctcaggagcttaacgctgtccttaccgaactgaca  
gggcttatcagtaagtgggggaaatttccagtatgatagcgcagaaaacgtactcaGAGCTCatgaatcgaattcacagtaata  
gcgacagcgccgaggagttaaccgccttaacacatcatcacttaagcaatgtcagttgcgtttcctcgggttcgctgggaaagcgcca  
gcatcgtgtgaattctacttttggcgatggcaacgcgcgtgtctgctatccgggaaaattagctcttcaggaggcaagcaatgcgttgaa  
gcaactgcttgatgcctgacccggaaatcataagcgtccatcattgcctgactttttgcagaccaatcccgcggttttatcaatgatgatg  
acgtcattaataactcaacgtctttggtaataacgctcaatcgttatgccaacagcttgagcgggcaactgaggtgcaaaatgcattacgta  
ataagcaggtaaaggagatcaggagcagatccagaagcgatagagcaggaggataaagcgcgtaaagcgggtatttttggcgct  
atttttgactggattaccggcatatttgaaaccgtgattggcgctttaaagttgtggaagggtttctgtccggaaatcccgagaaatggc  
tagcggcgtagcttatatggcgcgaggttgtgcaggaatggttaagccggagccgaaacggcaatgatgtgcggtgctgaccacga  
tacctgtcaggcaattattgacgtgacaagtaagattcaatttgggtgtgaagccgtcgcgtggcactggatgtttccagattggcctg  
gcttttatggcgacgagaggtttatctggcgacgtgcaaaagtgcttgactccggttttggcgaggaagtgggtgagcgtatggtaggt  
gcaggggaagcagaaatagaggagtggctgaaaagtttggcgaagaagtgcgcaaaagttttccaaacaatttgagccgcttgaa  
cgtgaaatggctatggcgaatgagatggcagaggaggtgccgagttttctcgtaacgtagaaaaataatgacgcgaagcgcggga  
aaaagcctttacgaaagaggggtgaaagcaatggcaaaagaagcggcaaaagaagccctggaaaaatgtgtgcaagaaggtgga  
aagttcctgttaaaaaatccgtaataaagttctcttcaatatgttcaaaaaatcctgtatgccttactgagggattgttcatttaaaggctt  
acaggtatcatgatgtgcaaccgagggcgccagtcagatgaatactggcatggttaacacagaaaaagcgaagatcgaaaagaaaa  
tagagcaattaataactcagcaacgggtttctggatttcataatgcaacaaacagaaaaccagaaaaagatagaacaaaaacgcttagag  
gagctttataaggggagcggtgccgcgcttagagatgtattagataccattgatcactatagtagcgttcaggcgagaatagctggctat  
cgcgcttaa

S2 amino acid sequence

SEQ ID NO: 66

MSSGNILWGSQNPVIFKNSFGVSNADTGSQDDLSQQNPFAEGYGVLLILLMVI

QAIANNKFIEVQKNAERARNTQEKSNEMDEVIKAAKGDAKTKEEVPEDVIKYMRD

NGILIDGMTIDDMAYKGDHGLDKGGLQAIKAALDNDANRNTDLSQGGITIQK

MSQELNAVLTLTGLISKWGEISSMIAQKTYSELMNRIHSNSDSAAGVTALTHHLS

NVSCVSSGSLGKRQHRVNSFTGDGNAACLLSGKISLQEASNALKQLLDVAVPNHHR

PSLPDFLQTNPAVLSMMMTSLILNVFGNNAQSLCQQLERATEVQNALRNKQVKEYQ

EQIQKAIEQEDKARKAGIFGAIFDWITGIFETVIGALKVVEGFLSGNPAEMASGVAYM

AAGCAGMVKGAETAAMCGADHDTQCAIIDVTSKIQFGCEAVALLDVFQIGRAF

MATRGLSGAAKVLDSGFEEVVERVMVGAGEAEIEELAEKPGEEVSEFSKQFEPL

REMAMANEMAEAEFSRNVENNMTRSAGKSFTKEGVKAMAKEAAKEALEKCVQ

EGGKFLKKFRNKVLFNMFKILYALLRDCSFKGLQAIRCATEGASQMNTGMVNTE

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KAKIEKKIEQLITQQRFLDFIMQQTENQKKIEQKRLEELYKGS GAALRDVLDITDHYS

SVQARIAGYRA

LTA1-GSAAS-S2 nucleic acid sequence

SEQ ID NO: 67

CATatggacaatggcgatcggtttataccgtgccgactcgcgtccccagatgagattaacgtagcggagggttaatgcc  
acgtgggcacaaatgagatattttgaccgtggaacacagatgaacattaacctttacgatcatgccgtgggacccagaccgggtttgtcc  
gttatgatgacgggtatgttagtacgagtttgccttacgctccgcacaccttgccgggacaaagtattttatcaggctacagcacatattac  
atattgtgatcgccactgccccaaacatgttcaatgtgaacgatgtgttgggggtttacagcccccatccatatgaacaagaagtctcg  
gcccttgggggatcccatatagccagatttatggttggtaccgcgtaaaattttggtgtgatgtatgaacgtttgcatcgtaaccgtgaata  
ccgcatcgctactaccgtaacttgaacattgcacctgccgaggcaggtatcggttagcgggattcccacccgatcatcaggcgtgg  
cgtgaggaaccggtggatccatcacgcccctcaggggtgcgggaacagtagtcgcggtccgcgcatccatgtcttcaggaacat  
cttatggggaagtcaaaacctattgtgttataaaatagcttcggcgctcagcaacgctgataccgggagccaggatgacttatcccagc  
aaaatccgtttgccaagggtatggtgttttgccttattctccttatggttattcaggctatcgcaataataaatttatgaagtcagaagaa  
cgctgaacgtgccagaaataccaggaagaatcaaatgagatggatgaggtgattgctaaagcagccaaaggggatgctaaaacca  
aagaggaggtgctcgaggtgtaattaaatacatgcgtgataatggtattctcatcgatggtatgaccattgatgattatatggttaaatat  
ggcgatcatgggaagctggataaaaggtggcctacaggcgatcaaagcggcttggataatgacgccaaccggaataccgatcttatg  
agtcaggggcagataaacaattcaaaaaatgtctcaggagcttaacgctgtccttaccgaactgacagggttatcagtaagtgggggg  
aaatttccagtgatgatagcgcagaaaacgtactcaGAGCTCATgaatcgaattcacagtaatagcgacagcgccgcaggagtaa  
ccgccttaacacatcatcacttaagcaatgtcagttgcgtttcctcggggttcgctgggaaagcgccagcatcggtgtaattctacttttggc  
gatggcaacgccgctgtctgctatccgggaaaattagctctcaggaggcaagcaatgcgttgaagcaactgcttgatgacctaccg  
gaaatcataagcgctccatcattgcctgactttttgcagaccaatcccgcgggttttatcaatgatgatgacgtcattaatactcaacgtctttg  
gtaataacgctcaatcggttatgccaacgcttgagcgggcaactgaggtgcaaatgcattacgtaataagcaggtaaaggagtatca  
ggagcagatccagaaagcgatagagcaggaggtataagcgcgtaaagcgggtatttttggcgctattttgactggattaccggcata  
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gcaggtgtgtgcaggaatgggttaaagccggagccgaaacggcaatgatgtgcggtgctgaccacgatacctgtcaggcaattattgac  
gtgacaagtaagattcaatttgggtgtgaagcgtcgcgctggcactggatgtttccagattggcgtgcttttatggcgacgagaggtt  
tatctggcgacgtgcaaaagtgtctgactccggttttggcgaggaagtggttgagcgtatggtagggtgcaggggaagcagaaatag  
aggagtgggtgaaaagtgttgcgagaagtgagcgaaagtttttccaaacaatttgagcgcgttgaaacgtgaaatggctatggcgaat  
gagatggcagaggagggtgccgagttttctcgtaacgtagaaaaataatgatgcgcaagcgcggaagaaagcctttacgaaagaggg  
ggtgaaagcaatggcaaaagaagcggcaaaagaagccctggaaaaatgtgtgcaagaaggtggaaagtctctgttaaaaaattcc  
gtaataaagttctcttcaatatgttcaaaaaatcctgtatgccttactgagggattgttcatttaaggcttacaggctatcagatgtgcaac  
cgaggggcccgatcagatgaatactggcatggttaacacagaaaaagcgaagatcgaaaagaaatagagcaattataactcagc  
aacggtttctggatttcatatgaacaaacagaaaaaccagaaaaagatagaacaaaaacgcttagaggagctttataaggggagcgg  
tgccgcgcttagagatgtattagataccattgatcactatagtagcgttcaggcgagaatagctggctatcgcgcttaactCGAG

LTA1-GSAAS-S2 Amino acid sequence

SEQ ID NO: 68

MDNGDRLYRADSRPPDEIKRSGGLMPRGHNEYFDRGTQMNINLYDHARGTQ  
TGFVRYDDGYVSTSLSLRSALHLAGQSILSGYSTYYIYVIATAPNMFNVNDVLGVYSP  
HPYEQEVSA LGGIPYSQIYGWYRVNFGVIDERLHRNREYDRYRNLIAPAEDGYR  
LAGFPDPDHQAWREEPWIHHAPQCGNSSRGSAA SMSSGNILWGSQNP I VFKN SFGVS  
NADTGSQDDL SQQNPFAEGYGVLLILLMVIQAIANNKFIEVQKNAERARNTQKSNE

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MDEVIAKAAKGAKTKEEVPEDVIKYM RDNGILIDGMTIDDYMAKYGDHGKLDKG  
GLQAIKAALDNDANRNTDLMSQGQITIQKMSQELNAVLTLTGLISKWGEISSMIAQ  
KTYSELMNRIHSNSDSAAGVTALTHHLSNVSCVSSGSLGKRQHRVNSTFGDGNAA  
CLLSGKISLQEASNAKQLLDVPGNHKRPSPDFLQTNPAVLSMMMTSLILNVFGN  
NAQSLCQQLERATEVQNALRNKQVKEYQEIQKAIEQEDKARKAGIFGAIFDWITGI  
FETVIGALKVVEGFLSGNPAEMASGVAYMAGCAGMVKAGAEAMMCGADHDC  
QAIIDVTSKIQFGCEAVALALDVFIQIRAFMATRGLSGAAAKVLD SGFGEEVVERMV  
GAGEAEIEELAEKFGEEVSESFSKQFEPLEREMAMANEMAEAAEF SRNVENNMTRS  
AGKSFTKEGVKAMAKEAAKEALEKCVQEGGKFLKKPRNKVLFNPFKKILYALLR  
DCSFKGLQAIRCATEGASQMNTGMVNTAKAKIEKKIEQLITQQRFLDFIMQQTENQK  
KIEQKRLEELYKGSGAALRDVLDTIDHYSSVQARIAGYRA\*

LTA1-SseB Nucleic acid sequence

SEQ ID NO: 69

ATGGGCAGCAGCCATCATCATCATCATCACAGCAGCGCCTGGTGCCGCG  
CGGCAGCCATatggacaatggcgatcggttataccgtgcccactcgctccccagatgagattaaacgtagcgggtgggtta  
atgccacgtgggcacaaatgagtagtttgaccgtggaacacagatgaacattaacctttacgatcatgcccgtgggaccagaccgggtt  
tgtccgttatgatgacgggtatgttagtagcaggtttgtccttacgctccgcacaccttgccgggacaaagtattttatcaggctacagcacat  
attacatttatgtgatcgccactgccccaaacatgttcaatgtgaacgatgtgtgggggtttacagccccatccatattgaacaagaagt  
ctcgcccttggggggatcccatatagccagatttatgggttggtaccgcgtaaattttggtgtgattgatgaacgtttgcatcgtaaccgtg  
aataccgcgatcgctactaccgtaactgaacattgcacctgcccaggacggctatcggttagcgggatccccccgatcatcaggcg  
tggtcgtaggaaacgtggatccatcacgccctcaggggtgcccgaacagtagtcgcccgtccgcggcatccatgtcttcaggaaa  
catcttatggggaagtcaaaacctattgtgtttaaaaatagcttcggcgtcagcaacgctgataccgggagccaggatgacttatccca  
gcaaaatccgtttgccaagggtatgggtgtttgcttattctccttatgggtattcaggctatcgcaataataaattttattgaagtcagaag  
aacgctgaacgtgccagaatccccaggaaaagtcaaatgagatggatgaggtgattgctaaagcagccaaaggggatgctaaaaac  
caaaggagggtgctgaggatgtaataatacatgcgtgataatggtattctcatcgatgggtatgaccattgatgatttatatggctaaat  
atggcgatcatgggaagctggataaaaggtggcctacaggcgatcaaagcggccttgataatgacgccaaccggaataccgatcttat  
gagtcaggggcagataacaattcaaaaaatgtctcaggagcttaacgctgtccttaccgaactgacagggcttatcagtaagtggggg  
gaaatttccagtagtagcgcagaaaacgtactcataaGGATCC

LTA1-SseB Amino acid sequence

SEQ ID NO: 70

MGSSHHHHHSSGLVPRGSHMDNGDRLYRADSRPPDEIKRSGGLMPRGHNE  
YFDRGTQMNINLYDHARGTQTGFVRYDDGYVSTSLSLRSAHLAQGISILSGYSTYYIY  
VIATAPNMFVNVDLVGVSPHPYEQEVSALGGIPYSQIYGWYRVNFGVIDERLHRNR  
EYRDYRYRNLIAPAEDGYRLAGFPDPHQAWREEPWIIHAPQCGGNSSRGSAASMS  
SGNILWGSQNPIVFKNSFGVSNADTGSQDDLQNPFAEGYGVLLILLMVIQAIANN  
KFIEVQKNAERARNTQEKSNEMDEVIKAAKGAKTKEEVPEDVIKYM RDNGILID  
GMTIDDYMAKYGDHGKLDKGLQAIKAALDNDANRNTDLMSQGQITIQKMSQELN  
AVLTQLTGLISKWGEISSMIAQKTY\*

HisScc2 chaperone for LTA1-CT053-CopB nucleic acid sequence

SEQ ID NO: 71

ATGGGCAGCAGCCATCACCATCATCACCACAGCCAGGATCCGatgagcactccat  
cttctaataattctaaaaaaccttcggcctcttttaataaaaaatcacgtagccgcttgcccgagattgctgcacaaaaaaagcaaaagc

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tgaggatttggaaacaaaataatcctgttctacggaagaggagacaaaacaagttctcatggacatcctacaggggttaagcaacgga  
ttaactcttcagcaaatatttaggtctctccgacgtctccttgaagagatctacaccgtagcatataccttctactcccaagggaatatcg  
ggaagctatcggtcttttccaaatcttaacagcctccaaacctcaatgctacaaaatacattcttaggtcttagctcttgctatcaccagctaaa  
aatgtatgatgaagcgcgttttgggttcttctagcttctcgatgctcaaccgaaaaccccatcctcctactacatcgccgatagcttgat  
gaagctaaaccaaccogaagaatctcaagacttctcgatattacgatcgatatgtgtagaacaagccggaatataaagttcttaaga  
tcgctgcagcattatgaagcaatctttagatgccgtgctgaaaaagagaaatctgcaaaaggctctgaaacacaagcctcctctcctaa  
aaacacaaaagctaaaaaagctgcttctaacaagaaaaagcaagtaagCGGCCGC

HisScc2 chaperone for LTA1-CT053-CopB Amino acid sequence

SEQ ID NO: 72

MGSSHHHHHSQDPMSTPSSNNSKKPSASFNKKSRSLAEIAAQKKAEDL  
EQKYPVPTEETKQVLMIDLQGLSNGLTQQILGLSDVLEEIYTVAYTFYSQGYRE  
AIGLFQILTASKPQCYKYLGLSSCYHQLKMYDEAAFGLAFDAQPENPIPPYYIADS  
LMKLNQPEESQDPLDITIDMCKNKPEYKVLKDRCSIMKQSLDAVLKKEKSAKGSQTQ  
ASSPKNTKAKKAASNNKKAK\*

CT053 nucleic acid sequence

SEQ ID NO: 73

Aaaagtgagcgtttaaaaaattagaatcagagcttcatgatcttaccagtggtgcaacttggccttgctcctaaaaaaga  
aatcgagagacaccaggaagaatacgtctgctagaaagcaaatccttgaagagaaagacgtctacaacttctcaagaaagcgg  
tgagatcaagagtagtaacccctcgagaactccagctaaaaccatttaccagatggcccgagcgtttcagacgttgagtttgtag  
aatcctcggtacagaaagtggtctcgatgccggtgacacaattgagattgacctaggtgatgaggcaagagaagaaagcggaaac  
gaactcgactactctagtgaaagcagatgaggatccttccagcgatcgcaatcggtggcgccgaggaggcatcatagatcctgacgcga  
atgaatgg

CT053 amino acid sequence

SEQ ID NO: 74

MKSERLKKLESELHDLTQWMQLGLVPKKEIERHQEEIRLLESKILEEKERLQL  
LKESGEIKEYVTPRRTPAKTIYPDGPSVSDVEFVSSDTEVDLDAGDTIEIDLGDEARE  
ESGNELDYSSDEDEDPFSDRNRWRRGGIIDPDANEW

CopB nucleic acid sequence

SEQ ID NO: 75

atgagcttgatccagcagcagctcggtatggtcgaatctgaaaaatgtgttatctcaggtcatcggtctacaccAcagg  
gggttctaatgctgacaaattaaccgacaatcaggtaaaacaagtccagcagaccgctcaaaaccgtgatgatctgtccatggagag  
cgacgtcgcggtggcggaacagccggaagatcggtgctcgctcggtccagatcgagggaacagagctgatgagcaaca  
gggacttgcggtgggaaagagacgggttctgctgatgctacatcattgaccagtcggcatccaaaggcgcttcagtcagcagtg  
attgaggataccagtaagtcctcgagctttcttcgcttccgagcctgtcaagcgtagatgcgacacatttgaggaaatccaatcgatc  
gtgtcttcagcaatggcgccaccaacgaattgtcattgacgaacttagagacaccgggattacaaagccgagtaccactccAcgc  
caggaaagtattggagatcagccttgcttagcgaaggccatcactgcattgggtgagagcactcaggctgccttggaataatttcagtc  
cactcagagtcagtcgcgaacatgaataagatgagtttggaatcccaaggcttgaaaaatcgacaaggagcgtgaagaatttaagaaa  
atgcaggagattcagcaaaagagcggcacaattcaaccatggatactgtgaataaagttatgattggcgtgacagtggaattacag  
aatctctgtgtttcagcattgtttacctgcggtttgggttgatggcacagccgctgctgggtgccacagccgccaccgctggggaac  
ggccgcgccacagcgcgtacctctgtgacgaccacagtcgctaccaggtgacgatgcaagcgggtgggtccaagtcgttaagcagg  
ctattatccaagcagtaaaacgcgccatcgctccaagcgattaaacaggggattaaagcaaggcattaaacaagcgatcaaacaggcag  
tcaaggcaagcgtgaagacacttgccaaaaatgtaggcaagattttcagcgcaggcaagaacgctgtgagtaagtccttcccAaaatt  
gtctaaggtgattaatcacttggttccaaatgggttactcttgccgtggggcccttacagcgggtgccgcagttagtcagtgccattac

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ctcccttcaattgtctgatatgcaaaaagaacttgcacaaatccaaaaggaagtgggtgcacttacggcgagagtgagatgatgaaa  
gcgtttacactgttctggcagcaagcttcgaaaatcgcggccaaaacaacggaatcaccttcagagacgcaacaacaggcagctaag  
accggcgcccatcgctaagcggtgtccgccatttcgggtgctttagctgctgctgctTAG

CopB amino acid sequence

SEQ ID NO: 76

MSLSSSSSDSSNLKNVLSQVIASTPQGVPNADKLTDNQVKQVQQTRQNRDD  
LSMESDVAVAGTAGKDRAASASQIEGQELIEQQGLAAGKETASADATSLTQSASKG  
ASSQQCIEDTSKSLELSSLSLSSVDATHLQEIQSIVSSAMGATNELSLTNLETPGLPKP  
STTPRQEVMEISLALAKAITALGESTQAALNFQSTQSQSANMNKMSLESQGLKIDK  
EREEFKMQEIQQSGTNSMTDVTNKMIGVTVAITVISVVSALFTCGLGLIGTAA  
GATAATAGATAAATTATSVTTTVATQVTMQAVVQVVKQAIQAVKRAIVQAIKQGI  
KQGIKQAIQAVKASVKTLAKNVGKIFSAGKNAVSKSPKLSKVINTLGSKWVTLGV  
GALTAVPQLVSGITSLSLQSDMQKELAQIQKEVGALTAQSEMMKAFTLFWQQASKIA  
AKQTESPSETQQAAKTGAQIAKALSAISGALAAAA

CT053-CopB nucleic acid sequence

SEQ ID NO: 77

aaaagtgagcggttataaaaaattagaatcagagcttcatgatcttaccagtggtgcaacttggccttgttctctaaaaagaa  
atcgagagacaccaggaagaaatccgtctgctagaaagcaaaatccttgaagagaaagaacgtctacaacttctcaaaagaaagcggt  
gagatcaagagtagctaacccctcgaagaactccagctaaaaccatttaccagatggccccagcggttcagacgttgagttttaga  
atcctcggatacagaagtgatctcgatgccggtgacacaattgagattgacctaggtgatgaggcaagagaagaagcggaacg  
aactcgactactctagtgaagcagtagaggatcctttcagcgatcgcaatcggttggcgccgaggagcatcatagatcctgacgcgaa  
tgaatggGGTTCAGCTGCTCAatgagcttgtcatccagcagcagctcggatagttcgaatctgaaaaatgtgttatctca  
ggtcatcgctctacaccAcaggggggttctaatgctgacaaattaaccgacaatcaggtaaaacaagtccagcagacccgtcaaaa  
ccgtgatgatctgtccatggagagcgacgtcgcggtggcggaacagccggaagatcggtgctgcgtcgcggtcccagatcgag  
ggacaagagctgattgagcaacagggaacttcgggtgggaaagagacggcttctgctgatgctacatcattgacccagtcggcatcc  
aaaggcgcttccagtcagcagtgattgaggataccagtaagtccctggagctttcttcgcttccagcctgtcaagcgtagatgcgaca  
catttgcaggaaatccaatcgatcggtgtcttcagcaatggcgccaccaacgaattgtcattgacgaacttagagacacgggattacc  
aaagccgagtagcactccAcgccaggaagttagggatcagccttgccttagcgaaggccatcactgcattgggtgagagcactca  
ggctgccttggaaaattttcagtcactcagagtcagtcgcgaacatgaataagatgagtttggaatcccaaggcttgaaaatcgaca  
aggagcgtgaagaatttaagaaaaatgcaggagattcagcaaaagagcggcacaattcaaccatggatactgtgaataaagttatgat  
tggcgtgacagtggaattacagtaatctctgttgtttcagcattgtttacctgcggtttgggcttgattggcacagccgctgcgggtgcc  
acagccgccaccgctggggcaacggccgccacgacgcctacctctgtgacgaccacagtcgctaccaggtgacgatgcaag  
cgggtggtccaagtctgtaagcaggctattatccaagcagtaaaacgcgccatcgctcaagcgattaaacaggggattaaagcaaggcat  
taaacaagcgatcaaacaggcagtcagggaagcgtagaacacttgccaaaaatgtaggcaagattttcagcgcaggcaagaacg  
ctgtgagtaagtccttcccAaaattgtctaaagtgattaatcaccttgggttccaaatgggttactcttggcgtgggggcccttacagcggt  
gccgcagtttagtcagtggttacctcccttcaattgtctgatatgcaaaaagaacttgcacaaatccaaaaggaagtggtgaccttac  
ggcgagagtgagatgatgaaagcggtttacactgttctggcagcaagcttcgaaaatcgcgggccaaacaaacggaatcaccttcaga  
gacgcaacaacaggcagctaagaccggcgccagatcgctaaagcggtgttcgccatttcgggtgctttagctgctgctgctTAG

CT053-CopB amino acid sequence

SEQ ID NO: 78

MKSERLKKLESELHDLTQWMQLGLVPKKEIERHQEEIRLLESKILEEKERLQL  
LKESGEIKEYVTPRRTPAKTIYPDGPSVSDVEFVESDTEVDLDAGDTIEIDLGDEARE



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ESGNELDYSEDEDEPFSDRNRWRRGGIIDPDANEWGSAASMSLSSSSSSDSSNLKN  
VLSQVIAS TPQGVPNADKLTDNQVKVQQTRQNRDDLMESEDVAVAGTAGKDRAA  
SASQIEGQELIEQQGLAAGKETASADATSLTQSASKGASSQQCIEDTSKSLELSSLSL  
SSVDATHLQEIQSIVSSAMGATNELSLTNLETPGLPKPSTTPRQEVMEISLALAKAITA  
LGESTQAALENFQSTQSQSANMNKMSLESQGLKIDKEREFPKMQEIQKSGTNST  
MDTVNKMIGVTVAITVISVVSALFTCGLGLIGTAAAGATAATAGATAAATTATSVT  
TTVATQVTMQAVVQVVKQAI IQAVKRAIVQAIKQGIKQGIKQAIKQAVKASVKTLA  
KNVGKIPISAGKNAVSKSFPKLSKVINTLGSKNVTLGVGALTAVPQLVSGITSLQLSD  
MQKELAQIQKEVGALTAQSEMMKAPTLEWQQASKIAAKQTESPSETQQQAAKTGA  
QIAKALSAISGALAAAA

LTA1-CT053-CopB nucleic acid sequence

SEQ ID NO: 79

CATatggacaatggcgatcggtttataccgtgccgactcgcggtccccagatgagattaaacgtagcgggtgggttaatgcc  
acgtgggcacaatgagatattttgacctgggaacacagatgaacattaacctttacgatcatgccgtgggacccagaccgggtttgtcc  
gttatgatgacgggtatgttagtacgagtttgccttacgctccgcacaccttgccgggacaaagtattttatcaggctacagcacatattac  
atattgtgatcgccactgccccaaacatgttcaatgtgaacgatgtgttgggggtttacagccccatccatataaacaagaagctctcg  
gccttgggggatcccatatagccagatattatggttggtaccgcgtaaattttgggtgatgatgaacgtttgcatcgtaaccgtgaata  
ccgcatcgctactaccgtaacttgaacattgcacctgccgaggacggctatcggttttagcgggattcccacccgatcatcaggcggtgg  
cgtgaggaacgtggatccatcacgccctcagggtgcggaacagtagtcgcCATatgaaaagtgagcgtttaaaaaatttag  
aatcagagcttcatgatcttaccagtggtgcaacttggccttggttcctaaaaaagaatcgagagacaccaggaagaaatccgtctg  
ctagaaagcaaaatccttgaagagaagaacgtctacaacttctcaaagaaagcgtgagatcaaagagtagctaaccctcgaaga  
actccagctaaaaccatttaccagatggccccagcgtttcagacgttgagttttagaatoctcggatcacagaagtgatctcgtatgcc  
ggtgacacaattgagattgacctagggtgatgaggcaagagaagaagcggaaacgaactcgactactctagtgaagacgatgagga  
tcctttcagcgatcgcaatcgttggcgccgaggaggcatcatagatcctgacgcgaatgaatggGGTTCAGCTGCTTCA  
atgagcttgcacccagcagcagctcgatagttcgaatctgaaaaatgtgttatctcaggtcatcggtctacaccAcaggggggtcct  
aatgtgcacaattaacccgacaatcaggtaaaacaagtcagcagaccggtcaaacctgtgatctgtccatggagagcgcgtc  
gcggtggcgggaacagccgaaaagatcgtgctgcgtcggcgtcccagatcgagggacaagagctgattgagcaacagggactt  
gcggttgggaaagagacggccttctgctgatgctacatcattgaccagtcggcatccaaaggcgttccagtcagcagtgattgagg  
ataccagtaagtccttgagctttcttcgctttcgagcctgtcaagcgtagatgcgacacatttgaggaaatccaatcgatcgtgtcttc  
agcaatggcgccaccaacgaattgtcattgacgaacttagagacaccgggattaccaagccgagtagtaccactccAcgccaggaa  
gttatggagatcagccttgcttagcgaaggccatcactgcatgggtgagagcactcaggctgccttggaattttcagtcactcag  
agtcagtcgcgaacatgaataagatgagtttgaatcccaaggcttgaaaatcgacaaggagcgtgaagaatttaagaaaaatgcag  
gagattcagcaaaagagcggcacaattcaaccatggatactgtgaataaagttatgattggcgtgacagtggaattacagtaattctct  
gttgtttcagcattgtttacctgcggtttggcgttgattggcacagccgctgcgggtgccacagccgccaccgctggggcaacggccg  
ccgccacgaccgctacctctgtgacgaccacagtcgctaccaggtgacgatgcaagcgggtgggtccaagtcgttaagcaggctattat  
ccaagcagtaaaacgcgccatcgtccaagcattaaacagggttaagcaaggcattaaacaagcgatcaaacaggcagtaagg  
caagcgtgaagacacttgccaaaaatgtaggcaagattttcagcgcaggcaagaacgctgtgagtaagtccttcccAaaattgtctaa  
ggtgatataacttggttccaaatgggttactcttggcgtggggcccttacagcgggtgcgcagtttagtcagtggttacctcctt

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caattgtctgatatgcaaaaagaacttgcacaaatccaaaaggaagtgggtgcacttacggcgagagtgagatgatgaaagcgttta  
cactgttctggcagcaagcttcgaaaatcgcgggccaaacaacggaatcaccttcagagacgcaacaacaggcagctaagaccggc  
gcccagatcgctaaagcggtgtccgccatttcgggtgcttttagctgctgctgctTAGCTCGAG

LTA1-CT053-CopB Amino acid sequence

SEQ ID NO: 80

MDNGDRLYRADSRPPDEIKRSGGLMPRGHNEYFDRGTQMNINLYDHARGTQ  
TGFVRYDDGYVSTSLSLRSAGLAGQSILSGYSTYYIYVIATAPNMFNVNDVLGVYSP  
HPYEQEVSA LGGIPYSQIYGWYRVNFGVIDERLHRNREYDRYRNLIAPAEDGYR  
LAGFPDPDHQAWREEPWIHHAPQGGNSRMKSERLKKLESELHDLTQWMQLGLVP  
KKEIERHQEEIRLLESKILEEKERLQLLKESGEIKEYVTPRRTPAKTIYPDGPSVSDVEF  
VESSDTEVDLDAGDTIEIDLGDEAREESGNELDYSSDEDDPFSDRNRRWRRGGIIDPD  
ANEWGSAA SMSLSSSSSSDSSNLKNVLSQVIASTPQGVPNADKLTDNQVKVQQTR  
QNRDLSMESDVA VAGTAGKDRAASASQIEGQELIEQQGLAAGKETASADATSLTQ  
SASKGASSQQCIEDTSKSLSSLSSLSVDATHLQEIQSIVSSAMGATNELSLTNLETP  
GLPKPSTTPRQEVMEISLALAKAITALGESTQAALENFQSTQSQSANMNKMSLESQG  
LKIDKEREFEFKMQEIQKSGTNSTMDTVNKMIGVTVAITVISVVSALFTCGLGLIG  
TAAAGATAATAGATAAAATTATSVTTTQVATQVAVVQVVKQAIQAVKRAIVQAI  
KQGIKQGIKQAIQAVKASVKTAKNVGKIFSA GKNAVSKSPKLSKVINTLGSKWV  
TLGVGALTAVPQLVSGITSLQLSDMQELAQIQKEVGALTAQSEMMKAPTFLFWQQA  
SKIAAKQTESPSETQQQAAKTGAQIAKALSAISGALAAAA\*

HisScc2 chaperone for LTA1-CT668-CopB nucleic acid sequence

SEQ ID NO: 81

ATGGGCAGCAGCCATCACCATCATCACCACAGCCAGGATCCGatgagcactccat  
cttctaataattctaaaaaaccttcggcctcttttaataaaaaatcacgtagccgcttggccgagattgctgcacaaaaaaaaagcaaaagc  
tgaggatttggaacaaaaatatcctgttcctacggaagaggagacaaaaaagttctcatggacatcctacaggggttaagcaacgga  
ttaactcttcagcaaatcttaggtctctccgacgtcctccttgaaagagatctacaccgtagcatataccttctactcccaagggaaatatcg  
ggaagctatcggtcttttccaaatcttaacagcctccaaacctcaatgctacaaatacatcttaggtcttagctcttgctatcaccagctaaa  
aatgtatgatgaagccgcttttggtttcttcctagcttttcgatgctcaaccgaaaaccccatccctccttactacatcgccgatagcttgat  
gaagctaaaccaaccggaagaatctcaagacttcctcgatattacgatcgatatgtgtaagaacaagccggaatataaagttcttaaga  
tcgctgcagcattatgaagcaatctttagatgccgtgctgaaaaagagaaatctgcaaaaggctctgaaacacaagcctcctctcctaa  
aaacacaaaagctaaaaaagctgcttctaacaagaaaaaagcaaaagtaagCGGCCGC

HisScc2 chaperone for LTA1-CT668-CopB Amino acid sequence

SEQ ID NO: 82

MGSSHHHHHSQDPMSTPSSNNSKKPSASFNKKRSRLAEIAAQKKAEDL  
EQKYPVPTEETKQVLM DILQGLSNGLT LQQILGLSDV LLEEIYTVAYTFYSQGKYRE  
AIGLFQILTASKPQCYKIYILGLSSCYHQLKMYDEAAFGLFAFDAQPENPIPPYIADS  
LMKLNQPEESQDFLDITIDMCKNKPEYKVLKDRCSIMKQSLDAVLKKEKSAKGETQ  
ASSPKNTKAKKAASNKKKAK\*

CT668 nucleic acid sequence

SEQ ID NO: 83

Atagatcctcttaagctttttccaaatcttgatggggataaggagagtgctgcggtgaataaaccttcagcatctcctatgcc  
agcgaattaagtaaaaatgttgccctattctcttttaggggtggaggtgctgcggttgattcgacagtgctccacagaaaagctatcggtga  
tggtatgatgcaggataaaaattcgagttgatcgatcctgagttggaggaagctctgaactctgaagagttacaagagcagatccatt

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tggtaaaaagtcggtttgtgggatgcacaaacgcagatgcaaatgcaagatcccgacaagttggcctctgagcatgtagatgctttagga  
gtcattgttgatttaataatggggattttcaagcgatagctgaacatacacacagacgggtcaagcagggtaatggtagcaagaaaa  
atctgttacacgcaagatagtcgattgggtctcttcaggagaagaaattttgaatcgtgctttgttgatttctccgatcgtaatggagaaag  
agaaacatttagccgattttctaaaagttcagtagtccggttcaaagagctacacaacgcgcgaggtatttggcagtagttaggtgccacg  
gtgagtagtgtaaaaacgattatgacaaccagtttaggt

CT668 amino acid sequence

SEQ ID NO: 84

MIDPLKLFNFDGDKESA AVNKPSASPMFSELSKNVASFSLGGGGAALDSTVS  
TEKLSLMAMMQDKNSQLIDPELEEALNSEELQEQIHLLKSRLWDAQTQMOMQDPD  
KLASEHVDALGVIVDLINGDFQAI AEHTQQTVKQNGDEEKS VTRKIVDWVSSGEEI  
LNRALLYFSRNGERETLADFLKVQYAVQRATQRAELFASILGATVSSVKTIMTTQL

CT668-CopB nucleic acid sequence

SEQ ID NO: 85

atagatcctcttaagctttttccaaattttgatggggataaggagagtgctgcggtgaataaaccttcagcatctcctatgccca  
gcgaattaagtaaaaaatgttgccctattctctttagggggtggaggtgctgcttggttcgacagtggtccacagaaaagctatcggtgat  
ggctatgatgcaggataaaaattcgcagttgatcgatcctgagttggaggaagctctgaactctgaagagtacaagagcagatccattt  
gttaaaaagtcggttggggatgcacaaacgcagatgcaaatgcaagatcccgacaagttggcctctgagcatgtagatgctttaggag  
tcattgttgatttaataatggggattttcaagcgatagctgaacatacacacagacgggtcaagcagggtaatggtagcaagaaaaat  
ctgttacacgcaagatagtcgattgggtctcttcaggagaagaaattttgaatcgtgctttgttgatttctccgatcgtaatggagaaaaa  
gaaacatttagccgattttctaaaagttcagtagtccggttcaaagagctacacaacgcgcgaggtatttggcagtagttaggtgccacgg  
tgagtagtgtaaaaacgattatgacaaccagtttaggtGGTTCAGCTGCTTCAatgagcttgatccagcagcagctcg  
gatagttcgaatctgaaaaatgtgttatctcaggtcatcgctctacaccAcaggggggttcctaatgctgacaaattaaccgacaatcag  
gtaaaaaagtcacgacagcccgctcaaaacggtgatgatctgtccatggagagcgacgtcgcggtggcgggaacagcgggaaaag  
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gatgctacatcattgaccagtcggcatccaaaggcgcttcagtcagcagtgatttaggataaccagtaagtccctggagcttctctcg  
ctttcgagcctgtcaagcgtagatgcgacacatttgaggaaatccaatcgatcggtgtcttcagcaatggcgccaccaacgaattgtc  
attgacgaacttagagacaccgggattaccaagccgagtagcactccAcgccaggaagttatggagatcagccttgcttagcgaa  
ggcccatcactgcatgggtgagagcactcaggtgccttgaaaaattttcagtcactcagagtcagtcgcgaacatgaataagatga  
gtttggaatcccaaggcttgaaaatcgacaaggagcgtgaagaatttaagaaaatgcaggagattcagcaaaagagcggcacaaaat  
caaccatggatactgtgaataaagtattgattggcggtgacagtggcaatatcagtaactctgtgtttcagcatgtttacctcggttttg  
gcttgattggcacagccgctgcggtgcccacagccgcccagctggggcaacggccgcccacgacccgctacctctgtgacgac  
cacagtcgctaccaggtgacgatgcaagcgggtggtccaagtcgttaagcaggtattatccaagcagtaaaacgcgccaatcgatcca  
agcgattaaacaggggattaaagcaaggcattaaacaagcgatcaaacaggcagtcagggaagcgtgaagacacttgccaaaaatg  
taggcaagattttcagcgcaggcaagaacgctgtgagtaagtcctcccAaaattgtctaagggtgattaatacacttggttccaaatggg  
ttactcttggtgggtggggcccttacagcgggtgccgagtttagtcagtggttacctcccttcaattgtctgatatgcaaaaagaacttgc  
acaaatccaaaaggaagtggtgacacttacggcgagagtgagatgatgaaagcgtttacactgttctggcagcaagcttcgaaaatc  
gcggccaaacaaacggaatcaccttcagagacgcaacaacaggcagctaaagaccggcgcccagatcgctaaagcgttggtccgcca  
tttcgggtgcttttagctgctgctTAG

CT668-CopB amino acid sequence

SEQ ID NO: 86

MIDPLKLFNFDGDKESA AVNKPSASPMFSELSKNVASFSLGGGGAALDSTVS  
TEKLSLMAMMQDKNSQLIDPELEEALNSEELQEQIHLLKSRLWDAQTQMOMQDPD

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KLASEHVDALGVIVDLINGDFQAI AEHTQQTVKQNGDEEKS VTRKIVDWVSSGEEI  
LNRALLYFSDRNGERETLADFLKVQYAVQRATQRAELFASILGATVSSVKTIMTTQL  
GSSAASMSLSSSSSSSNLKNVLSQVIAS TPQGVPNADKLTDNQVKVQQTRQNRD  
DLSMESDVAVAGTAGKDRAASASQIEGQELIEQQGLAAGKETASADATSLTQSASK  
GASSQQCIEDTSKSLELSSLSLSSVDATHLQEIQSIVSSAMGATNELSLTNLETPGLPK  
PSTTPRQEVMEISLALAKAITALGESTQAALENFQSTQSQSANMNKMSLESQGLKIDK  
EREEFKMQEIQQKSGTNSMTDTVNKVMIGVTVAITVISVVSALFTCGLGLIGTAAA  
GATAATAGATAAAATTATSVTTTVATQVMTQAVVQVVKQAIQAVKRAIVQAIKQGI  
KQGIKQAIKQAVKASVKT LAKNVGKIFSAGKNAVSKSPFKLSKVINTLGSKWVTLGV  
GALTAVPQLVSGITSLQLSDMQKELAQIQKEVGALTAQSEMMKAFTLFWQQASKIA  
AKQTESPSETQQQAAKTGAQIAKALSAISGALAAAA

LTA1-CT668-CopB nucleic acid sequence

SEQ ID NO: 87

CATatggacaatggcgatcggtttataccgtgccgactcgcgccccagatgagattaacgtagcggtagggtaatgcc  
acgtgggcacaatgagatattttgacctggaacacagatgaacattaacctttacgatcatgccgtgggacccagaccgggtttgtcc  
gttatgatgacgggtatgttagtacgagtttgccttaacgctccgcacaccttgccgggacaaagtattttatcaggctacagcacatattac  
atttatgtgatcgccactgccccaaacatgttcaatgtgaacgatgtgttgggggtttacagcccccatccatagaacaagaagtctcg  
gccccctgggggatcccatatagccagatttatggttggtaccgcgtaaattttggtgtgatgatgaacgtttgcatcgtaaccgtgaata  
ccgcatcgctactaccgtaactgaacattgcacctgcccaggagcggtatcggttagcgggattcccacccgatcatcaggcgtgg  
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tcaggagaagaaattttgaatcgctgctttgtgtatttctccgatcgtaaatggagaaagagaaacattagccgattttctaaaagttcagtat  
gccgttcaagagctacacaacgcgccaggtattttgccagttattctaggtgccacggtagtagtgtaaaaacgattatgacaaccca  
gttaggtGGTTACGTGCTTCAatgagcttgctcatccagcagcagctcggatagttcgaatctgaaaaatgtgttatctcag  
gtcatcgctctacaccAcaggggggttcctaatgctgacaaattaaccgacaatcaggtaaaacaagtcacgacacccgtcaaaac  
cgtgatgatctgtccatggagagcgacgtcgcggtggcggaacagccggaagagatcgctgctcggtcggtcccagatcgagg  
gacaagagctgattgagcaacagggaacttgccggtgggaaagagacggctctgctgatgctacatcattgacccagtcggcatcca  
aaggcgtctccagtcagcagtgatttgaggataccagtaagtccttgagctttcttcgctttcgagcctgtcaagcgtagatgcgacac  
atttgcaggaaatccaatcgatcggtgtcttcagcaatgggcgcccaaacgaattgtcatgacgaacttagagacacccgggtatacca  
aaggcagtagtaccactccAcgccaggaagtattggagatcagccttgcccttagcgaaggccatcactgcattgggtgagagcactca  
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tggcgtgacagtggaattacagtaaatctctgtgtttcagcattgtttacctgcggtttgggcttgattggcacagccgctgcgggtgccc  
acagccgccacccgtgggggcaacggccgcccagcaccgtacctctgtgacgaccacagtcgctaccaggtgacgatgcaag  
cgggtggtccaagtcgtaagcaggctattatccaagcagtaaaacgcgccatcgccaagcgattaaacaggggattaaagcaaggcat  
taacaagcgatcaaacaggcagtcagggaagcgtgaagacacttgccaaaaatgtaggcaagattttcagcgcaggcaagaacg

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ctgtgagtaagtccttcccAaaattgtctaaggtgattaatacacttggttccaaatgggttactcttggcgtgggggcccttacagcgg  
gcccagtttagtcagtggcattacctcccttcaattgtctgatatgcaaaaagaacttgcacaaatccaaaaggagtggtgaccttac  
ggcgcagagtgcagatgatgaaagcggtttactgttctggcagcaagcttcgaaaatcgcgcccaacaaacggaatcaccttcaga  
gacgcaacaacaggcagctaagaccggcgccagatcgctaaagcgttgctcgccatttcgggtgctttagctgctgctgctTAG  
CTCGAG

LTA1-CT668-CopB Amino acid sequence

SEQ ID NO: 88

MDNGDRLYRADSRPPDEIKRSGGLMPRGHNEYFDRGTQMNINLYDHARGTQ  
TGFVRYDDGYVSTSLSLRSAHLAQQSILSGYSTYYIYVIATAPNMFNVNDVLGVYSP  
HPYEQEVSALGGIPYSQIYGWYRVNFGVIDERLHRNREYRDRYYRNLNIAPAEDGYR  
LAGFPPDHQAWREEPWIHHAPQCGNSSRMIDPLKLPNFDGDKESA AVNKPSASP  
MPSELSKNVASFSLGGGAALDSTVSTEKLSLMAMMQDKNSQLIDPELEEALNSEEL  
QEQIHLKSLRLWDAQTMQMDPKLASEHVDALGVIVDLINGDFQAI AEHTQQTV  
KQNGDEEKSVTRKIVDWSSGEEILNRALLYFSRNGERETLADFLKVQYAVQRA  
TQRAELFASILGATVSSVKTIMTTQLGGSAA SMLSSSSSSDSSNLKNVLSQVIAS TPQ  
GVPNADKLTDNVKQVQQTQRNRDDL SMESDVAVAGTAGKDRAASQIEGQELI  
EQQGLAAGKETASADATSLTQSASKGASSQQCIEDTSKSLELSSLSLSSVDATHLQEI  
QSI VSSAMGATNELSLTNLETPGLPKPSTTPRQEVMEISLALAKAITALGESTQAALEN  
FQSTQSQSANMNKMSLESQGLKIDKERE EFKMQEIQQKSGTNSTMDTVNKVMIGV  
TVAITVISVVSALFTCGLGLIGTAAAGATAATAGATAAATTATSVTTTVATQVTMQA  
VVQVVKQAI IQAVKRAIVQAIKQGIKQGIKQAIKQAVKASVKTLAKNVGKIF SAGKN  
AVSKSFPKLSKVINTLGSKWVTLGVGALTAVPQLVSGITSLQLSDMQKELAQIQKEV  
GALTAQSEMMKAFTLFWQQASKIAAKQTESPSETQQQA AKTGAQIAKALSAISGAL  
AAAA\*

HisScc3 chaperone for CT053-CopB2 Nucleic acid sequence

SEQ ID NO: 89

ATGGGCAGCAGCCATCACCATCATCACACAGCCAGGATCCGatgccaccaagc  
aagatccaatgtcttgaaacttttgaaagaacttatggacacctttatctacaacatgcgtccctaatgcgtcatttagcctatctactcgata  
aaattgctcgctcttaccctcatatgtgtccgcttcccgcataatatggaagcgtactttgagaattatatccccaataaagatatccctctgg  
acacctatcaaaaaattttcaaaactgtcctcagaagatcttgaacaagctctacaaggaaggatacaacgcctatttacaaggagactatg  
aggaaagttctaccgctttttactggttgattttctttaaccatttgtgtctaaattttggttttcattaggagcttcgctccatcgcccaaaa  
atatcaacaagctcttcatgcttatggtgtagctgctttgctaagagaaaaagacccttatectcattactatgctacatctgctacacctt  
gctcaataatcctgaagaagctgaaaaagctcttgatcttgcttggtgcaaaaagtataaacaagctctgcctatagctctttaaagaaga  
aat ttttagcgatcaaatcgtagcctaaGCGGCCGC

HisScc3 chaperone for CT053-CopB2 amino acid sequence

SEQ ID NO: 90

MGSSHHHHHSQDPMPPSKIQCLETFFERTYGHLYLQHASLMRHLAYLLDKIA  
RSYPHMCPLPDNMEAYFENYIPNKDIPLDITYQKIFKLSSSEDLQVYKEGYNAYLQGD  
YEESSTAFYWLIFPNPFVSKFWFSLGASLHMRQKYQQALHAYGVAALLREKDPYPH  
YYAYICYTLLNPEEA EKALDLAWQKVKTSSAYSSLKEEILAIKSYA\*

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CopB2 nucleic acid sequence

SEQ ID NO: 91

atgagctcttggtttgcacagcgacgacgtcgctttgagccagacccttgatctgcctgacgcttcattggcggttcaaac  
cgaaaaatttccAtacagctgttcaatctctaaggaatccgccccAtcatgtattcgtaaaatcttcgccatttagcatctcagaaggaa  
agtgtccgctgtctttttctcggttacaaccgactactccgaaagaacgcatcctgttttcgggtcatcgcttccctcccaattgtcctcga  
ctgtccgcaccacaaacctcttctccatggaatcttttagcaactcccaggcacgcaactcgaccgtaaattgtcgagaagcttcattt  
gagctcagagttatccgcccgtgactccactaagccttcgtcgagcgaaccgggttaaaccatcggaatatctttgcacaccctgagc  
atcataaggaatccttctcaagtgtgaaaaaggataacttatctcctatcatggaggagatcgactcattctctgcagagacagagtcctt  
gaagagcgctttggtcaccacagaaaaggaggagacgggtggccaggagcaaaagcaccctAttgctgcgtacatctactccgccat  
caaagggcagcggggaatcacaagattctagcgaacacagctcaaaggaagatccttatagtcaacaaccgagccataaaatccaac  
gccgtaaaagagcgtgctaagcgcgctgctcccAattattactccgccaacgggtgggtatctttagtttgagctaccttcttataaaacagg  
ggatcttagcggtatttcagcgctatttcggcatacaaggataatttagaaacaactcagcaagagctgacctgttgcatcaagaacgta  
tcgagcaagtcacaaaGatcgtggataaaagtaagacaatcgcgcttttgggattcattagcatccattgtggccacaatcatctccatgga  
tcgaaatgggtgttgagtaaacatcatcgactgggaggtggaatccttctcgtgctctcttttgcgtgcgcttatcatgattgtaatttc  
attattggaagcattcgacgggtggcggtgcaatcgctaagcattaccaggtaacgatcttgaaaagaagatgcggttatcttaggttacgta  
aagttggccttaactgtgttctcgtgcttactgagtttaagcgcttctgtatgtagcaaaattaggaatgagtcgcttttggaggggggtgtg  
aagagtatcgaccAgcattaagtggtatgctgggtttgactcaaggcgtagcactgtatttacaatcttcacgcaaaagattcggtcc  
cgctgcactcagatcgacgcacgcattgaattgattaactgggaacgcgatgagttattcttgcgtgctgaacaactctcttgattcaatgc  
aaacgtccttcgaacaacttactgaacattacagttacaacgtgaaattgatcagacatttacagacgctttgcgcTAG

CopB2 amino acid sequence

SEQ ID NO: 92

MSSWFAQATDVALSQTLDLPDASLAVQTEKFPYSCSISKESAPSCIRKIFAHLA  
SQKESAPLSFSRLQPTTPKERILFFGSSPSSQLSSTVTRTTTSSPWNLFNSQARNSTRKL  
SEKLHLSSSELSARDSTKPSSSEPVKPSENLLHTPEHHKESFSSLKKNLSPIMEIDSFS  
AETESLEERLVTQKKEETVAQEQQHPLLRSTPPSKASGESQDSSEHSSKEDPYSQQP  
SHKIQRKRERAKRVVPIITPPTVGIFSLSYLLTKQGILADFSAYSAYKDNLETTQQELT  
MLHQERIEQVQKIVDKSKTMRFWDSLASIVATIIPIWEMGVAVTIIALGGGILSWCSLF  
AALIMIVISLLEAFDGWRAIAKHLPGNDLEKKMRYLGYVKLALTVFSCLLSLSALYV  
AKLGMSPLEGVVKSIPALSGMLGLTQGVALYLQSSSQKIRARCTQIDARIELINWE  
RDEYFLRAEQLLDSMQTSFEQLTETLQLQREIDQTFTDALR

CT053-CopB2 nucleic acid sequence

SEQ ID NO: 93

aaaagtgcgctttaaataaattagaatcagagcttcatgatcttaccagtggtgcaactggccttgcttccataaaagaa  
atcgagagacaccaggaagaaatccgtctgctagaaagcaaatccttgaaagagaagaacgtctacaacttctcaaagaagcggt  
gagatcaaagagtacgtaacccctcgaagaactccagctaaaaccatttaccagatggccccagcgcttcagacgcttgagttgtaga  
atcctcggatcacagaagtgatctcgatgccggtgacacaattgagattgacctaggtgatgaggcaagagaagaaagcggaacg  
aactcgactactctagtgaagcagatgaggatcctttcagcagatcgcaatcggttggcgccgaggagcatcatagatcctgacgcgaa  
tgaatggGGTTCAGCTGCTCAatgagctcttggtttgcacaggcgacggacgtcgctttgagccagacccttgatctgc  
ctgacgcttcattggcggttcaaaccgaaaaatttccAtacagctgttcaatctctaaggaatccgccccAtcatgtattcgtaaaatctt  
cgccatttagcatctcagaaggaaagtgtccgctgtctttttctcgcttacaaccgactactccgaaagaacgcatcctgttttcgggt  
catcgcttctcccaattgtcctcgactgtccgcaccacaaacctcttctccatggaatcttttagcaactcccaggcacgcaactcgac  
ccgtaaatgtcgagaagcttcatttgagctcagagttatccgcccgtgactccactaagccttcgtcgagcgaaccgggttaaacctac

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ggaaaaatccttttgacacccctgagcatcataaggaatccttctcaagtttgaaaaaggataacttatctcctatcatggaggagatcgac  
tcattctctgcagagacagagtcccttgaagagcgtttggtcaccagaaaaaggaggagacggtggcccaggagcaaaagcacc  
Attgtgcgtacatctactccgccatcaaaaggccagcggggaatcacaagattctagcgaacacagctcaaaggaagatccttatagt  
caacaaccgagccataaaaatccaacgcgtaaaagagcgtgctaagcgcgtcgtcccAattattactccgccaacggtgggtatcttta  
gtttgagctaccttcttcaaaaacaggggatccttagcggatttcagcgccatttcggcatacaaggataatttagaacaactcagcaag  
agctgaccatgttgcatcaagaacgtatcgagcaagtcacaaaGatcgtggataaaaagtaagacaatgcgcttttgggattcattagca  
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aaagaagatgcgttatcttaggttacgtaaaagtggccttaactgtgtctcgtgcttactgagtttaagcgcttgtatgtagcaaaattagg  
aatgagtcgcgttttggagggggtgtgaagagtatcgaccAgcattaagtgggtatgctgggttgactcaaggcgtagcactgtattt  
acaatcttcacgcaaaagattcgtgcccgtgcactcagatcgacgcacgattgaattgattaactgggaacgcgatgagtatttcttg  
cgtgctgaacaactcttctgattcaatgcaaacgtccttcgaacaacttactgaaacattacagttacaacgtgaaattgatcagacatttac  
agacgctttgcgcTAG

CT053-CopB2 amino acid sequence

SEQ ID NO: 94

MKSERLKKLESELHDLTQWMQLGLVPKKEIERHQEEIRLLESKILEEKERLQL  
LKESGEIKEYVTPRRTPAKTIYPDGPSVSDVEFVSSDTEVDLDAGDTIEIDLGDEARE  
ESGNELDYSSDEDDPFSDRNRRRGIIIDPDANEWGSAASSWFAQATDVALSQ  
TLDLPDASLAVQTEKFPYSCSISKESAPSCIRKIFAHLASQKESAPLSFSRLQPTTPKERI  
LFFGSSPSSQLSSTVRTTSSPWNLFNSQARNSTRKLSEKLHLSSELSARDSTKPSSE  
PVKPSENLLHTPEHHKESFSSLKKNLSPIMEEIDSFSAETESLEERLVTQKKEETVAQ  
EQKHPLLRTSTPPSKASGESQDSSEHSSKEDPYSQQPSHKIQRRKERAKRVVPIITPPT  
VGIFSLSYLLTKQILADFSAYSAYKDNLETTQQLTMLHQERIEQVQKIVDKSKTM  
RFWDSLASIVATIIPIWEMGVAVTIIALGGGILSWCSLFAALIMIVISLLEAFDGWRAIA  
KHLPGNDLEKKMRYLGYVKLALTVFSCLLSLSALYVAKLGMSPLEGVVKSIPALS  
GMLGLTQGVLYLQSSSQKIRARCTQIDARIELINWERDEYFLRAEQLLDSMQTSFE  
QLTETLQLQREIDQTFTDALR

LTA1-CT053-CopB2 nucleic acid sequence

SEQ ID NO: 95

CATatggacaatggcgatcggttataccgtgccgactcgcgtccccagatgagattaaacgtagcgggtgggttaatgcc  
acgtgggcacaaatgagtattttgaccgtggaacacagatgaacattaaccttacgatcatgccgtgggaccagaccgggtttgtcc  
gttatgatgacgggtatgttagtacgagttgtccttaacgctccgcacaccttgccgggacaaagtattttatcaggctacagcacatattac  
atttatgtgatcgccactgccccaaacatgttcaatgtgaacgatgtgttgggggtttacagcccccacatgaacaagaagtctcg  
gcccttggggggatcccatatagccagatttatggttggtaccgcgtaaattttggtgtgatgtatgaacgtttgcatcgtaaccgtgaata  
ccgcgatcgctactaccgtaacttgaacattgcacctgccgaggacggctatcgtttagcgggattcccacccgatcatcaggcgtgg  
cgtgaggaaccgtggatccatcacgcccctcaggggtgcgggaacagtagtcgcCATatgaaaagtgagcgtttaaaaaattag  
aatcagagcttcgatcttaccagtggtgcaacttggccttggtccttaaaaaaagaatcgagagacaccaggaagaagaatccgtctg  
ctagaaagcaaaatccttgaagagaaagacgtctacaacttctcaaaagaaagcgtgagatcaaagagtacgtaacccctcgaaga  
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gggtgacacaattgagattgacctaggtgatgaggcaagagaagaagcggaacgaactcgactactctagtgaagacgatgagga  
tcctttcagcgatcgcaatcgttggcgccgaggaggcatcatagatcctgacgcgaatgaatggGGTTCAGCTGCTTCA

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ccgtgtctttttctcgtttacaaccgactactcgaagaacgcacatcctgtttttcgggteatcgcttcctcccaatgtcctcgactgtcc  
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cgtttggtcaccagaaaaaggaggagacggtggccaggagcaaaagcaccAttgctgctacatctactccgcatcaaaggc  
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agtccaaaGatcgtggataaaagtaagacaatgcgcttttgggattcattagcatccattgtggccacaatcatccatggatcgaaat  
gggtgtgcagtaaacatcatcgactgggaggtggaatccttctcgtgctctcttttgcgcttatcatgattgtaatttcattattgg  
aagcatcgcaggggtgggtgcaatcgctaagcatttaccaggtaacgatcttgaaaagaagatgcgttatttaggttacgtaagttgg  
ccttaactgtgttctcgtgcttactgagtttaagcgcttgtagtagcaaaattaggaatgagtcgcttttggaggggttgtagaagag  
atcgaccAgcattaagtggtatgctgggttgactcaaggcgtgactgtatttacaatcttcacgcaaaagattcgtgcccgctgc  
actcagatcgacgcacgcattgaattgattaactgggaacgcgatgagtatcttgcgctggaacaaactcttgattcaatgcaaacgt  
ccttcgaacaacttactgaaacattacagttacaacgtgaaattgatcagacattacagacgcttttgcgcTAGCTCGAG

LTA1-CT053-CopB2 Amino acid sequence

SEQ ID NO: 96

MDNGDRLYRADSRPPDEIKRSGGLMPRGHNEYFDRGTQMNINLYDHARGTQ  
TGFVRYDDGYVSTSLSLRSAHLAQQSILSGYSTYYIYVIATAPNMFNVNDVLGVYSP  
HPYEQEVSALGGIPYSQIYGWYRVNFGVIDERLHRNREYDRYYRNLNIIAPAEDGYR  
LAGFPDPDQAWREEPWIHAPQCGNSSRMKSERLKKLESELHDLTQWMQLGLVP  
KKEIERHQEEIRLLESKILEEKERLQLLKESGEIKEVTPRRTPAKTIYPDGPSVSDVEF  
VESSDTEVDLDAGDTIEIDLGEAREESGNELDYSEDEDDPFSDRNRWRGGIIDPD  
ANEWGSAA SMSWFAQATDVALSQTLDLPDASLAVQTEKFPYSCSISKESAPSCIRKI  
FAHLASQKESAPLSFSLRLOPTTPKERILFFGSSPSSQLSSTVTRTTSSPWNLFNSNQARN  
STRKLSEKLHLSSELSARDSTKPSSEPVKPSENLLHTPEHHKESFSSLKKNLSPIMEE  
IDSFSAETESLEERLVTQKKEETVAQEQQHPLLRSTPPSKASGESQDSSEHSSKEDPY  
SQQPSHKIQRRKERAKRVVPIITPPTVGIFSLSYLLTKQGILADFSAYSAYKDNLETTQ  
QELTMLHQERIEQVQKIVDKSKTMRFWDSLASIVATIIPIWIEMGVAVTIIALGGGILS  
WCSLFAALIMIVISLLEAFDGWRAIAKHLPGNDLEKKMRYLGYVKLALTVPFSCLLSL  
SALYVAKLGMSPLLEGVVKSIAPALSGMLGLTQGVALYLQSSSQKIRARCTQIDARIE  
LINWERDEYFLRAEQLLDSMQTSFEQLTETLQLQREIDQTFDALR\*

HisScc3 chaperone for CT668-CopB2 nucleic acid sequence

SEQ ID NO: 97

ATGGGCAGCAGCCATCACCATCATCACCACAGCCAGGATCCGatgccaccaagc  
aagatccaatgtcttgaaacttttgaaagaacttatggacacctttatctacaacatcgctccctaagtcgctcatttagcctatctactcgata  
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aatcttagcgatcaaatcgtagcctaaGCGGCCGC

HisScc3 chaperone for CT668-CopB2 Amino acid sequence

SEQ ID NO: 98

MGSSHHHHHSQDPMPPSKIQCLETFFERTYGHLYLQHASLMRHLAYLLDKIA  
RSYPHMCPLPDNMEAYFENYIPNKDIPLDITYQKIFKLSSDLEQVYKEGYNAYLQGD  
YEESSTAFYWLIFNPFVSKFWFSLGASLHMRQKYQQALHAYGVAALLREKDPYPH  
YYAYICYTLNNPEEAELDLAWQKVKTSAYSSLKEEILAIKSYA\*

CT668-CopB2 nucleic acid sequence

SEQ ID NO: 99

aaaagtgcgcgtttaaaaaaattagaatcagagcttcatgatcttaccagtggtgcaacttggccttgttcctaaaaaagaa  
atcgagagacaccaggaagaaatccgtctgctagaaagcaaaatccttgaagagaaagaagctctacaacttctcaagaaagcggt  
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acaatcttcacgcaaaagattcgtgcccgctgcactcagatcgacgcacgcattgaattgattaactgggaacgcgatgagtagtttcttg  
cgtgctgaacaacttcttgattcaatgcaaacgtccttcgaacaacttactgaaacattacagttacaacgtgaaattgatcagacatttac  
agacgctttgcgcTAG

CT668-CopB2 amino acid sequence

SEQ ID NO: 100

MKSERLKKLESELHDLTQWMQLGLVPKKEIERHQEEIRLLESKILEEKERLQL  
LKESGEIKEYVTPRRTPAKTIYPDGPSVSDVEFVSSDTEVDLDAGDTIEIDLGDEARE  
ESGNELDYSSDEDDPFSDRNRWRRGGIIDPDANEWGSAASSSWFAQATDVALSQ  
TLDLPDASLAVQTEKFPYSCSISKESAPSCIRKIFAHLASQKESAPLSFSLQPTTPKERI  
LFFGSSPSSQLSSTVTRTTTSSPWNLFNSQARNSTRKLSEKLHLSSELSARDSTKPSSE  
PVKPSENLLHTPEHHKESFSSLKKNLSPIMEEIDSFSAETESLEERLVTQKKEETVAQ

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EQKHPLLRSTPPSKASGESQDSSEHSSKEDPYSQQPSHKIQRRKERAKRVVPIITPPT

VGIFSLSYLLTKQILADFSAYSAYKDNLETTQQELTMLHQERIEQVQKIVDKSKTM

RFWDSLASIVATIIPIWEMGVAVTIIALGGGILSWCSLFAALIMIVISLLEAFDGWRAIA

KHLPGNDLEKKMRYLGIVKLALTVFSCLLSLSALYVAKLGMSPLLEGVVKSIAPALS

GMLGLTQGVALYLQSSSQKIRARCTQIDARIELINWERDEYFLRAEQLLDSMQTSFE

QLTETLQLQREIDQTFDRLR

LTA1-CT668-CopB2 nucleic acid sequence

SEQ ID NO: 101

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LTA1-CT668-CopB2Amino acid sequence

SEQ ID NO: 102

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 HPYEQEVSA LGGIPYSQIYGWYRVNFGVIDERLHRNREYDRYRNLNIAPAEDGYR  
 LAGFPDPDHQAWREEPWIHHAPQCGNSSRMKSERLKKLESELHDLTQWMQLGLVP  
 KKEIERHQEEIRLLESKILEEKERLQLLKESGEIKYVTPRRTPAKTIYPDGPSVSDVEF  
 VESSDTEVDLDAGDTIEIDLGDEAREESGNELDYSSDEDDPFSDRNRWRRGGIIDPD  
 ANEWGSAASMSSWFAQATDVALSQTLDLDPDASLAVQTEKFPYSCSISKESAPSCIRKI  
 FAHLASQKESAPLSFSRLQPTTPKERILFFGSSPSSQLSSTVRTTSSPWNLFNSQARN  
 STRKLSEKLHLSSELSARDSTKPSSEPVKPSENLLHTPEHHKESFSSLKKDNLSPIMEE  
 IDSFAETESLEERLVTQKKEETVAQEQQKHLRLRTSTPPSKASGESQDSSEHSSKEDPY  
 SQQPSHKIQRRKERAKRVVPIITPPTVGIFSLSYLLTKQGILADFSAYSAYKDNLETTQ  
 QELTMLHQERIEQVQKIVDKSKTMRFWDSLASIVATIIPWIEMGVAVTIIALGGGILS  
 WCSLFAALIMIVISLLEAFDGWRAIAKHLPGNDLEKKMRYLGYVKLALTVFSCLLSL  
 SALYVAKLGMSPLLEGVVKSIAPALSGMLGLTQGVALYLQSSSQKIRARCTQIDARIE  
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dmLT eltA (LTa) nucleic acid sequence

SEQ ID NO: 113

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 aacagaattc gggatgaatt atgaataaag taaaatgt

eltB (LTb) nucleic acid sequence

SEQ ID NO: 114

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 tcatatacgg aatcgatggc aggcacaaaga gaaatggta tcatatcatt taagagcggc gaaacatttc aggtcgaagt  
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 aaggcggtag agggaggcaat accgatcctt aaactgtaac actataacag cttccactac agggagctgt tatagcacac  
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## SEQUENCE LISTING

Sequence total quantity: 118

SEQ ID NO: 1                   moltype = DNA   length = 1833  
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 misc\_feature               1..1833  
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 source                     1..1833  
                             mol\_type = other DNA  
                             organism = synthetic construct

## SEQUENCE: 1

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SEQ ID NO: 2                   moltype = AA   length = 607  
 FEATURE                    Location/Qualifiers  
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                             note = synthetic construct  
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                             mol\_type = protein  
                             organism = synthetic construct

## SEQUENCE: 2

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VTAPGGRGTQ VVNGTGVMA G STTYKELESA YTTVKGMLDT ASNTQQMDMI RLQAASNKRN 180
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AVGFAAVASV ATGAAATPML LLSGMALVSA VTSLADQISQ EAGGPPIISLG GFLSGLAGRL 420
LTALGVDQSQ ADQIAKIVAG LAVPVVLLIE PQMLGEMAQG VARLAGASDA TAGYIAMAMS 480
IVAAIAVAAI NAAGTAGAGS ASAIKGAWDR AAATATQVLQ GGTAVAQGGV GVSMAVDRKQ 540
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SEQ ID NO: 3                   moltype = DNA   length = 618  
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                             organism = Bordetella spp

## SEQUENCE: 3

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atcaccaaca  acatgcgc 618

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SEQ ID NO: 4      moltype = AA  length = 205
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source           1..205
                 mol_type = protein
                 organism = Bordetella spp

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VTAPGGRGTQ VVNGTGVMAG STTYKELESA YTTVKGMLDT ASNTQQMDMI RLQAASNKRN 180
EAFEVMTNTE KRRSDLNSSI TNNMR 205

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SEQ ID NO: 5      moltype = DNA  length = 1209
FEATURE          Location/Qualifiers
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                 organism = Bordetella spp

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gccagcgctg  ccacccggcg  ggcggccaca  cccatgctgc  tgctcagcgg  catggcactg 540
gtcagcgccg  tgacatcgct  ggcgcgaccg  atatcgcaag  aggcgggagg  cccgcctatc 600
agcctgggcg  ggtttctctc  cgggctggcc  ggacgtctgc  tgacagcggt  ggggggtggc 660
cagtcgcagg  ccgaccaa  tgccaagatc  gtcgcgggcc  tggccgtgcc  cgtcgtcttg 720
ctgatcgaa  ccagatgct  gggcgaaatg  gcgcaaggcg  tggccagggt  ggctggcgcc 780
agcgatgcca  ccgcggggta  catagccatg  gcgatgtcca  tcgtggcggc  gatcgcggtc 840
gccgcgatca  atgcgcggcg  tacagccggc  gcgggtagcg  cttcgcgcat  caagggggccc 900
tgggatcggg  ccgcgcgggt  agccaccag  gtccttcaag  ggggtacggc  agtggcgcaa 960
ggcgcgctcg  gcgtgtcgat  ggcagtcgat  cgcaaacagg  ccgatctcct  ggtcgccgac 1020
aaggcggtat  tggcgggcag  cctgacaaaa  ctgcggggcg  ccatggagcg  tgaggcggtc 1080
gatataaaga  agatctggc  tcaattcgac  gaggcctatc  acatgatcgc  gaagatgatc 1140
agcgatatgg  cgagtacgca  cagccagggt  agcgccaacc  tcggggcggg  ccaggcggtg 1200
tagctcgag 1209

```

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SEQ ID NO: 6      moltype = AA  length = 400
FEATURE          Location/Qualifiers
source           1..400
                 mol_type = protein
                 organism = Bordetella spp

```

```

SEQUENCE: 6
MTVMSTTIST APSGAALAPS RIDMRAPEPG SAGEGAGILA PVTTLALAAG RPAFPASPSL 60
RTAPVLDPVP RDLSPADLAD LLRVLSRAV DGQLATAREN LQDAQVKAKQ NTQAQLDKLD 120
AWFRKAEEEE SKGWSKVFG WIGKVLAVVA SALAVGFAAV ASVATGAAAT PMLLLSGMAL 180
VSAVTSADQD ISQEAGGPPI SLGGFLSGLA GRLLTALGVD QSQADQIAKI VAGLAVPVVL 240
LIEPQMLGEM AQGVARLAGA SDATAGYIAM AMSIVAAIAV AAINAAGTAG AGSASAIKGA 300
WDRAAAVATQ VLQGGTAVAQ GGVGVSMADV RKQADLLVAD KADLAASLTK LRAAMEREA 360
DIKKILAQFD EAYHMIKMI SDMASTHSQV SANLGRQAV 400

```

```

SEQ ID NO: 7      moltype = DNA  length = 534
FEATURE          Location/Qualifiers
misc_feature     1..534
                 note = synthetic construct
source           1..534
                 mol_type = other DNA
                 organism = synthetic construct

```

```

SEQUENCE: 7
atgggcagca  gccatcacca  tcataccacc  agccaggatc  cgatgccaaa  gtcagccgag 60
cagggcggtc  ccccgcgctc  agcttcgcat  gaggcggtgc  gccatattct  cgacgcaggc 120
gcttcgatgg  gcagcttgca  ggggttgga  gaggtgcaac  agcaggcggt  gtacgcgatc 180
gctcatggcg  cctacgaaca  gggccgctat  gccgacgcgt  tgaaaatggt  ctgcctgctg 240
gtcgcgtgcg  atccgctgga  agcccggtat  ctgctggccc  tgggcgcgcg  gggccaggag 300
ctggggctgt  acgagcatgc  cttgcagcaa  tacgcggccg  cggcggtgtt  gcagttggac 360
tccccaggc  cctctgttga  tggcgccgag  tgctgtgatg  cggtgggtcg  tcgcccgcac 420
gccctggata  cgctcgacat  ggtgcttgag  ttgtgcgggt  cgccggagca  tcggcccttg 480
cgcaaacggg  ccgagtcgct  gcgcaggagc  tatgcacgtg  ccgactgaaa  gctt 534

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SEQ ID NO: 8      moltype = AA  length = 175

```

-continued

|                        |                                |                                      |
|------------------------|--------------------------------|--------------------------------------|
| FEATURE                | Location/Qualifiers            |                                      |
| REGION                 | 1..175                         |                                      |
|                        | note = synthetic construct     |                                      |
| source                 | 1..175                         |                                      |
|                        | mol_type = protein             |                                      |
|                        | organism = synthetic construct |                                      |
| SEQUENCE: 8            |                                |                                      |
| MGSSHHHHH S QDPMPKSAE  | QGGSPASASH EALRHILDAG          | ASMGSLQGLD EVQQQALYAI 60             |
| AHGAYEQGRY             | ADALKMFCLL VACDPLEARY          | LLALGAAQOE LGLYEHALQQ YAAAAALQLD 120 |
| SPRPLHGAE              | CLYALGRRRD ALDTLDMVLE          | LCGSPEHAAL RERAESLRRS YARAD 175      |
| SEQ ID NO: 9           | moltype = AA length = 161      |                                      |
| FEATURE                | Location/Qualifiers            |                                      |
| REGION                 | 1..161                         |                                      |
|                        | note = synthetic construct     |                                      |
| source                 | 1..161                         |                                      |
|                        | mol_type = protein             |                                      |
|                        | organism = synthetic construct |                                      |
| SEQUENCE: 9            |                                |                                      |
| MPKSAEQGGS PASASHEALR  | HILDAGASMG SLQGLDEVQQ          | QALYAIHGA YEQGRYADAL 60              |
| KMFCLLVACD             | PLEARYLLAL GAAQQLGLY           | EHALQQYAAA AALQLDSRP L LHGAECLYA 120 |
| LGRRRDALDT             | LDMVLELCGS PEHAALRERA          | ESLRRSYARA D 161                     |
| SEQ ID NO: 10          | moltype = DNA length = 544     |                                      |
| FEATURE                | Location/Qualifiers            |                                      |
| misc_feature           | 1..544                         |                                      |
|                        | note = synthetic construct     |                                      |
| source                 | 1..544                         |                                      |
|                        | mol_type = other DNA           |                                      |
|                        | organism = synthetic construct |                                      |
| SEQUENCE: 10           |                                |                                      |
| ccatgggcag cagccatcat  | catcatcatc acagcagcgg          | cctgggtgcc cgcggcagcc 60             |
| atatgctcga gatgtcttta  | aatatcaccg aaaatgaaag          | catctctact gcagtaattg 120            |
| atgcaattaa ctctggcgct  | acactgaaag atattaatgc          | aattcctgat gatgatgatg 180            |
| atgacattta ttcataatgct | tatgactttt acaacaaagg          | aagaatagag gaagctgaag 240            |
| ttttcttcag gtttttatgt  | ataacgact tttacaatat           | agactacatt atgggactcg 300            |
| cagctattta tcagataaaa  | gaacagttcc aacaagcagc          | agacctttat gctgtcgctt 360            |
| ttgcattagg aaaaaatgac  | tataccaccag tattccatac         | tggaacaatgc cagcttcggt 420           |
| tgaaagcccc cttaaaagct  | aaagagtgct tcgaactcgt          | aattcaacac agcaatgatg 480            |
| aaaaattaaa aataaaagca  | caatcatact tggacgcaat          | tcaggatatac aaggagtagg 540           |
| atcc                   |                                | 544                                  |
| SEQ ID NO: 11          | moltype = AA length = 155      |                                      |
| FEATURE                | Location/Qualifiers            |                                      |
| REGION                 | 1..155                         |                                      |
|                        | note = synthetic construct     |                                      |
| source                 | 1..155                         |                                      |
|                        | mol_type = protein             |                                      |
|                        | organism = synthetic construct |                                      |
| SEQUENCE: 11           |                                |                                      |
| MSLNITENES ISTAVIDAIN  | SGATLKDINA IPDDMMDDIY          | SYAYDFYNKG RIEEAFFFR 60              |
| FLCIYDFYNV DYIMGLAAIY  | QIKEQFQAA DLYAVAFALG           | KNDYTPVFHT GQCQLRLKAP 120            |
| LKAKECFELV IQHSNDEKLK  | IKAQSYLDAI QDIKE               | 155                                  |
| SEQ ID NO: 12          | moltype = DNA length = 585     |                                      |
| FEATURE                | Location/Qualifiers            |                                      |
| source                 | 1..585                         |                                      |
|                        | mol_type = genomic DNA         |                                      |
|                        | organism = Escherichia coli    |                                      |
| SEQUENCE: 12           |                                |                                      |
| catatggaca atggcgatcg  | tttataccgt gccgactcgc          | gtccccaga tgagattaaa 60              |
| cgtagcgggtg ggttaatgcc | acgtggggcac aatgagtatt         | ttgaccgtgg aacacagatg 120            |
| aacatttaacc tttacgatca | tgcccggtggg acccagaccg         | ggtttgtccg ttatgatgac 180            |
| gggtatgtta gtacgagttt  | gtccttacgc tccgcacacc          | ttgcgggaca aagtatttta 240            |
| tcaggctaca gcacatatta  | catttatgtg atcgccactg          | ccccaaacat gttcaatgtg 300            |
| aacgatgtgt tgggggttta  | cagcccccat ccatatgaac          | aagaagtctc ggcccttggg 360            |
| gggatcccat atagccagat  | ttatgggttg taccgcgtaa          | atatttggtg gattgatgaa 420            |
| cgtttgcatc gtaaccgtga  | atacccgcat cgctactacc          | gtaacttgaa cattgcacct 480            |
| gccgaggacg gctatcggtt  | agcgggattc ccacccgata          | atcaggcggtg gcgtgaggaa 540           |
| ccgtgggatcc atcacgcccc | tcagggggtgc gggaacagta         | gtcgc 585                            |
| SEQ ID NO: 13          | moltype = AA length = 194      |                                      |
| FEATURE                | Location/Qualifiers            |                                      |
| source                 | 1..194                         |                                      |
|                        | mol_type = protein             |                                      |
|                        | organism = Escherichia coli    |                                      |

[illegible]

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ggaagtgtcg cttctgtctgt tttccagaac agcgcgtcga caaatctagc agacctgaca 3180
ttatcgaaat atcaagtgtga acaactgtca aaatatatca gtgaagcaat agaaaaattc 3240
ggccaattgc aggaagtaat tgcagatcta ttagcctcaa tgtccaactc tcaggctaata 3300
agaactgatg ttgcaaaagc aattttgcaa caaactactg cttgaggatc c 3351

```

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SEQ ID NO: 16      moltype = AA length = 1081
FEATURE
REGION            1..1081
                  note = synthetic construct
source            1..1081
                  mol_type = protein
                  organism = synthetic construct

```

```

SEQUENCE: 16
MDNGDRLYRA DSRPPDEIKR SGLMPRGHN EYFDRGTQMN INLYDHARGT QTGFVRYDDG 60
YVSTSLSLRS AHLAQQSILS GYSTYYIYVI ATAPNMFNVN DVLGVYSPHP YEQEVSALGG 120
IPYSQIYGWY RVNFGVIDER LHRNREYDRD YVRNLNIAPA EDGYRLAGFP PDHQAWAREEP 180
WIHHPQGGCG NSSRGSASAM NITTLTNSIS TSSFSFNNTN GSSTETVNSD IKTTTSSHPS 240
SLTMLNDTLH NIRTTNQALK KELSQKTLRN EYPINKDARE LLHSAPKEAE LDGQDMISHR 300
ELWAKIANSI NDINEQYLKV YEHAVSSYTQ MYQDFSAVLS SLAGWISPGG NDGNSVKLQV 360
NSLKKALEEL KEKYKDKPLY PANNTVSQEQ ANKWLTELG GYVVSINMTP 420
IDNMLKSLDN LGGNGEVVLD NAKYQAWNGF SAEDETMKNN LQTLVQKYSN ANSIFDNLVK 480
VLSSTISSCT DTDKLFHLFL EMHNVSTTTT GFPLAKILTS TELGNTIQA ANDAANKLFS 540
LTIADLTANQ NINTTNAHST SNILIPELKA PKSLNASSQL TLLIGNLIQI LGEKSLTALT 600
NKITAWKSQQ QARQQKNLEF SDKINTLLSE TEGLTRDYEK QINKLKNADS KIKDLENKIN 660
QIQTRLSNLD PESPEKKKLS REEIQLTIKK DAAVKDRTLI EQKTLSTHSK LTKSMQLEK 720
EIDSFSAFSN TASAEQLSTQ QKSLTGLASV TQLMATFIQL VGKNNESLK NDALFQSLQ 780
ESRKTEMERK SDEYAAVFRK AEELNRVMGC VGKILGALLT IVSVVAAAFS GGASLALAAV 840
GLALMVTDAI VQAATGNSFM EQALNPIMKA VIEPLIKLLS DAFTKMLEG L GVDKSKAKMI 900
GSILGAIAGA LVLVAAVVLV ATVGKQAAAK LAENIGKIIG KTLTDLIPKF LKNFSSQLDD 960
LITNAVARLN KFLGAAGDEV ISKQIISTHL NQAVLLGESV NSATQAGGSV ASAVFQNSAS 1020
TNLADLTLSK YQVEQLSKYI SEAIKFGQL QEVIADLLAS MSNSQANRTD VAKAILQQT 1080
A 1081

```

```

SEQ ID NO: 17      moltype = DNA length = 2415
FEATURE
misc_feature      1..2415
                  note = synthetic construct
source            1..2415
                  mol_type = other DNA
                  organism = synthetic construct

```

```

SEQUENCE: 17
catatggaca atggcgatcg tttataccgt gccgactcgc gtcccccaga tgagattaaa 60
cgtagcgggtg ggttaatgcc acgtggggcac aatgagattt ttgaccgtgg aacacagatg 120
aacattaaacc tttacgatca tgcccgtggg acccagaccg gggtttgtccg ttatgatgac 180
gggtatgtta gtacgagttt gtccctacgc tccgcacacc ttgctgggaca aagtatttta 240
tcaggctaca gccacatata catttatgtg atcgccactg ccccaaacat gttcaatgtg 300
aacgatgtgt tgggggttta cagcccccac ccatatgaac aagaagtctc ggccttggg 360
gggatcccat atagccagat ttatggttgg taccgcgtaa attttgggt gattgatgaa 420
cggttgcctc gtaaccgtga ataccgcgat cgctactacc gtaacttga cattgcacct 480
gccgaggacg gctatcgttt agcgggattc ccaccgcgat atcaggcgtg gcgtgaggaa 540
ccgtggatcc atcacgcccc tcagggggtg gggaacagta gtccgatgac cattgatctc 600
ggagtttccac tcacgtcgca ggcggcgccg ctgcaaggca tgcacctcaa gagcatggat 660
atccagactc tcattgggtg tgtaggggtt cgtcgccgag aactcctcac ggctcaaat 720
cagacccagg ccgaagtggg gcagaaggcc aatgaacgca tggcgagct caacgaggtc 780
ctgtccgcgc tgtcccgggc caaggccgag ttccgcccc atccgaagcc gggcgacacc 840
atcccgggct gggacaacca gaaggtcagc cggatcgagg ttctctcaa tgatgcgctg 900
cgcgctgccg gcctgacggg catgttcgaa gcgcgcgatg gccaaagtac cgcggcgccg 960
ggcgggggta cgcaggtcgt gaacggcacg ggcgtcatgg ccggttccac gacctataag 1020
gaactcgaaa gtgcctacac caccgtaaag gggatgctgg atacggcgct caatacgcaa 1080
cagatggaca tgatcagggt gcaggccgcc agcaacaagc gcaacgaggg ttccgaggtc 1140
atgaccaaca ccgagaagcg gcgcagcgac ctgaacagtt ccatcaccaa caacatcgcg 1200
aagcttatga ccgtcatgag tacgaccata tccacagccc cgagcggcgc cgcgcttgcg 1260
ccgtctcgca tagatatgcg gccaccggag cccgggagtg ccggcgcaagg cgcggcgatc 1320
ctggcgccgg tgacgacgct ggctctggcg gcggggccgg ccggttttcc agcgtcaccg 1380
tcgtgcgca ccgcgcccgt cctggatccg ccagtgcgag atctcagccc cgcgacttg 1440
gccgacctgc tgcgcgtctt gcgatccagg gcggtggacg ggcagttggc caccggcgcg 1500
gagaacctgc aggaacgcga agtcaaggcg aagcagaaca cccaggccca gctcgacaag 1560
ctggacgcat ggtttcgga ggcgaagag gccgagagca aggatgggt gagcaagggt 1620
ttcggtgga tcggcaaggt gctggcggtc gtggcatcgg ccctggcggt gggctttgcc 1680
gccgtcgcca gcgtggccac cggcgcgggc gccacaccca tgctgctgct cagcggcatg 1740
gcactggtga ggcgcgtgac atcgctggcc gaccagatat cgcaaggagg gggaggcccc 1800
cctatcagcc tggggcggtt tctctccggg ctggcgggac gtctgctgac agcgttgggg 1860
gtggatcagt cgcaggccga ccaaatggc aagatcgctg ccggcctggc cgtgcccgct 1920
gtcttgctga tcgaacccca gatgctgggc gaaatggcgc aaggcgtggc caggctggct 1980
ggcgccagcg atgccaccgc ggggtacata gccatggcga tgtccatcgt ggcggcgatc 2040
gcggtcgccg cgatcaatgc cgcgggtaca gccggcgcgg gtagcgcttc gccgatcaag 2100

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ggggcctggg atcgggccgc cgcggtagcc acccaggtcc ttcaaggggg tacggcagtg 2160
gcgcaaggcg gcgtcgccgt gtcgatggca gtcgatcgca aacaggccga tctcctggtc 2220
gccgacaagg cggatctggc ggcgagcctg acaaaactgc gggcgcccat ggagcgtgag 2280
gcggacgata tcaagaagat cctggctcaa ttcgacgagg cctatcacat gatcgcgaa 2340
atgatcagcg atatggcgag tacgcacagc caggtcagcg ccaacctcgg gcggcgccag 2400
gcggtgtagc tcgag 2415

```

```

SEQ ID NO: 18      moltype = AA  length = 801
FEATURE           Location/Qualifiers
REGION            1..801
                  note = synthetic construct
source            1..801
                  mol_type = protein
                  organism = synthetic construct

```

```

SEQUENCE: 18
MDNGDRLYRA DSRPPDEIKR SGGLMPRGHN EYFDRGTQMN INLYDHARGT QTGFVRYDDG 60
YVSTSLSLRS AHLAQQSILS GYSTYYIYVI ATAPNMFNVN DVLGVYSPHP YEQEVSA LGG 120
IPYSQIYGWY RVNFGVIDER LHRNREYRDR YYRNLNIAPA EDGYRLAGFP PDHQA WREEP 180
WIHHAPQGCQ NSSRMTIDL G VSLTSQAGGL QGIDLKSM DI QTLMVYVQGR RAELLTAQM Q 240
TQAEVVQKAN ERMAQLNEVL SALSRAKAEF PPNPKPGDTI PGWDNQKVS R IEVPLNDAL R 300
AAGLTGMFEA RDGQVTAPGG RGTQVVNGTG VMAGSTTYKE LESAYTTVK G MLDTASNTQ Q 360
MDMIRLQAAS NKRNEAFEV M TNTEKR RSD L NSSITNNMR K LMTVMSTT IS TAPSGAALAP 420
SRIDMRAPEP GSAGEGAGIL APVTTLALAA GRPAFFASPS LRTAPVLDPP VRDLSPADLA 480
DLLRVLRSRA VDGQLATARE NLQDAQVKAK QNTQAQLDKL DAWFRKAEEA ESKGWL SKVF 540
GWIGKVLAVV ASALAVGF AA VASVATGAAA TPMLLLSGMA LVSAVTS LAD QISQEAGGPP 600
ISLGGFLSGL AGRLLTALGV DQSQADQIAK IVAGLAVPVV LLIEPQMLGE MAQGVARLAG 660
ASDATAGYIA MAMSIVAIAA VAAINAAGTA GAGSASAIKG AWDRAAAVAT QVLQGGTAVA 720
QGGVGVSMAV DRKQADLLVA DKADLAASLT KLRAMEREA DDIKKILAQF DEAYHMI AKM 780
ISDMASTHSQ VSANLGR RQA V 801

```

```

SEQ ID NO: 19      moltype = DNA  length = 566
FEATURE           Location/Qualifiers
misc_feature       1..566
                  note = synthetic construct
source            1..566
                  mol_type = other DNA
                  organism = synthetic construct

```

```

SEQUENCE: 19
atgggcagca gccatcacca tcatcaccac agccaggatc cgatgacgca acgcgacgtg 60
aacatagacg acatcgaggc gcaggaaatg gcggcgggcg tgctggacgc ggtccagaac 120
ggcgcgacgc tgaaggacct gcatcagggt cgcgaggacc tgatggacgc catctatgcg 180
ttcgcgtaac gcttctacca gcagggggcg ctcgacgacg cggaggtgtt cttccgcttt 240
ctgcgcactc acgacttcta caacgccgaa tacgcgatgg ggctcgcggc ggtgtgccag 300
ttgaagaagg agtacgcgcg ggcgatcgat ctgtatgcac tcgcgtattc gctgtcgaag 360
gacgaccacc ggccgatgtt ccacaccggc caatgccatc tgctgatggg caaggcgggc 420
ctcgcgcggc cctgcttcgg catcgctcgc gagcgctcgc gcgacgagcg cctcgcgcgag 480
aaggcgagct gctatctcga cgggctcgac gaagtgggcg cgcgacggcg gcccgcatcc 540
gccgggaacg accactgagc ggcgcg 566

```

```

SEQ ID NO: 20      moltype = AA  length = 185
FEATURE           Location/Qualifiers
REGION            1..185
                  note = synthetic construct
source            1..185
                  mol_type = protein
                  organism = synthetic construct

```

```

SEQUENCE: 20
MGSSHHHHHH SQDPM TQRDV NIDDIEAQEM AAALLDAVQN GATLKD LHVQ PQLMDGIYA 60
FAYRFYQQGR LDDAEVFFRF LRIYDFYNAE YAMGLAAVCQ LKKEYARAID LYALAYSLSK 120
DDHRPMPHTG QCHLLMGKAA LARRCFGIVV ERSRDERLAQ KAQSYLDGLD EVGADAAPAS 180
AGNDH 185

```

```

SEQ ID NO: 21      moltype = DNA  length = 930
FEATURE           Location/Qualifiers
source            1..930
                  mol_type = genomic DNA
                  organism = Burkholderia spp

```

```

SEQUENCE: 21
catatgaaca tgcacgtgga catgggtcgt gcgctgaccg ttctgtattg gccggcgctg 60
gaggcgctgg cgaaaacccat gccggcggat gcgggtgcgc gtgcgatgac cgatgatgac 120
ctgcgtgcgg cgggtgtgga ccgtcgtgtt ccggagcaga agctgggtgc ggcgattgat 180
gaattcgca gccctgcgtc gccggatcgt atcgacggtc gttctgtgga tggccgtcgt 240
gcgaacctga ccgtttttga tgatgcgcgt gttgcgggtc gtggtcatgc gcgtgcgcaa 300
cgtaacctgc tggagcgtct ggagaccgaa ctgctggggtg gcaccctgga taccgcgggt 360
gacgaagggt gcattcagcc ggacccgatc ctgcaaggcc tggtggtatg tatcggtcag 420
ggcaaaagcg atattgacgc gtacgcgacc atcgtggaag gtctgaccaa gtattttcaa 480

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agcgtggcgcg acgttatgag caaactgcag gattacatta gcgcgaagga tgacaaaaac 540
atgaagatcg acggtggcaa gatcaaagcg ctgattcagc aagtgatcga ccacctgccg 600
accatgcagc tgcgcaaggg tgcggatatt gcgcgttggc gtaaagagct gggcgacgcg 660
gttagcatca gcgatagcgg tgtggttacc attaacccgg acaactgat caagatgcgt 720
gatagcctgc cgcgggatgg caccgtttgg gataccgcgc gttaccaagc gtggaacacc 780
gcgttcagcg gtcagaaagg ccagcatccg gaacgtcgtg cggatgcgcg tcgtaaatat 840
agccaccaga acagcaactt tgataacctg gtgaaggttc tgagcgggtc gattagcacc 900
ctgaccgaca cccagagcta tctgcaaatc

```

```

SEQ ID NO: 22      moltype = AA length = 309
FEATURE           Location/Qualifiers
source            1..309
                  mol_type = protein
                  organism = Burkholderia spp

```

```

SEQUENCE: 22
MMMHVDMGRA LTVRDWPALE ALAKTMPADA GARAMTDDDL RAAGVDRRVP EQKLGAAIDE 60
FASLRLPDRI DGRFVDGRRR NLTVFDDARV AVRGHARAQR NLLERLETEL LGGTLDTAGD 120
EGGIQPDPII QGLVDVIGQG KSDIDAYATI VEGLTKYFQS VADVMSKLQD YISAKDDKNM 180
KIDGGKIKAL IQQVIDHLPT MQLPKGADIA RWRKELGDAV SISDSGVVTI NPDKLIKMRD 240
SLPPDGTVWD TARYQAWNTA FSGQKGQHPH RRADARRKYS HQNSNFDNLV KVLSGAISTL 300
TDTQSYLQI

```

```

SEQ ID NO: 23      moltype = DNA length = 1869
FEATURE           Location/Qualifiers
source            1..1869
                  mol_type = genomic DNA
                  organism = Burkholderia spp

```

```

SEQUENCE: 23
atgagcagcg gtgttcaagg tggcccgcg gcgaacgcga acgcgtacca gaccaccccg 60
ctgcgtgatg cggcgagcgc gctgggcacc ctgagcccg aggcgtatgt ggatgtggtt 120
agcgcggcgc aacgtaactt cctggagcgt atgagccaac tggcgagcga acagtgcgat 180
gcgcaaccgg cggcgcatga tgcgcgtctg gatgacgctc cggcgctgcg tgcgcgcgag 240
gaacgtgacg cgcgcgcgct ggggtgcgagc gataccggta gccgtgcgag cgggtgcggcg 300
aaactgacgg agctgctggg tgtgctgatg agcgttatta gcgcgagcag cctggacgaa 360
ctgaagcaac gtagcgatat ctggaaccag atgagcaaac cggcgcaaga caacctgagc 420
cgtctgagcg atcggtttca gcgtgcgacc gacgagcgga aagcggcgcg ggatgcggcg 480
gaacagggcg cggcgcgcg gcgaacagcg ggtgcggcag cgaagcgcg ggatgcggcg 540
gtggatgcgg cgcacaaacc ttacgatgac gcggttaagc agggcctgcc ggatgacctg 600
ctgcaaacgc tgaaagcgcg gctggagcag gcgcgtcagc aagcgggtga tgcgcatggt 660
cgtgcgggatg cgctgcaggc ggaatgcgacc aagaactgg acgcggcgag cgcgtggcg 720
acccaagcgc gtgcgtgcga acagcaagtg gatgacgcgg ttaaccaggc gaccacgcaa 780
tatggtgcga gcgcgagcct gcgtaccccg caaagcccg gtctgagcgg tgcggcgagg 840
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gaaagcaagc agaaactggt caccgagatg caagcgaaag gtgagggcga actgcacaaag 960
aaaagcgacg aatatacagg gcaagtgaag aaagcggagg aaatgcagaa aacgatgggt 1020
tgcatacgca agattgtggg ttgggttatt accgcggtta gctttgcggc ggcgcggttt 1080
accggtggcg cgagcctggc gctggcgcg gtgggcctgg cgctggcggt tggtagcagag 1140
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tgcggcggtg atcagcaaaa agcgggaactg gcgggtgcga ttctgggtgc ggttgttacc 1320
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cgtaccgcga ccgcgatgac ccgtatgcgt cgtgcgattg gcgttgaggc gaaggaagac 1560
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gtgagccaaag cggcggtggt cattgtggtt ggcgttgagc gtgcgaaagc gatgggtctg 1680
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gcggtggacg cggttgcgga gcacaaccgt gttctggcgc aactgatgca gcaaatgagc 1800
gatgcggcg gaaatgcagac cagcaccggc aagctgatcc tgcgtaacgc gcgtgcgggt 1860
taaggatcc

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SEQ ID NO: 24      moltype = AA length = 620
FEATURE           Location/Qualifiers
source            1..620
                  mol_type = protein
                  organism = Burkholderia spp

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SEQUENCE: 24
MSSGVQGGPA ANANAYQTHP LRDAASALGT LSPQAYVDV SAAQRNFLER MSQASEQCD 60
AQPAHADARL DDRPALRAPQ ERDAPPLGAS DTGSRASGAA KLTELLGVLM SVISASSLDE 120
LKQSDIWNQ MSKAAQDNLS RLSDAFQRA DEAKAAADAA EQAAAAAKQA GADAKAADAA 180
VDAAQKRYDD AVKQGLPDDR LQSLKALEQ ARQQAGDAH RADALQADAT KKLDAASALA 240
TQARACEQQV DDAVNQATQQ YGASASLRTP QSPRLSGAAE LTAVLGKLQE LISSGNVKEL 300
ESKQKLFTFM QAKREAELOK KSEYQAQVK KAEEMQKTMG CIGKIVGWVI TAVSFAAAAF 360
TGGASLALAA VGLALAVGDE ISRATTGVSF MDKLMQPVMD AILKPLMEMI SSLITKALVA 420
CGVDQKAEAL AGAILGAVVT GVALVAAAFV GASAVKAVAS KVIDAMAGQL TKLMDSAIGK 480
MLVQLIEKFS EKSGLQALGS RTATAMTRMR RAIGVEAKED GMLLANRFEK AGTVMNVGNQ 540

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VSQAAGGIVV GVERAKAMGL LADVKEAMYD IKLLGDLKQ AVDAFAEHRN VLAQLMQQMS 600  
DAGEMQTSTG KLILRNARAV 620

SEQ ID NO: 25      moltype = DNA    length = 2805  
FEATURE            Location/Qualifiers  
misc\_feature       1..2805  
                    note = synthetic construct  
source              1..2805  
                    mol\_type = other DNA  
                    organism = synthetic construct

SEQUENCE: 25  
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gaggcgctgg cgaaaaccat gccggcgcat gccgggtgcg gtgcgatgac cgatgatgac 120  
ctgcgtgctgg cgggtgtgga cegtctgtgt cggagcaga agctgggtgc ggcgattgat 180  
gaattcgctga gccctgcgtct gccggatcgt atcgacggtc gtttctgtga tggccgtcgt 240  
gcgaacctga cgttttttga tgaatgcgct gtgctgcttc gtggtcatgc gcgtgctgca 300  
cgtaacctgc tggagcgtct ggagaccgaa ctgctgggtg gcacctgga taccgcggt 360  
gacgaaggtg gcatttcagcc ggaccgcatc ctgcaaggcc tgggtgatgt tatcggtcag 420  
ggcaaaagcg atattgacgc gtacgcgacc atcgtggaag gtctgaccaa gtattttcaa 480  
agcgtggcgg acgtttatgag caaactgcag gattacatta gcgcgaagga tgacaaaaac 540  
atgaagatcg acggtggcaa gatcaaagcg ctgattcagc aagtgtatga ccacctgccg 600  
accatgcagc tgcgaagggt tgcggatatt gcgctgtggc gtaaaagagt gggcgacgag 660  
gttagcatca gcgatagcgg tgtggttacc attaacccgg acaactgat caagatgcgt 720  
gatagcctgc cgccggatgg caccgtttgg gataccgctg gttaccaagc gtggaacacc 780  
gcgttcagcg gtcagaaggc ccagcatccg gaacgtcgtg cggatgcgag tcgtaaatat 840  
agccaccaga acagcaactt tgataacctg gtgaagggtc tgagcgggtg gattagcacc 900  
ctgaccgaca cccagagcta tctgcaaatc aagcttatga gcagcgggtg tcaagggtggc 960  
ccggcgcgga acgcgaagcg gtaccagacc caccgctgc gtgatgcggc gagcgcgctg 1020  
ggcaccctga gcccgagcgg gtatgtggat gtggttagcg cggcgcaacg taacttctcg 1080  
gagcgtatga gccaaactgc gagcgaacag tgcgatgcgc aaccggcgcc gcgatgatgcg 1140  
cgtctggatg atcgtccggc gctgcgtgct cgcgaggaac gtgacgcgcc gccgctgggt 1200  
gcgagcgata ccggtatgac tgcgagcggg gccggcgaac tgaccgagct gctgggtgtg 1260  
ctgatgagcg ttatttagcg gagcagcctg gacgaactga agcaacgtag cgatatctgg 1320  
aaccagatga gcaaacgcgg gcaagacaac ctgagccgtc tgagcgatgc gtttcagcgt 1380  
gcgaccgacg aggcgaagcg gccggcgcat gccggcggaac agggcgcgcc gccggcggaag 1440  
caagcggtgg cggacgcgaa agcggcgcat gccggcggtg atcgcgcgca aaaacgttac 1500  
gatgacgcgg ttaagcaggg cctgcggatg gaccgtctgc aaagcctgaa agcggcgctg 1560  
gagcagcgcg gtcagcaagc ggggtgatgc catggtcgtg cggatgcgct gcaggcggt 1620  
gcgaccaaga aactggacgc gccgagcgcg ctggcgaccc aagcgcgtgc gtgcaaacag 1680  
caagtggatg acgcgggtta ccaggcgacc cagcaatatg gtgagcgccg gacccctgcg 1740  
accccgcaaa gcccgctctc gagcgtgctg gccgagctga ccgcggtgct gggcaagctg 1800  
caggaaactg ttagcagcgg caacgttaaa gagctggaaa gcaagcagaa actgttcacc 1860  
gagatgcaag caagcgtgta gccggaactg caaaagaaaa gcgacgaata tcaggcgcaa 1920  
gtgaagaaag cggaggaatg gcagaaaaag atgggttgca tcggcaagat tgtgggttgg 1980  
gttattaccg cggttagcgt tgcggcgcg cggtttaccg gtggcgcgag cctggcgctg 2040  
gccggcggtg gccctggcgt gccgggttgg gacgagatta gccgtgagc caccggtgtg 2100  
agcttcctgg acaagctgat gcagcgggt atggatgcga tcctgaaacc gctgatggag 2160  
atgattagca gccctgatcc caaggcgctg gttgctgctg gcgttgatga gcaaaaagcg 2220  
gaactggcgg gtgcgattct ggggtgcggtt gttaccggtg tggcgctggt tgcggcgctg 2280  
tttgttggtg gagcgcggtt gaaagcggtt gcgagcaagg ttatcgacgc gatggcggtg 2340  
cagctgacca agctgatgga tagcgcgatt ggcaaaatgc tgggtgcaact gatcgagaaa 2400  
ttcagcgaaa agagcgctct gcaggcgctg ggtagcccta ccgagaccgc gatgaccctg 2460  
atgcgtcgtg cgattggcgt tgaggcgaa gaagacggtg tgctgctggc gaaccgtttt 2520  
gaaaaagcgg gcaccgtgat gaacgttggt aaccaagtga gccaaagcgg ggggtggcat 2580  
gtggttgagg ttgagcgtgc gaaagcgatg ggtctgctgg cggatgtgaa agaagcgatg 2640  
tatgacatca agctgctggg tgatctgctg aaacagggcg tggacgcgtt tgcgggagac 2700  
aaccgtgttc tggcgcaact gatgcagcaa atgagcgatg cgggcgaaat gcagaccagc 2760  
accggcaagc tgatctcgtg taacgcgcgt gcggtttaag gatcc 2805

SEQ ID NO: 26      moltype = AA    length = 931  
FEATURE            Location/Qualifiers  
REGION              1..931  
                    note = synthetic construct  
source              1..931  
                    mol\_type = protein  
                    organism = synthetic construct

SEQUENCE: 26  
MNMHVDMGRA LTVRDWPALE ALAKTMPADA GARAMTDDDL RAAGVDRRVP EQKLGAIDE 60  
FASLRLPDRI DGRFVDGRR NLTVFDDARV AVRGHARAQR NLLERLETEL LGGTLDTAGD 120  
EGGIQPDPI L QGLVDVIGQG KSDIDAYATI VEGLTKYFQS VADVMSKLQD YISAKDDKNM 180  
KIDGGIKAL IQQVIDHLPT MQLPKGADIA RWRKELGDAV SISDSGVVTI NPKLIKMRD 240  
SLPPDGTVWD TARYQAWNTA FSGQKGQHP ERRADARRKYS HQNSNFDNLV KVLSGAISTL 300  
TDTQSYLQIK LMSSGVQGGP AANANAYQTH PLRDAASALG TLPQAYVDV VSAQRNPLE 360  
RMSQLASEQC DAQPAANDAR LDDRPALRAP QERDAPPLGA SDTGSRASGA AKLTELGLVL 420  
MSVTSASSLD ELKQRSIDWN QMSKAAQDNL SRLSDFQRA TDEAKAADA AEQAAAAAKQ 480  
AGADAKAADA AVDAAQKRYD DAVKQGLPDD RLQSLKAAL E QARQQAGDAH GRADALQADA 540

-continued

|             |            |            |            |            |             |     |
|-------------|------------|------------|------------|------------|-------------|-----|
| TKKLDAAASAL | ATQARACEQQ | VDDAVNQATQ | QYGASASLRT | PQSPRLSGAA | ELTAVLQKQLQ | 600 |
| ELISSGNVKE  | LESQKQLFTE | MQAKREAELO | KKSDEYQAQV | KKAEMMQKTM | GCIGKIVGWV  | 660 |
| ITAVSFAAAA  | FTGGASLALA | AVGLALAVGD | EISRATTGVS | FMDKLMQFVM | DAILKPLMEM  | 720 |
| ISSLITKALV  | ACGVDQQAEE | LAGAILGAVV | TGVALVAAAF | VGASAVKAVA | SKVIDAMAGQ  | 780 |
| LTKLMDSAIG  | KMLVQLIEKF | SEKSGLQALG | SRTATAMTRM | RRAGVEAKE  | DGMLLANRFE  | 840 |
| KAGTVMNVGN  | QVSQAAGGIV | VGVERAKAMG | LLADVKEAMY | DIKLLGDLK  | QAVDAFAEHN  | 900 |
| RVLALMQQM   | SDAGEMQTST | GKLLRNARA  | V          |            |             | 931 |

SEQ ID NO: 27                   moltype = DNA   length = 3390  
 FEATURE                   Location/Qualifiers  
 misc\_feature               1..3390  
                              note = synthetic construct  
 source                    1..3390  
                              mol\_type = other DNA  
                              organism = synthetic construct

SEQUENCE: 27

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| catatggaca  | atggcgatcg  | tttataccgt  | gccgactcgc  | gtcccccaga  | tgagattaaa  | 60   |
| cgtagcgggtg | gggttaatgcc | acgtggggcac | aatgagattt  | ttgaccgtgg  | aacacagatg  | 120  |
| aacattaacc  | tttacgatca  | tgcccgtggg  | accagaccgg  | gggttgcctg  | ttatgatgac  | 180  |
| gggtatgtta  | gtacgagttt  | gtccttacgc  | tccgcacacc  | ttgcgggaca  | aagtatttta  | 240  |
| tcaggctaca  | gcacatatta  | catttatgtg  | atcgccactg  | ccccaaacat  | gttcaatgtg  | 300  |
| aacgatgtgt  | tgggggttta  | cagcccccac  | ccatatgaac  | aagaagtctc  | ggccctgggg  | 360  |
| gggatcccat  | atagccagat  | ttatgggttg  | taccgcgtaa  | attttggtgt  | gattgatgaa  | 420  |
| cggttgcatc  | gtaaccgtga  | ataccgcgat  | cgctactacc  | gtaacctgaa  | cattgcacct  | 480  |
| gccgaggacg  | gctatcgttt  | agcgggattc  | ccaccgcgat  | atcaggcggtg | gcgtgaggaa  | 540  |
| ccgtggatcc  | atcacgcccc  | tcagggggtgc | gggaacagta  | gtcgccatgc  | gaacatgcac  | 600  |
| gtggacatgg  | gtcgtgcgct  | gaccgttcgt  | gattggccgg  | cgctggaggcg | gctggcgaaa  | 660  |
| accatgccgg  | cggatgcggg  | tgccgcgtgc  | atgaccgatg  | atgacctgcg  | tgccggcggt  | 720  |
| gtggaccgtc  | gtgttcgcga  | gcagaagctg  | gggtgcggcg  | ttgatgaatt  | cgcgagccctg | 780  |
| cgctcgccgg  | atcgatatga  | cggtcggttc  | gtggatggcc  | gtcgtgcgaa  | cctgaccggt  | 840  |
| tttgatgatg  | cgctggttgc  | gggtcggtgt  | catgcgcgtg  | cgcaacgtaa  | cctgctggag  | 900  |
| cgtctggaga  | cgaactgcgt  | gggtggcacc  | ctggataccg  | cggttgacga  | aggtggcatt  | 960  |
| cagccggacc  | cgatccgtga  | aggcctgggt  | gatgttatcg  | gtcaggcgaa  | aagcgatatt  | 1020 |
| gacgcgtacc  | gcaccatcgt  | ggaaggtctg  | accaagattt  | ttcaaagcgt  | ggcggacggt  | 1080 |
| atgagcaaac  | tgacaggata  | cattagcgcg  | aaggatgaca  | aaaacatgaa  | gatcgacggt  | 1140 |
| ggcaagatca  | aagcgctgat  | tcagcaagtg  | atcgaccacc  | tgccgaccat  | gcagctgccg  | 1200 |
| aaaggtgcgg  | atattgcggc  | ttggcgtaaa  | gagctggggc  | acgcggttag  | catcagcgat  | 1260 |
| agcgtgtgtg  | ttaccattaa  | cccggacaaa  | ctgatcaaga  | tgcgtagatg  | cctgcgcgcg  | 1320 |
| gatggcaccg  | tttgggatac  | cgcgcgttac  | caagcggtga  | acaccgcgtt  | cagcggctcag | 1380 |
| aaagccagcg  | atccggaaacg | tcgtgcggat  | gcgcgtcgta  | aatatagcca  | ccagaacagc  | 1440 |
| aaacttgata  | accttggtgaa | gggtctgagc  | gttgcgatta  | gcaccctgac  | cgacacccag  | 1500 |
| agctatctgc  | aaatcaagct  | tatgagcagc  | gggtgtcaag  | gtggcccggc  | ggcgaacgcg  | 1560 |
| aacgcgtacc  | agaccacacc  | gctgcgtgat  | gcggcgagcg  | cgctgggcac  | cctgagcccg  | 1620 |
| caggcgatg   | tggatgtggt  | tagcgcggcg  | caacgtaact  | tcctggagcg  | tatgagccaa  | 1680 |
| ctggcgagcg  | aacagtgcca  | tgccgaaccg  | gcggcgcatg  | atgcgcgtct  | ggatgatcgt  | 1740 |
| ccggcgctgc  | gtgcgcgtgc  | ggaacgtgac  | gcgcgcgcgc  | tgggtgcgag  | cgataccggt  | 1800 |
| agccgtgcga  | gcgggtgcgc  | gaaactgacc  | gagctgctgg  | gtgtgctgat  | gagcgttatt  | 1860 |
| agcgcgagca  | gcctggacga  | actgaagcaa  | cgtagcgata  | tctggaacca  | gatgagcaaa  | 1920 |
| gcggcgcaag  | acaacctgag  | cgctctgagc  | gatgcgtttc  | agcgtgcgac  | cgacgcggcg  | 1980 |
| aaagcggcgg  | cggatgcggc  | ggaacaggcg  | gcggcgggcg  | cgaaagcaag  | gggtgcggac  | 2040 |
| gcgaagcggg  | cggatgcggc  | gggtggtgcg  | gcgcaaaaac  | gttacgatga  | cgcggttaag  | 2100 |
| caaggccctgc | cggatgacgc  | ctcgcaaaag  | ctgaaagcgg  | cgctggagca  | ggcgcgtcag  | 2160 |
| caagcgggtg  | atgcgcgatg  | tcgtgcggat  | gcgctgcagg  | cggtatgcgc  | caagaaaactg | 2220 |
| gacgcggcga  | gcgcgctggc  | gacccaagcg  | cgtgcgtgcg  | aacagcaagt  | ggatgacgcg  | 2280 |
| gttaaccagg  | cgacccagca  | atatggtgcg  | agcgcgagcc  | tgctacccc   | gcaaaagccc  | 2340 |
| cgtctgagcg  | gtgcggcgga  | gctgaccgcg  | gtgctgggca  | agctgcagga  | actgattagc  | 2400 |
| agcggcaacg  | ttaaagagct  | ggaaagcaag  | cagaaactgt  | tcaccgagat  | gcaagcgaa   | 2460 |
| cgtgagcgcg  | aactgcacaaa | gaaaagcgac  | gaatatcagg  | cgcaagtga   | gaaagcgga   | 2520 |
| gaaatgcaga  | aaacgatggg  | ttgcatcggc  | aagattgtgg  | gttgggttat  | taccgcgggt  | 2580 |
| agctttgcgg  | cggcgcggtt  | taccggtggc  | gcgagccctg  | cgctggcggc  | gggtggccctg | 2640 |
| gcgctggcgg  | ttggtgacga  | gattagccgt  | gcgaccaccc  | gtgtgagctt  | catggacaag  | 2700 |
| ctgatgcagc  | cggttatgga  | tgcatcctg   | aaacgcgtga  | tggagatgat  | tagcagcctg  | 2760 |
| atcaccgaag  | cgctggttgc  | gtgcggcggt  | gatcagcaaa  | aagcggaact  | ggcgggtgcg  | 2820 |
| attctgggtg  | cggttgttac  | cgggtgtggc  | ctggttgccg  | cggcggtttg  | tggtgcgagc  | 2880 |
| gcggtgaaa   | cgggtgcgag  | caagggttat  | gacgcgatgg  | cgggtcagct  | gaccaagctg  | 2940 |
| atggatagcg  | cgattgcca   | aattgctggt  | caactgatcg  | agaaattcag  | cgaaaagagc  | 3000 |
| ggctgcagcg  | cgctgggtg   | ccgtaccgcg  | accgcgatga  | cccgtatgcg  | tcgtgcgatt  | 3060 |
| ggcgttgagg  | cgaagggaaga | cggtatgctg  | ctggcgaaac  | gttttgaaaa  | agcgggcacc  | 3120 |
| gtgatgaacg  | ttggttaacca | agtgcgcaaa  | gcggcggggtg | gcattgtggt  | tgccggttag  | 3180 |
| cgtgcgaag   | cgatgggtct  | gctggcggtg  | gtgaaagaa   | cgatgtatga  | catcaagctg  | 3240 |
| ctgggtgatc  | tgtgtaaaaca | ggcgggtggac | gcgtttgccc  | agcacaaccg  | tgttctggcg  | 3300 |
| caactgatgc  | agcaaatgag  | cgatgcgggc  | gaaatgcaga  | ccagcacccg  | caagctgatc  | 3360 |
| ctgcgtaacg  | cgcgtgcggt  | ttaaggtacc  |             |             |             | 3390 |

SEQ ID NO: 28                   moltype = AA   length = 1125  
 FEATURE                   Location/Qualifiers

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REGION 1..1125  
note = synthetic construct

source 1..1125  
mol\_type = protein  
organism = synthetic construct

SEQUENCE: 28

|            |            |            |            |             |             |      |
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| MDNGDRLYRA | DSRPPDEIKR | SGGLMPRGHN | EYFDRGTQMN | INLYDHARGT  | QTGFVRYDDG  | 60   |
| YVSTSLSLRS | AHLAQQSILS | GYSTYYIYVI | ATAPNMFNVN | DVLGVYSPHP  | YEQEVSAALGG | 120  |
| IPYSQIYGWY | RVNFGVIDER | LHRNREYRDR | YYRNLNIAPA | EDGYRLAGFP  | PDHQAWREEP  | 180  |
| WIHHAPQCGC | NSSRMNMHVD | MGRALTVRDW | PALEALAKTM | PADAGARAMT  | DDDLRAAGVD  | 240  |
| RRVPEQKLG  | AIDEFASLRL | PDRIDGRFVD | GRRANLTVFD | DARVAVRGHA  | RAQRNLLERL  | 300  |
| ETELLGGTLD | TAGDEGGIQP | DPILQGLVDV | IGQGKSDIDA | YATIVEGLTK  | YFQSVADVMS  | 360  |
| KLQDYISAKD | DKNMKIDGGK | IKALIQQVID | HLPTMQLPKG | ADIARWRKEL  | GDAVSISDSG  | 420  |
| VVTINPDKLI | KMRDSLPPDG | TVWDTARYQA | WNTAFSGQKG | QHPERRADAR  | RKYSHQNSNF  | 480  |
| DNLVKVLSGA | ISTLTDTQSY | LQIKLMSSGV | QGGPAANANA | YQTHPLRDAA  | SALGTLSPQA  | 540  |
| YVDVVSAAQR | NFLERMSQLA | SEQCDAQPAA | HDARLDDRPA | LRAPQERDAP  | PLGASDTGSR  | 600  |
| ASGAAKLTE  | LGVLMVISIA | SSLDELKQRS | DIWNQMSKAA | QDNLSRLSDA  | FQRATDEAKA  | 660  |
| AADAAEQAAA | AAKQAGADAK | AADAAYDAAQ | KRYDDAVKQG | LPDDRLLQSLK | AALEQARQQA  | 720  |
| GDAHGRADAL | QADATKKLDA | ASALATQARA | CEQQVDDAVN | QATQQYGASA  | SLRTPQSPRL  | 780  |
| SGAAELTAVL | GKLQELISSG | NVKELESKQK | LFTMQAKRE  | AELQKKSDEY  | QAQVKKAEEM  | 840  |
| QRTMGICIGI | VGWVITAVSF | AAAAFTGGAS | LALAAVGLAL | AVGDEISRAT  | TGVSPMDKLM  | 900  |
| QPVMDAILKP | LMEMISSLIT | KALVACGVDQ | QKAELAGAIL | GAVVTGVALV  | AAAFVGASAV  | 960  |
| KAVASKVIDA | MAGQLTKLMD | SAIGKMLVQL | IEKFSEKSGL | QALGSRATATA | MTRMRRAIGV  | 1020 |
| EAKEDGMLLA | NRFEKAGTVM | NVGNQVSQAA | GGIVVGVERA | KAMGLLADVK  | EAMYDIKLLG  | 1080 |
| DLKQAVDAF  | AEHNRVLAQL | MQQMSDAGEM | QTSTGKLILR | NARAV       |             | 1125 |

SEQ ID NO: 29 moltype = DNA length = 552  
FEATURE Location/Qualifiers  
misc\_feature 1..552  
note = synthetic construct  
source 1..552  
mol\_type = other DNA  
organism = synthetic construct

SEQUENCE: 29

|            |            |            |            |            |             |     |
|------------|------------|------------|------------|------------|-------------|-----|
| atgggcagca | gccatcacca | tcataccacc | agccaggatc | cgatgaacca | gccgaccacct | 60  |
| tccgacacgc | accagcaaca | ggcgcgtggg | gccttcctcg | gcgacggcgg | cacctgtggcg | 120 |
| atgcttcgcg | gactcagcga | ggacaccctg | gagcagctct | atgcgcgtgg | cttcaaccag  | 180 |
| taccaggcgg | gcaagtggga | cgacgcgcag | aagatcttcc | aggcactgtg | catgtctcgac | 240 |
| cactacgacg | cccgcctact | tctcggcctg | ggcgcctgcc | gccagtcctc | cggtctctat  | 300 |
| gaacaggccc | tcgacagcta | cagctacggc | gcgctgatgg | acatcaacga | gccgcgcttt  | 360 |
| cccttccatg | cgcgcgagtg | ccacctgcaa | ctgggtgatc | tcgacggagc | cgagagtggc  | 420 |
| ttctactcgg | cccgggccct | ggcgcgggca | cagccggcgc | acgagggcct | ggcgcgcgct  | 480 |
| gccggcgcca | tggtggaagc | cgtaacccgc | agaaaggatc | gagcctatga | atccgataac  | 540 |
| gcttgaaagc | tt         |            |            |            |             | 552 |

SEQ ID NO: 30 moltype = AA length = 181  
FEATURE Location/Qualifiers  
REGION 1..181  
note = synthetic construct  
source 1..181  
mol\_type = protein  
organism = synthetic construct

SEQUENCE: 30

|            |            |            |             |            |            |     |
|------------|------------|------------|-------------|------------|------------|-----|
| MGSSHHHHHH | SQDPMNQPTP | SDTDQQQALE | AFLRDGGTLA  | MLRGLSEDTL | EQLYALGFNQ | 60  |
| YQAGKWDDAQ | KIFQALCMLD | HYDARYFLGL | GACRQSLGLY  | EQALQSYSYG | ALMDINEPRF | 120 |
| PFHAAECHLQ | LGDLDGAESG | FYSARALAAA | QPAHEALAAAR | AGAMLEAVTA | RKDRAYESDN | 180 |
| A          |            |            |             |            |            | 181 |

SEQ ID NO: 31 moltype = DNA length = 885  
FEATURE Location/Qualifiers  
source 1..885  
mol\_type = genomic DNA  
organism = Pseudomonas spp

SEQUENCE: 31

|            |            |            |            |             |             |     |
|------------|------------|------------|------------|-------------|-------------|-----|
| catatggaag | tcagaaacct | taatgccgct | cgcgagctgt | tcctggacga  | gtcctctggcc | 60  |
| gcgtcggcgg | cgctcgccag | tgccgagcag | gaggaactgc | tggccctggt  | gcgcagcgag  | 120 |
| cggatcgtgc | tggcccacgc | cggccagcgc | ctgagcgagg | cgcaagtgtc  | caaggcgctc  | 180 |
| gcctgggttc | tcgcggccaa | tcctgtccgc | cctccggggc | agggcctcga  | ggtactccgc  | 240 |
| gaagtctctc | aggcacgtcg | gcagcccggt | gcgcagtggt | atctgcgtga  | gttctctggt  | 300 |
| tcggcctatt | tccagctgca | cgggcgtctc | gacgaggatg | tcctcggtgt  | ctacaaggat  | 360 |
| gtcctgcaga | cccaggacgg | caagcgcaag | gcgctgctcg | acgagctcaa  | ggcgcgtgac  | 420 |
| gcggagtgtg | aggtctacag | cgtgatccag | tcgcagatca | acgcccgcgt  | gtcggccagg  | 480 |
| cagggcatca | ggatcgacgc | tggcggatc  | gatctgggtc | accccacgct  | atatggtgat  | 540 |
| gccgtcggcg | atcccagggt | gaaggacagc | cccagagtat | cgctgtctgag | caatctggtat | 600 |
| accttcagcg | gcaagctgtc | gatcaaggat | tttctcagcg | gctcgccgaa  | gcagagcggg  | 660 |
| gaactcaagg | gcctcagcga | tgagtacccc | ttcgagaagg | acaacaaccc  | ggtcggcgaat | 720 |

-continued

|            |            |            |            |             |            |     |
|------------|------------|------------|------------|-------------|------------|-----|
| ttcgccacca | cggtgagcga | ccgctcgcgt | ccgctgaacg | acaagggtcaa | cgagaagacc | 780 |
| acctgtctca | acgacaccag | ctcccgtac  | aactcggcgg | tcgagggcgt  | caaccgcttc | 840 |
| atccagaaat | acgacagcgt | cctgagcgac | attctcagcg | cgatc       |            | 885 |

SEQ ID NO: 32                   moltype = AA   length = 294  
 FEATURE                    Location/Qualifiers  
 source                     1..294  
                           mol\_type = protein  
                           organism = Pseudomonas spp

SEQUENCE: 32

|            |            |            |            |            |            |     |
|------------|------------|------------|------------|------------|------------|-----|
| MEVRNLNAAR | ELFLDELLAA | SAAPASAEQE | ELLALLRSER | IVLAHAGQPL | SEAQVLKALA | 60  |
| WLLAANPSAP | PGQGLEVLRE | VLQARRQPGA | QWDLREFLVS | AYFSLHGRLD | EDVIGVYKDV | 120 |
| LQTQDGKRKA | LLDELKALTA | ELKVYSVIQS | QINAALSARQ | GIRIDAGGID | LVDPTLYGYA | 180 |
| VGDPRWKDSP | EYALLSNLDT | FSGKLSIKDF | LSGSPKQSGE | LKGLSDEYPP | EKDNPNVGNF | 240 |
| ATTVSDRSFP | LNDKVNKETT | LLNDTSSRYN | SAVEALNRFI | QKYDSVLSDI | LSAI       | 294 |

SEQ ID NO: 33                   moltype = DNA   length = 1179  
 FEATURE                    Location/Qualifiers  
 source                     1..1179  
                           mol\_type = genomic DNA  
                           organism = Pseudomonas spp

SEQUENCE: 33

|             |            |             |             |            |             |      |
|-------------|------------|-------------|-------------|------------|-------------|------|
| atgaaccgga  | ttacgctgga | acgtgctggt  | ctgcgctatg  | gtgttgccga | tgtcgtgtgac | 60   |
| atcccggctc  | tgggtcgccc | ggtcgcacgt  | gatgtggaaa  | gtctgcgtgt | tgaacgtctg  | 120  |
| gcagcaccgg  | cagctgcaag | cgcactctgg  | accggtgtcg  | ctctgacgcc | gccgtctgca  | 180  |
| gcaagtccag  | aacgtctgga | agttgtctaa  | cgcgcggaaa  | ttgcctcact | ggtccaggca  | 240  |
| gtgggtgaag  | acgtgggtct | ggcacgtcaa  | gtggttcttg  | caggtgcac  | gaccctgctg  | 300  |
| agcgcaggtc  | tgatgtcgcc | gcaggcgttc  | gaaattgaac  | tggccaaaat | caccggcgaa  | 360  |
| gttgaaaaatc | agcagaaaaa | actgaaaactg | acggaatcg   | aacaggcccg | taaacagaa   | 420  |
| ctgcaaaaaa  | tggagataa  | ccagcaaaaa  | atcccgcaat  | cggaagaagc | tcgcaagaa   | 480  |
| gcgcagaaaa  | gcggcctggc | cgcaaaaatt  | tttggttggg  | tttctgctat | cgcgagtatt  | 540  |
| atcgtgggtg  | caatcatggt | tgcaaccggt  | gtcgtgtctg  | cagcaggtgc | actgatgatt  | 600  |
| gctggcggtg  | tcatgggtgt | cgtgagtcag  | tccgtgcagc  | aagcagctgc | ggatggctctg | 660  |
| atctcaaaag  | aagtgatgga | aaaactgggc  | ccggccctga  | tgggtattga | aatggccgtg  | 720  |
| gcactgctgg  | ccgcagttgt | ctcctttggt  | gggttcagcag | ttggtggtct | ggcacgtctg  | 780  |
| ggtgcaaaaa  | tcggcggtga | agctgcggaa  | atgacggcat  | ccctggcttc | aaaagtggca  | 840  |
| gacctgggcg  | gtaaatctcg | ctctctggcg  | ggccagtcac  | tgtcgcatag | cctgaaactg  | 900  |
| ggtgtgcaag  | tttctgatct | gaccttgga   | gttgcaaacg  | gcgcgcacac | ggctacgcac  | 960  |
| agtgtgtttc  | aagcgaaagc | tcgcaatcgt  | caggccgatg  | ttcaagaatc | ccgtgcagac  | 1020 |
| ctgaccacgc  | tgcagggtgt | cattgaacgt  | ctgaaagaag  | aactgagccg | catgctggaa  | 1080 |
| gcctttcagg  | aaattatgga | acgcactctc  | gcaatgctgc  | aagcgaaagg | cgaaccctg   | 1140 |
| cacaatctgt  | cttcccgtcc | ggcggctatc  | tgaggatcc   |            |             | 1179 |

SEQ ID NO: 34                   moltype = AA   length = 390  
 FEATURE                    Location/Qualifiers  
 source                     1..390  
                           mol\_type = protein  
                           organism = Pseudomonas spp

SEQUENCE: 34

|            |            |            |            |            |            |     |
|------------|------------|------------|------------|------------|------------|-----|
| MNPITLERAG | LPYGVADAGD | IPALGRPVAR | DVESLRVERL | AAPAAASASG | TGVALTPPSA | 60  |
| ASQQRLEVAN | RAETASLVQA | VGEDVGLARQ | VVLAGASTLL | SAGLMSPOAF | EIELAKITGE | 120 |
| VENQQKLLKL | TEIQARKQN  | LQKMEDNQK  | IRESEEAKE  | AQKSGLAAKI | FGWISAIASI | 180 |
| IVGAIMVATG | VGAAAGALMI | AGGVMGVVSQ | SVQQAADGL  | ISKEVMEKLG | PALMGIEMAV | 240 |
| ALLAAVVSFG | GSVVGGLARL | GAQIGGKAAE | MTASLASKVA | DLGKFGSLA  | GQSLSHSLKL | 300 |
| GVQVSDLTLD | VANGAAQATH | SGFQAKAANR | QADVQESRAD | LTTLQGVIER | LKEELSRMLE | 360 |
| AFQEIMERIF | AMLQAKGETL | HNLSSRPAAI |            |            |            | 390 |

SEQ ID NO: 35                   moltype = DNA   length = 2070  
 FEATURE                    Location/Qualifiers  
 misc\_feature               1..2070  
                           note = synthetic construct  
 source                     1..2070  
                           mol\_type = other DNA  
                           organism = synthetic construct

SEQUENCE: 35

|             |             |             |             |            |             |     |
|-------------|-------------|-------------|-------------|------------|-------------|-----|
| catatggaag  | tcagaaacct  | taatgccgct  | cgcgagctgt  | tcctggacga | gctcctggcc  | 60  |
| gcgtcggcgg  | cgctcgccag  | tgccgagcag  | gaggaaactgc | tggccctgtt | gcgcagcgag  | 120 |
| cggatcgtgc  | tggcccacgc  | cgcccagccg  | ctgagcgagg  | cgcaagtgtc | caaggcgctc  | 180 |
| gcctgggttc  | tcgcggccaa  | tccgtccgcg  | cctccggggc  | agggcctcga | ggtactccgc  | 240 |
| gaagtccctgc | aggcacgtcg  | gcagcccggt  | gcgcagtggt  | atctgcgtga | gttccctggtg | 300 |
| tcggcctatt  | tcagcctgca  | cgggcgtctc  | gacgaggtatg | tcacggtgt  | ctacaaggat  | 360 |
| gtcctgcaga  | cccaggacgg  | caagcgcaag  | gcgctgctcg  | acgagctcaa | ggcgctgacc  | 420 |
| gcggagttag  | aggtctacag  | cgtgatccag  | tcgcagatca  | acgcccgcgt | gtcggccagg  | 480 |
| cagggcatca  | ggatgcagcg  | tggcgggtatc | gatctggtcg  | acccacgcgt | atatggctat  | 540 |
| gcgctcggcg  | atcccagggtg | gaaggacagc  | cccagtatg   | cgctgctgag | caatctggat  | 600 |
| accttcagcg  | gcaagctgtc  | gatcaaggat  | tttctcagcg  | gctcgccgaa | gcagagcggg  | 660 |

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gaactcaagg gcctcagcga tgagtacccc ttcgagaagg acaacaaccc ggtcggaat 720
ttcgccacca cggtagcaga ccgctcgcgt ccgctgaacg acaaggtcaa cgagaagacc 780
accctgctca acgacaccag ctcccgtac aactcggcgg tcgaggcgct caaccgcttc 840
atccagaaat acgacagcgt cctgagcgac attctcagcg cgatcggatc catgaaccgg 900
attacgctgg aacgtgctgg tctgcccgtat ggtgttgccg atgctgggtga catcccggct 960
ctgggtcgcc cggtcgcacg tgatgtggaa agtctgcgtg ttgaacgtct ggcagcaccg 1020
gcagctgcaa gcgcactggt caccggtgtc gctctgacgc cgcctgtctgc agcaagtcat 1080
caacgtctgg aagttgctaa ccgcccggaa attgcctcac tggccagggc agtgggtgaa 1140
gacgtgggtc tggcacgtca agtgggtctg gcaggtgcat gcacctgct gagcgcaggt 1200
ctgatgtcgc cgcaggcggt gcaaatgaa ctggccaaaa tcaccggcga agtggaaaat 1260
cagcagaaaa aactgaaact gacggaaatc gaacaggccc gtaaacagaa cctgcaaaaa 1320
atggaagata accagcaaaa aatcccggaa tcggaagaag ctgcgaaaga agcgcagaaa 1380
agcggcctgg ccgcaaaaa ttttggttgg atttctgcta tcgcgagtat tatcgtgggt 1440
gcaatcctgg ttgcaaccgg tgcggtgct gcagcaggtg cactgatgat tgctggcggt 1500
gtcatgggtg tcgtgagtcg gtcctgcag caagcagctg cggatggtct gatctcaaaa 1560
gaagtgatgg aaaaactggg cccggccctg atgggtattg aaatggccgt ggcactgctg 1620
gccgcagttg tctccttgg tggttcagca gttggtgtgc tggcacgtct ggggtcaaaa 1680
atcggcgcta aagctgcgga aatgacggca tcccctggctt caaaagtggc agacctgggc 1740
ggtaaatctg gctctctggc gggccagtcg ctgtcgcata gcctgaaact ggggtgtgca 1800
gtttctgata tcgacctgga cgttgcaaac ggcgcgcgac aggtctacga cagtgggttt 1860
caagcgaaag ctgcgaatcg tcaggccgat gttcaagaat cccgtgcaga cctgaccacg 1920
ctgcagggtg tcattgaacg tctgaaagaa gaactgagcc gcattgctga agcctttcag 1980
gaaattatgg aacgcattct cgcattgctg caagcgaaag gcgaaacccct gcacaatctg 2040
tcttcccctc cggcggttat ctgaggatcc 2070

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SEQ ID NO: 36      moltype = AA  length = 686
FEATURE           Location/Qualifiers
REGION            1..686
                  note = synthetic construct
source            1..686
                  mol_type = protein
                  organism = synthetic construct

```

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SEQUENCE: 36
MEVRNLNAAR ELFLDELHAA SAAPASAEQE ELLALLRSER IVLAHAGQPL SEAQVLKALA 60
WLLAANPSAP PGQGLEVLRE VLQARRQPGA QWDLREFLVS AYPFSLHGRDL EDVIGVYKDV 120
LQTQDGKRA LLDELKALTA ELKVYSVIQS QINAALSARQ GIRIDAGGID LVDPTLYGYA 180
VGDPWRKDSF EYALLSNLDT FSGKLSIKDF LSGSPKQSGE LKGLSDEYPF EKDNPNVGNF 240
ATTVSDRSRP LNDKVNEKTT LLNDTSSRYN SAVEALNRFI QKYDSVLSDI LSAIGSMNPI 300
TLERAGLPYG VADAGDIPAL GRPVARDVES LRVERLAAPA AASASGTGVA LTPPSAASQQ 360
RLEVANRAEI ASLVQAVGED VGLARQVLLA GASTLLSAGL MSPQAFIEEL AKITGEVENQ 420
QKKLKLTEIE QARKQNLQKM EDNQQKIRE EEAKEAQKS GLAAKIFGWI SAIASIIVGA 480
IMVATGVGAA AGALMIAGGV MGVSQSVQQ AAADGLISKE VMEKLGPALE GIEMAVALLA 540
AVVSGGSFV GSKARLGAKE GKAEMTAS LASKVADLGG KFGSLAGQSL SHSLKLGQV 600
SDLTLDVANG AAQATHSGFQ AKAANRQADV QESRADLTL QGVIERLKEE LSRMLEAFQE 660
IMERIFAMLO AKGETLHNL SRPAAI 686

```

```

SEQ ID NO: 37      moltype = DNA  length = 2655
FEATURE           Location/Qualifiers
misc_feature       1..2655
                  note = synthetic construct
source             1..2655
                  mol_type = other DNA
                  organism = synthetic construct

```

```

SEQUENCE: 37
catatggaca atggcgatcg tttataccgt gccgactcgc gtcccccaga tgagattaaa 60
cgtagcgggtg ggttaaatgcc acgtggggc aatgagtatt ttgaccgtgg aacacagatg 120
aacattaacc tttacgatca tgcctgtggg acccagaccg ggtttgtccg ttatgatgac 180
gggtatgtta gtacgagttt gtccttacgc tccgcacacc ttgcgggaca aagtatttta 240
tcaggctaca gcacatatta catttatgtg atcgccactg ccccaaacat gttcaatgtg 300
aacgatgtgt tgggggttta cagcccccat ccatatgaac aagaagtctc ggccttggg 360
gggatcccat atagccagat ttatggttgg taccgcgtaa attttggtg gattgatgaa 420
cgtttgcata gtaaccgtga ataccgcgat cgctactacc gtaacttgaa cattgcacct 480
gccgaggacg gctatcgttt agcgggattc ccaaccgata atcaggcggt gcgtgaggaa 540
ccgtggatcc atcacgcccc tcagggggtg gggaaacagta gtcgccatat ggaagtcaga 600
aaccttaagt ccgctcgcca gctgttcctg gacgagctcc tggccgcgtc ggcggcgct 660
gccagtgccc agcaggagga actgctggcc ctgttgcgca gcgagcggt cgtgctggcc 720
cacgcccggc agcgcgtgag cgaggcgcaa gtgctcaagg cgctcgctgt gttgctcgcg 780
gccaatccgt ccgcgcctcc cggggcaggc ctcgaggtac tccgcgaagt cctgcaggca 840
cgtcggcagc cgggtgcgca gtgggatctg cgtgagttcc tgggtgtcggc ctatttcagc 900
ctgcacgggc gtctcgacga ggtatgcatc ggtgtctaca aggatgtcct gcagaccag 960
gacggcaagc gcaaggcgct gctcgacgag ctcaaggcgc tgaccgcgga gttgaaggtc 1020
tacagcgtga tccagtcgca gatcaacgcc gcgctgtcgg ccaggcaggg catcaggatc 1080
gacgctggcg gtatcgatct gtcgaccccc acgctatatg gctatgcctg cggcgatccc 1140
aggtggaagg acagccccga gtatgcgctg ctgagcaatc tggatacctt cagcggcaag 1200
ctgtcgatca aggattttct cagcggctcg ccgaagcaga gcggggaaact caagggcctc 1260
agcgatgagt accccttcga gaaggacaac aaccgggtcg gcaatttcgc caccacgggt 1320

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agcgaccgct cgcgtccgct gaacgacaag gtcaacgaga agaccaccct gctcaacgac 1380
accagctccc gctacaactc ggcgggtcgag gcgctcaacc gcttcatcca gaaatacgcac 1440
agcgtcctga gcgacattct cagcgcgatc ggatccatga acccgattac gctggaacgt 1500
gctgggtctgc cgtatggtgt tgcgatgctt ggtgacatcc cggctctggg tcgcccggtc 1560
gcacgtgatg tggaaagtct gcgtgttgaa cgtctggcag caccggcagc tgcaagcgca 1620
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gctaaccgcg cggaaattgc tctactggtc caggcagtgg gtgaagacgt gggctctggca 1740
cgtcaagtgg ttctggcagg tgcctcgacc ctgctgagcg caggtctgat gtcgcccag 1800
gcgttcgaaa ttgaactggc caaaatcacc ggcgaaagtg aaaaatcagca gaaaaaactg 1860
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caaaaaatcc gcgaatcgga agaagctgag aaagaagcgc agaaaagcgg cctggccgca 1980
aaaatttttg gttggatttc tgcctatcgc agtattatcg tgggtgcaat catggttgca 2040
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agtcagtgct gcagcgaagc agctcggtat ggtctgatct caaaagaagt gatggaaaaa 2160
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gcggaatgga cggcatccct ggcttcaaaa gtggcagacc tgggcggtaa attcggtct 2340
ctggcgggcc agtcactgtc gcatagcctg aaactgggtg tgcaagtttc tgatctgacc 2400
ctggacgttg caaacggcgc cgcacaggct acgcacagtg gttttcaagc gaaagctgcg 2460
aatcgtcagg ccgatgttca agaatcccgt gcagacctga ccacgtgca ggggtgtcatt 2520
gaacgtctga aagaagaact gagccgcgat ctggaagcct ttcaggaaat tatggaacgc 2580
atcttcgcaa tgctgcaagc gaaaggcgaa accctgcaca atctgtcttc ccgtccggcg 2640
gctatctgag gatcc 2655

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SEQ ID NO: 38      moltype = AA length = 880
FEATURE           Location/Qualifiers
REGION            1..880
                  note = synthetic construct
source            1..880
                  mol_type = protein
                  organism = synthetic construct

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SEQUENCE: 38
MDNGDRLYRA DSRPPDEIKR SGGMLPRGHN EYFDRGTQMN INLYDHARGT QTGFVRYDDG 60
YVSTSLSLRS AHLAQSLLS GYSTYIYIVI ATAPNMFNVN DVLGVYSPHP YEQEVSA LGG 120
IPYSQIYGWY RVNFGVIDER LHRNREYRDR YYRNLNIAPA EDGYRLAGFP PDHQAWREEP 180
WIHHA PQCG NSSRMEVRNL NAARELFLDE LLAASAPAS AEQEBELLALL RSERIVLAHA 240
GQPLSEAQVL KALAWLLAAN PSAPPGQGLE VLREVLQARR QPGAQWDLRE FLVSAYFSLH 300
GRLDEDVIGV YKDLVLTQDG KRKALLDELK ALTAELKVYS VIQSQINAAL SARQGI RIDA 360
GGIDLVDPTL YGYAVGDPRW KDSPEYALLS NLDTFSGKLS IKDFLSGSPK QSGELKGLSD 420
EYPFEKDNPN VGNFATTYSD RSRPLNDKVN EKTLLNDTS SRYNSAVEAL NRFIQYDSV 480
LSDILSAIGS MNPITLERAG LPYGVADAGD IPALGRPVAR DVESLRVERL AAPAAASASG 540
TGVALTPPSA ASQQRLEVAN RAETIASLVQA VGEDVGLARQ VVLGASTLL SAGLMS PQAF 600
ETELAKITGE VENQKKLKL TEIEQARKQN LQKMEDNQOK IRESEEAKE AQKSGLA AKI 660
FGWISAIASI IVGAIMVATG VGAAAGALMI AGGVMGVVSQ SVQQAADGL ISKEVMEKLG 720
PALMGIEMAV ALLAAVVSFG GSAVGLLARL GAKIGGKAAE MTASLASKVA DLGGKFGSLA 780
GQSLSHSLKL GVQVSDLTLD VANGAAQATH SGFQAKAANR QADVQESRAD LTTLQGVIER 840
LKEELSRMLE AFQEIMERIF AMLQAKGETL HNLSSRPAAI 880

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SEQ ID NO: 39      moltype = DNA length = 555
FEATURE           Location/Qualifiers
misc_feature       1..555
                  note = synthetic construct
source            1..555
                  mol_type = other DNA
                  organism = synthetic construct

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```

SEQUENCE: 39
atgggcagca gccatcacca tcatcaccac agccaggatc cgatgcaaca agagacgaca 60
gacactcaag aataccagct ggcaatggaa tccttcctaa aaggaggggg aactatcgcc 120
atgctcaacg aaatttcaag tgacacttta gagcaactct actctcttgc gtttaaccaa 180
taccagtcag gaaaatacga ggaatgctac aaggtcttcc aagctctctg tgtgctagac 240
cactatgatt cactgtttctt tttagggcta ggcgcttgct gtcaagccat ggggcaatac 300
gacttagcga ttcatagcta cagctatggc gccataatgg atataaaaaga acctcgtttt 360
ccgtttcatg ctgccgaatg ttactgcaa aaggagagac ttgctgaagc agaaagtggc 420
ttgttcttgg ctcaagagct tatcgcagac aaacctgagt ttaaggagct ttccaccgca 480
gttagctcaa tgttagaagc aattaaattg aaaaaggaga tggaacatga gtgcgctgat 540
aaccatgaa agctt 555

```

```

SEQ ID NO: 40      moltype = AA length = 188
FEATURE           Location/Qualifiers
REGION            1..188
                  note = synthetic construct
source            1..188
                  mol_type = protein
                  organism = synthetic construct

```

```

SEQUENCE: 40
MGSSHHHHHH SSGLVPRGSH MQQETTD TQE YQLAMESFLK GGGTIAM LNE ISSDTLEQLY 60

```



-continued

|            |            |            |            |            |            |     |
|------------|------------|------------|------------|------------|------------|-----|
| SLAFNQYQSG | KYEDAHKVFO | ALCVLDHYDS | RFFLGLGACR | QAMGQYDLAI | HSYSYGAIMD | 120 |
| IKEPRFPFHA | AECLLQKGEL | AEAESGLFLA | QELIADKPEF | KELSTRVSSM | LEAIKLLKEM | 180 |
| EHECVDNP   |            |            |            |            |            | 188 |

SEQ ID NO: 41                   moltype = DNA   length = 981  
 FEATURE                    Location/Qualifiers  
 source                      1..981  
                             mol\_type = genomic DNA  
                             organism = Yersinia spp

SEQUENCE: 41

|             |             |             |             |             |             |     |
|-------------|-------------|-------------|-------------|-------------|-------------|-----|
| catatgatta  | gagcctacga  | acaaaaccca  | caacatttta  | ttgaggatct  | agaaaaagtt  | 60  |
| aggggtggaac | aacttactgg  | tcattggttct | tcagtttttag | aagaattggg  | tcagtttagtc | 120 |
| aaagataaaa  | atatagatat  | ttccattaaa  | tatgatccca  | gaaaagattc  | ggagggtttt  | 180 |
| gccaatagag  | taattactga  | tgatatcgaa  | ttgctcaaga  | aaatcctagc  | ttattttcta  | 240 |
| cccgaggatg  | ccattcttaa  | aggcggtcac  | tatgacaacc  | aactgcaaaa  | tggcatcaag  | 300 |
| cgagtaaaag  | agttccttga  | atcatcgccg  | aatacacaat  | gggaattgcg  | ggcgttcacg  | 360 |
| gcagtaatgc  | atttctcttt  | aaccgcgcag  | cgtatcgatg  | atgatatttt  | gaaagtgatt  | 420 |
| gttgattcaa  | tgaatcatca  | tggtgatgcc  | cgtagcaagt  | tgctggaaga  | attagctgag  | 480 |
| cttaccgccc  | aattaaagat  | ttattcagtt  | attcaagccg  | aaattaataa  | gcattctgtc  | 540 |
| agtagtgcca  | ccataaaatat | ccatgataaa  | tccattaatc  | tcattggataa | aaatttatat  | 600 |
| ggttatacac  | atgaagagat  | ttttaagcc   | agcgcagagt  | acaaaattct  | cgagaaaatg  | 660 |
| cctcaaacca  | ccattcaggt  | ggatggggag  | gagaaaaaaa  | tagtctcgat  | aaaggacttt  | 720 |
| cttgggaagt  | agaataaaa   | aaccggggcg  | ttgggtaatc  | tgaaaaaact  | atactcttat  | 780 |
| aataaagata  | ataatgaatt  | atctcacttt  | gccaccacct  | gctcggataa  | gtccaggccg  | 840 |
| ctcaacgact  | tggttagcca  | aaaaacaact  | cagctgtctg  | atattacatc  | acgttttaat  | 900 |
| tcagctattg  | aagcactgaa  | ccgttttcatt | cagaaaatag  | attcagtgat  | gcaacgtctg  | 960 |
| ctagatgaca  | cgtctggtaa  | a           |             |             |             | 981 |

SEQ ID NO: 42                   moltype = AA   length = 326  
 FEATURE                    Location/Qualifiers  
 source                      1..326  
                             mol\_type = protein  
                             organism = Yersinia spp

SEQUENCE: 42

|            |            |            |            |            |            |     |
|------------|------------|------------|------------|------------|------------|-----|
| MIRAYEQNPQ | HFIEDLEKVR | VEQLTGHGSS | VLEELVQLVK | DKNIDISIKY | DPRKDSEVFA | 60  |
| NRVITDDIEL | LKKILAYFLP | EDAILKGGHY | DNQLQNGIKR | VKEFLESSPN | TQWELRAFMA | 120 |
| VMHFLSTADR | IDDDILKVIV | DSMNHGDDAR | SKLREELAE  | TAEIKIYSVI | QAEINKHLSS | 180 |
| SGTINIHDKS | INLMDKNLYG | YTDEEIFKAS | AEYKILEKMP | QTTIQVDGSE | KKIVSIKDFL | 240 |
| GSENKRTGAL | GNLKNSYSYN | KDNNELSHFA | TTCSDKSRPL | NDLVSQKTTQ | LSDITSRFNS | 300 |
| AIEALNRFIQ | KYDSVMQRLL | DDTSGK     |            |            |            | 326 |

SEQ ID NO: 43                   moltype = DNA   length = 1212  
 FEATURE                    Location/Qualifiers  
 source                      1..1212  
                             mol\_type = genomic DNA  
                             organism = Yersinia spp

SEQUENCE: 43

|             |             |             |            |            |            |      |
|-------------|-------------|-------------|------------|------------|------------|------|
| atgagtcgct  | tgataaccca  | tgatcgctca  | acgccagtaa | ctggaagtct | acttcctcac | 60   |
| gtcagagcac  | cagcgcccg   | cccccttcag  | actcaacaag | tcgcgggaga | actgaaggat | 120  |
| aaaaatgggt  | gggtgagttc  | tcaggggcgt  | cagctccctg | caccactagc | agtggttgcc | 180  |
| agccaaagtca | ctgaaggaca  | acagcaagaa  | atcactaaat | tattggagtc | ggtaacccgc | 240  |
| ggcacggcag  | gattctcaact | gattatcaat  | tatgtttcag | tgtaacgcaa | ttttacgctc | 300  |
| gcttcacctg  | atacatttga  | gattgagtta  | ggtaagctag | ttttctaatt | agaagaagta | 360  |
| cgcaaaagaca | taaaaatcgc  | tgatattcag  | cgtcttcacg | aacaaaacat | gaagaaaatt | 420  |
| gaagagaatc  | aagagaaaaat | caaagaaaca  | gaagagaatg | ccaagcaagt | caagaaatcc | 480  |
| ggcatggcat  | caaagatttt  | tggctgggtc  | agcgccatag | cctcagtggt | tatcggtgcc | 540  |
| atcatgggtg  | cctcaggggt  | aggagccgtt  | gccggtgcaa | tgatgattgc | ctcaggcgta | 600  |
| attgggatgg  | cgaatatgct  | tgtgaaacaa  | gcggcggaag | atggcctgat | atcccaagag | 660  |
| gcaatgcaag  | tattaggggc  | gatactcact  | gcgattgaag | tcgcattgac | tgtagtttca | 720  |
| accgtaaatga | cctttggcgg  | ttcggcacta  | aaatgcctgg | ctgatattgg | cgcaaaactc | 780  |
| gggtgctaaca | cgcgaagtct  | tgctgcta    | ggagccgagt | tttcggccaa | agttgcccc  | 840  |
| atttcgacag  | gcataatcaa  | cactgtcggg  | aatgcagtga | ctaaattagg | gggcagtttt | 900  |
| ggtagtttaa  | caatgagcca  | tgtaatccgt  | acaggatcac | aggcaacaca | agtcgcgctt | 960  |
| gggtgtggga  | gcggaataac  | tcagaccatc  | aataataaaa | aacaagctga | tttacaacat | 1020 |
| aataacgctg  | atttggcctt  | gaacaaggca  | gacatggcag | cggtacaaag | tattattgac | 1080 |
| cgactcaaaag | aagagtattc  | ccatttgtca  | gagtcacatc | aacaagtgat | ggaactgatt | 1140 |
| ttccagatga  | ttaatgcaaa  | agggtgacatg | ctgcataatt | tggccggcag | accccatact | 1200 |
| gtttaaggta  | cc          |             |            |            |            | 1212 |

SEQ ID NO: 44                   moltype = AA   length = 401  
 FEATURE                    Location/Qualifiers  
 source                      1..401  
                             mol\_type = protein  
                             organism = Yersinia spp

SEQUENCE: 44

|            |            |            |            |            |            |    |
|------------|------------|------------|------------|------------|------------|----|
| MSALITHDRS | TPVTGSLLPY | VETPAPAPLQ | TQQVAGELKD | KNGGVSSQGV | QLPAPLAVVA | 60 |
|------------|------------|------------|------------|------------|------------|----|

-continued

|            |            |            |            |            |            |     |
|------------|------------|------------|------------|------------|------------|-----|
| SQVTEGQQQE | ITKLESVTR  | GTAGSQLISN | YVSVLTNFTL | ASPDTFEIEL | GKLVSNLEEV | 120 |
| RDKIADIQ   | RLHEQNMKKI | EENQEKIKET | EENAKQVKKS | GMASKIFGWL | SAIASVVIGA | 180 |
| IMVASGVGAV | AGAMMIASGV | IGMANMAVKQ | AAEDGLISQE | AMQVLGPILT | AIEVALTVVS | 240 |
| TVMTFGGSAL | KCLADIGAKL | GANTASLAAK | GAEFSAKVAQ | ISTGISNTVG | NAVTKLGGSF | 300 |
| GSLTMSHVIR | TGSQATQVAV | GVGSGITQTI | NNKQADLQH  | NNADLALNKA | DMAALQSIID | 360 |
| RLKEELSHLS | ESHQQVMELI | FQMINAKGDM | LHNLAGRPH  | V          |            | 401 |

SEQ ID NO: 45                   moltype = DNA   length = 2199  
 FEATURE                   Location/Qualifiers  
 misc\_feature               1..2199  
                           note = synthetic construct  
 source                    1..2199  
                           mol\_type = other DNA  
                           organism = synthetic construct

SEQUENCE: 45

|             |             |             |             |             |             |      |
|-------------|-------------|-------------|-------------|-------------|-------------|------|
| catatgatta  | gagcctacga  | acaaaaccca  | caacatttta  | ttgaggatct  | agaaaaagtt  | 60   |
| aggggtggaac | aacttactgg  | tcattggttct | tcagtttttag | aagaattgggt | tcagttagtc  | 120  |
| aaagataaaa  | atatagatat  | ttccattaaa  | tatgatccca  | gaaaagattc  | ggagggtttt  | 180  |
| gccaatagag  | taattactga  | tgatatcgaa  | ttgctcaaga  | aaatcctagc  | ttattttcta  | 240  |
| cccgaggatg  | ccattcttaa  | aggcggtcat  | tatgacaacc  | aactgcaaaa  | tggcatcaag  | 300  |
| cgagtaaaag  | agttccttga  | atcatcgccg  | aatacacaat  | gggaattgcg  | ggcggtcatg  | 360  |
| gcagtaatgc  | atttctcttt  | aaccgcccgt  | cgtatcgatg  | atgatatttt  | gaaagtgtat  | 420  |
| gttgattcaa  | tgaatcatca  | tggtgatgcc  | cgtagcaagt  | tgctgaaga   | attagctgag  | 480  |
| cttaccgccc  | aattaaagat  | ttattcagtt  | attcaagccg  | aaattaataa  | gcattctgtc  | 540  |
| agtagtgcca  | ccataaatat  | ccatgataaa  | tccattaatc  | tcattgataa  | aaatttatat  | 600  |
| ggttatcacg  | atgaagagat  | ttttaagacc  | agcgagagat  | acaaaattct  | cgagaaaaatg | 660  |
| cctcaaacca  | ccattccaggt | ggatgggagc  | gagaaaaaaa  | tagtctcgat  | aaaggacttt  | 720  |
| cttgggaagt  | agaataaaa   | aaccggggcg  | ttgggtaatc  | tgaaaaactc  | atactcttat  | 780  |
| aataaagata  | ataatgaatt  | atctcacttt  | gccaccacct  | gctcggataa  | gtccaggccg  | 840  |
| ctcaacgact  | tggttagcca  | aaaaacaact  | cagctgtctg  | atattacatc  | acgttttaat  | 900  |
| tcagctattg  | aagcactgaa  | ccgtttcatt  | cagaaatatg  | attcagtgat  | gcaacgtctg  | 960  |
| ctagatgaca  | cgtctggtta  | aggatccatg  | agtgcggtga  | taacccatga  | tcgctcaacg  | 1020 |
| ccagtaactg  | gaagtctact  | tcctcactgc  | gagacaccag  | cgcccgcccc  | ccttcagact  | 1080 |
| caacaagtgc  | cggggaactg  | gaaggataaa  | aatggtgggg  | tgagttctca  | gggcgtacag  | 1140 |
| ctcccgtcac  | cactagcagt  | ggttgccagc  | caagtcactg  | aaggacaaca  | gcaagaaatc  | 1200 |
| actaaattat  | tggagtcggt  | caccgcggcg  | acggcaggat  | ctcaactgat  | atcaaatat   | 1260 |
| gtttcagtcg  | taacgaattt  | tacgctcgct  | tcacctgata  | catttgagat  | tgagttaggt  | 1320 |
| aagctagttt  | ctaatttaga  | agaagtacgc  | aaagacataa  | aaatcgctga  | tattcagcgt  | 1380 |
| cttcatgaac  | aaaacatgaa  | gaaaattgaa  | gagaatcaag  | agaaaatcaa  | agaaacagaa  | 1440 |
| gagaatgccca | agcaactgaa  | gaaatccggc  | atggcatcaa  | agatttttgg  | ctggcctcagc | 1500 |
| gccatagcct  | cagtggttat  | gggtgccatc  | atgggtggcct | caggggttagg | agccggttgc  | 1560 |
| ggtgcaatga  | tgattgcttc  | aggcgtaatt  | gggtggtgga  | atatggctgt  | gaaacaagcg  | 1620 |
| gcggaagatg  | gcctgatata  | ccaagaggca  | atgcaagtat  | tagggccgat  | actcactgcg  | 1680 |
| attgaagtgc  | catcgactgt  | agtttcaacc  | gtaatgacct  | ttggcggttc  | ggcactaaaa  | 1740 |
| tgccgtggctg | atattggcgc  | aaaactcggt  | gctaaccaccg | caagttctgc  | tgctaaggga  | 1800 |
| gccaggtttt  | cggccaaaagt | tgccaaaatt  | tcgacaggca  | tatcaaacac  | tgctgggaat  | 1860 |
| gcagtgacta  | aatagggggg  | cagttttggt  | agtttaacaa  | tgagccatgt  | aatccgtaca  | 1920 |
| ggatcacagg  | caacacaagt  | cgcggttggt  | gtgggcagcg  | gaataactca  | gaccatcaat  | 1980 |
| aataaaaaaac | aagctgattt  | acaacataat  | aacgctgatt  | tgcccttgaa  | caaggcgagac | 2040 |
| atggcagcgt  | tacaaagtat  | tattgaccga  | ctcaagaaga  | agttatccca  | tttgtcagag  | 2100 |
| tcacatcaac  | aagtgatgga  | actgattttc  | cagatgatta  | atgcaaaagg  | tgacatgctg  | 2160 |
| cataatttgg  | ccggcagacc  | ccatactggt  | taaggtacc   |             |             | 2199 |

SEQ ID NO: 46                   moltype = AA   length = 729  
 FEATURE                   Location/Qualifiers  
 REGION                    1..729  
                           note = synthetic construct  
 source                    1..729  
                           mol\_type = protein  
                           organism = synthetic construct

SEQUENCE: 46

|            |            |            |             |            |            |     |
|------------|------------|------------|-------------|------------|------------|-----|
| MIRAYEQNPQ | HFIEDLEKVR | VEQLTGHGSS | VLEELVQLVK  | DKNIDISIKY | DPRKDSVFFA | 60  |
| NRVITDDIEL | LKKILAYFLP | EDAILKGGHY | DNQLQNGIKR  | VKEFLESSPN | TQWELRAFMA | 120 |
| VMHFSLTADR | IDDDILKVIV | DSMNHGDDAR | SKLREELAE   | TAEKLIYSVI | QAEINKHLSS | 180 |
| SGTINIHDKS | INLMDKNLYG | YTDDEIFKAS | AEYKILEKMP  | QTTIQVDGSE | KKIVSIKDFL | 240 |
| GSENKRTGAL | GNLKNSYSYN | KDNNELSHFA | TTCSDKSRPL  | NDLVSQKTQ  | LSDITSRFNS | 300 |
| AIEALNRFIQ | KYDSVMQRL  | DDTSKGKMS  | ALITHDRSTP  | VTGSLLPYVE | TPAPAPLQTQ | 360 |
| QVAGELKDN  | GGVSSQGVQL | PAPLAVVASQ | VTEGQQQEIT  | KLLESVTRGT | AGSQLISNVV | 420 |
| SVLTNFTLAS | PDTFEIELGK | LVSNLEEVK  | DIKIADIQRL  | HEQNMKKIEE | NQEKIKETEE | 480 |
| NAKQVKKSGM | ASKIFGWL   | IASVVIGAIM | VASGVGAVAG  | AMMIASGVIG | MANMAVKQAA | 540 |
| EDGLISQEAM | QVLGPILTAI | EVALTVVSTV | MTFGGSALKC  | LADIGAKLGA | NTASLAAKGA | 600 |
| EFSAKVAQIS | TGISNTVGNA | VTKLGGSGFS | LTMMSHVIRTG | SQATQVAVGV | GSGITQTINN | 660 |
| KKQADLQHNN | ADLALNKADM | AALQSIIDRL | KEELSHLSES  | HQQVMELIFQ | MINAKGDM   | 720 |
| NLAGRPHTV  |            |            |             |            |            | 729 |

SEQ ID NO: 47                   moltype = DNA   length = 2784

-continued

|              |                                |
|--------------|--------------------------------|
| FEATURE      | Location/Qualifiers            |
| misc_feature | 1..2784                        |
|              | note = synthetic construct     |
| source       | 1..2784                        |
|              | mol_type = other DNA           |
|              | organism = synthetic construct |

SEQUENCE: 47

|            |             |             |             |             |            |      |
|------------|-------------|-------------|-------------|-------------|------------|------|
| catatggaca | atggcgatcg  | tttataccgt  | gccgactcgc  | gtcccccaga  | tgagattaaa | 60   |
| cgtacgggtg | gggttaatgcc | acgtgggcac  | aatgagtatt  | ttgaccgtgg  | aacacagatg | 120  |
| aacattaacc | tttaacgatca | tgcctgtggg  | accagaccg   | gggttgccg   | ttatgatgac | 180  |
| gggtatgtta | gtacgagttt  | gtccttacgc  | tccgcacacc  | ttgcgggaca  | aagtatttta | 240  |
| tcaggctaca | gcacatatta  | catttatgtg  | atcgccactg  | ccccaaacat  | gttcaatgtg | 300  |
| aacgatgtgt | tgggggttta  | cagcccccac  | ccatatgaac  | aagaagtctc  | ggcccttggg | 360  |
| gggatcccat | atagccagat  | ttatggttgg  | taccgcgtaa  | attttgggtg  | gattgatgaa | 420  |
| cgtttgcatc | gtaaccgtga  | ataccgcgat  | cgctactacc  | gtaacttgaa  | cattgcacct | 480  |
| gccgaggacg | gctatcgttt  | agcgggattc  | ccaccggatc  | atcaggcggtg | gcgtgaggaa | 540  |
| ccgtggatcc | atcacgcccc  | tcagggggtg  | gggaacagta  | gtcgccatat  | gattagagcc | 600  |
| tacgaacaaa | accacacaaca | ttttattgag  | gatctagaaa  | aagttagggt  | ggaacaactt | 660  |
| actggtcatg | gttcttcagt  | tttagaagaa  | ttgggtcagt  | tagtcaagaa  | taaaaatata | 720  |
| gatattttca | ttaaatatga  | tcccagaaaa  | gattcggagg  | tttttgccaa  | tagagtaatt | 780  |
| ctgatgata  | tcgaattgct  | caagaaaaatc | ctagcttatt  | ttctaccgga  | ggatgccatt | 840  |
| cttaaggcgg | gtcattatga  | caaccaactg  | caaaatggca  | tcaagcgagt  | aaaagagttc | 900  |
| cttgaatcat | atccgaatac  | acaatgggaa  | ttgcggggcg  | tcattggcagt | aatgcatttc | 960  |
| tctttaaccg | ccgatcgatg  | cgatgatgat  | attttgaag   | tgattgttga  | ttcaatgaat | 1020 |
| catcatgggt | atgcccgtag  | caagttgcgt  | gaagaattag  | ctgagcttac  | cgccgaatta | 1080 |
| aagattttat | cagttattca  | agccgaaatt  | aataagcatc  | tgtctagtag  | tggcaccata | 1140 |
| aatatccatg | ataaatccat  | taattctcatg | gataaaaaat  | tatatgggta  | tacagatgaa | 1200 |
| gagattttta | agccagcgcg  | agagtacaaa  | attctcgaga  | aaatgcctca  | aaccaccatt | 1260 |
| caggtggatg | ggagcgagaa  | aaaaatagtc  | tcgataaagg  | actttcttgg  | aagtgagaat | 1320 |
| aaaagaaccg | gggctgtggg  | taattctgaaa | aactcatact  | cttaataata  | agataataat | 1380 |
| gaattatctc | actttgcccac | cacttgctcg  | gataagttcca | ggcgcgtcaa  | cgacttggtt | 1440 |
| agccaaaaaa | caactcagct  | gtctgatatt  | acatcacgtt  | ttaattcagc  | tattgaagca | 1500 |
| ctgaaccggt | tcatttcagaa | atatgattca  | gtgatgcaac  | gtctgctaga  | tgacacgtct | 1560 |
| ggtaaggatg | ccatgagttg  | gttgataacc  | catgatcgct  | caacgccagt  | aactggaagt | 1620 |
| ctacttccct | acgtcgagac  | accagcgccc  | gcccccttc   | agactcaaca  | agtcgcgga  | 1680 |
| gaactgaagg | ataaaaatgg  | tggggtgagt  | tctcagggcg  | tacagctccc  | tgccaccata | 1740 |
| gcagtggttg | ccagccaaagt | cactgaaggga | caacagcaag  | aaatcactaa  | attattggag | 1800 |
| tcggtcaccg | gcggcacggc  | aggatctcaa  | ctgatataca  | attatgtttc  | agtgttaacg | 1860 |
| aattttacgc | tcgcttcacc  | tgatacattt  | gagattgagt  | taggtaaagt  | agtttcta   | 1920 |
| ttagaagaag | tacgcacaaga | cataaaaatc  | gctgatattc  | agcgtcttca  | tgaacaaaac | 1980 |
| atgaagaaaa | ttgaagagaa  | tcaagagaaa  | atcaagaaaa  | cagaagagaa  | tgccaagcaa | 2040 |
| gtcaagaaat | ccggcatggc  | atcaaaagatt | tttgctggcg  | tcagcgccat  | agcctcagtg | 2100 |
| gttatcggtg | ccatcaggtg  | ggcctcaggg  | gtaggagccg  | ttgcgggtgc  | aatgatgatt | 2160 |
| gcctcaggcg | taattgggat  | ggcgaaatg   | gctgtgaac   | aagcggcgga  | agatggcctg | 2220 |
| atatcccaag | aggaatgca   | agtattaggg  | ccgatactca  | ctcgatttga  | agtcgcatg  | 2280 |
| actgtagttt | caaccgtaat  | gacctttggc  | ggttcggcac  | taaaatgcct  | ggctgatatt | 2340 |
| ggcgcaaaac | tcggtgtcta  | caccgcaagt  | cttgetgcta  | aaggagccga  | gttttcggcc | 2400 |
| aaagttggcc | aaatttcgac  | aggcatatca  | aacactgtcg  | ggaatgcagt  | gactaaatta | 2460 |
| gggggcagtg | ttggttatgt  | aaacaatgagc | catgtaatcc  | gtacaggatc  | acaggcaaca | 2520 |
| caagtcgccc | ttggtgtggg  | cagcgggaata | actcagacca  | tcaataataa  | aaaacaagct | 2580 |
| gatttacaac | ataataacgc  | tgatttggcc  | ttgaacaagg  | cagacatggc  | agcgttaca  | 2640 |
| agttattatg | accgactcaa  | agaagagtta  | tccattttgt  | cagagtcaca  | tcaacaagtg | 2700 |
| atggaactga | ttttccagat  | gattaatgca  | aaaggtgaca  | tgctgcataa  | tttggccggc | 2760 |
| agaccccata | ctgtttaagg  | tacc        |             |             |            | 2784 |

SEQ ID NO: 48

|         |                                |
|---------|--------------------------------|
| FEATURE | Location/Qualifiers            |
| REGION  | 1..923                         |
|         | note = synthetic construct     |
| source  | 1..923                         |
|         | mol_type = protein             |
|         | organism = synthetic construct |

SEQUENCE: 48

|            |            |            |            |            |             |     |
|------------|------------|------------|------------|------------|-------------|-----|
| MDNGDRLYRA | DSRPDEIKR  | SGGLMPRGHN | EYFDRGTQMN | INLYDHARGT | QTGFVRYDDG  | 60  |
| YVSTSLSLRS | AHLAQQSILS | GYSTYYIYVI | ATAPNMFNVN | DVLGVYSPHP | YEQEVSAALGG | 120 |
| IPYSQIYGMW | RVNPGVIDER | LHRNREYRDR | YRNLNIAPA  | EDGYRLAGFP | PDHQAWREEP  | 180 |
| WIHHAPQCGC | NSSRMIRAYE | QNPQHFIEDL | EKVRVEQLTG | HGSSVLEELV | QLVKDKNIDI  | 240 |
| SIKYDPRKDS | EVFANRVITD | DIELLKKILA | YFLPEDAILK | GHHYDNQLQN | GIKRVKEFLE  | 300 |
| SSPNTQWELR | AFMAVMHPSL | TADRIDDIL  | KVIVDSMNH  | GDARSKLREE | LAELTAELEKI | 360 |
| YSVIAEINK  | HLSSSGTINI | HDKSINLMDK | NLYGYTDEEI | FKASAEYKIL | EKMPQTITQV  | 420 |
| DGSEKKIVSI | KDFLGSENKR | TGALGNLKNS | YSYNKDNNEL | SHFATTCSDK | SRPLNDLVSQ  | 480 |
| KTTQLSDITS | RFNSAIEALN | RFIQKYDSVM | QRLDDTSGK  | GSMSALITHD | RSTPVTGSL   | 540 |
| PVETPAPAP  | LQTQVAGEL  | KDKNGGVSSQ | GVQLPAPLAV | VASQVTEGQQ | QEITKLEESV  | 600 |
| TRGTAGSQLI | SNYVSVLTNF | TLASPDTFEI | ELGLKLVNLE | EVRKDIKIAD | IQLRHEQNMK  | 660 |
| KIEENQEKIK | ETENAKQVK  | KSGMASKIFG | WLSAIASVVI | GAIMVASGVG | AVAGAMMIAS  | 720 |
| GVIGMANMAV | KQAAEDGLIS | QEAMQVLGPI | LTAIEVALTV | VSTVMTFGGS | ALKCLADIGA  | 780 |

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|            |            |            |            |            |            |     |
|------------|------------|------------|------------|------------|------------|-----|
| KLGANTASLA | AKGAEFSAKV | AQISTGISNT | VGNAVTKLGG | SFGSLTMSHV | IRTSQATQV  | 840 |
| AVGVGSGITQ | TINNKKQADL | QHNNADLALN | KADMAALQSI | IDRLKEELSH | LSESHQQVME | 900 |
| LIFQMINAKG | DMLHNLGRP  | HTV        |            |            |            | 923 |

SEQ ID NO: 49                      moltype = DNA   length = 546  
 FEATURE                           Location/Qualifiers  
 misc\_feature                      1..546  
                                     note = synthetic construct  
 source                              1..546  
                                     mol\_type = other DNA  
                                     organism = synthetic construct

SEQUENCE: 49

|             |            |            |            |             |            |     |
|-------------|------------|------------|------------|-------------|------------|-----|
| atgggcagca  | gccatcacca | tcatcaccac | agccaggatc | cgatgggacta | ccagaacaac | 60  |
| gtcagcggaag | aacgtgttgc | ggaaatgatt | tgggatgccg | ttagtgaagg  | cgccacgcta | 120 |
| aaagacgttc  | atggaatccc | tcaagatatg | atggacggtt | tatatgctca  | tgcttatgag | 180 |
| ttttataacc  | agggacgact | ggatgaagct | gagacgttct | ttcgtttctt  | atgcatttat | 240 |
| gattttttaca | atcccgatta | caccatggga | ctggcggcag | tatgccaaact | gaaaaaacaa | 300 |
| tttcagaaaag | catgtgacct | ttatgcagta | gcgtttacgt | tacttaaaaa  | tgattatcgc | 360 |
| cccggttttt  | ttaccgggca | gtgtcaatta | ttaatgcgta | aggcagcaaa  | agccagacag | 420 |
| tgttttgaac  | ttgtcaatga | acgtactgaa | gatgagcttc | tgccgggcaaa | agcgttggtc | 480 |
| tatctggagg  | cgtaaaaaac | ggcggagaca | gagcagcaca | gcgagcagga  | gaaggagtaa | 540 |
| aaagctt     |            |            |            |             |            | 546 |

SEQ ID NO: 50                      moltype = AA   length = 179  
 FEATURE                           Location/Qualifiers  
 REGION                              1..179  
                                     note = synthetic construct  
 source                              1..179  
                                     mol\_type = protein  
                                     organism = synthetic construct

SEQUENCE: 50

|            |            |            |            |            |            |     |
|------------|------------|------------|------------|------------|------------|-----|
| MGSSHHHHH  | SQDPMYQNN  | VSEERVAEMI | WDAVSEGATL | KDVHGIPQDM | MDGLYAHAYE | 60  |
| FYNQGRLEDE | ETFFRFLCIY | DFYNPDYTMG | LAAVCQLKKQ | FQKACDLYAV | AFTLLKNDYR | 120 |
| PVFFTGQCQL | LMRKAAKARQ | CFELVNERTE | DESLRAKALV | YLEALKTAET | EQHSEQEKE  | 179 |

SEQ ID NO: 51                      moltype = DNA   length = 1029  
 FEATURE                           Location/Qualifiers  
 source                              1..1029  
                                     mol\_type = genomic DNA  
                                     organism = Salmonella spp

SEQUENCE: 51

|             |             |            |            |            |            |      |
|-------------|-------------|------------|------------|------------|------------|------|
| atgcttaata  | ttcaaaatta  | ttccgcttct | cctcatccgg | ggatcggtgc | cgaacggcgc | 60   |
| cagactccct  | cggcgagcga  | gcacgtcgag | actgcgctgg | taccgtctac | cacagaacat | 120  |
| cgcggtacag  | atatcatttc  | attatcgtag | gcggctacta | aaatccacca | ggcacagcag | 180  |
| acgctgcagt  | caacgccacc  | gatctctgaa | gagaataatg | acgagcgcac | gctggcgcgc | 240  |
| cagcagttga  | ccagcagcct  | gaatgcgctg | gcgaagtccg | gcgtgtcatt | atccgcagaa | 300  |
| caaaatgaga  | acctgcggag  | cgcgttttct | gcgcgcagct | cggccttatt | tagcgttcgc | 360  |
| cctatggcgc  | agccgagaaac | aaccatttct | gatgctgaga | tttgggatat | ggtttcccaa | 420  |
| aatatatcgg  | cgataggtga  | cagctatctg | ggcgtttatg | aaaacgttgt | cgcagtctat | 480  |
| accgattttt  | atcaggcctt  | cagtgatatt | ctttccaaaa | tgggaggtcg | gttattacca | 540  |
| ggtaaggacg  | gtaataccgt  | taagctagat | gttacctcac | tcaaaaatga | tttaaacagt | 600  |
| ttagtcaata  | aatataatca  | aataaacagt | aataccgttt | tatttccagc | gcagtcaggc | 660  |
| agcggcggtta | aagtagccac  | tgaagcggaa | gcgagacagt | ggctcagtga | attgaattta | 720  |
| ccgaatagct  | gcctgaaatc  | ttatggatcc | ggttatgtcg | tcaccgttga | tctgacgcca | 780  |
| ttacaaaaaa  | tggttcagga  | tattgatggt | ttaggcgcgc | cgggaaaaga | ctcaaaactc | 840  |
| gaaatggata  | acgccaataa  | tcaagcctgg | cagtcggggt | ttaaaagcga | ggaagaaaat | 900  |
| atgaaaacca  | cattacagac  | gctgacgcaa | aaatatagca | atgccaatcc | attgtacgac | 960  |
| aacctggtta  | aagtgtctgag | cagtagcaga | agtagcagcc | tggaaaccgc | caaaagcttc | 1020 |
| ctgcaaggga  |             |            |            |            |            | 1029 |

SEQ ID NO: 52                      moltype = AA   length = 343  
 FEATURE                           Location/Qualifiers  
 source                              1..343  
                                     mol\_type = protein  
                                     organism = Salmonella spp

SEQUENCE: 52

|             |            |            |            |            |            |     |
|-------------|------------|------------|------------|------------|------------|-----|
| MLNIQNSAS   | PHPGIVAERP | QTPSASEHVE | TAVVPSTTEH | RGTDIISLSQ | AATKIHQAQQ | 60  |
| TLQSTPPISE  | ENNDERFLAR | QQLTSSLNAL | AKSGVSLSAE | QENLRSFAFS | APTSALFSAS | 120 |
| PMAQPRTTIS  | DAEIWMVVSQ | NISAIGDSYL | GVYENVVAVY | TDFYQAFSDI | LSKMGGWLLP | 180 |
| GKDGNVTKLID | VTSLKNDLNS | LVNKYNQINS | NTVLFPAQSG | SGVKVATEAE | ARQWLSELNL | 240 |
| PNSCLKSYGS  | GYVTVDLTP  | LQKMQDIDG  | LGAPGKDSKL | EMDNAKYQAW | QSGFKAQEEN | 300 |
| MKTTLTQTLTQ | KYSNANSLYD | NLVKVLSTTI | SSSLETAKFS | LQG        |            | 343 |

SEQ ID NO: 53                      moltype = DNA   length = 1782  
 FEATURE                           Location/Qualifiers  
 source                              1..1782

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mol_type = genomic DNA
organism = Salmonella spp

SEQUENCE: 53
atggttaaatg acgcaagtag cATTAGccgt agcggatata cccaaaatcc gcgcctcgct 60
gaggcggtt ttgaaggcgt tcgtaagaac acggactttt taaaagcggc ggataaagct 120
tttaaatgatg tgggtggcaac gaaagcgggc gaccttaag ccggaacaaa gtccggcgag 180
agcgctatta atacggtggg tctaaagccg cctacggacg ccgcccggga aaaactctcc 240
agcgaaggcg aattgacatt actgcttggt aagttaatga cctactggg cgatgtttcg 300
ctgtctcaac tggagtctcg tctggcggta tggcaggcga tgattgagtc acaaaaagag 360
atgggggattc aggtatcgaa agaattccag acggctctgg gagaggctca ggaggcgacg 420
gatctctatg aagccagtat caaaaagacg gataccgcca agagtgttta tgacgctgcg 480
acccaaaaac tgacgcaggc gcaaaaataa ttgcaatcgc tggaccgggc tgacccgggc 540
tatgcacaag ctgaagccgc ggtagaacag gccgaaaaa agcgcacaga ggcgaagag 600
gccttagata aggccacgga tgcgacggtt aaagcaggca cagacgcaa agcgaaagcc 660
gagaagcggc ataacattct gaccaaattc cagggaaacgg ctaatgcgcg ctctcagaat 720
caggtttccc agggtagcga ggataatctg tcaaatgtcg ccgcctcac tatgctcatg 780
gccatgttta ttgagattgt gggcaaaaat acggaagaaa gcctgcaaaa cgatcttgcg 840
cttttcaacg ccttgacagga agggcgctcag gcggagatgg aaaaagaatc ggctgaattc 900
caggaagaga cgcgcaaaac cgaggaaacg aaccgcatta tgggatgtat cgggaaagtc 960
ctcgcgcgcg tgctaaccat tgtcagcggt gtggcgcgtg tttttacggg tggggcgagt 1020
ctggcgctgg ctggcggtgg acttgcggtg atggtggccg atgaaattgt gaaggcgcg 1080
acgggagtggt cgtttattca gcaggcgcta aaccgatta tggagcatgt gctgaagccg 1140
ttaatggagc tgattggcaa ggcgattacc aaagcgctgg aaggattagg cgtcgataag 1200
aaaacggcag agatggccgg cagcattgtt ggtgcgattg tcgcccgtat tgcctaggtg 1260
gcggtcattg ttggtgtcgc agttgtcggg aaagcgcgcg cggcgaaaact gggtaacgcg 1320
ctgagcaaaa tgatgggcga aacgattaag aagttgggtc ctaacgtgct gaaacagttg 1380
gcgcaaaacg gcagcaaaact ctttaccagg gggatgcaac gtattactag cggctctgggt 1440
aatgtgggta gcaagatggg cctgcaaacg aatgccttaa gtaaaagagt ggtaggtaat 1500
accctaaata aagtggcggt gggcatggaa gtcacgaata ccgcagccca gtcagccggt 1560
gggtttgccc agggcgattt tattaaaaat gccacgcagg cgcttgctga ttttatgctc 1620
gcccgttttg cattggatca gattcagcag tgggttaaac aatccgtaga aatatttggt 1680
gaaaaccaga aggtaacggc ggaactgcaa aaagccatgt cttctgcggt acagcaaaat 1740
gcggatgctt cgcgttttat tctgcgccag agtcgcgcgt aa 1782

SEQ ID NO: 54      moltype = AA  length = 593
FEATURE           Location/Qualifiers
source            1..593
                  mol_type = protein
                  organism = Salmonella spp

SEQUENCE: 54
MVNDASSISR SGYTQNRLA EAFEGVRKN TDFLKAADKA PKDVVATKAG DLKAGTKSGE 60
SAINTVGLKP PTDAAREKLS SEQQLTLLLG KLMTLLGDVS LSQLESRLAV WQAMIESQKE 120
MGIQVSKEFQ TALGEAQEAT DLYEASIKKT DTAKSVDAA TKKLTQQONK LQSLDPADPG 180
YAQAEAAVEQ AGKEATEAKE ALDKATDATV KAGTDAKAKA EKADNITKF QGTANAASQN 240
QVSQGEQDNL SNVARLTMLM AMFIEIVGKN TEESLQNDLA LFNALQEGRQ AEMEKKSAEF 300
QEETRKAET NRIMCIGKV LGALLTIVSV VAAVFTGGAS LALAAGVLAV MVADEIVKAA 360
TGVSFIIQAL NPIMEHVLKP LMELIGKAIT KALEGLGVDK KTAEMAGSIV GAIVAAIAMV 420
AVIVVVAVVG KGAAAKLGNA LSKMMGETIK KLVPNVLKQL AQNGSKLFTQ GMQRITSGLG 480
NVGSKMGLQT NALSKELVGN TLNKVALGME VTNTAAQSAG GVAEGVFIKN ASEALADFML 540
ARFAMDQIQ WLKQSVEIFG ENQKVTAELQ KAMSSAVQON ADASRFILRQ SRA 593

SEQ ID NO: 55      moltype = DNA  length = 2817
FEATURE           Location/Qualifiers
misc_feature      1..2817
                  note = synthetic construct
source            1..2817
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 55
atgcttaata ttcaaaatta ttccgcttct cctcatccgg ggatcgttgc cgaacggcgg 60
cagactccct cggcgagcga gcacgtcgag actgcgctgg tacggtctac cacagaacat 120
cgcggtacag atatcatttc attatcgag cgggctacta aaatccacca ggcacagcag 180
acgctgcagt caacgccacc gatctctgaa gagaataatg acgagcgcac gctggcgcg 240
cagcagttga ccagcagcct gaatgcgctg gcgaagtcgg cggtgtcatt atccgcagaa 300
caaaatgaga acctgcggag cgcgttttct gcgcgcgacg cgcccttatt tagcgcttcg 360
cctatggcgc agccgagaaac aaccatttct gatgctgaga tttgggatat ggtttcccaa 420
aataatcagg cgataggtga cagctatctg ggcgtttatg aaaaactgtg cgcagtcctat 480
accgattttt atcaggccctt cagtgatatt ctttccaaaa tgggaggctg gttattacca 540
ggtaaggacg gtaataccgt taagctagat gttacctcac tcaaaaatga tttaaacagt 600
tagtcaata aatataatca aataaacagt aataccgttt tatttcacgc gcagtcaggc 660
agcggcggtt agctagccac tgaagcggaa gcgagacagt ggctcagtga attgaattta 720
ccgaatagct ggcctgaatc ttatggatcc ggttatgtcg tcaccgttga tctgacgcca 780
ttacaaaaaa tggttcagga tattgatggt ttaggcgcgc cgggaaaaga ctcaaaactc 840
gaaatggata acgccaataa tcaagcctgg cagtcggggt ttaaagcgca ggaagaaaat 900
atgaaaacca cattacagac gctgacgcaa aaatatagca atgccaattc attgtacgac 960
aacctggtaa aagtgtctgag cagtacgata agtagcagcc tggaaacccg caaaagcttc 1020

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|             |             |            |             |             |             |      |
|-------------|-------------|------------|-------------|-------------|-------------|------|
| ctgcaaggag  | tgcacatggt  | aaatgacgca | agtagcatta  | gccgtagcgg  | ataatcccaa  | 1080 |
| aatccgcgccc | tgcgtgaggg  | ggcttttgaa | ggcgttcgta  | agaacacgga  | ctttttaaaa  | 1140 |
| gcggcgggata | aagcttttaa  | agatgtggtg | gcaacgaaag  | cgggcgacct  | taaagccgga  | 1200 |
| acaaagtcgg  | gcgagagcgc  | tattaatacg | gtgggtctaa  | agccgcctac  | ggaagccggc  | 1260 |
| cgggaaaaac  | tctccagcga  | agggcaattg | acattactgc  | ttggcaagtt  | aatgacccta  | 1320 |
| ctgggcgatg  | tttcgctgtc  | tcaactggag | tctcgtctgg  | cggtatggca  | ggcgatgatt  | 1380 |
| gagtcacaaa  | aagagatggg  | gattcaggtg | tcgaaagaat  | tccagacggc  | tctgggagag  | 1440 |
| gctcaggagg  | cgacggatct  | ctatgaagcc | agatcaaaa   | agacgggata  | cgccaagagt  | 1500 |
| gtttatgacg  | ctgcgaccaa  | aaaactgacg | caggcgcaaa  | ataaattgca  | atcgctggac  | 1560 |
| ccggctgacc  | ccggctatgc  | acaagctgaa | gccgcggtag  | aacaggccgg  | aaaagaagcg  | 1620 |
| acagaggcga  | aagaggcctt  | agataaggcc | acggatgcga  | cggttaaagc  | aggcacagac  | 1680 |
| gccaagcga   | aagccgagaa  | agcggataac | attctgacca  | aattccaggg  | aacggctaata | 1740 |
| gccgcctctc  | agaatcagggt | ttcccagggt | gagcaggata  | atctgtcaaa  | tgtcgcccg   | 1800 |
| ctcactatgc  | tcatggccat  | gtttattgag | attgtgggca  | aaaatacgga  | agaaagccctg | 1860 |
| caaaacgatc  | ttgcgctttt  | caacgccttg | caggaaagggc | gtcaggcgga  | gatggaaaag  | 1920 |
| aaatcggctg  | aattccagaa  | agagacgcgc | aaagccgagg  | aaacgaaccg  | cattatggga  | 1980 |
| tgtatcggga  | aagtcctcgg  | cgcgctgcta | accattgtca  | gcgttgtggc  | cgctgttttt  | 2040 |
| accggtgggg  | cgagtcctgc  | gctggctgcg | gtgggacttg  | cggttaaggt  | ggccgatgaa  | 2100 |
| attgtgaagg  | cggcgacggg  | agtgtcgttt | attcagcagg  | cgctaaaccg  | gattatggag  | 2160 |
| catgtgctga  | agcgcgtta   | ggagctgatt | ggcaaggcga  | ttaccaaaagc | gctgggaagg  | 2220 |
| ttaggcgtcg  | ataagaaaac  | ggcagagatg | gccgcgagca  | tgttgtgtgc  | gattgtcgcc  | 2280 |
| gctattggca  | tggtggcggt  | cattgtggtg | gtcgcagttg  | tcgggaaagg  | cgcggcgggc  | 2340 |
| aaactgggta  | acgcgctgag  | caaaatgatg | ggcgaacga   | ttaagaagtt  | ggtgcctaac  | 2400 |
| gtgctgaaac  | agttggcgca  | aaacgcgacg | aaactcttta  | cccaggggat  | gcaacgtatt  | 2460 |
| actagcggtc  | tggtgaatgt  | gggtagcaag | atgggcctgc  | aaacgaatgc  | cttaagtaaa  | 2520 |
| gagctggtag  | gtaataccct  | aaataaagtg | gcgttgggca  | tggaagtcac  | gaataccgca  | 2580 |
| gcccagtcag  | ccggtgtggt  | tgccgagggc | gtatttatta  | aaaatgccag  | cgaggcgctt  | 2640 |
| gctgatttta  | tgctcgcccg  | ttttgcatg  | gatcagattc  | agcagtggtt  | taaaccaatcc | 2700 |
| gtgaaaatat  | ttggtgaaaa  | ccagaaggta | acggcggaac  | tgcaaaaagc  | catgtcttct  | 2760 |
| gcggtacagc  | aaaatgcgga  | tgcttcgcgt | tttattctgc  | gccagagtcg  | cgcataaa    | 2817 |

SEQ ID NO: 56                   moltype = AA   length = 938  
 FEATURE                       Location/Qualifiers  
 REGION                        1..938  
                               note = synthetic construct  
 source                        1..938  
                               mol\_type = protein  
                               organism = synthetic construct

SEQUENCE: 56

|             |            |             |             |             |             |     |
|-------------|------------|-------------|-------------|-------------|-------------|-----|
| MLNIQNYAS   | PHPGIVAERP | QTPSASEHVE  | TAVVPSTTEH  | RGTDIISLSQ  | AATKIHQAQQ  | 60  |
| TLQSTPPISE  | ENNDERTLAR | QQLTSSLNAL  | AKSGVSLSAE  | QNNENLRSFAS | APTSALFSAS  | 120 |
| PMAQPRTTIS  | DAEIDWMSQ  | NISAIGDSYL  | GVYENVVAVY  | TDFYQAFSDI  | LSKMGGWLLP  | 180 |
| GKDGNTVVKLD | VTSLKNDLNS | LVNKYNQINS  | NTVLFPAQSG  | SGVKVATEAE  | ARQWLSSELNL | 240 |
| PNSCLKSYGS  | GYVVTVDLTP | LQKMVQDIDG  | LGAPGKDSKL  | EMDNAKYQAW  | QSGFKAQEEN  | 300 |
| MKTTTLQTLTQ | KYSNANSLYD | NLVKVLSSSTI | SSSLETAKSF  | LQGVDMVNDA  | SSISRSRGYTQ | 360 |
| NPRLEAAAFE  | GVRKNTDFLK | AADKAFKDVV  | ATKAGDLKAG  | TKSGESAINT  | VGLKPPTDAA  | 420 |
| REKLSSESQL  | TLLLGKLMTL | LGDVLSLSQLE | SRLAVWQAMI  | ESQKEMGIQV  | SKEFQTLALGE | 480 |
| AQEATDLYEA  | SIKKTDTAKS | VYDAATKKLT  | QAQNKQLQSLD | PADPGYAQAE  | AAVEQAGKEA  | 540 |
| TEAKEALDKA  | TDAIVKAGTD | AKAKAEKADN  | ILTKFQGTAN  | AASQNVQSQG  | EQDNLSNVAR  | 600 |
| LTMLMAMFIE  | IVGKNTESL  | QNDLALFNAL  | QEGRQAEEMEK | KSAEFQEBTR  | KAEETNRIMG  | 660 |
| CTGKVLGALL  | TIVSVVAADF | TGGASLALAA  | VGLAVMVADE  | IVKAATGVSF  | IQQALNPIME  | 720 |
| HVLKPLMELI  | GKATIKALEG | LGVDKKTAE   | AGSIVGAIVA  | AIAMVAVIVV  | VAVVVGKGA   | 780 |
| KLGNALSMM   | GETIKKLVPN | VLKQLAQNGS  | KLFTQGMQRI  | TSGLGNVGSK  | MGLQTNALS   | 840 |
| ELVGNLNLKV  | ALGMEVNTA  | AQSAGGVAEG  | VFIKNASEAL  | ADFMRLARFAM | DQIQQWLKQS  | 900 |
| VEIFGENQKV  | TAELOKAMSS | AVQQNADASR  | FILRQSR     |             |             | 938 |

SEQ ID NO: 57                   moltype = DNA   length = 3423  
 FEATURE                       Location/Qualifiers  
 misc\_feature                  1..3423  
                               note = synthetic construct  
 source                        1..3423  
                               mol\_type = other DNA  
                               organism = synthetic construct

SEQUENCE: 57

|             |             |             |            |             |            |     |
|-------------|-------------|-------------|------------|-------------|------------|-----|
| catatggaca  | atggcgatcg  | tttataccgt  | gccgactcgc | gtcccccaga  | tgagattaaa | 60  |
| cgtagcgggtg | gggttaatgcc | acgtggggcac | aatgagtatt | ttgaccgtgg  | aacacagatg | 120 |
| aacattaacc  | tttacgatca  | tgcccgtggg  | accagacggc | gggtttgcgc  | ttatgatgac | 180 |
| gggtatggtta | gtacgagttt  | gtccttacgc  | tccgcacacc | ttgcgggaca  | aagtatttta | 240 |
| tcaggctaca  | gcacatatta  | catttatgtg  | atcgccactg | ccccaaacat  | gttcaatgtg | 300 |
| aacgatgtgt  | tggggggttta | cagcccccac  | ccatatgaac | aagaagcttc  | ggcccttggg | 360 |
| gggatcccat  | atagccagat  | ttatggttgg  | taccgcgtaa | attttggtgt  | gattgatgaa | 420 |
| cgtttgcac   | gtaaccgtga  | ataccgcgat  | cgctactacc | gtaacttgaa  | cattgcacct | 480 |
| gccgaggacg  | gctatcgttt  | agcgggattc  | ccaccgcgat | atcaggcggtg | gcgtgaggaa | 540 |
| ccgtggatcc  | atcacgcccc  | tcagggggtgc | gggaacagta | gtcgcggggtc | cgcggcatcc | 600 |
| atgcttaata  | ttcaaaaatta | ttccgcttct  | cctcatccgg | ggatcggttg  | cgaacggcgc | 660 |
| cagactccct  | cggcgagcga  | gcacgtcgag  | actgccgttg | taccgtctac  | cacagaacat | 720 |

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cgcggtacag atatcatttc attatcgcag gcggctacta aaatccacca ggcacagcag 780
acgctgcagt caacgccacc gatctctgaa gagaataatg acgagcgcac gctggcgcg 840
cagcagttga ccagcagcct gaatgcgctg gcgaagtcgc gcgtgtcatt atccgcagaa 900
caaaatgaga acctgcccag cgcgctttct gcgcgcgacgt cggccttatt tagcgcttcg 960
cctatggcgc agccgagaac aaccatttct gatgctgaga tttgggatat ggtttcccaa 1020
aatatatcgg cgatagtgga cagctatctg ggcgctttatg aaaacgttgt cgcagcttat 1080
accgattttt atcaggccctt cagtgatatt ctttccaaaa tgggaggtcg gttattacca 1140
ggtaaggacg gtaataccgt taagctagat gttacctcac tcaaaaatga tttaaacagt 1200
ttagtcaata aatataatca aataaacagt aataccgctt tattccagc gcagtcagcg 1260
agcggcgctta aagtagccac tgaagcggaa gcgagacagt ggctcagtga attgaattta 1320
ccgaatagct gcctgaaatc ttatggatcc ggttatgtcg tcaccggtga tctgacgcca 1380
ttacaaaaaa tgggttcagga tattgatggt ttaggcgcgc cgggaaaaga ctcaaaactc 1440
gaaatggata acgccaataa tcaagcctgg cagtcggggt ttaaagcgca ggaagaaaat 1500
atgaaaacca cattacagac gctgacgcaa aaatatagca atgccaattc attgtacgac 1560
aacctggtaa aagtgcgtgag cagtacgata agtagcagcc tggaaaacgc caaaagcttc 1620
ctgcaaggag tcgacatggt aaatgacgca agtagcatta gccgtagcgg atatacccaa 1680
aatccgcgcc tcgctgagcg ggcttttgaa ggcttctgta agaacacgga ctttttaaaa 1740
cgcgcgagata aagcttttaa agatgtggtg gcaacgaaag cgggcgacct taaagccgga 1800
acaaagtcgc gcgagagcgc tattaatacg gtgggtctaa agccgcctac ggaagccgcc 1860
cgggaaaaac tctccagcga agggcaattg acattactgc ttggcaagt aatgacccta 1920
ctggcgcatg tttcgctgtc tcaactggag tctcgtctgg cggtatggca ggcatgatt 1980
gagtcacaaa aagagatggg gattcaggtt tcgaaagaat tccagacggc tctgggagag 2040
gctcaggagg cgacggatct ctatgaagcc agtatcaaaa agacgggata cgccaagagt 2100
gtttatgacg ctgcgaccaa aaaactgacg caggcgcaaa ataaattgca atcgctggac 2160
ccggctgacc ccggctatgc acaagctgaa gccgcggtag aacaggccgg aaaagaagcg 2220
acagaggcga aagaggcctt agataaggcc acggatgcga cggttaaagc aggcacagac 2280
gccaagcgaa aagccgagaa agcggataac attctgacca aattccaggg aacggcta 2340
gccgcctctc agaatacagg ttcccagggt gagcaggata atctgtcaaa tgtcgcccg 2400
ctcaactatg tcatggccat gtttattgag attgtgggca aaaatacgga agaaagcctg 2460
caaaacgatc ttgcgctttt caacgccttg caggaaaggg gtcaggcgga gatggaaaag 2520
aaatcggctg aattccagga agagacgcgc aaagccgagg aaacgaaacc cattatggga 2580
tgtatcggga aagtcctcgg cgcgctgcta accattgtca gcgttgtggc cgtgttttt 2640
accggtgggg cgagtcctgc gctggctgcg gtgggacttg cggtaattgt ggccgatgaa 2700
attgtgaagg cggcgacggg agtgcgtttt attcagcagg cgtaaaacc gattatggag 2760
catgtgtcga agcgtgtaat ggagctgatt ggcaaggcga ttaccaaagc gctggaaagg 2820
ttagcgctgc ataagaaaac ggcagagatg gccgcgagca ttgttggtgc gattgtcgcc 2880
gctattgcga tcgtggcggt cattgtggtg gtcgcagttg tcgggaaagg cgcggcgggc 2940
aaactgggta aagcgctgag caaaatgatg ggcgaaacga ttaagaagt ggtgcctaac 3000
gtgctgaaac agttggcgca aaacgcgacg aaactcttta ccaggggat gcaacgtatt 3060
actagcggtc tgggtaqeenm gggtagcaag atgggcctgc aaacgaaatg cttaaagtaa 3120
gagctggtag gtaataccct aaataaagt gcgttgggca tggaaagcac gaataccgca 3180
gcccagtcag ccggtggtgt tgcgaggggc gtatttatta aaaatgccag cgaggcgctt 3240
gctgatttta cgtcgcccg ttttgccatg gatcagattc agcagtggtt taaacaatcc 3300
gtagaaaat ttggtgaaaa ccagaaggta acggcggaac tgcaaaaagc catgtcttct 3360
gcggtacagc aaaatgcgga tgcttcgcgt tttattctgc gccagagtcg cgcataactc 3420
gag 3423

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SEQ ID NO: 58      moltype = AA length = 1137
FEATURE           Location/Qualifiers
REGION            1..1137
                  note = synthetic construct
source            1..1137
                  mol_type = protein
                  organism = synthetic construct

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```

SEQUENCE: 58
MDNGDRLRYA DSRPPDEIKR SGLMPRGHN EYFDRGTQMN INLYDHARGT QTGFVRYDDG 60
YVSTSLSLRS AHLAQQSILS GYSTYYIYVI ATAPNMFNVN DVLGVYSPHP YEQEVSA LGG 120
IPYSQIYGWY RVNFGVIDER LHRNREYRDR YYRNLNIAPA EDGYRLAGFP PDHQA WREEP 180
WIHHAPOGCG NSSRGSASMS LNIQNYASAP HPGIVAERPQ TPSASEHVET AVVPSTTEHR 240
GTDIISLSQA ATKIHQAQQT LQSTPPISEE NNDERTLARQ QLTSSLNALA KSGVSLSAEQ 300
NENLRSAFSA PTSALFSASP MAQPRTTISD AEIWDMSVQN ISAIGDSYLG VYENVVAVYT 360
DFYQAFSDIL SKMGGWLLPG KDGN TVKLDV TSLKNDLNSL VNKNYQINSN TVLFPAQSGS 420
GVKVATEAEA RQWLSENLPL NSCLKSYGSG YVVTVDLTPL QKMVDIDGL GAPGKDSKLE 480
MDNAKYQAWQ SGFKAQEENM KTTLQTLTQK YSNANSLYDN LVKVLSSSTIS SLETAKSFL 540
QGVDMVNDAS SISRSQYTON PRLAEAAFEF VRKNTDFLKA ADKAFKDVVA TKAGDLKAGT 600
KSGESAINTV GLKPPTDAAR EKLSSSEQLT LLLGLKMTLL GDVSLSQLES RLAVWQAMIE 660
SQKEMGIQVS KEPQTALGEA QEATDLYEAS IKKTD TAKSV YDAATKKLTQ AQNK LQSLDP 720
ADPGYAQAEA AVEQAGKEAT EAKEALDKAT DATVKAGTDA KAKAEKADNI LTKFQGTANA 780
ASQNVQSQGE QDNLSNVARL TMLMAMFIEI VGKNTESLQ ND LALFNALQ EGRQAEMEKK 840
SAEFQEETRK AEETNRIMGC IGVVLGALLT IVSVVAAVFT GGASLALAAV GLAVMVADEI 900
VKAATGVSPFI QALNPSIMEH VLKPLMELIG KAITKALEGL GVDKKTAEMA GSVIGAI VAA 960
IAMVAVIVVV AVVVGKAAAK LGNALSMMMG ETIKKLVPNV LKQLAQNGSK LFTQGMQRIT 1020
SGLGNVGSKM GLQTNALSKE LVGNTLNKVA LGMEVTNTAA QSAGGVAEGV FIKNASEALA 1080
DFMLARFAMD QIQQWLKQSV EIFGENQKVT AELOKAMSSA VQQNADASRF ILRQSR A 1137

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SEQ ID NO: 59      moltype = DNA length = 522

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FEATURE                      Location/Qualifiers
misc_feature                  1..522
                               note = synthetic construct
source                        1..522
                               mol_type = other DNA
                               organism = synthetic construct

SEQUENCE: 59
atgggcagca gccatcacca tcataccacc agccaggatc cgatgaaaaa agaccgcacc 60
ctacaacagg cacatgacac gatgcgggtt ttccggcggt gcggtcgcgt gcgtatgttg 120
ttggatgacg atgttacaca gccgcttaat actctgtatc gctatgccac gcagcttatg 180
gaggtaaaaa aattcgccgg cgcagcgcca ctttttcaat tgcgtacgat atatgatgcc 240
tggtcatttg actactgggt tcggttaggg gaatgctgcc aggcctcaaaa acattggggg 300
gaagcgatat acgcttatgg acgcgcggca caaattaaga ttgatgcgcc gcaggcgcca 360
tgggcgcgag cggaatgcta tctcgcggtg gataacgtct gttatgcaat caaagcgta 420
aaggccgttg tgcgtatttg cggcgaggtc agtgaacatc aaattctccg acagcgtgca 480
gaaaagatgt tacagcaact ttctgacagg agctaaaagc tt                    522

SEQ ID NO: 60                moltype = AA length = 171
FEATURE                      Location/Qualifiers
REGION                        1..171
                               note = synthetic construct
source                        1..171
                               mol_type = protein
                               organism = synthetic construct

SEQUENCE: 60
MGSSHHHHHH SQDPMKKDPT LQQAHDTMRF FRRGGSLRML LDDDVTTQPLN TLYRYATQLM 60
EVKEFAGAAAR LFQLLLIYDA WSPDYWFRLG ECCQAQKHVG EAIYAYGRAA QIKIDAPQAP 120
WAAAECYLAC DNVCYAIKAL KAVVRICGEV SEHQILRQRA EKMLQQQLSDR S          171

SEQ ID NO: 61                moltype = DNA length = 588
FEATURE                      Location/Qualifiers
source                        1..588
                               mol_type = genomic DNA
                               organism = Salmonella spp

SEQUENCE: 61
atgtcttcag gaaacatctt atggggaagt caaaacccta ttgtgtttaa aaatagcttc 60
ggcgtaagca acgctgatac cgggagccag gatgacttat ccagcaaaaa tccgtttgcc 120
gaagggtatg gtgttttgcg tatctcctt atggttatcc aggcctatgc aaataataaa 180
tttattgaag tcagagaaga cgctgaacgt gccagaaaaa gccagaaaaa gtcaaatgag 240
atggatgagg tgaattgctaa agcagccaaa ggggatgcta aaaccaaaaga ggaggtgcct 300
gaggatgtaa ttaataacat gcgtgataat ggtattctca tcgatgggat gaccattgat 360
gattatatgg ctaaatatgg cgatcatggg aagctggata aagtgggcct acaggcgatc 420
aaagcggcgt tggataatga cgccaaccgg aataccgata ttatgagtca ggggcagata 480
acaattcaaa aaatgtctca ggagcttaac gctgtcctta cccaactgac agggccttatc 540
agtaagtggg gggaaatttc cagtatgata gcgcagaaaa cgtactca                    588

SEQ ID NO: 62                moltype = AA length = 196
FEATURE                      Location/Qualifiers
source                        1..196
                               mol_type = protein
                               organism = Salmonella spp

SEQUENCE: 62
MSSGNILWGS QNPVIFKNSF GVSNADTGSQ DDLSSQNPFA EGYGVLLILL MVIQAIANNK 60
FIEVQKNAER ARNTQEKSENE MDEVIKAAK GDAKTKEEVP EDVIKYMEDN GILIDGMTID 120
DYMAKYGDHG KLDKGGQLQAI KAALDNDANR NTDLMSQGGI TIQMSQELN AVLTQLTGLI 180
SKWGEISSMI AQKTYT                    196

SEQ ID NO: 63                moltype = DNA length = 1455
FEATURE                      Location/Qualifiers
source                        1..1455
                               mol_type = genomic DNA
                               organism = Salmonella spp

SEQUENCE: 63
atgaatcgaa ttcacagtaa tagcgacagc gccgcaggag taaccgcctt aacacatcat 60
cacttaagca atgtcagttg cgtttcctcg ggttcgctgg gaaagcgcca gcacgtgtg 120
aattctactt ttggcgatgg caacgcgcgcg tgtctgctat ccgggaaaaa tagtcttcag 180
gaggcaagca atgcgttgaa gcaactgctt gatgcggtac ccggaaaaat taagcgtcca 240
tcattgcctg actttttgca gaccaatccc gcggttttat caatgatgat gacgtcatta 300
atactcaacg tcttttgtaa taacgctcaa tcgttatgcc aacagcttga gcgggcaact 360
gaggtgcaaa atgcattacg taataagcag gtaaaaggag atcaggagca gatccagaaa 420
gcgatagagc aggaggataa agcgcgtaaa gcgggtattt ttggcgctat ttttgactgg 480
attaccggca attttgaaac cgtgattggc gccttaaaag ttgtggaagg ttttctgtcc 540
ggaaatcccg cagaaatggc tagcggcgta gcttatatgg ccgcaggttg tcaggaatg 600
gttaaagccg gagccgaaac ggcaatgatg tgcggtgctg accacgatac ctgtcaggca 660
attattgacg tgacaagtta gattcaattt ggttgatgaa ccgtcgcgct ggcactggat 720
gttttcacga ttggccgtgc ttttatggcg acgagagggt tatctggcgc agctgcaaaa 780

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gtgcttgact ccggtttttg cgaggaagtg gttgagcgta tggtaggtgc aggggaagca 840
gaaatagagg agttgggtga aaagttttgg gaagaagtga gcgaaagtgt ttccaaacaa 900
tttgagccgc ttgaacgtga aatggctatg gcgaatgaga tggcagagga ggctgccgag 960
ttttctcgta acgtagaaaa taatatgacg cgaagcgcg gaaaaagctt tacgaaagag 1020
ggggtgaaaag caatggcaaa agaagcggca aaagaagccc tggaaaaatg tgtgcaagaa 1080
ggtggaaagt tcctgttaaa aaaattccgt aataaagttc tcttcaatat gttcaaaaaa 1140
atcctgtatg ccttactgag ggattgttca tttaaaggct tacaggctat cagatgtgca 1200
accgagggcg ccagtcagat gaatactggc atgggttaaca cagaaaaagc gaagatcgaa 1260
aagaaaaatg agcaattaat aactcagcaa cggtttcttg atttcataat gcaacaaaca 1320
gaaaaccaga aaaagataga acaaaaacgc ttagaggagc ttataagggt gagcgggtgc 1380
gcgcttagag atgtattaga taccattgat cactatagta gcgttcaggc gagaatagct 1440
ggctatcgcg cttaa

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SEQ ID NO: 64      moltype = AA  length = 484
FEATURE           Location/Qualifiers
source            1..484
                  mol_type = protein
                  organism = Salmonella spp

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```

SEQUENCE: 64
MNRHSNSDS AAGVTALTHH HLSNVSCVSS GSLGKRQHRV NSTFGDGNA CLLSGKISLQ 60
EASNALKQLL DAVPGNHKRP SLPDFLQTNP AVLSMMMTSL ILNVFGNNAQ SLCQQLERAT 120
EVQNALRNKQ VKEYQEIQIK AIEQEDKARK AGIFGAIFDW ITGIFETVIG ALKVVVEGFLS 180
GNPAEMASGV AYMAAGCAGM VKAGAETAMM CGADHDTCAQ IIDVTSKIQF GCEAVALALD 240
VFQIGRAFMA TRGLSGAAAK VLDSGFGEEV VERMVGAGEA EIEELAEKFG EEVSESFQSK 300
FEPLEREMAM ANEMAEEAAE FSRNVENNMT RSAGKSFTKE GVKAMAKEAA KEALEKCVQE 360
GGKFLKKKFR NKVLFNMFKK ILYALLRDCS FKGLQAIRCA TEGASQMNTG MVNTEKAKIE 420
KKIEQLITQQ RFLDFIMQQT ENQKKIEQKR LEELYKGSQA ALRDVLDITD HYSSVQARIA 480
GYRA

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SEQ ID NO: 65      moltype = DNA  length = 2049
FEATURE           Location/Qualifiers
misc_feature       1..2049
                  note = synthetic construct
source            1..2049
                  mol_type = other DNA
                  organism = synthetic construct

```

```

SEQUENCE: 65
atgtcttcag gaaacatctt atgggggaagt caaaacccta ttgtgtttaa aaatagcttc 60
ggcgctcagca acgctgatac cgggagccag gatgacttat cccagcaaaa tccgttttgc 120
gaagggtatg gtgttttggc tattctcctt atgggtatct aggctatcgc aaataataaa 180
tttattgaag tccagaagaa cgctgaacgt gccagaaaaa cccaggaaaa gtcaaatgag 240
atggatgagg tgattgctaa agcagccaaa ggggatgcta aaacccaaaga ggagggtgct 300
gaggatgtaa ttaaatcaat cgctgataat ggtattctca tcgatgggat gaccattgat 360
gattatatgg ctaaatatgg cgtatcatgg aagctggata aaggtggcct acaggcgatc 420
aaagcggctt tggataatga cgcacaacgg aataccgacg ttatgagtcg ggggcagata 480
acaattcaaa aaatgtctca ggagcttaac cgtgtcctta cccaactgac agggcctatc 540
agtaagtggg ggggaatttc cagtatgata gcgcagaaaa cgtactcaga gctcatgaat 600
cgaattcaca gtaatagcga cagcgccgca ggagtaaccg ccttaacaca tcatcactta 660
agcaatgtca gtgtcgcttc ctggggaaagc gccagcatcg tgtgaattct 720
acttttggcg atggcaacgc cgcgtgtctg ctatccggga aaattagtct tcaggaggca 780
agcaatgcgt tgaagcaact gcttgatgcc gtaccgggaa atcataagcg tccatcattg 840
cctgactttt tgcagaccaa tcccgcggtt ttatcaatga tgaatgacgtc attaatactc 900
aacgtctttg gtaataacgc tcaatcgtaa tgccaacagc ttgagcgggc aactgagggtg 960
caaatgcgat tacgtaataa gcaggtaaag gagtatcagg agcagatcca gaaagcgata 1020
gagcaggagg ataaagcgcg taaagcgggt atttttggcg ctatttttga ctggattacc 1080
ggcatatttg aaaccgtgat tggcgctcta aaagtgtggg aaggttttct gtccggaaat 1140
cccgcagaaa tggctagcgg cgtagcttat atggccgcag gttgtgcagg aatgggttaa 1200
gccggagccg aaacggcaat gatgtgcggt gctgaccacg atacctgtca ggcaattatt 1260
gacgtgacaa gtaagattca atttggttgt gaagccgctg cgtctggcact ggatgttttc 1320
cagattggcc gtgcttttat ggcgacgaga ggtttatctg gcgcagctgc aaaagtgtct 1380
gactccggtt ttgscgagga agtgggttag cgtatggtag gtgcagggga agcagaaaa 1440
gaggagtgtg ctgaaaagtt tggcgaagaa gtgagcgaaa gtttttccaa acaatttgag 1500
ccgcttgaa gtgaaatggc tatggcgaat gagatggcag aggaggtctg caggttttct 1560
cgtaacgtag aaaataatat gacgcgaagc gcgggaaaaa gctttacgaa agaggggggtg 1620
aaagcaatgg caaaagaagc ggcaaaagaa aatgtgtgca agaaggtgga 1680
aagttcctgt taaaaaaatt ccgtaataaa gttctcttca atatgttcaa aaaaatcctg 1740
tatgccttac tgagggtatg ttcatttaaa ggcttacagg ctatcagatg tgcaaccgag 1800
ggcgccagtc agatgaatac tggcatgggt aacacagaaa aagcgaagat cgaagagaaa 1860
atagagcaat taataactca gcaacgggtt ctggatttca taatgcaaca aacagaaaac 1920
cagaaaaaga tagaacaata acgcttagag gagctttata aggggagcgg tgccgcgctt 1980
agagatgtat tagatactat tgatcactat agtagcgttc aggcgagaat agctggctat 2040
cgcgcttaa

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SEQ ID NO: 66      moltype = AA  length = 682
FEATURE           Location/Qualifiers
REGION            1..682

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note = synthetic construct  
source 1..682  
mol\_type = protein  
organism = synthetic construct

SEQUENCE: 66

|             |             |            |             |            |             |     |
|-------------|-------------|------------|-------------|------------|-------------|-----|
| MSSGNILWGS  | QNPIVFKNSF  | GVSNADTGSQ | DDLSSQONPFA | EGYGVLLILL | MVIQAIANNK  | 60  |
| FIEVQKNAER  | ARNTOEKSNE  | MDEVIKAAK  | GDAKTKEEVP  | EDVIKMYMRD | GILIDGMTID  | 120 |
| DYMAKYGDHG  | KLDKGGQLQAI | KAALDNDANR | NTDLMSQGGI  | TIQKMSQELN | AVLTQLTGLI  | 180 |
| SKWGEISSMI  | AQKTYSELN   | RIHSNSDSAA | GVTALTHHHL  | SNVSCVSSGS | LGKRQHRVNS  | 240 |
| TFGDGNAACL  | LSGKISLQEA  | SNALKQLLDA | VPGNHKRPSL  | PDFLQTNPAV | LSMMMTSLIL  | 300 |
| NVFGNNAQSL  | CQQLERATEV  | QNALRNKQVK | EYQEQIQKAI  | EQEDKARKAG | IFGAIQDWIT  | 360 |
| GIFETVIGAL  | KVVEGFLSGN  | PAEMASGVAY | MAAGCAGMVK  | AGAETAMMCG | ADHDTQCAII  | 420 |
| DVTSKIQFGC  | EAVALLDVF   | QIGRAFMATR | GLSGAAAKVL  | DSGPGEEVVE | RMVGAGEAEI  | 480 |
| EELAEKFGE   | VSEFSKQFE   | PLEREMAMAN | EMAEAAEF    | RNVENNMTRS | AGKSFTKEGV  | 540 |
| KAMAKEAAKE  | ALEKCVQEGG  | KFLKKFRNK  | VLFNMFKKIL  | YALLRDCSFK | GLQAIKRCATE | 600 |
| GASQMMTGMV  | NTEKAKIEKK  | IEQLITQQR  | LDFIMQQTEN  | QKKIEQKRLE | ELYKSGGAAL  | 660 |
| RDVLDITIDHY | SSVQARIAGY  | RA         |             |            |             | 682 |

SEQ ID NO: 67 moltype = DNA length = 2655  
FEATURE Location/Qualifiers  
misc\_feature 1..2655  
note = synthetic construct  
source 1..2655  
mol\_type = other DNA  
organism = synthetic construct

SEQUENCE: 67

|             |             |             |             |             |             |      |
|-------------|-------------|-------------|-------------|-------------|-------------|------|
| catatggaca  | atggcgatcg  | tttataccgt  | gccgactcgc  | gtccccaga   | tgagattaaa  | 60   |
| cgtagcgggtg | gggttaatgcc | acgtggggcac | aatgagattt  | ttgaccgtgg  | aacacagatg  | 120  |
| aacattaaacc | tttacgatca  | tgcccggtgg  | accagacccg  | ggtttgcctg  | ttatgatgac  | 180  |
| gggtatgtta  | gtacgagttt  | gtccttacgc  | tccgcacacc  | ttgcgggaca  | aagtatttta  | 240  |
| tcaggtacaca | gcacatatta  | catttatgtg  | atcgccactg  | ccccaaacat  | gttcaatgtg  | 300  |
| aacgatgtgt  | tggtgggttta | cagcccccac  | ccatatgaac  | aagaagtctc  | ggcccttggg  | 360  |
| gggatcccat  | atacgcagat  | ttatggttgg  | taccgcgtaa  | atattgtgtg  | gattgatgaa  | 420  |
| cgtttgcac   | gtaacgcgtga | ataccgcgat  | cgctactacc  | gtaacttgaa  | cattgcacct  | 480  |
| gccgaggacg  | gctatcggtt  | agcgggattc  | ccaccgcgac  | atcaggcggtg | gcgtgaggaa  | 540  |
| ccgtggatcc  | atcacgcccc  | tcagggggtgc | gggaacagta  | gtcgcgggtc  | cgccggcatcc | 600  |
| atgtcttcag  | gaaacattct  | atgggggaagt | caaaacccta  | ttgtgtttaa  | aaatagcttc  | 660  |
| ggcgctcagca | acgctgtatc  | cggggagccag | gatgacttat  | cccagcaaaa  | tcctgtttgc  | 720  |
| gaaggggatg  | tgtgttttgc  | tattctcctt  | atggttatcc  | aggctatcgc  | aaataataaa  | 780  |
| tttattgaag  | ttcagaagaa  | gcgtgaacgt  | gccagaaata  | cccaggaaaa  | gtcaaatgag  | 840  |
| atggatgagg  | tgattgctaa  | agcagccaaa  | ggggatgcta  | aaaccaaaga  | ggaggtgcct  | 900  |
| gaggatgtaa  | ttaaatcatc  | cgctgataat  | ggtattctca  | tcgatgggat  | gaccattgat  | 960  |
| gattatatgg  | ctaaatattg  | cgatcatggg  | aagctgggata | aagtggtgct  | acaggcgatc  | 1020 |
| aaagcggtct  | tggtataatga | cgccaaccgg  | aataccgac   | ttatgagtca  | ggggcgagata | 1080 |
| ccaattcaaa  | aaatgtctca  | ggagcttaac  | gctgtcctta  | cccaactgac  | agggtctatc  | 1140 |
| agtaagtggg  | gggaaatttc  | cagtatgata  | gcgcagaaaa  | cgtactcaga  | gctcatgaat  | 1200 |
| cgaattcaca  | gtaatagcga  | cagcgccgca  | ggagtaaccg  | ccttaacaca  | tcatacctta  | 1260 |
| agcaatgtca  | gttgcggttc  | ctcggtttcg  | ctgggaaagc  | gccagcagtc  | tgtgaattct  | 1320 |
| acttttggcg  | atggcaacgc  | cgcgtgtctg  | ctatccggga  | aaattagtct  | tcaggaggca  | 1380 |
| agcaatcggt  | tgaagcaact  | gcttgatgcc  | gtaccgggaa  | atcataagcg  | tcctcattg   | 1440 |
| cctgactttt  | tgacagccaa  | tcccgcggtt  | ttatcaatga  | tgatgacgtc  | attaatactc  | 1500 |
| aacgtctttg  | gtaataacgc  | tcaatcgtta  | tgccaacagc  | ttgagcgggc  | aactgagggtg | 1560 |
| caaaatgcac  | tacgtataaa  | gcaggtaaa   | gagtatcagg  | agcagatcca  | gaaagcgata  | 1620 |
| gagcaggagg  | ataaagcgcg  | taaagcgggt  | atttttggcg  | ctatttttga  | ctggattacc  | 1680 |
| ggcatatttg  | aaaccggtat  | tgccgcctta  | aaagtgtgtg  | aaggttttct  | gtccggaaat  | 1740 |
| ccgcagaaaa  | tggttagcgg  | cgtagcttat  | atggccgcag  | gttgtgcagg  | aatgggttaa  | 1800 |
| gccggagccg  | aaacggcaat  | gatgtgcggt  | gctgaccacg  | atacctgtca  | ggcaattatt  | 1860 |
| gacgtgacaa  | gtaagattca  | atttggttgt  | gaagccgtcg  | cgtgggcact  | ggatgttttc  | 1920 |
| cagattggcc  | gtgcttttat  | ggcgacgaga  | ggtttatctg  | gcgcagctgc  | aaaagtgtct  | 1980 |
| gactccggtt  | ttggcgaggga | agtgggttag  | cgtatggtag  | gtgcaggggga | agcagaaata  | 2040 |
| gaggagttgg  | ctgaaaagtt  | tgccgaagaa  | gtgagcgaaa  | gtttttccaa  | acaatttgag  | 2100 |
| ccgcttgaa   | gtgaaatggc  | tatggcgaa   | gagatggcag  | aggaggctgc  | cgagttttct  | 2160 |
| cgtaacgtag  | aaaataatat  | gacgcgaagc  | gcgggaaaaa  | gctttacgaa  | agaggggggtg | 2220 |
| aaagcaatgg  | caaaagaagc  | ggcaaaagaa  | gccctggaaa  | aatgtgtgca  | agaaggtgga  | 2280 |
| aagttcctgt  | taaaaaaatt  | ccgtaataaa  | gttctctcca  | atatgttcaa  | aaaaatcctg  | 2340 |
| tatgccttac  | tgagggatgt  | ttcatttaaa  | ggcttacagg  | ctatcagatg  | tgcaaccgag  | 2400 |
| ggcgccagtc  | agatgaatac  | tggtcatggt  | aacacagaaa  | aagcgaagat  | cgaaaagaaa  | 2460 |
| atagagcaat  | taataactca  | gcaacggttt  | ctggatttca  | taagtgaaca  | aacagaaaaa  | 2520 |
| cagaaaaaga  | tagaacaata  | acgcttagag  | gagctttata  | aggggagcgg  | tgccgcgctt  | 2580 |
| agagatgtat  | tagataccat  | tgatcactat  | agtagcggtc  | aggcgagaat  | agctggctat  | 2640 |
| cgcgcttaac  | tcgag       |             |             |             |             | 2655 |

SEQ ID NO: 68 moltype = AA length = 881  
FEATURE Location/Qualifiers  
REGION 1..881  
note = synthetic construct

-continued

```

source          1..881
                 mol_type = protein
                 organism = synthetic construct

SEQUENCE: 68
MDNGDRLYRA DSRPPDEIKR SGGMLPRGHN EYFDRGTQMN INLYDHARGT QTGFVRYDDG 60
YVSTSLSLRS AHLAQQSILS GYSTYYIYVI ATAPNMFNVN DVLGVYSPHP YEQEVSA LGG 120
IPYSQIYGWY RVNFGVIDER LHRNREYRDR YYRNLNIAPA EDGYRLAGFP PDHQA WREEP 180
WIHHAPQGC G NSSRGSAA SM SSGNILWGSQ NPIVFKNSFG VSNADTGSQD DLSQQNPFAE 240
GYGVLLILLM VIQAIANNKF IEVQKNAERA RNTQEKSNEM DEVIKAAKAG DAKTKEEVPE 300
DVIKYM RDNG ILIDGMTIDD YMAKYGDHGK LDKGGLQAIK AALDNDANRN TDLMSQQTIT 360
IQKMSQELNA VLTQLTGLIS KWGEISSMIA QKTYSELMNR IHSNSDSAAG VTALTHHHLS 420
NVSCVSSGSL GKRQHRVNST FGDGNAACLL SGKISLQEAS NALKQLLDVAV PGNHKRPSLP 480
DFLQTNPAVL SMMMTSLILN VFGNNAQSLC QQLERATEVQ NALRNKQVKE YQEQIQKAI E 540
QEDKARKAGI FGAIFDWITG IFETVIGALK VVEGFLSGNP AEMASGVAYM AAGCAGMVKA 600
GAETAMMCGA DHDTCQAIID VTSKI QFGCE AVALALDVFO IGRAFMATRG LSGAAAKVLD 660
SGFGEEVVER MVGAGEAIE ELAEKFGEEV SESFSKQFEP LEREMAMANE MAEEAAEF SR 720
NVENNMTRSA GKSFTKEGVK AMAKEAAKEA LEKCVQEGGK FLLKKFRNKV LFNMFKKILY 780
ALLRDCSPFG LQARCATGEG ASQMNMTGMVN TEKAKIEKKI BQLITQQRFL DFIMQQTENQ 840
KKIEQKRLEE LYKGS GAALR DVLDTIDHYS SVQARIAGYR A 881

SEQ ID NO: 69      moltype = DNA length = 1254
FEATURE            Location/Qualifiers
misc_feature        1..1254
                    note = synthetic construct
source              1..1254
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 69
atgggcagca gccatcatca tcatcatcac agcagcggcc tgggtgccg cggcagccat 60
atggacaatg gcgatcggtt ataccgtgcc gactcgcgtc cccagatga gattaaacgt 120
agcgggtgggt taatgccacg tgggcacaat gagtattttg accgtggaac acagatgaac 180
attaaaccctt acgcatcatgc cegtgggacc cagaccgggt ttgtccgtta tgatgacggg 240
tatgttagta cgagtttgct cttacgctcc gcacaccttg cgggacaaag tattttatca 300
ggctacagca catattacat ttatgtgatc gccactgccc caaacatgtt caatgtgaac 360
gatgtgttgg gggtttacag cccccatcca tatgaacaag aagtctcggc ccttgggggg 420
atcccatata gccagattta tggttgggtac cgcgtaaat ttggtgtgat tgatgaacgt 480
ttgcatcgta acagtgaata ccgcgatcgc tactaccgta acttgaacat tgcacctgcc 540
gaggacggct atcgtttagc gggattccca cccgatcatc aggcgtggcg tgaggaaacc 600
tggtatccatc acgcccctca ggggtgcggg aacagtagtc gcgggtccgc ggcatccatg 660
tcttcaggaa acatcttatg ggggaagtcaa aaccctattg tgtttaaaaa tagcttcggc 720
gtcagcaacg ctgataccgg gagccaggat gacttatccc agcaaaatcc gtttgccgaa 780
gggtatggtg ttttgcttat tctccttatg gttattcagg ctatcgcaa taataaattt 840
attgaagtcg agaagaaacg tgaacgtgcc agaaataacc aggaaaagtc aaatgagatg 900
gatgaggtga ttgctaaagc agccaaaggg gatgctaaaa ccaaagagga ggtgcctgag 960
gatgtaatta aatacatgcg tgataatggt attctcatcg atggtagac cattgatgat 1020
tatatggcta aatatggcga tcatgggaag ctggataaag gtggcctaca ggcgatcaaa 1080
gcggcttttg ataatgacgc caaccggaat accgatctta tgagtcaggg gcagataaca 1140
attcaaaaaa tgtctcagga gcttaacgct gtccttacc aactgacagg gcttatcagt 1200
aagtgggggg aaatttccag tatgatagcg cagaaaacgt actcataagg atcc 1254

SEQ ID NO: 70      moltype = AA length = 415
FEATURE            Location/Qualifiers
REGION              1..415
                    note = synthetic construct
source              1..415
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 70
MGSSHHHHHH SSGLVPRGSH MDNGDRLYRA DSRPPDEIKR SGGMLPRGHN EYFDRGTQMN 60
INLYDHARGT QTGFVRYDDG YVSTSLSLRS AHLAQQSILS GYSTYYIYVI ATAPNMFNVN 120
DVLGVYSPHP YEQEVSA LGG IPYSQIYGWY RVNFGVIDER LHRNREYRDR YYRNLNIAPA 180
EDGYRLAGFP PDHQA WREEP WIHHAPQGC G NSSRGSAA SM SSGNILWGSQ NPIVFKNSFG 240
VSNADTGSQD DLSQQNPFAE GYGVLLILLM VIQAIANNKF IEVQKNAERA RNTQEKSNEM 300
DEVIKAAKAG DAKTKEEVPE DVIKYM RDNG ILIDGMTIDD YMAKYGDHGK LDKGGLQAIK 360
AALDNDANRN TDLMSQQTIT IQKMSQELNA VLTQLTGLIS KWGEISSMIA QKTY S 415

SEQ ID NO: 71      moltype = DNA length = 749
FEATURE            Location/Qualifiers
misc_feature        1..749
                    note = synthetic construct
source              1..749
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 71
atgggcagca gccatcacca tcatcaccac agccaggatc cgatgagcac tccatcttct 60
aataattcta aaaaaccttc ggcctctttt aataaaaaa cactagaccg cttggccgag 120

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attgtgtcac aaaaaaaagc aaaagctgag gatttggaaac aaaaatatcc tgttctctacg 180
gaagaggaga caaaacaagt tctcatggac atcctacagg ggtaagcaa cggattaact 240
cttcagcaaa ttttaggtct ctccgacgtc ctccctgaag agatctacac cgtagcatat 300
accttctact cccaaggga atacgggaa gctatcggtc ttttccaaat cttacagacc 360
tccaaacctc aatgtacaa atacatctta ggtcttagct cttgctatca ccagctaaaa 420
atgtatgatg aagcgcgttt tggtttcttc cttagctttcg atgctcaacc cgaaaacccc 480
atccctcctt actacatcgc cgatagcttg atgaagctaa accaaccgga agaactctca 540
gaacttctcg atattacgat cgatatgtgt aagaacaagc cggaatataa agttctttaa 600
gatcgtgtga gcattatgaa gcaatcttta gatgccgtgc tgaaaaaaga gaaatctgca 660
aaaggctctg aaacacaagc ctctctcct aaaaacacaa agctaaaaa agctgcttct 720
aacaagaaaa aagcaaaagta agcgccgcg

```

SEQ ID NO: 72                   moltype = AA   length = 246  
FEATURE                        Location/Qualifiers  
REGION                         1..246  
                               note = synthetic construct  
source                         1..246  
                               mol\_type = protein  
                               organism = synthetic construct

SEQUENCE: 72

```

MGSSHHHHHH SQDPMSTPSS NNSKKPSASF NKKSRSLAE IAAQKKAKAE DLEQKYPVPT 60
EETKQVLMD ILQGLSNGLT LQQLGLSDV LLEIYTVAY TFYSQGKYRE AIGLFQILTA 120
SKPQCYKYIL GLSSCYHQLK MYDEAAFQFP LAFDAQPENP IPPYYIADSL MKLNQPEESQ 180
DFLDITIDMC KNKPEYKVLK DRCSIMKQSL DAVLKKEKSA KGSETQASSP KNTKAKKAAS 240
NKKKAK

```

SEQ ID NO: 73                   moltype = DNA   length = 441  
FEATURE                        Location/Qualifiers  
source                         1..441  
                               mol\_type = genomic DNA  
                               organism = Chlamydia spp

SEQUENCE: 73

```

aaaagtgtgagc gtttaaaaaa attagaatca gagcttcgat atcttaccga gtggatgcaa 60
cttggccttg ttcctaaaaa agaaatcgag agacaccagg aagaaatccg tctgctagaa 120
agcaaaatcc ttgaagagaa agaactgcta caacttctca aagaaagcgg tgagatcaaa 180
gagtacgtaa cccctcgaa aactccagct aaaaccattt acccagatgg ccccgagcgt 240
tcagacgttg agttgttaga atcctcggat acagaagtgg atctcgatgc cggtagacaa 300
attgagattg acctagtgta tgaggcaaga gaagaaagcg gaaacgaact cgactactct 360
agtgaagacg atgaggatcc tttcagcgat cgcaatcgtt ggcgcggagg aggcacata 420
gatcctgacg cgaatgaatg g

```

SEQ ID NO: 74                   moltype = AA   length = 148  
FEATURE                        Location/Qualifiers  
source                         1..148  
                               mol\_type = protein  
                               organism = Chlamydia spp

SEQUENCE: 74

```

MKSERLKKLE SELHDLTQWM QLGLVPKKEI ERHQEEIRLL ESKILEEKER LQLLKESGEI 60
KEYVTPRRTP AKTIYPDGPS VSDVEFVSS DTEVDLDAGD TIEDLDGEA REESGNELDY 120
SSEDDDPFS DRNRWRRGGI IDPDANEW

```

SEQ ID NO: 75                   moltype = DNA   length = 1464  
FEATURE                        Location/Qualifiers  
source                         1..1464  
                               mol\_type = genomic DNA  
                               organism = Chlamydia spp

SEQUENCE: 75

```

atgagcttgt catccagcag cagctcggat agttcgaatc tgaaaaatgt gttatctcag 60
gtcatcgcgt ctacaccaca ggggggttct aatgtgtaca aattaaccga caatcaggta 120
aaacaagtcc agcagaccgg tcaaaaccgt gatgatctgt ccatggagag cgacgtcgcg 180
gtggcggaag cagccggaag agatcgtgct gcgtcggcgt ccagatcga gggacaagag 240
ctgattgagc aacagggact tgcggctggg aaagagacgg cttctgctga tgctacatca 300
ttgacccagt cggcatccaa aggcgcttcc agtcagcagt gtattgagga taccagtaag 360
tccctggagc tttcttcgtt ttcgagcctg tcaagcgtag atgcgacaca ttgacaggaa 420
atccaatcga tcgtgtcttc agcaatgggc gccaccaacg aattgtcatt gacgaactta 480
gagacaccgg gattaccaaa gccgagtacc actccacgcc aggaagtat ggagatcagc 540
cttgccctag cgaaggccat cactgcattg ggtgagagca ctcagggtgc cttggaaaat 600
tttcagttca ctcagagtcg gtcgcggaac atgaataaga tgagtttgga atcccaaggc 660
tgaaaaatcg acaaggagcg tgaagaattt aagaaaatgc aggagattca gcaaaagagc 720
ggcacaaatt caaccatgga tactgtgaat aaagtattga ttggcgtgac agtggcaatt 780
acagtaattc ctgtgttttc agcattgttt acctgcggtt tgggcttgat tggcacagcc 840
gtcgcgggtg ccacagcgcg caccgctggg gcaacggccg ccgccaagc cgtacacct 900
gtgacgacca cagtcgtac ccaggtagc atgcaagcgg tggccaagt cgttaagcag 960
gctattatcc aagcagtaaa acgcgccatc gtccaagcga ttaaacaggg gattaagcaa 1020
ggcattaaac aagcgatcaa acaggcagtc aaggcaagcg tgaagacact tgccaaaaat 1080
gtaggcaaga ttttcagcgc aggcgaagaac gctgtgagta agtccttccc aaaatgtct 1140

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aagggtgatta atacacttgg ttccaaatgg gttactcttg gcgtgggggc ccttacagcg 1200
gtgccgcagtg tagtcagtggt cattacctcc cttcaattgt ctgatatgca aaaagaactt 1260
gcacaaatcc aaaaggaagt ggggtgcactt acggcgcgaga gtgagatgat gaaagcgttt 1320
acactgttct ggcagcaagc ttcgaaaatc gcggccaaac aaacggaatc accttcagag 1380
acgcaacaac aggcagctaa gaccggcgcc cagatcgcta aagcgttgct cgccatttcg 1440
ggtgcttttag ctgctgctgc ttag
1464

```

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SEQ ID NO: 76      moltype = AA  length = 487
FEATURE           Location/Qualifiers
source            1..487
                  mol_type = protein
                  organism = Chlamydia spp

```

```

SEQUENCE: 76
MSLSSSSSSD SSNLKNVLSQ VIASTPQGVN NADKLTNDQV KQVQQTRQNR DDLMSMEDVA 60
VAGTAGKDRS ASASQIEGQE LIEQQGLAAG KETASADATS LTQSASKGAS SQQCIEDTSK 120
SLELSSLSL SSV DATHLQE IQSIVSSAMG ATNELSLTNL ETPGLPKPST TPRQEVMEIS 180
LALAKAITAL GESTQAALEN FQSTQSQSAN MNKMSLESQG LKIDKEREFF KKMQEIQQKS 240
GTNSTMDTVN KVMIGVTVAI TVISVVSALF TCGGLIGLTA AAGATAATAG ATAAATTATS 300
VTTTATQVT MQAVVQVVKQ AIIQAVKRAI VQAIKQGIKQ GIKQAIKQAV KASVKTLAKN 360
VGKIFSAGKN AVSKSFPKLS KVINTLGSKW VTLGVGALTA VPQLVSGITS LQLSDMQKEL 420
AQIQKEVGAL TAQSEMMKAF TLFWQQASKI AAKQTESPSE TQQQAAKTGA QIAKALSAIS 480
GALAAAA
487

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SEQ ID NO: 77      moltype = DNA  length = 1920
FEATURE           Location/Qualifiers
misc_feature       1..1920
note = synthetic construct
source            1..1920
                  mol_type = other DNA
                  organism = synthetic construct

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```

SEQUENCE: 77
aaaagtgagc gtttaaaaaa attagaatca gagcttcacg atcttaccca gtggatgcaa 60
cttgcccttg ttccataaaa agaaatcgag agacaccagg aagaaatccg tctgctagaa 120
agcaaaaatcc ttgaagagaa agaactgtct caacttctca aagaaagcgg tgagatcaaa 180
gagtagcgtaa cccctcgaaag aactccagct aaaaccattt acccagatgg ccccgagcgt 240
tcagacgttg agtttgtaga atcctcggat acagaagtgg atctcgatgc cggtagacaca 300
attgagattg acctaggtga tgaggcaaga gaagaaagcg gaaacgaaat cgactactct 360
agtgaagacg atgaggatcc ttccagcgat cgcaatcggt ggccgcccagg aggcacata 420
gatcctgacg cgaatgaatg ggggttcagct gcttcaatga gcttgctcat cagcagcagc 480
tcggatagtt cgaatctgaa aaatgtgtta tctcaggtca tcgctgtctac accacagggg 540
gttctcaatg tgcacaaatt aaccgacaat caggtaaaac aagtcacaga gaccgcgtcaa 600
aacggtgatg atctgtccat ggagagcgac gtccgcggtg cggaacacag cggaagaagat 660
cgtgctgcgt cgccgtccca gatcgaggga caagagctga ttgagcaaca gggacttgcg 720
gctgggaaag agacggcttc tgctgatgct acatcattga cccagtcggc atccaaaggg 780
gcttccagtc agcagtgatg tgaggatacc agtaagtcct tggagctttc ttcgctttcg 840
agcctgtcaa gcgtagatgc gacacatttg caggaaatcc aatcgatcgt gtcttcagca 900
atggggcgcca ccaacgaatt gtcattgacg aacttagaga caccgggatt accaaagcgg 960
agtaccactc cagcccgagg agttatggag atcagccttg ccttagcgaa ggccatcact 1020
gcattgggtg agagcactca ggtgccttg gaaaattttc agtccactca gagtcagtc 1080
gccaacatga ataagatgag ttggaatcc caaggtctga aaatcgacaa ggagcgtgaa 1140
gaatttaaga aaatgcagga gattcagcaa aagagcggca caaattcaac catggatact 1200
gtgaataaag ttatgattgg cgtgacagtg gcaattacag taatctctgt tgtttcagca 1260
ttgtttacct gcggtttggg cttgattggc acagccgctg cgggtgccac agccgccacc 1320
gctggggcaa cgcccgccgc caccgaccgt acctctgtga cgaccacagt cgctaccag 1380
gtgacgatgc aagcgggtgt ccaagtcggt aagcaggcta ttatccaaag agtaaaacgc 1440
gcatcgtcc aagcgattaa acaggggatt aagcaaggca ttaaaacagc gatcaaacag 1500
gcagtcgaag caagcgtgaa gacacttgcc aaaaatgtag gcaagatttt cagcgcagcg 1560
aagaacgctg tgagtaagtc cttcccaaaa ttgtctaagg tgattaatac acttggttcc 1620
aaatgggtta ctcttgccgt gggggccctt acagcggtgc cgcagttagt cagtggcatt 1680
acctcccttc aattgtctga tatgcaaaaa gaacttgcac aaatccaaaa ggaagtgggt 1740
gcacttacgg ccagagattg gatgatgaaa gcgtttacac tgtctgggca gcaagcttcg 1800
aaaatcgcg ccaaacaaac ggaatcacct tcagagacgc aacaacaggg agctaagacc 1860
ggcgcccaga tcgctaaagc gttgtccgcc attcgggtg ctttagctgc tgctgcttag 1920

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```

SEQ ID NO: 78      moltype = AA  length = 640
FEATURE           Location/Qualifiers
REGION           1..640
note = synthetic construct
source            1..640
                  mol_type = protein
                  organism = synthetic construct

```

```

SEQUENCE: 78
MKSERLKKLE SELHDLTQWM QLGLVPKKEI ERHQEEIRLL ESKILEEKER LQLLKESGEI 60
KEYVTPRRTP AKTIYDPGPS VSDVEFVSS DTEVDLDAGD TIEIDLGDEA REESGNELDY 120
SSEDEDPFS DRNRWRGQI IDPDANEWGS AASMSLSSSS SSDSSNLKNV LSQVIASSTPQ 180
GVFNADKLTND NQVKQVQQTR QNRDDLMSMES DVAVAGTAGK DRAASASQIE GQELIEQQGL 240

```

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|            |            |            |            |            |            |     |
|------------|------------|------------|------------|------------|------------|-----|
| AAGKETASAD | ATSLTQSASK | GASSQQCIED | TSKSLELSSL | SSLSSVDATH | LQEIQSIVSS | 300 |
| AMGATNELSL | TNLETPGLPK | PSTTPRQEV  | EISLALAKAI | TALGESTQAA | LENFQSTQSQ | 360 |
| SANMNKMSLE | SQGLKIDKER | EEFKMQEIQ  | QKSGTNSTMD | TVNKVMIGVT | VAITVISVVS | 420 |
| ALFTCGGLGI | GTAAGATAA  | TAGATAAATT | ATSVTTTVAT | QVTMQAVVQV | VKQAIQAVK  | 480 |
| RAIVQAIKQG | IKQGIKQAIK | QAVKASVKTL | AKNVGKIFSA | GKNAVSKSFP | KLSKVINTLG | 540 |
| SKWVTLGVGA | LTAVPQLVSG | ITSLQLSDMQ | KELAQIQKEV | GALTAQSEMM | KAFTLFWQQA | 600 |
| SKIAAKQTES | PSETQQQAAK | TGAQIAKALS | AISGALAAAA |            |            | 640 |

SEQ ID NO: 79                   moltype = DNA   length = 2517  
 FEATURE                   Location/Qualifiers  
 misc\_feature               1..2517  
                              note = synthetic construct  
 source                    1..2517  
                              mol\_type = other DNA  
                              organism = synthetic construct

SEQUENCE: 79

|             |             |             |            |             |             |      |
|-------------|-------------|-------------|------------|-------------|-------------|------|
| catatggaca  | atggcgatcg  | tttataccgt  | gccgactcgc | gtcccccaga  | tgagattaaa  | 60   |
| cgtagcgggtg | gggttaatgcc | acgtggggcac | aatgagattt | ttgaccgtgg  | aacacagatg  | 120  |
| aacattaacc  | tttacgatca  | tgcccggtggg | accagaccgg | gggttggtccg | ttatgatgac  | 180  |
| gggtatgtta  | gtacgagttt  | gtccttacgc  | tccgcacacc | ttgcgggaca  | aagtatttta  | 240  |
| tcaggctaca  | gcacatatta  | cattttatgtg | atcgccactg | ccccaaacat  | gttcaatgtg  | 300  |
| aacgatgtgt  | tgggggttta  | cagcccccac  | ccatatgaac | aagaagtctc  | ggccctgggg  | 360  |
| gggatcccat  | atagccagat  | ttatgggttg  | taccgcgtaa | attttggtgt  | gattgatgaa  | 420  |
| cgtttgcatc  | gtaaccgtga  | ataccgcgat  | cgctactacc | gtaacttgaa  | cattgcacct  | 480  |
| gccgaggacg  | gctatcgttt  | agcgggattc  | ccaccgcgat | atcaggcggtg | gcgtgaggaa  | 540  |
| ccgtggatcc  | atcacgcccc  | tcagggggtgc | gggaacagta | gtcgccatat  | gaaaagtggag | 600  |
| cgtttaaaaa  | aattagaatc  | agagcttcat  | gatcttacct | agtgatgca   | acttgccctt  | 660  |
| gttcctaaaa  | aagaaatcga  | gagacaccag  | gaagaaatcc | gtctgctaga  | aagcaaaatc  | 720  |
| cttgaaagaga | aagaacgtct  | acaacttctc  | aaagaaagcg | gtgagatcaa  | agagtacgta  | 780  |
| acccctcgaa  | gaactccagc  | taaaaccatt  | taccagatg  | gccccagcgt  | ttcagacggt  | 840  |
| gagtttgtag  | aatcctcgga  | tacagaaagt  | gatctcgatg | ccggtgacac  | aattgagatt  | 900  |
| gacctagggtg | atgaggcaag  | agaagaaagc  | ggaaacgaac | tcgactactc  | tagtgaagac  | 960  |
| gatgaggatc  | ctttcagcga  | tcgcaatcgt  | tggcgccgag | gaggcatcat  | agatccctgac | 1020 |
| gcgaatgaat  | gggggttcagc | tgcttcaatg  | agcttgctat | ccagcagcag  | ctcgatagtt  | 1080 |
| tcgaatctga  | aaaatgtgtt  | atctcaggtc  | atcgcgctca | caccacaggg  | ggttcctaat  | 1140 |
| gctgacaaat  | taaccgacaa  | tcaggtaaaa  | caagtccagc | agaccgcgtca | aaaccgtgat  | 1200 |
| gatctgtcca  | tggagagcga  | cgtcgcggtg  | gcgggaacag | ccggaagaaga | tcgtgctgcg  | 1260 |
| tcggcgctccc | agatcgaggg  | acaagagctg  | attgagcaac | agggacttgc  | ggctggggaaa | 1320 |
| gagacgggctt | ctgctgatgc  | tacatcattg  | accagtcgg  | catccaaagg  | cgcttccagt  | 1380 |
| cagcagtgta  | ttgaggatac  | cagtaagtcc  | ctggagcttt | cttcgctttc  | gagcctgtca  | 1440 |
| agcgtagatg  | cgacacattt  | gcaggaaatc  | caatcgatcg | tgtcttcagc  | aatgggcgcc  | 1500 |
| accaaagcaat | tgtcattgac  | gaacttagag  | acaccgggat | taccaaagcc  | gagtaccact  | 1560 |
| ccacgccagg  | aagttatgga  | gatcagcctt  | gccttagcga | aggccatcac  | tgcatgggtg  | 1620 |
| gagagcactc  | aggctgcctt  | ggaaaatttt  | cagtccactc | agagtcagtc  | cgcgaaacatg | 1680 |
| aaaaagatga  | gtttggaatc  | ccaaggcttg  | aaaatcgaca | aggagcgtga  | agaatttaag  | 1740 |
| aaaatgcagg  | agattcagca  | aaagagcgcc  | acaaattcaa | ccatggatag  | tgtgaataaa  | 1800 |
| gttatgattg  | gcgtgacagt  | ggcaattaca  | gtaatctctg | ttgtttcagc  | attgtttacc  | 1860 |
| tcgggtttgg  | gcttgattgg  | cacagccgct  | gcgggtgcca | cagcccgccac | cgctggggca  | 1920 |
| acggccgccc  | ccacgcccgc  | tacctgtgtg  | acgaccacag | tcgctaccga  | ggtgacgatg  | 1980 |
| caagcgggtg  | tccaagtcgt  | taagcaggct  | attatccaa  | cagtaaaacg  | cgccatcgct  | 2040 |
| caagcgatta  | aacaggggat  | taagcaaggc  | attaacaacg | cgatcaaaac  | ggcagtcacg  | 2100 |
| gcaagcgtga  | atagcacaac  | caaaaatgta  | ggcaagattt | tcagcgcagg  | caagaacgct  | 2160 |
| gtgagtaagt  | ccctcccaaa  | attgtctaag  | gtgattaata | cacttggttc  | caaatgggtt  | 2220 |
| actcttgccg  | tgggggcccc  | tacagcggtg  | ccgcagttag | tcagtggcat  | tacctccctt  | 2280 |
| caattgtctg  | atatgcaaaa  | agaacttgca  | caaatccaaa | aggaagtggg  | tgcaacttacg | 2340 |
| gcgcagagtg  | agatgatgaa  | agcgtttaca  | ctgttctggc | agcaagcttc  | gaaaatcgcg  | 2400 |
| gccaacaaaa  | cggaatcacc  | ttcagagacg  | caacaacagg | cagctaagac  | cggcgcccag  | 2460 |
| atcgctaaag  | cggtgtccgc  | catttcgggt  | gctttagctg | ctgctgctta  | gctcgag     | 2517 |

SEQ ID NO: 80                   moltype = AA   length = 834  
 FEATURE                   Location/Qualifiers  
 REGION                    1..834  
                              note = synthetic construct  
 source                    1..834  
                              mol\_type = protein  
                              organism = synthetic construct

SEQUENCE: 80

|            |            |            |            |            |             |     |
|------------|------------|------------|------------|------------|-------------|-----|
| MDNGDRLYRA | DSRPPEIKR  | SGGLMPRGHN | EYFDRGTQMN | INLYDHARGT | QTGFVRYDDG  | 60  |
| YVSTSLSLRS | AHLAQQSILS | GYSYYIYVI  | ATAPNMFNVN | DVLGVYSPHP | YEQEVSAALGG | 120 |
| IPYSQIYGWY | RVNPGVIDER | LHRNREYDR  | YRNLNIAPA  | EDGYRLAGFP | PDHQAWREEP  | 180 |
| WIHHAPQGGC | NSSRMKSERL | KKLESELHDL | TQWMLGLVP  | KKEIERHQEE | IRLLESKILE  | 240 |
| EKERLQLLKE | SGEIKYVTP  | RRTPAKTIYP | DGPSVSDVEF | VESDTEVDL  | DAGDTIEIDL  | 300 |
| GDEAREESGN | ELDYSSDEDE | DPFSDRNRR  | RGGIIDPDAN | EWGSAASMSL | SSSSSDSSN   | 360 |
| LKNVLSQVIA | STPQGVNAD  | KLTDNQVKQV | QOTRONRDDL | SMESDVAVAG | TAGKDRAASA  | 420 |
| SQIEGQELIE | QQGLAAGKET | ASADATSLTQ | SASKGASSQQ | CIEDTSKSLE | LSSLSSLSSV  | 480 |
| DATHLQEIQS | IVSSAMGATN | ELSLTNLETP | GLPKPSTTPR | QEVMEISLAL | AKAITALGES  | 540 |

```

SEQ ID NO: 81      moltype = DNA   length = 749
FEATURE            Location/Qualifiers
misc_feature       1..749
                   note = synthetic construct
source             1..749
                   mol_type = other DNA
                   organism = synthetic construct

```

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SEQ ID NO: 82      moltype = AA  length = 246
FEATURE            Location/Qualifiers
REGION            1..246
                  note = synthetic construct
source            1..246
                  mol_type = protein
                  organism = synthetic construct
```

```
SEQ ID NO: 83      moltype = DNA  length = 666
FEATURE           Location/Qualifiers
source            1..666
                  mol_type = genomic DNA
                  organism = Chlamydia spp
```

```
SEQ ID NO: 84      moltype = AA   length = 222
FEATURE            Location/Qualifiers
source             1..222
                   mol_type = protein
                   organism = Chlamydia spp
```

```
SEQ ID NO: 85      moltype = DNA length = 2145
FEATURE            Location/Qualifiers
misc_feature       1..2145
                   note = synthetic construct
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-continued

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source          1..2145
                 mol_type = other DNA
                 organism = synthetic construct

SEQUENCE: 85
atagatcctc ttaagctttt tccaaatttt gatggggata aggagagtgc tgcgggtgaat 60
aaaccttcag catctcctat gccacgcgaa ttaagtaaaa atgttgccctc attctcttta 120
gggggtggag gtgctgcggt ggattcgaca gtgtccacag aaaagctatc gttgatggct 180
atgatgcagg ataaaaattc gcagttgatc gatcctgagt tggaggaagc tctgaactct 240
gaagagttac aagagcagat ccatttggtt aaaagtcggt tgtgggagtc acaaacgcag 300
atgcaaatgc aagatcccca caagttggcc tctgagcatg tagatgcttt aggagtcatt 360
gttgatttaa tcaatgggga ttttcaagcg atagctgaac atacacaaca gacggtcagg 420
cagggtaatg gtgacgaaga aaaatctggt acacgcaaga tagtcgattg ggtctcttca 480
ggagaagaaa ttttgaatcg tgccttggtg tatttctccg atcgtaatgg agaaagagaa 540
acattagccg atttctttaa agttcagtat gccgttcaaa gagctacaca acgcgccgag 600
ttatttgcca gtattctagg tgcacgggtg agtagtgtaa aaacgattat gacaacccag 660
ttagggtggt cagctgcttc aatgagcttg tcatccagca gcagctcgga tagttcgaat 720
ctgaaaaatg tcttatctca ggtcatcgcg tctacaccac aggggggttc taatgctgac 780
aaattaaccc acaatcaggt aaacaagtc cagcagaccc gtcaaaaccc tgatgatctg 840
tccatggaga gcgacgtcgc ggtggcggga acagccggaa aagatcgctg tgcgtcggcg 900
tcccagatcg agggacaaga gctgattgag caacagggac ttgctgctgg gaaagagacg 960
gcttctgctg atgctacatc attgacccag tcggcatcca aaggcgcttc cagtcagcag 1020
tgtattgagg ataccagtaa gtcctgggag ctttcttccg tttcagacct gtcaagcgta 1080
gatgcgacac atttgcagga aatccaatcg atcgtgtctt cagcaatggg cgccaccaac 1140
gaattgtcat tgacgaactt agagacaccg ggattaccaa agccgagtag cactccaccg 1200
caggaagtta tggagatcag ccttgcctta gcgaaggcca tcaactgatt gggtagagac 1260
actcaggtcg ccttggaaaa ttttcagtc actcagagtc agtccgcgaa catgaataag 1320
atgagtttgg aatcccaagg cttgaaaaat gacaaggagc gtgaagaatt taagaaaaatg 1380
caggagattc agcaaaaagag cgccacaatg tcaacctagg atactgtgaa taaagtatat 1440
attggcgtga cagtggtcaat tacagtaatc tctgtgtttt cagcattggt taacctcggt 1500
ttgggtctga ttggcacagc cgtcgcgggt gccacagccg ccaccgctgg ggcaacggcc 1560
gccgccacga ccgtacacctc tgtgacgacc acagtcgcta ccagggtgac gatgcaagcg 1620
gtggtccaaag tcgttaagca ggctattatc caagcagtaa aacgcgccat cgtccaagcg 1680
attaacacgg ggattaaagca aggcattaaa caagcgatca aacagggcag caaggcaagc 1740
gtgaagacac ttgccaaaata ttaggcaagc attttcagcg caggcaagaa cgctgtgagt 1800
aagtccttcc caaaattgtc taagtgattt aatacacttg gttccaaatg gggtactctt 1860
ggcgtggggg cccttacagc ggtgcgcgag ttagtcagtg gcattacctc ccttcaattg 1920
tctgatatgc aaaaagaaatc tgacaaaatc caaaagggaag tgggtgcact tacggcgcgag 1980
agtgaagtga tgaagcggtt tacactgttc tggcagcaag cttcgaaaaa cgcggccaaa 2040
caaacgggaat caccttcaga gacgcaacaa caggcagcta agaccggcgc ccagatcgct 2100
aaagcgttgt ccgccatttc ggggtgctta gctgctgctg cttag 2145

```

```

SEQ ID NO: 86      moltype = AA length = 715
FEATURE           Location/Qualifiers
REGION            1..715
                  note = synthetic construct
source            1..715
                  mol_type = protein
                  organism = synthetic construct

```

```

SEQUENCE: 86
MIDPLKLPFN FDGDKESAAV NKPSASPMPS ELSKNVASFS LGGGGAALDS TVSTEKLSLM 60
AMMQDKNSQL IDPLEEALN SEELQEQIHL LKSLRLWDAQT QMQMQDPDKL ASEHVDALGV 120
IVDLINGDFQ AIAEHTQQT VTRKIVDWVS SGEEILNRL LYFSDRNGER 180
ETLADFLKVQ YAVQRATQRA ELFASILGAT VSSVKTIMTT QLGGSAASMS LSSSSSSDSS 240
NLKNVLSQVI ASTPQGVFNA DKLTNDQVKQ VQQTRQNRDD LSMESDVAVA GTAGKDRAAS 300
ASQIEGQELI EQQLAAGKE TASADATSLT QSASKGASSQ QCIEDTSKSL ELSSLSLSLS 360
VDATHLQEIQ SIVSSAMGAT NELSLTNLET PGLPKPSTTP RQEVMEISLA LAKAITALGE 420
STQAALENFQ STQSQSANMN KMSLESQGLK IDKEREFPK MQEIQQKSGT NSTMDTVNKV 480
MIGVTVAITV ISVVSALFTC GLGLIGTAAA GATAATAGAT AAATTATSVT TTVATQVTMQ 540
AVVQVVQKAI IQAVKRAIVQ AIKQGIKQGI KQAIKQAVKA SVKTLAKNVG KIFSAGKNAV 600
SKSFPKLSKV INTLGSKWVT LGVGALTAVP QLVSGITSLQ LSDMQKELAQ IQKEVGALTA 660
QSEMMKAFTL FWQQASKIAA KQTESPSETQ QQAAKTGAQI AKALSAISGA LAAAA 715

```

```

SEQ ID NO: 87      moltype = DNA length = 2742
FEATURE           Location/Qualifiers
misc_feature       1..2742
                  note = synthetic construct
source            1..2742
                  mol_type = other DNA
                  organism = synthetic construct

```

```

SEQUENCE: 87
catatggaca atggcgatcg tttataccgt gccgactcgc gtcccccaga tgagattaaa 60
cgtagcgggt ggtaaatgcc acgtggggc aatgagtatt ttgaccgtgg aacacagatg 120
aacattaacc tttacgatca tgcctgtggg acccagaccg ggtttgtccg ttatgatgac 180
gggatgttta gtacgagttt gtccttacgc tccgcacacc ttgctgggaca aagtatttta 240
tcaggctaca gcacatatga catttatgtg atcgccactg ccccaaacat gttcaatgtg 300
aacgatgtgt tgggggttta cagcccccac ccatatgaac aagaagtctc ggccttggg 360

```



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gggatcccat atagccagat ttatggttgg taccgcgtaa attttgggtg gattgatgaa 420
cgtttgcacg gtaaccgtga ataccgcgat cgctactacc gtaacttgaa cattgcacct 480
gccgaggacg gctatcgttt agcgggattc ccaccgcgat atcaggcggtg gcgtgaggaa 540
ccgtggatcc atcacgcccc tcaggggtgc ggggaacagta gtcgccatat gatagatcct 600
cttaagcttt ttccaaatth tgatggggat aaggagagtg ctgcggtgaa taaaccttca 660
gcatctccta tgcccagcga attaagtaaa aatgttgcct cattctcttt aggggggtgga 720
ggtgctgcgt tggattcgac agtgtccaca gaaaagctat cgttgatggc tatgatgcag 780
gataaaaaatt cgcagttgat cgatccctgag ttggagggaag ctctgaactc tgaagagtta 840
caagagcaga tccatttgtt aaaaagtcgt ttgtggggatg cacaacgcga gatgcaaatg 900
caagatcccg acaagttggc ctctgagcat gtagatgctt taggagtcct tgttgattta 960
atcaatgggg attttcaagc gatagctgaa catcacacac agacgggtcaa gcagggtaat 1020
ggtgacgaag aaaaatctgt tacacgcaag atagtcgatt gggtctcttc aggagaagaa 1080
atthtgaatc gtgctttggt gtatttctcc gatcgtaatg gagaaagaga aacattagcc 1140
gatttcttaa aggttcagta tgccgttcaa agagctacac aacgcgcga gttatttgc 1200
agtattctag gtgccacggt gagtagtgta aaaacgatta tgacaaccca gttagggtgt 1260
tcagctgctt caatgagctt gtcattccagc agcagctcgg atagtctgaa tctgaaaaat 1320
gtgttatctc agtctatcgc gctacacca caggggggttc ctaatgctga caaatatacc 1380
gacaatcagg taaaacaagt ccagcagacc cgtcaaaacc gtgatgatct gtccatggag 1440
agcgcagctc cggtggcggg aacagccgga aaagatcgtg ctgcgtcggc gtcccagatc 1500
gagggacaa agctgattga gcaacagggg cttgcggctg ggaagagac ggcttctgct 1560
gatgtacat cattgaccca gtccgcatcc aaaggcgctt ccagtcagca gtgtattgag 1620
gataccagta agtccctgga gctttcttcg ctttcgagcc tgtcaagcgt agatgcgaca 1680
catttgcagg aaatccaatc gatcgtgtct tcagcaatgg gcgccacca cgaattgtca 1740
tgacgaact tagagacacc gggattacca aagccgagta ccactccacg ccagggaagt 1800
atggagatca gcccttgcctt agcgaaggcc atcactgcat tgggtgagag cactcaggct 1860
goccttgaaa attttcagtc cactcagagt cagtcgcgca acatgaataa gatgagtttg 1920
gaatcccaag gcttgaaaat cgacaaggag cgtgaagaat ttaagaaaat gcaggagatt 1980
cagcaaaaaga gcggcacaaa ttcaaccatg gatactgtga ataaagtat gattggcgtg 2040
acagtggcaa ttacagtaat ctctgttgtt tcagcattgt ttacctgcgg ttggggcttg 2100
attggcacag ccgctgcggg tgcccacagc gccaccgctg gggcaacggc cgcgcgccag 2160
accgctacct ctgtgacgac cacagtgcgt acccaggtga cgatgcaagc ggtggtccaa 2220
gtcgttaagc aggtatttat ccaagcagta aaacgcgcca tcgtccaagc gattaaacag 2280
gggattaagc aaggcattaa acaagcgatc aaacaggcag tcaaggcaag cgtgaagaca 2340
cttgccaaaa atgtaggcaa gattttcagc gcaggcaaga acgctgtgag taagtccctc 2400
ccaaaattgt ctaagggtgat taatacactt ggttccaaat gggttactct tggcgtgggg 2460
gcctttacag cgttgccgca gttagtcagt ggcattacct cccttcaatt gtctgatatg 2520
caaaaagaac ttgcacaaat ccaaaaggaa gtgggtgcac ttacggcgca gagtgagatg 2580
atgaaagcgt ttacactggt ctggcgacaa gcttcgaaaa tcgcggccaa acaaacggaa 2640
tcaccttcag agacgcaaca acaggcagct aagaccggcg ccagatcgc taaagcgttg 2700
tccgccattt cgggtgcttt agctgctgct gcttagctcg ag 2742

```

```

SEQ ID NO: 88      moltype = AA length = 909
FEATURE
REGION            1..909
                    note = synthetic construct
source            1..909
                    mol_type = protein
                    organism = synthetic construct

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```

SEQUENCE: 88
MDNGDRLYRA DSRPPDEIKR SGLMPRGHN EYFDRGTQMN INLYDHARGT QTGFVRYDDG 60
YVSTSLSLRS AHLAQQSILS GYSTYYIYVI ATAPNMFNVN DVLGVYSPHP YEQEVSA LGG 120
IPYSQIYGWY RVNFGVIDER LHRNREYRDR YVRNLNIAPA EDGYRLAGFP PDHQA WREEP 180
WIHHAPOGCG NSSRMIDPLK LFPNFDGDKS SAAVNKPSAS PMPSELSKNV ASFSLGGGGA 240
ALDSTVSTEK LSLMAMMQDK NSQLIDPELE EALNSEELQE QIHLKSLRLW DAQTQMOMQD 300
PDKLASEHVD ALGVIVDLIN GDFQAI AEHT QQTVKQNGD EEKSVTRKIV DWVSSGEEIL 360
NRALLYFSDR NGERETLADF LKVQYAVQRA TQRAELFASI LGATVSSVKT IMTTQLGGS A 420
ASMSLSSSSS SDSSNLKNVL SQVIAS TPQG VPNA DKLTDN QVKVQQTRQ NRDDL SMESD 480
VAVAGTAGKD RAASASQIEG QELIEQQGLA AGKETASADA TSLTQSASKG ASSQQCIEDT 540
SKSLELSSLS SLSSVDATHL QEIQSIVSSA MGATNELSLT NLETPGLPKP STTPRQEVME 600
ISLALAKAIT ALGESTQAAL ENFQSTQSQS ANMNKMSLES QGLKIDKERE EFKKMQEIQQ 660
KSGTNS TMDT VNKVMIGVTV AITVISVSSA LFTCGLGLIG TAAAGATAAT AGATAAATTA 720
TSVTTT VATQ VTMQAVVQVV KQAI IQAVKR AIVQAIKQGI KQGIKQAIKQ AVKASVKTLA 780
KNVGKIFSA GKNVYSKFPK LSKVINTLGS KWTLVGVGAL TAVPQLVSGI TSLQLSDMQK 840
ELAQIQKEVG ALTAQSEMMK AFTLFWQQAS KIAAKQTESP SETQQQA AKT GAQIAKALSA 900
ISGALAAAA

```

```

SEQ ID NO: 89      moltype = DNA length = 647
FEATURE
misc_feature      1..647
                    note = synthetic construct
source            1..647
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 89
atgggcagca gccatcacca tcatcaccac agccaggatc cgatgccacc aagcaagatc 60
caatgtcttg aaacttttga aagaacttat ggacaccttt atctacaaca tgcgtcccta 120

```

-continued

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atgcgctcatt tagcctatct actcgataaa attgctcgct cttaccctca tatgtgtccg 180
cttcccgata atatggaagc gtactttgag aattatatcc ccaataaaga tatccctctg 240
gacacctatc aaaaaatttt caaactgtcc tcagaagatc ttgaacaagt ctacaaggaa 300
ggatacaacg cctatttaca aggagactat gaggaagatt ctaccgcttt ttactgggtg 360
attttccttta acccatttgt gtctaaattt tgggtttcat taggagcttc gctccatag 420
cgccaaaaat atcaacaagc tcttcacgct tatggtgtag ctgctttgct aagagaaaaa 480
gacccttata ctcattacta tgcctacatc tgctacaccc tgctcaataa tcctgaagaa 540
gctgaaaaag ctcttgatct tgcttggcaa aaagtaaaaa caagctctgc ctatagctct 600
ttaaaagaag aaattttagc gatcaaatcg tacgcctaag cggccgcg 647

```

```

SEQ ID NO: 90      moltype = AA length = 212
FEATURE           Location/Qualifiers
REGION           1..212
                  note = synthetic construct
source           1..212
                  mol_type = protein
                  organism = synthetic construct

```

```

SEQUENCE: 90
MGSSHHHHHH SODPMPPSKI QCLETFFERTY GHLYLQHASL MRHLAYLLDK IARSYPHMCP 60
LPDNMEAYFE NYIPNKDIPL DTYQKIFKLS SEDLEQVYKE GYNAYLQGDY EESSTAFYWL 120
IFFNPFVSKF WFSLGASLHM RQKYQQALHA YGVAALLREK DPYPHYIYAI CYTLNNPEE 180
AEKALDLAWQ KVKTSAYSS LKEEILAIKS YA 212

```

```

SEQ ID NO: 91      moltype = DNA length = 1521
FEATURE           Location/Qualifiers
source           1..1521
                  mol_type = genomic DNA
                  organism = Chlamydia spp

```

```

SEQUENCE: 91
atgagctctt gggttgacac ggcgacggac gtcgctttga gccagaccct tgatctgcct 60
gacgcttcat tggcggttca aaccgaaaaa ttccataca gctgttcaat ctctaaggaa 120
tccgccccat catgtattcg taaaatcttc gccatttag catctcagaa ggaaagtgtc 180
ccgctgtctt ttctcgtttt acaaccgact actccgaaag aacgcacccg gtttttcggg 240
tcacgccttc cttcccaatt gtccctcgact gtccgcacca caacctcttc tccatggaat 300
cttttttagc actcccaggc acgcaactcg acccgtaaat tgtcggagaa gcttcatttg 360
agctcagagt tatccgcccg tgactccact aagccttcgt cgagcgaacc ggttaaacca 420
tcggaaaaatc ttttgacac cctgagcat cataaggaaat cctctcaag tttgaaaaag 480
gataacttat ctctatcat ggaggagac gactcattct ctgcagagac agagtccctt 540
gaagagcgtt tggtcaccca gaaaaaggag gagacgggtg cccaggagca aaagcaccca 600
ttgctgcgta catctactcc gccatcaaag gccagcgggg aatcacaaaga ttctagcgaa 660
cacagctcaa aggaagatcc ttatagtcaa caaccgagcc ataaaatcca acgccgtaaa 720
gagcgtgcta agcgcgtcgt cccaattatt actccgcca cgggtgggtat ctttagtttg 780
agctaccttc ttacaaaaa ggggatctta gcggatttca gcgcctattc ggcatacaag 840
gataatttag aaacaactca gcaagagctg accatgttgc atcaagaacg tatcgagcaa 900
gtccaaaaga tcgtggataa aagtaagaca atgcgctttt gggattcatt agcatccatt 960
gtggccacaa tcattccatg gatcgaaatg ggtgttgca taaccatcat cgcactggga 1020
ggtggaatcc tttcctgggt ctctcttttt gctgcgctta tcatgattgt aatttcatta 1080
ttggaagcat tcgacgggtg gcgtgcaatc gctaagcatt taccaggtaa cgatcttgaa 1140
aagaagatgc cttatttagg ttacgtaaag ttggccttaa ctgtgttctc gtgcttactg 1200
agttaaagcg gcttgatagt agcaaaaatta ggaatgagtc cgcttttgga ggggggttg 1260
aagagtatcg caccagcatt aagtgtgatg ctgggtttga ctcaaggcgt agcactgtat 1320
ttacaatctt tcagcaaaa gattcgtgcc cgctgcactc agatcgacgc acgcattgaa 1380
ttgattaact gggaaacgca tgagtatttc ttgctgtgct aacaacttct tgattcaatg 1440
caaacgtcct tcgaacaact tactgaaaca ttacagttac aacgtgaaat tgatcagaca 1500
ttacagacg ctttgcgcta g 1521

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SEQ ID NO: 92      moltype = AA length = 506
FEATURE           Location/Qualifiers
source           1..506
                  mol_type = protein
                  organism = Chlamydia spp

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SEQUENCE: 92
MSSWFAQATD VALSQTLDLP DASLAVQTEK FPYSCSISKE SAPSCIRKIF AHLASQKESA 60
PLSFSRLQPT TPKERILFFG SSPSSQLSST VRTTSSPWN LFSNSQARNs TRKLSEKLHL 120
SSELsARDST KPSSSEVPKP SENLLHTPEH HKESFSSLKK DNLSPIMEEI DSFSaETESL 180
EERLVTQKKE ETVAQEQKHP LLRTSTPPSK ASGESQDSSE HSSKEDPYSQ QPSHKIQRRK 240
ERAKRVVPII TPPTVGIFSL SYLLTKQGIL ADFSAYSAYK DNLETTQOEL TMLHqERIEQ 300
VQKIVDKSKT MRFWDsLASI VATIIPWIEM GVAVTIIALG GGILSWCSLF AALIMIVISL 360
LEAFDGWRAI AKHLPgNDLE KKMRYLGYVK LALTvFSCLL SLSALYVAKL GMSPLLEGV 420
KSIAPALSGM LGLTQGVAlY LQSSSQKIRA RCTQIDARIE LINWERDEYF LRaEQLLDSM 480
QTSFEQLTET LQLQREIDQT FTDALR 506

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SEQ ID NO: 93      moltype = DNA length = 1977
FEATURE           Location/Qualifiers
misc_feature     1..1977
                  note = synthetic construct

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source          1..1977
                mol_type = other DNA
                organism = synthetic construct

SEQUENCE: 93
aaaagtgagc gtttaaaaaa attagaatca gagcttcacg atcttaccca gtggatgcaa 60
cttgcccttg ttccataaaa agaaatcgag agacaccagg aagaaatccg tctgctagaa 120
agcaaaatcc ttgaagagaa agaacgtcta caacttctca aagaaagcgg tgagatcaaa 180
gagtacgtaa cccctcgaag aactccagct aaaaccattt acccagatgg cccagcggt 240
tcagacgttg agttttaga atcctcggat acagaagtgg atctcgatgc cggtgacaca 300
attgagattg acctaggtga tgaggcaaga gaagaaagcg gaaacgaact cgactactct 360
agtgaagacg atgaggatcc ttccagcgat cgcaatcggt ggcgccgagg aggcatacata 420
gatcctgacg cgaatgaatg gggttcagct gcttcaatga gctcttggtt tgcacaggcg 480
acggacgtcg ctttgagcca gacccttgat ctgctgacg cttcattggc ggttcaaacc 540
gaaaaatttc catacagctg ttcaatctct aaggaatccg ccccatcatg tattcgtaaa 600
atcttcgccc atttagatcc tcagaaggaa agtgctcccg tgtctttttc tcgtttataa 660
ccgactactc cgaaagaacg catcctggtt ttccgggtcat cgccttcctc ccaattgtcc 720
tcgactgtcc gcaccacaac ctcttctcca tggaaatctt ttagcaactc ccaggcacgc 780
aactcgaccc gtaaatgtgc ggagaagcct catttgagct cagagttatc cgcctgtgac 840
tccactaagc cttcgtcgag cgaacgggtt aaaccatcgg aaaatctttt gcacaccct 900
gagcatcata aggaatcctt ctcaagtttg aaaaaggata acctatctcc tatcatggag 960
gagatcgact cattctctgc agagacagag tcccttgaag agcgtttggt caccagaaa 1020
aaggaggaga cgggtggccc ggagcaaaag caccattgc tgcgtacatc tactccgcca 1080
tcaaaaggca gcggggaatc acaagattct agcgaacaca gctcaaaagg agatccttat 1140
agtcaacaac cgagccataa aatccaacgc cgtaaagagc gtgtaagcgc cgtcgtccca 1200
attattactc cgccaacggg ggggtatctt agtttgagct accttcttac aaaacagggg 1260
atcttagcgg atttcagcgc ctattcggca tacaaggata atttagaaac aactcagcaa 1320
gagctgacca tgttgcatca agaactgac gagcaagtc aaaagatcgt ggataaaagt 1380
aagacaatgc gcttttggga ttcattagca tccattgtgg ccacaatcat tccatggatc 1440
gaaatgggtg ttgagctaac catcatcgca ctgggaggtg gaatccttcc ctgggtgctct 1500
ctttttgctg cgcttatcat gattgtaatt tcattattgg aagcattcga cgggtggcgt 1560
gcaatcgcta agcattttacc ggttaacgat cttgaaaaga agatgcgtta tttaggttac 1620
gtaaaagtgg ccttaactgt gttctcgtgc ttactgagtt taagcgcctt gtatgtagca 1680
aaattagga ttagtcgcgt ttgggagggg gttgtgaaga gtatcgacc agcattaagt 1740
ggtatgctgg gtttgactca aggcgtagca ctgtatttac aatcttcac gcaaaagatt 1800
cgtgccgcgt gcaactcagat cgacgcacgc attgaattga ttaactggga acgcgatgag 1860
tatttcttgc gtgctgaaca actctctgat tcaatgcaaa cgtccttcga acaacttact 1920
gaaacattac agttacaacg tgaattgat cagacattta cagacgcttt gcgctag 1977

SEQ ID NO: 94      moltype = AA length = 659
FEATURE           Location/Qualifiers
REGION            1..659
                  note = synthetic construct
source            1..659
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 94
MKSERLKKLE SELHDLTQWM QLGLVPKKEI ERHQEEIRLL ESKILEEKER LQLLKESGEI 60
KEYVTPRRTP AKTYPDGPS VSDVEFVSS DTEVDLDAGD TIEIDLDEA REESGNELDY 120
SSEDEDPFS DRNRWRGGI IDPDANEWS AASMSSWFAQ ATDVALSOTL DLPDASLAVQ 180
TEKFPYSCSI SKESAPSCIR KIFAHLASQK ESAPLSFSRL QPTTPKERIL FFGSSPSSQL 240
SSTVRTTSS PWNLFNSQA RNSTRKLSK LHLSELARS DSTKPSSEP VKPSENLLHT 300
PEHHKESFSS LKKDNLSPIM EEIDSFSAET ESLEERLVTQ KKEETVAQEQ KHPLLRTSTP 360
PSKASGESQD SSEHSSKEDP YSQQPSHKIQ RRKERAKRVV PIITPPTVGI FSLSYLLTKQ 420
GILADFSAYS AYKDNLETTQ QELTMLHQR IEQVQKIVDK SKTMRFWDSL ASIVATIIPW 480
IEMGVAVTII ALGGGILSWC SLFAALIMIV ISLLEAFDGV RAIKHLPGN DLEKKMRYLG 540
YVKLALTVPFS CLLSLSALYV AKLGMSPLLE GVVKSIAPAL SGMGLTLQGV ALYLQSSSQK 600
IRARCTQIDA RIELINWERD EYFLRAEQLL DSMQTSFEQL TETLQLQREI DQFTFDALR 659

SEQ ID NO: 95      moltype = DNA length = 2574
FEATURE           Location/Qualifiers
misc_feature      1..2574
                  note = synthetic construct
source            1..2574
                  mol_type = other DNA
                  organism = synthetic construct

SEQUENCE: 95
catatggaca atggcgatcg ttatataccgt gccgactcgc gtccccaga tgagattaaa 60
cgtagcggtg ggttaattgcc acgtgggcac aatgagtatt ttgaccgtgg aacacagatg 120
aacattaacc tttagcatca tgcccgtggg acccagaccg gggttgcctg ttatgatgac 180
gggatgttta gtacgagttt gtccttacgc tccgcacacc ttgcccggca aagtatttta 240
tcaggtacac gccacatata catttatgtg atcgccactg ccccaaacat gttcaatgtg 300
aacgatgtgt tgggggttta cagcccccac ccatatgaac aagaagtctc ggccttggg 360
gggatcccat atagccagat ttatggttgg taccgcgtaa atttgtgtg gattgatgaa 420
cgtttgcatc gtaaccgtga ataccgcgat cgctactacc gtaacttgaa cattgcacct 480
gccagggacg gctatcggtt agcgggattc ccaccgcgac atcaggcggt gcgtgaggaa 540
ccgtggatcc atcacgcccc tcagggggtgc ggggaacagta gtcgccatat gaaaagttag 600

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cgtttaaaaa aattagaatc agagcttcat gatcttacc agtgatgca acttgccct 660
gttcctaaaa aagaaatcga gagacaccag gaagaaatcc gtctgctaga aagcaaaatc 720
cttgaagaga aagaacgtct acaacttctc aaagaaagcg gtgagatcaa agagtacgta 780
acccctcgaa gaactccagc taaaaccatt taaccagatg gcccagcgt ttcagacgtt 840
gagtttgtag aatcctcgga tacagaagtg gatctcgatg ccggtgacac aattgagatt 900
gacctaggtg atgaggaagc agaagaaagc ggaaacgaac tgcactactc tagtgaagac 960
gatgaggatc ctttcagcga tcgcaatcgt tggcgccgag gaggcacatc agatcctgac 1020
gcgaatgaat ggggttcagc tgcttcaatg agctcttggg ttgcacaggg gacggacgtc 1080
gctttgagcc agacccttga tctgcctgac gcttcattgg cggttcaaac cgaaaaaattt 1140
ccatacagct gttcaatctc taaggaatcc gcccatcatc gtattcgtaa aatcttcgcc 1200
catttagcat ctcagaagga aagtgcctcg ctgtcttttt ctggtttaca accgactact 1260
ccgaagaagc gcctcctggt ttccgggtca tcgccttcct cccaattgtc ctgcactgtc 1320
cgcaccacaa cctcttctcc atggaatctt tttagcaact ccagggcacg caactcgacc 1380
cgtaaatgtt cgggaagct tcatttgagc tcagagtta ccgcccgtga ctccactaag 1440
ccttcgtcga gcgaaccggt taaccatcgc gaaaatcttt tgcacacccc tgagcatcat 1500
aaggaatcct ctcaagttt gaaaaaggat aacttatctc ctatcatgga ggagatcgac 1560
tcattctctg catgacagaga gtcccttgaa gagcgtttgg tcaccacagaa aaaggaggag 1620
acgggtggccc agggacaaaa gcaccatttg ctgcgtacat ctactccgcc atcaaaagcc 1680
agcgggggaat cacaagatcg tagcgaacac agctcaaagg aagatcctta tagtcaacaa 1740
ccgagccata aaatcccaag ccgtaaaagag cgtgctaagc gcgtcgtccc aattattact 1800
ccgccaacgg tgggtatctt tagtttgagc taccttctta caaaacaggg gatcttagcg 1860
gatttcagcg cctatctggc atacaaggat aatttagaaa caactcagca agagctgacc 1920
atgtgtgcatc aagaacgtat cgagcaagtc caaaagatcg tggataaaa taagacaatg 1980
cgcttttggg attcattagc atccattgtg gccacaatca ttccatggat cgaatagggt 2040
gttgccagtaa ccatcatcgc actgggaggt ggaatccttt cctggtgctc tcttttggct 2100
gcgcttatca tgattgtaat ttcatatttg gaagcattcg acgggtggcg tgcaatcgct 2160
aagcatttac caggtaacga tcttgaaaag aagatgcggt atttaggtta cgtaaagttg 2220
gccttaactg tttctctggt ctactgagt ttaagcgcct tgtatgtagc aaaattagga 2280
atgagtccgc ttgtggaggg ggtgtgtaag agtatcgac cagcattaag tggtagctg 2340
ggtttgactc aagcgctagc actgatttta caatcttcat cgcaaaagat tcgtgcccgc 2400
tgcactcaga tcgcgcacg cattgaattg attaactggg aacgcgatga gtatttctt 2460
cgtgctgaac aacttcttga ttcaatgcaa acgtccttcg aacaacttac tgaacatta 2520
cagttacaac gtgaaattga tcagacattt acagacgctt tgcgctagct cgag 2574

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SEQ ID NO: 96      moltype = AA length = 853
FEATURE           Location/Qualifiers
REGION            1..853
                  note = synthetic construct
source            1..853
                  mol_type = protein
                  organism = synthetic construct

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SEQUENCE: 96
MDNGDRLYRA DSRPPDEIKR SGGLMPRGHN EYFDRGTQMN INLYDHARGT QTGFVRYDDG 60
YVSTSLSLRS AHLAQQSILS GYSTYYIYVI ATAPNMFNVN DVLGVYSPHP YEQEVSAALGG 120
IPYSQIYGWY RVNPGVIDER LHRNREYDR YRNLNIAPA EDGYRLAGFP PDHQAWREEP 180
WIHHAPQCGC NSSRMKSERL KLESELHDL TQWMQLGLVP KKEIERHQEE IRLLESKILE 240
EKERLQLLKE SGEIKEYVTP RRTPAKIYYP DGPSVSDVEF VESSDTEVDL DAGDTIEIDL 300
GDEAREESGN ELDYSSDEDE DPFSDRNRWR RGGIIDPDAN EWGSAASMS WFAQATDVAL 360
SQTLDPDAS LAVQTEKFPY SCSISKESAP SCIRKIFAHL ASQKESAPLS FSRLQPTTPK 420
ERILFFGSSP SSQLSSTVRT TTSSPWNLFS NSQARNSTRK LSEKLHLSSE LSARDSTKPS 480
SSEPVKPSEN LLHTPEHKE SFSSLKIDNL SPIMEEIDSF SAETESLEER LVTQKKEETV 540
AQEQKHPLLR TSTPPSKASG ESQDSSEHSS KEDPYSQQPS HKIQRRKERA KRVVPIITPP 600
TVGIFSLSYL LTKQIGILADF SAYSAYKDNL ETTQOELTML HQERIEQVQK IVDKSKTMRP 660
WDSLASIVAT IIPWIEMGVA VTIIALGGGI LSWCSLFAAL IMIVISLLEA FDGWRRAIAKH 720
LPGNDLEKKM RYLVGYVKLAL TVFSCLLSLS ALYVAKLGMS PLLEGVVKSI APALSGMLGL 780
TQGVVALYQS SSQKIRARCT QIDARIELIN WERDEYFLRA EQLLDSMQTS FEQLTETLQL 840
QREIDQTFD ALR 853

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SEQ ID NO: 97      moltype = DNA length = 647
FEATURE           Location/Qualifiers
misc_feature       1..647
                  note = synthetic construct
source            1..647
                  mol_type = other DNA
                  organism = synthetic construct

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SEQUENCE: 97
atgggcagca gccatcacca tcataccacc agccaggatc cgatgccacc aagcaagatc 60
caatgtcttg aaacttttga aagaacttat ggacaccttt atctacaaca tgcgtcccta 120
atgcgtcatt tagcctatct actcgataaa attgctcgct cttaccctca tatgtgtccg 180
cttcccgata atatggaagc gtactttgag aattatatcc ccaataaaga tatccctctg 240
gacacctatc aaaaaatttt eaaactgtcc tcagaagatc ttgaacaagt ctacaaggaa 300
ggatacaacg cctattttaca aggagactat gaggaaaagt ctaccgcttt ttactggttg 360
attttcttta acccatttgt gtctaaattt tggttttcat taggagcttc gctccatatg 420
cgccaaaaat atcaacaagc tcttcatgct tatggtgtag ctgctttgct aagagaaaaa 480
gacccttata ctcattacta tgcctacatc tgctacaccc tgctcaataa tcttgaagaa 540
gctgaaaaag ctcttgatct tgcttggcaa aaagtaaaaa caagctctgc ctatagctct 600

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 ttaaaagaag aaatttttagc gatcaaatcg tacgcctaag cggccgc 647

SEQ ID NO: 98 moltype = AA length = 212  
 FEATURE Location/Qualifiers  
 REGION 1..212  
 note = synthetic construct  
 source 1..212  
 mol\_type = protein  
 organism = synthetic construct

SEQUENCE: 98  
 MGSSHHHHH S QDPMPPSKI QCLETFFERTY GHLYLQHASL MRHLAYLLDK IARSYPHMC 60  
 LPDNMEAYFE NYIPNKDIPL DTYQKIFKLS SEDLEQVYKE GYNAYLQGDY EESSTAFYWL 120  
 IFNPFVSKF WFSLGASLHM RQKYQALHA YGVAALLREK DPYPHYIYAI CYTLNNPPE 180  
 AEKALDLAWQ KVKTSAYSS LKEEILAIKS YA 212

SEQ ID NO: 99 moltype = DNA length = 1977  
 FEATURE Location/Qualifiers  
 misc\_feature 1..1977  
 note = synthetic construct  
 source 1..1977  
 mol\_type = other DNA  
 organism = synthetic construct

SEQUENCE: 99  
 aaaagtgaagc gtttaaaaaa attagaatca gagcttcagc atcttaccga gtggatgcaa 60  
 cttggccttg ttcctaaaaa agaatcgcag agacaccagg aagaaatccg tctgctagaa 120  
 agcaaaatcc ttgaagagaa agaacttcta caacttctca aagaaagcgg tgagatcaaa 180  
 gagtacgtaa cccctcgaa aactccagct aaaaccattt acccagatgg ccccgagcgt 240  
 tcagacgttg agttttaga atcctcggat acagaagtgg atctcgatgc cggtagacaca 300  
 attgagattg acctaggtga tgaggcaaga gaagaaagcg gaaacgaact cgactactct 360  
 agtgaagacg atgaggatcc ttccagcgat cgcaatcggt ggcgcgaggg aggcacata 420  
 gatcctgacg cgaatgaatg ggggttcagct gcttcaatga gctcttggtt tgacacggcg 480  
 acggacgtcg ctttgagcca gaccttgat ctgcctgacg cttcattggc ggttcaaac 540  
 gaaaaatttc catacagctg ttcaatctct aaggaatccg ccccatcatg tattcgtaaa 600  
 atcttcgccc atttagcatc tcagaaggaa agtgctccgc tgtctttttc tcgtttacaa 660  
 ccgactactc cgaagaagc catcctggtt ttcgggtcat cgccttctc ccaattgtcc 720  
 tcgactgtcc gcaccacaac ctcttctcca tggaaatctt ttagcaactc ccaggcacgc 780  
 aactcgaccc gtaaatgtgc ggagaagcct catttgagct cagagttatc cgcccgtagc 840  
 tccactaagc cttcgtcgag cgaacgggtt aaaccatcgg aaaatctttt gcacaccctc 900  
 gagcatcata aggaatcctt ctcaagtttg aaaaaggata acttatctcc tatcatggag 960  
 gagatcgact cattctctgc agagacagag tcccttgaag agcgtttggt caccagaaa 1020  
 aaggaggaga cgggtggcca ggagcaaaa caccatttgc tgcgtacatc tactcggcca 1080  
 tcaaggcca cgggggaatc acaagattct agcgaacaca gctcaaggaa agatccttat 1140  
 agtcaacaac gcagccataa aatccaacgc cgtaaagagc gtgctaagcg cgtgctccca 1200  
 attattactc cgccaacggt ggggtatctt agtttgagct accttcttac aaaacagggg 1260  
 atcttagcgg atttcagcgc ctattcggca tacaaggata atttagaaac aactcagcaa 1320  
 gagctgacca tgttgcatca agaactatc gagcaagtc aaaagatcgt ggataaaagt 1380  
 aagacaatgc gcttttggga ttcattagca tccattgtgg ccacaatcat tccatggatc 1440  
 gaaatgggtg ttgcagtaac catcatcgca ctgggaggtg gaatccttcc ctgggtgctc 1500  
 ctttttgctg agcttatcatt tcatattgg aagcattcga cgggtggcgt 1560  
 gcaatcgcta agcatttacc aggtaacgat cttgaaaaga agatgcgtta tttaggttac 1620  
 gtaaatgttg ccttaactgt gttctcgtgc ttaactgagtt taagcgcctt gtatgtagca 1680  
 aaattaggaa tgagtccgct tttggagggg gttgtgaaga gtatgcgacc agcattaagt 1740  
 ggtatgctgg ttgtgactca aggcgtagca ctgtatttac aatcttcatc gcaaaagatt 1800  
 cgtgcccgcg gcaactcagc cgacgcacgc attgaattga ttaactggga acgcgtagag 1860  
 tatttcttgc gtgctgaaca acttcttgat tcaatgcaaa cgtccttcga acaacttact 1920  
 gaaacattac agttacaacg tgaattgat cagacattta cagacgcttt gcgctag 1977

SEQ ID NO: 100 moltype = AA length = 659  
 FEATURE Location/Qualifiers  
 REGION 1..659  
 note = synthetic construct  
 source 1..659  
 mol\_type = protein  
 organism = synthetic construct

SEQUENCE: 100  
 MKSERLKKLE SELHDLTQWM QLGLVPKKEI ERHQEIRLL ESKILEEKER LQLLKESGEI 60  
 KEYVTPRRTP AKTIYPDGPS VSDVEFVSS DTEVDLDAGD TIEIDLGDEA REESGNELDY 120  
 SSEDEDPFS DRNRWRGGI IDPDANEWGS AASMSSWFAQ ATDVALSQTL DLPDASLAVQ 180  
 TEKFPYSCSI SKESAPSCIR KIFAHLASQK ESAPLSFSRL QPTTPKERIL FFGSSPSSQL 240  
 SSTVRTTSS PWNLFNSQA RNSTRKLSEK LHLSELSEAR DSTKPSSEEP VKPSENLLHT 300  
 PEHHKESFSS LKKNLSPIM EEIDSPSAET ESLEERLVTQ KKEETVAEQE KHPLLRTSTP 360  
 PSKASGESQD SSEHSSKEDP YSQQPSHKIQ RRKERAKRVV PIITPPTVGI FSLSYLLTKQ 420  
 GLADFSAYS AYKNLLETTQ QELTMLHQER IEQVQKIVDK SKTMRFWDSL ASIVATIIIPW 480  
 IEMGVAVTII ALGGGILSWC SLFAALIMIV ISLLEAFDGW RAIKHLPGN DLEKKMRYLG 540  
 YVKLALTVFS CLLSLALYV ALKLGMSPLLE GVVKSIPAL SGMLGLTQGV ALYLQSSSQK 600  
 IRARCTQIDA RIELINWERD EYFLRAEQLL DSMQTSFEQL TETLQLQREI DQFTFDALR 659

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SEQ ID NO: 101          moltype = DNA length = 2574
FEATURE                Location/Qualifiers
misc_feature            1..2574
                        note = synthetic construct
source                  1..2574
                        mol_type = other DNA
                        organism = synthetic construct

SEQUENCE: 101
catatggaca atggcgatcg tttataccgt gccgactcgc gtccccaga tgagattaaa 60
cgtagcgggtg gggttaatgcc acgtggggcac aatgagtagt ttgaccgtgg aacacagatg 120
aacattaacc tttacgatca tgcccgtggg acccagaccg gggttgcctg ttatgatgac 180
gggtatgtta gtacgagttt gtccttacgc tccgcacacc ttgcgggaca aagtatttta 240
tcaggctaca gcacatatata catttatgtg atcgccactg ccccaaacat gttcaatgtg 300
aacgatgtgt tgggggttta cagcccccat ccatatgaac aagaagctct ggcccttggg 360
gggatcccat atagccagat ttatggttgg taccgcgtaa atttgggtg gattgatgaa 420
cgtttgcacg gtaaccgtga ataccgcgat cgctactacc gtaacttgaa cattgcacct 480
gccgaggacg gctatcgttt agcgggattc ccaccgcgat atcaggcgtg gcgtgaggaa 540
ccgtggatcc atcacgcccc tcagggggtgc gggaacagta gtcgccatat gaaaagtga 600
cgtttaaaaa aattagaatc agagcttcat gatcttacc agtggtatgca acttggcctt 660
gttcctaaaa aagaaatcga gagacaccag gaagaaatcc gtctgctaga aagcaaaatc 720
cttgaagaga aagaactgct acaacttctc aaagaaagcg gtgagatcaa agagtacgta 780
accctcgaag gaactccagc taaaaccatt taccagatg gcccagcgt ttccagcgtt 840
gagttttagt aatcctcgga tacagaagtg gatctcgatg ccggtgacac aattgagatt 900
gacctagggt atgaggcaag agaagaaagc ggaaacgaac tcgactactc tagtgaagac 960
gatgaggatc ctttcagcga tcgcaatcgt tggcgccgag gaggcacat agatcctgac 1020
gccaatgaat ggggttcagc tgcctcaatg agctcttggg ttgcacagcg gacggacgtc 1080
gctttgagcc agacccttga tctgcctgac gcttcattgg cggttcaaac cgaataattt 1140
ccatacagct gttcaatctc taaggaaatc gcccatcat gtattcgtaa aatcttcgcc 1200
catttagcat ctcagaagga aagtgcctcg ctgtcttttt ctgctttaca accgactact 1260
ccgaagaagc gccctctgtt ttccgggtca tcgccttcct ccaattgtc ctgactgtc 1320
cgcaccacaa cctcttctcc atgggaatct tttagcaact ccagggcacg caactcgacc 1380
cgtaaatgtt cggagaagct tcatttgagc tcagagttat ccgcccgtga ctccactaag 1440
ccttcgtcga gcgaaccggt taaaccatcg gaaaatcttt tgcacacccc tgagcatcat 1500
aagggaatcct tctcaagtgt gaaaaaggat aacttatctc ctatcatgga ggagatcgac 1560
tcattctctg cagagacaga gtcccttgaa gagcgtttgg tcaccacaga aaaggaggag 1620
acggtggccc aggagcaaaa gcaccattg ctgcgtacat ctactccgcc atcaaggccc 1680
agcggggaat cacaagatgc tagcgaacac agctcaaaag aagatcctta tagtcaacaa 1740
ccgagccata aaatccacag ccgtaaaagag cgtgctaagc gcgtcgtccc aattattact 1800
ccgccaacgg tgggtatctt tagtttgagc taccttctta caaaacaggg gatcttagcg 1860
gatttcagcg cctattcggc atacaaggat aatttagaaa caactcagca agagctgacc 1920
atgttgcatc aagaacgtat cgagcaagtc caaaagatcg tggataaaag taagacaatg 1980
cgcttttggg ttcatctgtg atccattgtg gccacaatca ttccatggat cgaatgggt 2040
gttgagtaaa ccatcatcgc actgggaggt ggaatccttt cctgggtgct tctttttgct 2100
gcgcttatca tgatttgaat ttcatattg gaagcattcg acgggtggcg tgcaatcgct 2160
aagcatttac caggtaacga tcttgaaaag aagatgcgtt atttaggtta cgtaaagtgt 2220
gccttaactg tgtctcgtg cttactgagt ttaagcgctt tgtagtagc aaaattagga 2280
atgagtcctg ttttggaggg ggttgtgaag agtatcgcac cagcattaag tggtagctg 2340
ggtttgactc aaggcgtagc actgtattta caatcttcat cgcaaaagat tctgtcccgc 2400
tgcaactcaga tcgacgcagc cattgaattg attaaactgg aacgcgatga gtatttcttg 2460
cgtgctgaac aacttcttga ttcaatgcaa acgtccttcg aacaacttac tgaacatta 2520
cagttacaac gtgaatttga tcagacattt acagacgctt tgcgctagct cgag 2574

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SEQ ID NO: 102          moltype = AA length = 853
FEATURE                Location/Qualifiers
REGION                  1..853
                        note = synthetic construct
source                  1..853
                        mol_type = protein
                        organism = synthetic construct

SEQUENCE: 102
MDNGDRLYRA DSRPPDEIKR SGLMPRGHN EYFDRGTQMN INLYDHARGT QTGFVRYDDG 60
YVSTSLSLRS AHLAQQSILS GYSTYYIYVI ATAPNMFNVN DVLGVVSPHP YEQEVSAALGG 120
IYPSQIYGWY RVNFGVIDER LHRNREYRDR YRNLNIAPA EDGYRLAGFP PDHQAWAREEP 180
WIHHAPQGGC NSSRMKSERL KLESELHDL TQWMQLGLVP KKEIERHQEE IRLLESKILE 240
EKERLQLLKE SGEIKEYVTP RRTPAKTIYP DGPSVSDVEF VESDTEVDL DAGDTIEIDL 300
GDEAREESGN ELDYSEDDE DPFSDRNRWR RGGIIPDAN EWGSAASMSS WFAQATDVAL 360
SQTLDLPDAS LAVQTEKFPY SCSISKESAP SCIRKIFAHL ASQKESAPLS FSRLQPTTPK 420
ERILFFGSSP SSQLSSTVRT TTSSPWNLF S NSQARNSTRK LSEKLHLSSE LSARDSTKPS 480
SSEPVPKSEN LLHTPEHHKE SFSSLKNDL SPIMEEIDSF SAETESLEER LVTQKKEETV 540
AQEQKHPLLR TSTPPSKASG ESQDSSEHSS KEDPYSQQPS HKIQRRKERA KRVVPITTP 600
TVGIFSLSYL LTKQGILADF SAYSAKYDNL ETTQQELTML HQERIEQVQK IVDKSKTMRP 660
WDSLASIVAT IIPWIEGVA VTIIALGGGI LSWCSLFAAL IMIVISLLEA FDGWRAIAKH 720
LPGNDLEKKM RYLGIVKLAL TVFSCLLSLS ALYVAKLGMS PLLEGVVKSI APALSGMLGL 780
TQGVALYLQS SSQKIRARCT QIDARIELIN WERDEYFLRA BQLLDMSQTS FEQLTETLQL 840
QREIDQTFDT ALR 853

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|                |                                |             |             |            |            |     |
|----------------|--------------------------------|-------------|-------------|------------|------------|-----|
| SEQ ID NO: 103 | moltype = DNA length = 810     |             |             |            |            |     |
| FEATURE        | Location/Qualifiers            |             |             |            |            |     |
| misc_feature   | 1..810                         |             |             |            |            |     |
|                | note = synthetic construct     |             |             |            |            |     |
| source         | 1..810                         |             |             |            |            |     |
|                | mol_type = other DNA           |             |             |            |            |     |
|                | organism = synthetic construct |             |             |            |            |     |
| SEQUENCE: 103  |                                |             |             |            |            |     |
| atgcgttgca     | ctcgggccaat                    | tgcgcaaaacc | gcaagaacag  | gctggctgac | gtggctggcg | 60  |
| attcttgcg      | tcacggcgcc                     | cgtgacttcg  | ccggcatggg  | ccgacgatcc | tcccgccacc | 120 |
| gtataccgct     | atgactcccg                     | cccgccggag  | gaagctttcc  | agaacggatt | cacggcgctg | 180 |
| ggaaacaacg     | acaatgtgct                     | cgaccatctg  | accggacgtg  | ctgcgcaggt | cggcagcagc | 240 |
| aacagcgctt     | tcgtctccac                     | cagcagcagc  | cggcgctata  | ccgaggtcta | tctcgaaact | 300 |
| cgcatacgag     | aagcggtcga                     | ggccgaacgc  | gcccgcaggg  | gcaccggcca | cttcacgcgg | 360 |
| tacatctacg     | aagtcgcgcg                     | cgacaacaat  | tctacgcggc  | ccgccagctc | gtacttcgaa | 420 |
| tacgtcgaca     | cttatgtgcga                    | caatgccggc  | cgtatcctcg  | ccggcgcgct | ggccacacct | 480 |
| cagagcgcaat    | atcttggcaca                    | ccggcgcatc  | ccgcccgaaa  | acatccgcag | ggtaacgcgg | 540 |
| gtctatcaca     | acggcatcac                     | cggcgagacc  | acgaccacgc  | agtattccaa | cgctcgctac | 600 |
| gtcagccagc     | agactcgcgc                     | caatcccaac  | ccctacacat  | cgcaaggctc | cgtagcgtcg | 660 |
| atcgtcggca     | cattgggtgcy                    | catggcgccg  | gtgataggcg  | cttgcatggc | gcggcaggcc | 720 |
| ggaagctccg     | aggccatggc                     | agcctgggtc  | gaaccgcgcg  | gcgaggcgat | ggttctcgtg | 780 |
| tactacgaaa     | gcatecgcta                     | ttcgtttctg  |             |            |            | 810 |
|                |                                |             |             |            |            |     |
| SEQ ID NO: 104 | moltype = AA length = 269      |             |             |            |            |     |
| FEATURE        | Location/Qualifiers            |             |             |            |            |     |
| REGION         | 1..269                         |             |             |            |            |     |
|                | note = synthetic construct     |             |             |            |            |     |
| source         | 1..269                         |             |             |            |            |     |
|                | mol_type = protein             |             |             |            |            |     |
|                | organism = synthetic construct |             |             |            |            |     |
| SEQUENCE: 104  |                                |             |             |            |            |     |
| MRCFTRAIQT     | ARTGWLTLWA                     | ILAVTAPVTS  | PAWADDPAPT  | VYRYSRPPPE | DVFQNGFTAW | 60  |
| GNNDNVLDHL     | TGRSCQVGSS                     | NSAFVSTSSS  | RRYTEVYLEH  | RMQEAWEAER | AGRGTGHPFI | 120 |
| YIYEVRADNN     | FYGAASSYFE                     | YVDTYGDNAG  | RILAGALATY  | QSEYLAHRRR | PPENIRRVTR | 180 |
| VYHNGITGET     | TTTEYSNARY                     | VSQOTRANPN  | PYTSRRRSVAS | IVGTLVRMAP | VIGACMARQA | 240 |
| ESSEAMAAWS     | ERAGEAMVLV                     | YYESIAYSF   |             |            |            | 269 |
|                |                                |             |             |            |            |     |
| SEQ ID NO: 105 | moltype = DNA length = 681     |             |             |            |            |     |
| FEATURE        | Location/Qualifiers            |             |             |            |            |     |
| misc_feature   | 1..681                         |             |             |            |            |     |
|                | note = synthetic construct     |             |             |            |            |     |
| source         | 1..681                         |             |             |            |            |     |
|                | mol_type = other DNA           |             |             |            |            |     |
|                | organism = synthetic construct |             |             |            |            |     |
| SEQUENCE: 105  |                                |             |             |            |            |     |
| atgcgcatcg     | accgcaagac                     | gctctgccat  | ctcctgtccg  | ttctgcggtt | ggccctctct | 60  |
| ggatctcacg     | tggcgcgggc                     | ctccacgcga  | ggcatcgctc  | ttccgcgcga | ggaacagatt | 120 |
| accacgcatg     | gcagccccta                     | tggacgctgc  | gcgaacaaga  | cccgtgccct | gaccgtggcg | 180 |
| gaattgcgcg     | gcagcggcga                     | ctctgaggag  | tacctgcgtc  | atgtgacgcg | cggctggtca | 240 |
| atatttgcgc     | tctacgatgg                     | cacctatctc  | ggcgcgcaat  | atggcggcgt | gatcaaggac | 300 |
| ggaacaccgc     | gcggcgcatc                     | cgacctgaaa  | acgacgttct  | gcatcatgac | cacgcgcaat | 360 |
| acgggtcaac     | ccgcaacgga                     | tctaactact  | agcaacgtca  | ccgcccaact | cctgctctcc | 420 |
| agcacaaca      | gcaggctatg                     | cgcgctcttc  | ctcagaagcg  | gcgaaccggt | cattggcgcc | 480 |
| tgaccagcc      | cgtatgacgg                     | caagtactgg  | agcatgtaca  | gccggctgcy | gaaatgctt  | 540 |
| tacctgatct     | agctggcgcg                     | catctccgta  | cgcgtccatg  | tcagcaagga | agaacagtat | 600 |
| tacgatcatg     | aggagccaac                     | gttcgagact  | tacgcacctt  | ccggcatctc | catctgcaat | 660 |
| cctggatcat     | ccttatgctg                     | a           |             |            |            | 681 |
|                |                                |             |             |            |            |     |
| SEQ ID NO: 106 | moltype = AA length = 226      |             |             |            |            |     |
| FEATURE        | Location/Qualifiers            |             |             |            |            |     |
| REGION         | 1..226                         |             |             |            |            |     |
|                | note = synthetic construct     |             |             |            |            |     |
| source         | 1..226                         |             |             |            |            |     |
|                | mol_type = protein             |             |             |            |            |     |
|                | organism = synthetic construct |             |             |            |            |     |
| SEQUENCE: 106  |                                |             |             |            |            |     |
| MPIDRKTLCH     | LLSVLPLALL                     | GSHVARASTP  | GIVIPPQEQI  | TQHGGPYGRC | ANKTRALTVA | 60  |
| ELRGSGLDQE     | LYRHVTRGWS                     | IFALYDGTYL  | GGEYGGVIKD  | GTPGGAFDLK | TTFCIMTRTN | 120 |
| TGQPATDHYY     | SNVTATRLLS                     | STNSRLCAVF  | VRSGQPVIGA  | CTSPYDGKYW | SMYSRLRKML | 180 |
| YLIYVAGISV     | RHVHKEEQY                      | YDYEDATFET  | YALTGISICN  | PGSSLC     |            | 226 |
|                |                                |             |             |            |            |     |
| SEQ ID NO: 107 | moltype = DNA length = 684     |             |             |            |            |     |
| FEATURE        | Location/Qualifiers            |             |             |            |            |     |
| misc_feature   | 1..684                         |             |             |            |            |     |
|                | note = synthetic construct     |             |             |            |            |     |

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source          1..684
                mol_type = other DNA
                organism = synthetic construct

SEQUENCE: 107
atgctgatca acaacaagaa gctgcttcat cacattctgc ccatacctggt gctcgccctg 60
ctgggcatgc gcacggccca ggccgttgcg ccaggcatcg tcatcccgcc gaaggcactg 120
ttcaccacaac agggcgggcgc ctatggacgc tgcccgaacg gaacccgcgc cttgaccgtg 180
gccgaactgc gcggcaacgc cgaattgcag acgtatttgc gccagataac gcccggtctg 240
tccatatacg gtctctatga cggtagctac ctggggccagg cgtacggcgg catcatcaag 300
gacgcgcgcg caggcgcggg gttcatttat cgcgaaactt tctgcacac gaccatatac 360
aagaccgggc aaccggctgc ggatcactac tacagcaagg tcacggccac gcgcctgctc 420
gccagcacca acagcaggct gtgcgcggtt ttcgtcaggg acgggcaatc ggcatcgga 480
gcctgcgcca gcccgatatg aggcaggtag agagacatgt acgacgcgct gcggcgctg 540
ctgtacatga tctatatgtc cggccttgcc gtacgcgtcc acgtcagcaa ggaagagcag 600
tattacgact acgaggacgc cacattccag acctatgcc tcacccgcat tccctctgc 660
aaccggcgag cgtcgatatg ctga 684

SEQ ID NO: 108      moltype = AA length = 227
FEATURE            Location/Qualifiers
REGION              1..227
                    note = synthetic construct
source              1..227
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 108
MLNNKKLLH HILPILVLAL LGMRTAQAVA PGIVIPPKAL FTQQGGAYGR CPNGTRALTV 60
AELRGNALQ TYLRQITPGW SIYGLYDGTI LGQAYGGIIK DAPPGAGFIY RETFCITTIY 120
KTGQPAADHY YSKVTATRLI ASTNSRLCAV FVRDQGSVIG ACASPYEGRY RDMYDALRRL 180
LYMIYMSGLA VRVHVSKEEQ YYDYEDATFQ TYALTGISLC NPAASIC 227

SEQ ID NO: 109      moltype = DNA length = 459
FEATURE            Location/Qualifiers
misc_feature        1..459
                    note = synthetic construct
source              1..459
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 109
atgctgagac gcttccccac tcgaaccacc gccccgggac agggcgggcg cggcggtg 60
cgctgcgcgc cctggcggtg gttgctggca tccggcgcgga tgacgcattt tcccccgcc 120
ctggccgacg ttccttatgt ctgggtgaag accaatatgg tggtcaccag cgtagccatg 180
aagccgatg aagtcacccc gacgcgcacg ctgggtctgc gcacgcgcgc caaactgggc 240
gcgcggcgca gcaccccgga cgcgcacgtg ccgttctgct tcggcaagga tctcaagcgt 300
cccggcagca gtcctcatgga agtcacgttg cgcgcgcgtc tcatgcaaca acggcgctg 360
cgcatgttct tgggtcccaa gcaactcact ttcgaaggca agcccgcgct cgaactgatc 420
cggatggtcg aatgcagcgg caagcaggat tgccctga 459

SEQ ID NO: 110      moltype = AA length = 152
FEATURE            Location/Qualifiers
REGION              1..152
                    note = synthetic construct
source              1..152
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 110
MLRRFPTRTT APGQGGARRS RVRALAWLLA SGAMTHLSPA LADVPYVLVK TNMVVTSVAM 60
KPYEVTPTRM LVCGIAAKLG AAASSPDAHV PFCFGKDLKR PGSSPMEVML RAVFMQQRPL 120
RMFLGPKQLT FEGKPALELI RMVECSGKQD CP 152

SEQ ID NO: 111      moltype = DNA length = 402
FEATURE            Location/Qualifiers
misc_feature        1..402
                    note = synthetic construct
source              1..402
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 111
atgcagcggc aagcaggatt gccctgaag gcgaacccca tgcataccat cgcattccatc 60
ctgttgtccg tgctcgcatc atacagcccg gctgacgtcg ccggttgcc gaccatctg 120
tacaagaact tcaactgtcca ggagctggcc ttgaaactga agggcaagaa tcaggagttc 180
tgctgacccg ccttcattgtc gggcagaagc ctgggtccgg cgtgcctgtc cgacgcggga 240
cacgagcagc acacgtggtt cgacaccatg cttggctttg ccatatccgc gtatgcgctc 300
aagagccgga tcgcgctgac ggtggaagac tcgccgtatc cgggcactcc cggcgatctg 360
ctcgaactgc agatctgccc gctcaacgga tattcggaat ga 402

SEQ ID NO: 112      moltype = AA length = 133

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FEATURE                Location/Qualifiers
REGION                1..133
                        note = synthetic construct
source                1..133
                        mol_type = protein
                        organism = synthetic construct

SEQUENCE: 112
MQRQAGLPLK ANPMHTIASI LLSVLGIYSP ADVAGLPTHL YKNFTVQELA LKLKGNQEF 60
CLTAFMSGRS LVRACLS DAG HEHDTWFDTM LGFAISAYAL KSRIALTVED SPYPGTPGDL 120
LELQICPLNG YCE 133

SEQ ID NO: 113        moltype = DNA length = 868
FEATURE                Location/Qualifiers
misc_feature          1..868
                        note = synthetic construct
source                1..868
                        mol_type = other DNA
                        organism = synthetic construct

SEQUENCE: 113
atgattgaca tcatgttgca tatagggttag ataaaaacaag tggttatctt tccggattgt 60
cttcttgtat gatataataag ttttcctcga tgaaaaaatat aactttcatt ttttttattt 120
tattagcatc gccattatat gcaaatggcg acagattata cgtgctgac tctagacccc 180
cagatgaaat aaaacggttc cggagtctta tgcccagagg taatgagtag ttcgatagag 240
gaactcaaat gaatattaat ctttatgatc acgcgagagg aacacaaacc ggctttgtca 300
gatatgatga cggatatggt tccacttctc ttagtttgag aagtgtcac ttagcaggac 360
agtatatatt atcaggatat tcacttacta tatatatcgt tatagcaaat atgtttaatg 420
ttaatgatgt aattagcgta tacagccctc acccatatga acaggagggt tctgcgttag 480
gtggaatacc atattctcag atatatggat ggtatcgtgt taattttggt gtgattgatg 540
aacgattaca tcgtaacagg gaatatagag accggtatta cagaaatctg aatatagctc 600
cggcagagga tggttacaga ttagcagggt tcccaccgga tcaccaagct tggagagaag 660
aaccctggat tcatcatgca ccacaagggt gtggagattc atcaggaaca atcacagggt 720
atacttgaat tgaggagacc cagaatctga gcacaatata tgccagggaa tatcaatcaa 780
aagttaagag gcagataatt tcagactatc agtcagagggt tgacatatat aacagaattc 840
gggatgaatt atgaataaag taaaatgt 868

SEQ ID NO: 114        moltype = DNA length = 604
FEATURE                Location/Qualifiers
misc_feature          1..604
                        note = synthetic construct
source                1..604
                        mol_type = other DNA
                        organism = synthetic construct

SEQUENCE: 114
gttgacatat ataacagaat tcgggatgaa ttatgaataa agtaaaatgt tatgttttat 60
ttacggcggt actatcctct ctatatgcac acggagctcc ccagactatt acagaactat 120
gttcggaata tcgcaacaca caaatatata cgataaatga caagatacta tcatatacgg 180
aatcgatggc agggcaaaaga gaaatgggta tcattacatt taagagcggc gaaacatttc 240
aggtcgaagt cccgggcagt caacatatag actcccagaa aaaagccatt gaaaggatga 300
aggacacatt aagaatcaca tatctgaccg agaccaaaat tgataaatta tgtgtatgga 360
ataataaaac ccccaattca attgcggcaa tcagtatgaa aaactagtgt gctttaaaag 420
catgtctaata gctaggaacc tatataacaa ctactgtact tatactaagt agccttatgc 480
tgcatttgaa aaggcgggta aggaggcaat accgacacct aaactgtaac actataacag 540
cttcactac agggagctgt tatagcacac agaaaaaact aagctaggct ggaggggcaa 600
gctt 604

SEQ ID NO: 115        moltype = DNA length = 777
FEATURE                Location/Qualifiers
source                1..777
                        mol_type = genomic DNA
                        organism = Vibrio cholerae

SEQUENCE: 115
atggtaaaaga taatatttgt gttttttatt ttcttatcat cattttcata tgcaaatgat 60
gataagttat atcgggcaga ttctagacct cctgatgaaa taaagcagtc aggtggtcct 120
atgccaagag gacagagtga gtactttgac cgaggtactc aaatgaatat caacctttat 180
gatcatgcaa gaggaactca gacgggattt gttaggcacg atgatggata tgtttccacc 240
tcaattagtt tgagaagtgc ccacttagtg ggtcaaaacta tattgtctgg tcattctact 300
tattatatat atgttatagc cactgcaccc aacatgttta acgttaatga tgtattaggg 360
gcatacagtc ctcatacaga tgaacaagaa gtttctgctt taggtgggat tccatactcc 420
caaatatag gatggtatcg agttcatttt ggggtgcttg atgaacaatt acatcgtaat 480
aggggctaca gagatagata ttacagtaac ttagatatgt ctccagcagc agatggttat 540
ggattggcag gtttccctcc ggagcataga gcttggaggg aagagccgtg gattcatcat 600
gcaccgcccgt gttgtgggaa tgctccaaga tcatcgatca gtaatacttg cgatgaaaaa 660
acccaaagtc taggtgtaaa attccttgac gaataccaat ctaaagttaa aagacaaata 720
ttttcaggct atcaatctga tattgatata cataatagaa ttaaggatga attatga 777

SEQ ID NO: 116        moltype = AA length = 70

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FEATURE                      Location/Qualifiers
source                        1..70
                               mol_type = protein
                               organism = Vibrio cholerae

SEQUENCE: 116
HRAWREEPWI HHAPPGCGNA PRSSMSNTCD EKTQSLGVKF LDEYQSKVKR QIFSGYQSDI 60
DTHNRIKDEL                                     70

SEQ ID NO: 117                moltype = DNA length = 375
FEATURE                      Location/Qualifiers
source                        1..375
                               mol_type = genomic DNA
                               organism = Vibrio cholerae

SEQUENCE: 117
atgattaaat taaaatttgg tgtttttttt acagttttac tatcttcagc atatgcacat 60
ggaacacctc aaaatattac tgatttgtgt gcagaatacc acaacacaca aatatatacg 120
ctaaatgata agatattttc gtatacagaa tctctagctg gaaaaagaga gatggctatc 180
attactttta agaatggtgc aatttttcaa gtagaagtac caggtagtca acatatagat 240
tcacaaaaaa aagcgattga aaggatgaag gataccctga ggattgcata tcttactgaa 300
gctaaagtcg aaaagttatg tgtatggaat aataaaacgc ctcatgcgat tgcgcgaatt 360
agtatggcaa attaa                                     375

SEQ ID NO: 118                moltype = AA length = 124
FEATURE                      Location/Qualifiers
source                        1..124
                               mol_type = protein
                               organism = Vibrio cholerae

SEQUENCE: 118
MIKLKFGVFF TVLLSSAYAH GTPQNITDLC AEYHNTQIYT LNDKIFSYTE SLAGKREMAI 60
ITFKNGAIFQ VEVPGSQHID SQKKAIERMK DTLRIAYLTE AKVEKLCVWN NKTPHAIAAI 120
SMAN                                             124

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What is claimed is:

1. A fusion polypeptide comprising a fusion of a needle tip protein or an antigenic fragment thereof and a translocator protein or an antigenic fragment thereof from a Type III secretion system (T3SS) of a Gram negative bacteria; wherein the gram negative bacteria is not a *Salmonella enterica*, or *Shigella* spp.

2. The polypeptide of claim 1, wherein the fusion polypeptide is arranged so that the needle tip protein is 5' of the translocator protein.

3. The polypeptide of claim 1, wherein the gram negative bacteria comprises *Bordetella* spp., *Burkholderia* spp., *Chlamydia* spp., *Pseudomonas* spp., *Vibrio* spp., or *Yersinia* spp.

4. The polypeptide of claim 1, wherein the needle tip protein comprises Bsp22, LcrV, BipD, PcrV, CT053, or CT668.

5. The polypeptide of claim 1, wherein the translocator protein comprises BopB, YopB, BipB, PopB, CopB, or CopB2.

6. The polypeptide of claim 1, wherein the fusion comprises the *Bordetella* spp. needle-tip protein (Bsp22) and translocator protein (BopB), or antigenic fragments thereof.

7. The polypeptide of claim 1, wherein the fusion comprises the *Yersinia* spp. needle-tip protein (LcrV) and translocator protein (YopB), or antigenic fragments thereof.

8. The polypeptide of claim 1, wherein the fusion comprises the *Burkholderia* spp. needle-tip protein (BipD) and translocator protein (BipB), or antigenic fragments thereof.

9. The polypeptide of claim 1, wherein the fusion comprises the *Pseudomonas* spp. needle-tip protein (PcrV) and translocator protein (PopB), or antigenic fragments thereof.

10. The polypeptide of claim 1, wherein the fusion comprises the *Chlamydia* spp. needle-tip protein (CT053 or CT668) and translocator protein (CopB or CopB2), or antigenic fragments thereof.

11. The polypeptide of claim 1, wherein the fusion further comprises double mutant labile toxin (dmLT) or an antigenic fragment thereof from Enterotoxigenic *Escherichia coli* or cholera toxin or an antigenic fragment thereof.

12. The polypeptide of claim 1, wherein the dmLT comprises the active moiety LTA1.

13. The polypeptide of claim 1, wherein the dmLT retains its ADP ribosylation activity.

14. The polypeptide of claim 1, wherein the dmLT is 5' of the needle tip protein and translocator protein fusion.

15. A vaccine comprising one or more of the fusion polypeptides of any of claims 1-14.

16. The vaccine of claim 15, further comprising one or more components of an acellular pertussis vaccine.

17. The vaccine of claim 15, further comprising pertussis toxoid (PTd).

18. A method of treating, inhibiting, or preventing an infection of a Gram negative bacteria in a subject comprising administering to the subject the fusion polypeptide of any of claims 1-14 or vaccine of claims 15-17.

19. The method of claim 18, wherein the method further inhibits or prevents colony formation of the bacteria and/or transmission of the bacteria to another subject.

20. A method of eliciting an immune response in a subject to a Gram negative bacteria comprising administering to the subject the fusion polypeptide of any of claims 1-14 or vaccine of claims 15-17.

21. The method of claim 20, wherein the immune response comprises a sterilizing immune response.

22. The method of claims 18-21, wherein the bacteria comprises *Bordetella* spp., *Burkholderia* spp., *Chlamydia* spp., *Pseudomonas* spp., *Vibrio* spp., or *Yersinia* spp.

23. A method of eliciting an immune response against at least one Gram negative bacteria serovar in a subject in need thereof, comprising administering to the subject a compo-

sition comprising at least one needle tip protein or an antigenic fragment thereof and/or at least one translocator protein or an antigenic fragment thereof; wherein said composition is administered in an amount sufficient to elicit an immune response to said at least one Gram negative bacteria serovar in said subject; and wherein the Gram negative bacteria is not a *Shigella* spp. or *Salmonella enterica*.

**24.** The method of claim **23**, wherein the bacteria comprises *Bordetella* spp., *Burkholderia* spp., *Chlamydia* spp., *Pseudomonas* spp., *Vibrio* spp., or *Yersinia* spp.

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